



Oracle Linux 8: System Administration I

Activity Guide
D1102121GC10

Author

Darryl Balaski

**Technical Contributor
and Reviewer**

Matt Slingsby

Graphic Designer

Lakhan Mandar

Editor

Raj Kumar

Publishers

Syed Imtiaz Ali

Sujatha Nagendra

Pavithran Adka

Veena Narasimhan

1011152022

Copyright © 2022, Oracle and/or its affiliates.

Disclaimer

This document contains proprietary information and is protected by copyright and other intellectual property laws. The document may not be modified or altered in any way. Except where your use constitutes "fair use" under copyright law, you may not use, share, download, upload, copy, print, display, perform, reproduce, publish, license, post, transmit, or distribute this document in whole or in part without the express authorization of Oracle.

The information contained in this document is subject to change without notice and is not warranted to be error-free. If you find any errors, please report them to us in writing.

Restricted Rights Notice

If this documentation is delivered to the United States Government or anyone using the documentation on behalf of the United States Government, the following notice is applicable:

U.S. GOVERNMENT END USERS: Oracle programs (including any operating system, integrated software, any programs embedded, installed or activated on delivered hardware, and modifications of such programs) and Oracle computer documentation or other Oracle data delivered to or accessed by U.S. Government end users are "commercial computer software" or "commercial computer software documentation" pursuant to the applicable Federal Acquisition Regulation and agency-specific supplemental regulations. As such, the use, reproduction, duplication, release, display, disclosure, modification, preparation of derivative works, and/or adaptation of i) Oracle programs (including any operating system, integrated software, any programs embedded, installed or activated on delivered hardware, and modifications of such programs), ii) Oracle computer documentation and/or iii) other Oracle data, is subject to the rights and limitations specified in the license contained in the applicable contract. The terms governing the U.S. Government's use of Oracle cloud services are defined by the applicable contract for such services. No other rights are granted to the U.S. Government.

Trademark Notice

Oracle and Java are registered trademarks of Oracle and/or its affiliates. Other names may be trademarks of their respective owners.

Intel and Intel Inside are trademarks or registered trademarks of Intel Corporation. All SPARC trademarks are used under license and are trademarks or registered trademarks of SPARC International, Inc. AMD, Epyc, and the AMD logo are trademarks or registered trademarks of Advanced Micro Devices. UNIX is a registered trademark of The Open Group.

Third-Party Content, Products, and Services Disclaimer

This documentation may provide access to or information about content, products, and services from third parties. Oracle Corporation and its affiliates are not responsible for and expressly disclaim all warranties of any kind with respect to third-party content, products, and services unless otherwise set forth in an applicable agreement between you and Oracle. Oracle Corporation and its affiliates will not be responsible for any loss, costs, or damages incurred due to your access to or use of third-party content, products, or services, except as set forth in an applicable agreement between you and Oracle.

Table of Contents

Practices for Lesson 1: Course Introduction	7
Course Practice Environment: Security Credentials.....	8
Practices for Lesson 1: Overview	9
Practice 1-1: Exploring the Gateway Machine.....	10
Practice 1-2: Exploring the Host02 VM.....	14
Practice 1-3: Exploring the Host03 VM.....	17
Practice 1-4: Logging Off from Your Gateway Machine	19
Practices for Lesson 2: Introduction to Oracle Linux.....	21
Practices for Lesson 2: Overview	22
Practice 2-1: Quiz – Introduction to Oracle Linux	23
Solution 2-1: Quiz – Introduction to Oracle Linux	25
Practice 2-2: Viewing Kernel Information.....	26
Practices for Lesson 3: Installing Oracle Linux 8	29
Practices for Lesson 3: Overview	30
Practices for Lesson 4: OpenSSH	31
Practices for Lesson 4: Overview	32
Practice 4-1: Connecting to a Remote Server by Using <code>ssh</code>	33
Practice 4-2: Configuring OpenSSH to Connect Without a Password	37
Practices for Lesson 5: The GRUB Bootloader	41
Practices for Lesson 5: Overview	42
Practice 5-1: Exploring the GRUB 2 Bootloader.....	43
Practice 5-2: Booting Different Kernels.....	47
Practice 5-3: Rebuilding the <code>initramfs</code>	50
Practices for Lesson 6: Introduction to <code>systemd</code>	53
Practices for Lesson 6: Overview	54
Practice 6-1: Exploring <code>systemd</code> Units	55
Practice 6-2: Working with <code>systemd</code> Service Units.....	61
Practices for Lesson 7: Managing Time.....	69
Practices for Lesson 7: Overview	70
Practice 7-1: Configuring Date and Time.....	71
Practice 7-2: Configuring Chrony.....	74
Practice 7-3: Automating Tasks with <code>cron</code>	80
Practice 7-4: Automating Tasks Using <code>systemd</code> Timers	83

Practices for Lesson 8: Kernel and Device Configuration	89
Practices for Lesson 8: Overview	90
Practice 8-1: Using Loadable Kernel Modules.....	91
Practice 8-2: Exploring the <code>/proc</code> File System.....	95
Practice 8-3: Exploring the <code>sysfs</code> File System	99
Practice 8-4: Using the <code>sysctl</code> Utility.....	101
Practice 8-5: Exploring and Configuring <code>udev</code> Rules	104
Practices for Lesson 9: Package Management	109
Practices for Lesson 9: Overview	110
Practice 9-1: Using the <code>rpm</code> Utility	111
Practice 9-2: Accessing the Oracle Linux Yum Server	115
Practice 9-3: Creating a Local Yum Repository.....	119
Practice 9-4: Using the <code>dnf</code> Utility	124
Practice 9-5: Using <code>dnf</code> Modules	131
Practices for Lesson 10: Oracle Ksplice	137
Practices for Lesson 10: Overview	138
Practice 10-1: Installing the Ksplice Offline Client and Kernel Updates	139
Practice 10-2: Demonstration of Ksplice Online Updates.....	144
Practices for Lesson 11: User and Group Administration	145
Practices for Lesson 11: Overview	146
Practice 11-1: Administering User Accounts	147
Practice 11-2: Administering Group Accounts	152
Practice 11-3: Implementing User Private Groups	153
Practice 11-4: Configuring Password Aging	156
Practice 11-5: Restricting the Use of the <code>su</code> Command	158
Practice 11-6: Allowing the Use of the <code>sudo</code> Command	160
Practices for Lesson 12 Partitions, File Systems, and Swap	165
Practices for Lesson 12: Overview	166
Practice 12-1: Listing the Current Disk Partitions	167
Practice 12-2: Partitioning a Storage Device	169
Practice 12-3: Creating <code>ext4</code> File Systems.....	174
Practice 12-4: Increasing Swap Space	179
Practice 12-5: Removing Partitions and Additional Swap Space	181
Practices for Lesson 13: Network Configuration	185
Practices for Lesson 13: Overview	186
Practice 13-1: Configuring the <code>eth1</code> Network Interface	187
Practice 13-2: Using the <code>nmcli</code> Utility.....	192

Practice 13-3: Using the <code>nmtui</code> Utility	203
Practice 13-4: Using the <code>ip</code> Utility	206
Practices for Lesson 14: Network Security	213
Practices for Lesson 14: Overview	214
Practice 14-1: Exploring <code>firewalld</code>	215
Practice 14-2: Configuring <code>firewalld</code>	221
Practice 14-3: Configuring <code>nftables</code>	226
Practices for Lesson 15: Oracle on Oracle	233
Practices for Lesson 15: Overview	234
Practice 15-1: Installing and Running Oracle Database Preinstall	235
Practice 15-2: Preparing Disks for ASM Use	243
Practice 15-3 Installing and Configuring ASMLib	245
Practice 15-4 Reverting Changes Made to <code>host02</code>	249
Practices for Lesson 16: System Monitoring and Management	253
Practices for Lesson 16: Overview	254
Practice 16-1: Using <code>sosreport</code> to Collect System Information	255
Practice 16-2: Using Standard Linux Performance Monitoring Tools	259
Practice 16-3: Installing and Using OSWatcher	270
Practice 16-4: Using OSWatcher Analyzer	277
Practices for Lesson 17: System Logging	285
Practices for Lesson 17: Overview	286
Practice 17-1: Configuring System Logging	287
Practice 17-2: Using <code>rsyslog</code> Templates	292
Practice 17-3: Using <code>logwatch</code>	294
Practice 17-4: Using <code>systemd-journald</code>	297
Practices for Lesson 18: Troubleshooting	303
Practices for Lesson 18: Overview	304
Practice 18-1: Transferring Utilities from <code>gateway</code>	305
Practice 18-2: System Utility Commands Fail	306
Solution 18-2: System Utility Commands Fail	308
Practice 18-3: A <code>cron</code> Job Fails to Run	309
Solution 18-3: A <code>cron</code> Job Fails to Run	311
Practice 18-4: A User Cannot Log In	313
Solution 18-4: A User Cannot Log In	316
Practice 18-5: A File System Does Not Mount	318
Solution 18-5: A File System Does Not Mount	322
Practice 18-6: Network Connectivity Problem	326
Solution 18-6: Network Connectivity Problem	328

Practice 18-7: Remote Access Problem	331
Solution 18-7: Remote Access Problem	334
Practice 18-8: Log File Is Not Getting Updated	337
Solution 18-8: Log File Is Not Getting Updated	340
Appendix A: Source Code for Problem-Causing Executables.....	342

Practices for Lesson 1:
Course Introduction

Course Practice Environment: Security Credentials

For OS usernames and passwords, see the following:

- If you are attending a classroom-based or live virtual class, ask your instructor or LVC producer for OS credential information.
- If you are using a self-study format, refer to the communication that you received from Oracle University for this course.

For product-specific credentials used in this course, see the following table:

Product-Specific Credentials		
Virtual Machines/Application	Username	Password
host02/OS	root	oracle
host02/OS	oracle	oracle
host03/OS	root	oracle
host03/OS	oracle	oracle

Practices for Lesson 1: Overview

Practices Overview

In these practices, you:

- Log in to your classroom **gateway** system and become familiar with the configuration.
- Connect to the virtual machines used for the hands-on practices and become familiar with the VM guest configurations.

Assumptions

- A **gateway** system has been assigned to you
- The GNOME desktop is installed on **gateway**.
- Two additional guests (virtual machines) are created: **host02** and **host03**
- All the guest VMs are running Oracle Linux.

Practice 1-1: Exploring the Gateway Machine

Overview

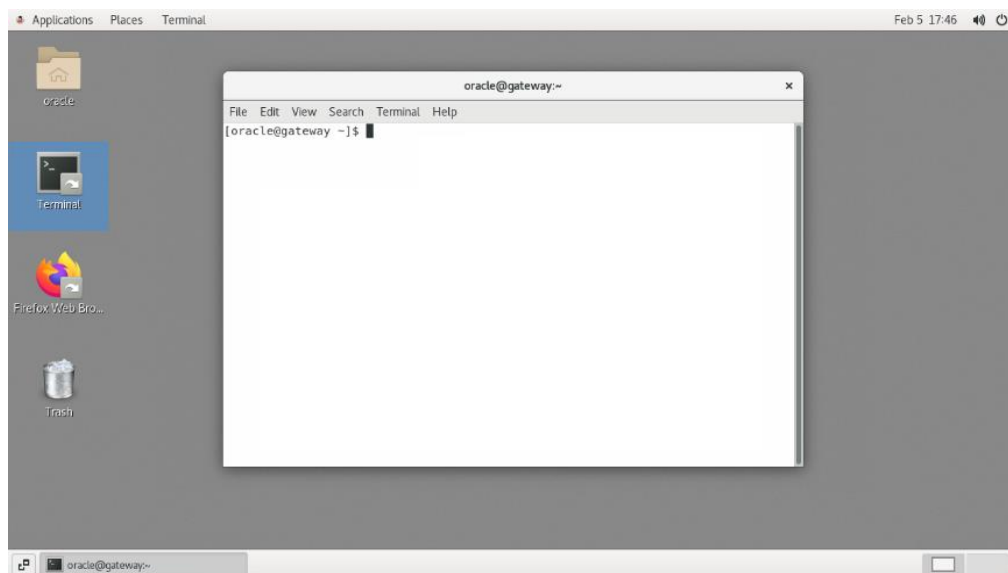
In this practice, you explore the **gateway** configuration and directory structure.

Assumptions

- You have been supplied with connection details for your **gateway** system.
- You are logged on to your **gateway** as `oracle` with the appropriate password.
- There are two additional guests (virtual machines): **host02** and **host03**.

Tasks

1. Open a terminal window.
Double-click the **Terminal** icon on the desktop.
 - A terminal window appears as shown:



2. Determine the operating system that is running on **gateway**.
 - a. Use the `cat` command to view the contents of the `/etc/ovs-release` file.

```
[oracle@gateway ~]$ cat /etc/oracle-release
Oracle Linux Server release 8.3
```

- Note: For subsequent steps, this guide will just use `$` for the prompt unless there is a change of host or user.

- b. Use the `uname -a` command to display the operating system version.

```
$ uname -a
Linux gateway.example.com 5.4.17-2036.102.0.2.el8uek.x86_64 #2
SMP Tue Jan 5 12:46:40 PST 2021 x86_64 x86_64 x86_64 GNU/Linux
```

- The operating system is GNU/Linux, specifically Oracle Linux 8.3.
- The Linux kernel is 5.4.17-2036.102.0.2.el8uek.x86_64.
- The host name is gateway.example.com.

3. View the network configuration of **gateway**.

- a. Use the `ip addr` command to display the network configuration.

- Some details may be different in your practice environment.

```
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state
UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq
state UP group default qlen 1000
    link/ether aa:bb:00:01:82:81 brd ff:ff:ff:ff:ff:ff
    inet 10.237.17.50/23 brd 10.237.17.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::a8bb:ff:fe01:8281/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq
state UP group default qlen 1000
    link/ether 00:21:f6:dc:01:01 brd ff:ff:ff:ff:ff:ff
    inet 192.0.2.1/24 brd 192.0.2.255 scope global eth1
        valid_lft forever preferred_lft forever
    inet6 fe80::221:f6ff:fedc:101/64 scope link
        valid_lft forever preferred_lft forever
4: eth2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq
state UP group default qlen 1000
    link/ether 00:21:f6:dc:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.1/24 brd 192.168.1.255 scope global eth2
        valid_lft forever preferred_lft forever
    inet6 fe80::221:f6ff:fedc:102/64 scope link
        valid_lft forever preferred_lft forever
```

- The address shown for eth0 is different for your system.
- The eth1 interface has an IP address of 192.0.2.1.
- The eth2 interface has an IP address of 192.168.1.1.

- b. Verify the `/etc/hosts` file.

```
$ cat /etc/hosts
127.0.0.1    localhost localhost.localdomain localhost4
localhost4.localhostdomain4
::1         localhost localhost.localdomain localhost6
localhost6.localhostdomain6

192.0.2.1    gateway.example.com    gateway
192.0.2.102  host02.example.com      host02
192.0.2.103  host03.example.com      host03

10.237.17.50 eg18281
```

- The entry with IP address 10.237.17.50 and host name eg18281 is different for each practice environment.

- c. Verify the `/etc/resolv.conf` file.

```
$ cat /etc/resolv.conf
search edu.oracle.com
nameserver 10.237.18.52
nameserver 152.68.154.3
```

- This file identifies DNS servers in the Oracle University network that can resolve external host names and IP addresses.

4. Explore the staged files and directories.

- a. View the contents of the `/stage` directory.

- There are four entries (three directories and one file) in the `/stage` directory.
- These files are used in various practices in this course.

```
$ ls -l /stage
total 5144
drwxr-xr-x. 3 root root    4096 Feb  7 14:58 extras
-rw-r--r--. 1 root root 5253120 Feb  7 14:58 oswbb840.tar
drwxr-xr-x. 2 root root    4096 Feb  9 11:22 ts_scripts
drwxr-xr-x. 2 root root    4096 Feb  9 11:22 ts_source
```

- b. View the contents of the `/stage/extras/Packages` directory.

These files will form a local Yum repository of packages, which will be installed in various practices in this course.

```
$ ls /stage/extras/Packages
kssplice-1.0.48-1.el8.x86_64.rpm
kssplice-core0-1.0.48-1.el8.x86_64.rpm
kssplice-known-exploit-detection-1.0.48-1.el8.x86_64.rpm
kssplice-offline-1.0.48-1.el8.x86_64.rpm
kssplice-tools-1.0.48-1.el8.x86_64.rpm
oracleasm-lib-2.0.17-1.el8.x86_64.rpm
oracleasm-support-2.1.12-1.el8.x86_64.rpm
oracle-database-preinstall-19c-1.0-1.el8.x86_64.rpm
oswatcher-8.3.0-5.el8.noarch.rpm
uptrack-1.2.71-0.el8.noarch.rpm
uptrack-offline-1.2.71.offline-0.el8.noarch.rpm
uptrack-updates-5.4.17-2011.7.4.el8uek.x86_64-20201217-0.noarch.rpm
```

5. View the status of a service that is running.

Show the DNS service status.

```
$ systemctl status dnsmasq
• dnsmasq.service - DNS caching server.
  Loaded: loaded (/usr/lib/systemd/system/dnsmasq.service;
  enabled; vendor preset: disabled)
  Active: active (running) since...
  ...
```

- Note this service is used to support some of the practices.

Practice 1-2: Exploring the Host02 VM

Overview

In this practice, you:

- Log on to **host02**
- View the Oracle Linux version on **host02**
- View the Unbreakable Enterprise Kernel version on **host02**
- View the storage devices available on **host02**
- View the network configuration on **host02**

Assumptions

- You are logged on to **gateway** as the `oracle` user.
- The **host02** VM guest is running.

Tasks

1. Explore the **host02** VM guest.

- a. Use the `ssh` command to log on to **host02**.

Because this is the first time you have logged on using `ssh`, the command checks to make sure you are connecting to the host you think you are connecting to. Enter **yes**. Provide the `root` password.

```
[oracle@gateway ~]$ ssh root@host02
The authenticity of host 'host02 (192.0.2.102)' can't be
established.
ECDSA key fingerprint is ...
Are you sure you want to continue connecting
(yes/no/[fingerprint])? yes
Warning: Permanently added 'host02,192.0.2.102' (ECDSA) to the
list of known hosts.
root@host02's password:
[root@host02 ~]# hostname
host02.example.com
```

- The `hostname` command confirms whether you have successfully logged on to **host02**.
- b. Use the `cat` command to view the contents of the `/etc/oracle-release` file.

```
# cat /etc/oracle-release
Oracle Linux Server release 8.3
```

- c. Use the `uname -r` command to determine your running kernel version.

```
# uname -r
5.4.17-2011.7.4.el8uek.x86_64
```


- d. Use the `fdisk` command to view the storage devices.

```
# fdisk -l | grep /dev
Disk /dev/xvda: 30.5 GiB, 32768000000 bytes, 64000000 sectors
/dev/xvda1 *          2048  2099199  2097152    1G 83 Linux
/dev/xvda2          2099200 52430847 50331648   24G 83 Linux
/dev/xvda3          52430848 56625151  4194304    2G 83 Linux
/dev/xvda4          56625152 63999999  7374848   3.5G  5 Extended
/dev/xvda5          56627200 62918655  6291456    3G 82 Linux swap /
Solaris
Disk /dev/xvdb: 5 GiB, 5299999744 bytes, 10351562 sectors
Disk /dev/xvdd: 5 GiB, 5299999744 bytes, 10351562 sectors
```

- Three devices are available: `/dev/xvda`, `/dev/xvdb`, and `/dev/xvdd`.
- The `/dev/xvda` device has Oracle Linux *Minimal Install* packages installed.

- e. Use the `ip addr` command to display the network interfaces.

Only a partial output is shown.

```
# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue ...
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet addr:127.0.0.1/8 scope host lo
    ...
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:01:01:01 brd ff:ff:ff:ff:ff:ff
    inet 192.0.2.102/24 brd 192.0.2.255 scope ... eth0
    ...
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:01:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.102/24 brd 192.168.1.255 scope ... eth1
    ...
```

- The system has two Ethernet network interfaces, `eth0` and `eth1`.
- The `eth0` interface is on the `192.0.2.0/24` subnet, has an IPv4 address of `192.0.2.102`, and provides access to **gateway** and the other VM guest systems.
- The `eth1` interface is on the `192.168.1.0/24` subnet, has an IPv4 address of `192.168.1.102`.

- f. View the `/etc/hosts` file on **host02**.

```
# cat /etc/hosts
127.0.0.1      localhost localhost.localdomain localhost4 ...
::1           localhost localhost.localdomain localhost6 ...
```

- This file contains default values.

2. Use the `exit` command to log off **host02**.

```
# exit
logout
Connection to host02 closed.
```

You are now the `oracle` user on **gateway**.

Practice 1-3: Exploring the Host03 VM

Overview

In this practice, you:

- Log on to **host03**
- View the Oracle Linux version on **host03**
- View the Unbreakable Enterprise Kernel version on **host03**
- View the storage devices available on **host03**
- View the network configuration on **host03**

Assumptions

- You are logged on to **gateway** as the `oracle` user.
- The **host03** VM guest is running.

Tasks

1. Explore the **host03** VM guest.

a. Use the `ssh` command to log on to **host03**.

Because this is the first time you have logged on using `ssh`, the command checks to make sure you are connecting to the host you think you are connecting to. Enter **yes**. Provide the `root` password.

```
[oracle@gateway ~]$ ssh root@host03
The authenticity of host 'host03 (192.0.2.103)' can't be
established.
ECDSA key fingerprint is ...
Are you sure you want to continue connecting
(yes/no/[fingerprint])? yes
Warning: Permanently added 'host03,192.0.2.103' (ECDSA) to the
list of known hosts.
root@host03's password:
[root@host03 ~]# hostname
host03.example.com
```

- The `hostname` command confirms whether you have successfully logged on to **host03**.
- b. Use the `cat` command to view the contents of the `/etc/oracle-release` file.

```
# cat /etc/oracle-release
Oracle Linux Server release 8.3
```

c. Use the `uname -r` command to determine your running kernel version.

```
# uname -r
5.4.17-2011.7.4.el8uek.x86_64
```

- d. Use the `fdisk` command to view the storage devices.

```
# fdisk -l | grep /dev
Disk /dev/xvda: 30.5 GiB, 32768000000 bytes, 64000000 sectors
/dev/xvda1 *          2048  2099199  2097152    1G 83 Linux
/dev/xvda2           2099200 52430847 50331648   24G 83 Linux
/dev/xvda3           52430848 56625151  4194304    2G 83 Linux
/dev/xvda4           56625152 63999999  7374848   3.5G  5 Extended
/dev/xvda5           56627200 62918655  6291456    3G 82 Linux swap /
Solaris
Disk /dev/xvdb: 5 GiB, 5299999744 bytes, 10351562 sectors
Disk /dev/xvdd: 5 GiB, 5299999744 bytes, 10351562 sectors
```

- Three devices are available: `/dev/xvda`, `/dev/xvdb`, and `/dev/xvdd`.
 - The `/dev/xvda` device has Oracle Linux Minimal Install packages installed, in addition to several other necessary packages.
- e. Use the `ip addr` command to display the network interfaces.

Only a partial output is shown.

```
# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue ...
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet addr:127.0.0.1/8 scope host lo
    ...
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:01:01 brd ff:ff:ff:ff:ff:ff
    inet 192.0.2.103/24 brd 192.0.2.255 scope ... eth0
    ...
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:01:02 brd ff:ff:ff:ff:ff:ff
```

- The system has two Ethernet network interfaces, `eth0` and `eth1`.
 - The `eth0` interface is on the `192.0.2.0/24` subnet, has an IPv4 address of `192.0.2.103`, and provides access to **gateway** and the other VM guest systems.
 - The `eth1` interface does not have an IP address assigned.
- f. View the `/etc/hosts` file on **host03**.

```
# cat /etc/hosts
127.0.0.1      localhost localhost.localdomain localhost4 ...
::1           localhost localhost.localdomain localhost6 ...
```

- This file contains default values.
2. Use the `exit` command to log off **host03**.

```
# exit
logout
Connection to host03 closed.
```

You are now the `oracle` user on **gateway**.

Practice 1-4: Logging Off from Your Gateway Machine

Overview

In this practice, you learn how to log off from your system.

Tasks

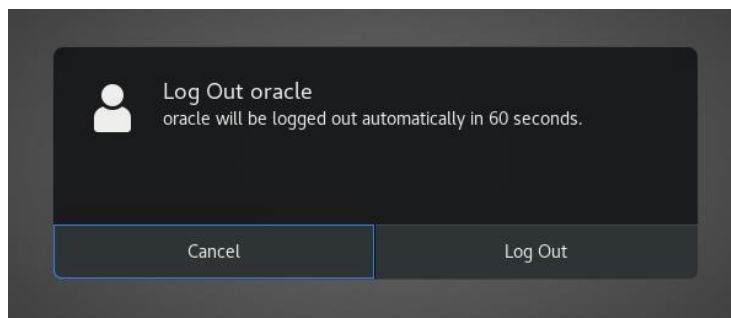
1. Learn how to log off from your student PC.
 - a. Click either of the icons in the top-right corner of the desktop.
 - b. Select **oracle** from the menu to open the user submenu.



- c. Select **Log Out** from the submenu.

You can click the **Log Out** button to log out.

 - However, do not log out until the end of each day of training.
 - d. Click the **Cancel** button to stay logged on.



Practices for Lesson 2:
Introduction to Oracle Linux

Practices for Lesson 2: Overview

Practices Overview

In these practices, you test your knowledge of Linux and of Oracle's contributions and commitment to Linux. You also view the different kernel packages that are installed with Oracle Linux.

Practice 2-1: Quiz – Introduction to Oracle Linux

Overview

In this quiz, you answer questions about the history of Linux in general, as well as Oracle's commitment and contributions to Linux.

Questions

Choose the best answer from those provided for each multiple choice or True/False question.

1. Which of the following statements are true? (Choose all that apply)
 - a. Linux is a UNIX-like operating system named after its developer, Linus Torvalds.
 - b. Richard Stallman released version 0.01 of the Linux kernel in September 1991.
 - c. The goal of the GNU Project to create a free, UNIX-like operating system was realized in 1992 with the addition of the Hurd kernel.
 - d. Distributors of GPL software must provide the binaries at no cost, but they can charge for the source code.
2. Thousands of programmers all over the world participate in the development of the Linux operating system.
 - a. True
 - b. False
3. Linus Torvalds releases a new version of the mainline kernel approximately:
 - a. Every month
 - b. Every three months
 - c. Every six months
 - d. Every year
4. The revision control system for the Linux kernel source code is called:
 - a. SCCS
 - b. RCS
 - c. BitKeeper
 - d. Git
5. A Linux distribution is a collection of software built on top of the Linux kernel and offered as a complete package.
 - a. True
 - b. False
6. Which of the following statements are true? (Choose all that apply)
 - a. All Oracle Linux code is available to the Linux community.
 - b. Oracle Ksplice allows you to apply kernel and select user space library updates on a running system.
 - c. Oracle has a dedicated Linux kernel development team.
 - d. Oracle has made many technical contributions to the Linux community.

7. Oracle Linux is fully compatible with which of the following Linux distributions?
 - a. Debian
 - b. Ubuntu
 - c. SUSE
 - d. RHEL
8. The Unbreakable Enterprise Kernel is optimized for Oracle applications and is the only kernel offered with Oracle Linux.
 - a. True
 - b. False

Solution 2-1: Quiz – Introduction to Oracle Linux

Solutions

1. Which of the following statements are true? (Choose all that apply)
 - a. Linux is a UNIX-like operating system named after its developer, Linus Torvalds.
2. Thousands of programmers all over the world participate in the development of the Linux operating system.
 - a. True
3. Linus Torvalds releases a new version of the mainline kernel approximately:
 - b. Every three months
4. The revision control system for the Linux kernel source code is called:
 - d. Git
5. A Linux distribution is a collection of software built on top of the Linux kernel and offered as a complete package.
 - a. True
6. Which of the following statements are true? (Choose all that apply)
 - a. True
 - b. True
 - c. True
 - d. True
7. Oracle Linux is fully compatible with which of the following Linux distributions?
 - d. RHEL
8. The Unbreakable Enterprise Kernel is optimized for Oracle applications and is the only kernel offered with Oracle Linux.
 - b. False. It is true that the Unbreakable Enterprise Kernel is optimized for Oracle applications, but Oracle Linux also includes a Red Hat-compatible kernel for customers requiring strict RHEL compatibility.

Practice 2-2: Viewing Kernel Information

Overview

In this practice, you connect to the **host02** VM, view the installed kernel packages, and determine which kernel is running. You also view the text file used by Oracle support teams to easily identify the active kernel. Note that the displayed sample output might not represent your own system.

Assumptions

- You are logged on to **gateway** as the `oracle` user.

Tasks

1. Connect to **host02** as the `root` user by using `ssh`.
 - a. Use the `ssh` command to connect to **host02** as `root`. Confirm that you have connected to **host02** by entering the `hostname` command:

```
$ ssh root@host02
root@host02's password:
# hostname
host02.example.com
```

2. List the kernel release that is currently running on your system.
 - a. Use the `uname -r` command to print the kernel release.

```
# uname -r
5.4.17-2011.7.4.el8uek.x86_64
```

- Note that the Unbreakable Enterprise Kernel is running by default.

3. List the kernel packages that are installed on your system.
 - a. Use the `rpm -qa` command to query the kernel packages installed on your system.

```
# rpm -qa | grep "^kernel"
kernel-uek-5.4.17-2011.7.4.el8uek.x86_64
kernel-core-4.18.0-240.el8.x86_64
kernel-modules-4.18.0-240.el8.x86_64
kernel-tools-4.18.0-240.el8.x86_64
kernel-4.18.0-240.el8.x86_64
kernel-tools-libs-4.18.0-240.el8.x86_64
```

- Note that the following two `kernel` packages are installed (associated `core`, `modules`, and `tools` packages are also installed):
 - Unbreakable Enterprise Kernel (`kernel-uek-5.4.17-2011.7.4.el8uek.x86_64`) package
 - Red Hat-compatible kernel (`kernel-4.18.0-240.el8.x86_64`) package

4. View the `/etc/*release` files on your system.
 - a. Change to the `/etc/` directory and view the “release” files.

```
# cd /etc
# ls -l *release
-rw-r--r-- ... oracle-release
-rw-r--r-- ... os-release
-rw-r--r-- ... redhat-release
lrwxrwxrwx ... system-release -> oracle-release
# cat oracle-release
Oracle Linux Server release 8.3
# cat redhat-release
Red Hat Enterprise Linux release 8.3 (Ootpa)
```

- Note that `/etc/system-release` is a symbolic link to `/etc/oracle-release`.

5. Log out from **host02**.

```
# logout
Connection to host02 closed.
```

You are now the `oracle` user on **gateway**.

Practices for Lesson 3: Installing Oracle Linux 8

Practices for Lesson 3: Overview

Practices Overview

There are no hands-on practices for this lesson.

Practices for Lesson 4:

OpenSSH

Practices for Lesson 4: Overview

Practices Overview

In these practices, you do the following:

- You verify that the OpenSSH packages are installed and that the `sshd` service is running.
- You use the `ssh` and `scp` utilities.
- You use the `ssh-keygen` utility to generate keys enabling connectivity without supplying a password.

Assumptions

The VMs used are configured for DNS or have host entries in `/etc/hosts`.

Practice 4-1: Connecting to a Remote Server by Using `ssh`

Overview

In this practice, you verify that the OpenSSH packages are installed, verify that the `sshd` service is started on the server, and use the `ssh` utility to establish a connection and execute a command on a remote system.

Assumptions

- You are logged in to **gateway** as the `oracle` user.
- This practice is performed on the **host02** and **host03** VMs.
- The prompts include either **host02** or **host03** to indicate which system to enter the command from.

Tasks

1. Log in to **host03**.

Use the `ssh` command to log in to **host03** as the `root` user.

- Provide the `root` password when prompted.

```
[oracle@gateway ~]$ ssh root@host03
root@host03's password:
Last login: ...
```

2. Verify that the `sshd` service is running on **host03**.

- a. Use the `rpm` command to verify that the `openssh` packages are installed.

```
[root@host03 ~]# rpm -qa | grep openssh
openssh-8.0p1-5.el8.x86_64
openssh-clients-8.0p1-5.el8.x86_64
openssh-server-8.0p1-5.el8.x86_64
```

- In this example, the packages are already installed.

- b. Use the `systemctl` command to verify that the `sshd` service is started.

```
[root@host03 ~]# systemctl status sshd
• sshd.service - OpenSSH server daemon
  Loaded: loaded (/usr/lib/systemd/system/ssh.service; enab...
  Active: active (running) since ...
  ...
```

- If necessary, press the Q key to return to the command prompt.
- In this example, the service is enabled and running.

3. Use the `logout` command to log off from **host03**.

```
[oracle@host03 ~]$ logout
Connection to host03 closed.
[oracle@gateway ~]$
```

4. Log in to **host02** and verify that the `ssh` client software is available.
 - a. From the terminal window on **gateway**, use the `ssh` command to log in to **host02** as the `root` user.

Provide the `root` password when prompted.

```
[oracle@gateway ~]$ ssh root@host02
root@host02's password:
Last login: ...
```

- b. On **host02**, use the `rpm` command to verify that the `openssh` packages are installed.

```
[root@host02 ~]# rpm -qa | grep openssh
openssh-8.0p1-5.el8.x86_64
openssh-clients-8.0p1-5.el8.x86_64
openssh-server-8.0p1-5.el8.x86_64
```

- In this example, the packages are already installed.

- c. Use the `exit` command to log off as `root` on **host02**.

```
[root@host02 ~]# exit
logout
Connection to host02 closed.
```

5. Log in to remote **host03** from **host02**.

- a. From **gateway**, use the `ssh` command to log back in to **host02** as user `oracle`.

- Provide the password when prompted.

```
[oracle@gateway ~]$ ssh oracle@host02
oracle@host02's password:
Last login:
[oracle@host02 ~]$
```

- b. Use the `ls` command to display a long listing of all files in the home directory of user `oracle`.

```
[oracle@host02 ~]$ ls -la
total 12
drwx-----. 2 oracle oracle 62 Feb 3 13:51 .
drwxr-xr-x. 3 root root 20 Feb 3 13:51 ..
-rw-r--r--. 1 oracle oracle 18 Aug 3 2020 .bash_logout
-rw-r--r--. 1 oracle oracle 141 Aug 3 2020 .bash_profile
-rw-r--r--. 1 oracle oracle 376 Aug 3 2020 .bashrc
```

- Notice that there is no `~/ .ssh` directory.

- c. Perform a remote login to **host03** by using the `ssh` command.

Answer “**yes**” when asked, “Are you sure ...”

Provide the password when prompted.

```
[oracle@host02 ~]$ ssh host03
The authenticity of host 'host03 (192.0.2.103)' can't be
established.
ECDSA key fingerprint is SHA256:...
Are you sure you want to continue connecting
(yes/no/[fingerprint])? yes
Warning: Permanently added 'host03,192.0.2.103' (ECDSA) to the
list of known hosts.
oracle@host03's password:
Last login: ...
[oracle@host03 ~]$
```

- When no username is specified `ssh` attempts to connect using the local username.

- d. Use the `hostname` command to display the host name to confirm that you successfully logged in to **host03**.

```
[oracle@host03 ~]$ hostname
host03.example.com
```

- e. Use the `logout` command to close the `ssh` connection to **host03**.
Use the `hostname` command to confirm that you are back to **host02**.

```
[oracle@host03 ~]$ logout
Connection to host03 closed.
[oracle@host02 ~]$ hostname
host02.example.com
```

6. View the `.ssh` directory in the `oracle` user's home directory.

- a. Use the `ls` command to display a long listing of all files in the home directory of user `oracle`.

```
[oracle@host02 ~]$ ls -la
...
drwx----- . 2 oracle oracle 25 date_time .ssh
```

- Notice that there is now a `~/.ssh` directory.

- b. Use the `cd` command to change to the `~/.ssh` directory and then use `ls` to view the contents of the directory.

```
[oracle@host02 ~]$ cd .ssh
[oracle@host02 .ssh]$ ls
known_hosts
```

- Notice that the `known_hosts` file was created.

- c. Use the `cat` command to view the contents of the `known_hosts` file.

```
[oracle@host02 .ssh]$ cat known_hosts
host03,192.0.2.103 ecdsa-sha2-nistp256 ...
```

- Notice that **host03** is now a “known host”.

7. Log in to remote **host03** from **host02**.

- a. Perform a remote login to **host03** using the `ssh` command.

Provide the password when prompted.

```
[oracle@host02 .ssh]$ ssh host03
oracle@host03's password:
Last login...
[oracle@host03 ~]$
```

- Notice that you still need to provide a password, but are not asked to confirm this time, because the `known_hosts` file identifies **host03** as a “known host.”

- b. Use the `logout` command to close the `ssh` connection to **host03**.

Use the `hostname` command to confirm that you are back to **host02**.

Use the `cd` command to change back to the user’s home directory.

```
[oracle@host03 ~]$ logout
Connection to host03 closed.
[oracle@host02 .ssh]$ hostname
host02.example.com
[oracle@host02 .ssh]$ cd
[oracle@host02 ~]$
```

- c. Log in to **host03** as user `root` and run the `ls` command with a single `ssh` command.

- Provide the password when prompted.

```
[oracle@host02 ~]$ ssh root@host03 ls
root@host03's password:
anaconda-ks.cfg
[oracle@host02 ~]$
```

- Note that the `ls` command ran on the remote system displaying the contents of the remote directory, and then the remote connection closed.

- d. Use the `hostname` command to confirm that you are back to **host02**.

```
[oracle@host02 ~]$ hostname
host02.example.com
```

Practice 4-2: Configuring OpenSSH to Connect Without a Password

Overview

In this practice, you use the `ssh-keygen` command to generate an RSA key pair and configure OpenSSH to connect to a remote system without supplying a password. You also use the `scp` command in this practice.

Assumptions

- This practice is performed on the **host02** and **host03** VMs.
- You are logged in to **host02** as the `oracle` user.
- The prompts include either **host02** or **host03** to indicate which system to enter the command from.

Tasks

1. Use the `ssh-keygen` command to create the public and private parts of an RSA key.
 - a. On **host02**, use the `whoami` command to confirm you are logged in as the `oracle` user.

```
[oracle@host02 ~]$ whoami
oracle
```

- b. Use the `ls` command to view the contents of the `~/.ssh` directory.

```
[oracle@host02 ~]$ ls ~/.ssh
known_hosts
```

- c. Use the `ssh-keygen -t rsa` command to create the public and private parts of an RSA key.

Accept all the defaults.

```
[oracle@host02 ~]$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/oracle/.ssh/id_rsa):
ENTER
Created directory '/home/oracle/.ssh'.
Enter passphrase (empty for no passphrase): ENTER
Enter same passphrase again: ENTER
Your identification has been saved in /home/oracle/.ssh/id_rsa.
Your public key has been saved in /home/oracle/.ssh/id_rsa.pub.
The key fingerprint is:
SHA256:... oracle@host02.example.com
The key's randomart image is:
...
```

- d. Use the `ls` command to view the contents of the `~/.ssh` directory.

```
[oracle@host02 ~]$ ls ~/.ssh
id_rsa      id_rsa.pub  known_hosts
```

- Note that the `ssh-keygen` command generated two key files.

2. Log in to remote **host03** from **host02**.

- a. Perform a remote login to **host03** using the `ssh` command.

Provide the password when prompted.

```
[oracle@host02 .ssh]$ ssh host03
oracle@host03's password:
Last login...
[oracle@host03 ~]$
```

- Notice that you still need to provide a password because **host03** does not recognize your key.

- b. Use the `mkdir` command to create the `.ssh` directory for the `oracle` user.

```
[oracle@host03 ~]$ mkdir -m 700 .ssh
```

- c. Use the `logout` command to close the `ssh` connection to **host03**.

Use the `hostname` command to confirm that you are back to **host02**.

```
[oracle@host03 ~]$ logout
Connection to host03 closed.
[oracle@host02 ~]$ hostname
host02.example.com
```

3. Use the `scp` command to copy `~/.ssh/id_rsa.pub` on the local system (**host02**) to `~/.ssh/authorized_keys` on the remote system (**host03**).

Answer “**yes**” to continue.

Provide the password when prompted.

```
[oracle@host02 ~]$ scp ~/.ssh/id_rsa.pub
host03:~/.ssh/authorized_keys
oracle@host03's password:
id_rsa.pub                                100% 579    1.2MB/s   00:00
[oracle@host02 ~]$
```

- Note that a password is required to make the connection.
- Note that the file is copied, but you are still connected to the local system (**host02**).
- Note that this command would overwrite any existing file, losing the existing authorized keys. In this case the file does not exist, so it is safe for us to copy it directly to **host03**.

4. Log in to remote **host03** from **host02**.

- a. Perform a remote login to **host03** by using the `ssh` command.

```
[oracle@host02 ~]$ ssh host03
Last login:...
```

- Note that you no longer need to enter a password.

- b. Use the `hostname` command to confirm that you successfully logged in to **host03**.

```
[oracle@host03 ~]$ hostname
host03.example.com
```

- c. Use the `ls` command to view the contents of the `~/.ssh` directory.

```
[oracle@host03 ~]$ ls ~/.ssh
authorized_keys
```

- Note the existence of the `authorized_keys` file, which allowed you to connect without supplying a password.

5. Remove files and log off.

- a. Use the `rm` command to remove the `authorized_keys` file on **host03**.

```
[oracle@host03 ~]$ rm ~/.ssh/authorized_keys
```

- b. Use the `logout` command to close the connection to **host03**.

Use the `hostname` command to confirm that you are back to **host02**.

```
[oracle@host03 ~]$ logout
Connection to host03 closed.
[oracle@host02 ~]$ hostname
host02.example.com
```

- c. Use the `rm` command to remove the `id_rsa` and `id_rsa.pub` files on **host02**.

```
[oracle@host02 ~]$ rm ~/.ssh/id_rsa
[oracle@host02 ~]$ rm ~/.ssh/id_rsa.pub
```

- d. Log off from **host02** in preparation for the next practice.

Use the `exit` command to close the `ssh` connection to **host02**.

```
[oracle@host02 ~]# exit
logout
Connection to host02 closed.
```

You are now the `oracle` user on the **gateway** VM.

Practices for Lesson 5: The GRUB Bootloader

Practices for Lesson 5: Overview

Practices Overview

In these practices, you:

- Explore the GRUB 2 bootloader
- Update the GRUB 2 environment to boot different kernels and use different options
- Rebuild the `initramfs` file to include an additional module

Practice 5-1: Exploring the GRUB 2 Bootloader

Overview

In this practice, you explore the `/boot` directory, the GRUB 2 configuration file, and kernel boot parameters. The displayed sample output might not exactly match what you see on your system. In some cases, only a partial output is shown.

Assumptions

- You are logged on to **gateway** as the `oracle` user.

Tasks

1. Connect to **host03** using the `ssh` command.

Provide the `root` password when prompted.

```
[oracle@gateway ~]$ ssh root@host03
root@host03's password:
Last login...
[root@host03 ~]# whoami
root
```

- The `whoami` command confirms that you are the `root` user.
2. Examine the partition and file system layout for **host03**.
The partition mounted on `/boot` contains the kernel and other files used to boot your system.

Use the `df` command to confirm the mounted partitions. Some of your numbers might vary.

```
# df -h
Filesystem      Size  Used Avail Use% Mounted on
...
/dev/xvda2      24G   1.7G   23G    7% /
/dev/xvda3      2.0G   47M   2.0G    3% /home
/dev/xvda1     1014M  210M   805M   21% /boot
...
```

- Note that `/boot` is a separate partition mounted on `/dev/xvda1`.
 - The `/boot` partition is always located on a standard partition.
 - Placing the `/boot` partition on an LVM volume is not supported.
3. Explore the `/boot` partition.

- a. Use the `ls` command to view the `/boot/grub2` directory.

```
# ls /boot/grub2
device.map  fonts  grub.cfg  grubenv  i386-pc
```

- Note the GRUB 2 configuration file, `grub.cfg`, in this directory.

- b. Use the `less` command to view the GRUB 2 configuration file.

`/boot/grub2/grub.cfg`.

Navigate through the file using the standard “`less`” navigation keys.

- The ENTER key advances one line.
- The SPACEBAR advances one screen.
- The up and down arrows scroll forward and backward one line at a time.
- Press “q” to exit the `less` command.

```
# less /boot/grub2/grub.cfg
#
# DO NOT EDIT THIS FILE
#
# It is automatically generated by grub2-mkconfig using templ...
# from /etc/grub.d and settings from /etc/default/grub
#

### BEGIN /etc/grub.d/00_header ###
...
### END /etc/grub.d/00_header ###
...
### BEGIN /etc/grub.d/10_linux ###
...
if [ -z "${kernelopts}" ]; then
    set kernelopts="root=UUID=... ro resume=UUID=... rhgb quiet "
fi

insmod blscfg
blscfg
### END /etc/grub.d/10_linux ###

### BEGIN /etc/grub.d/10_reset_boot_success ###
# Hiding the menu is ok if last boot was ok or if this is a
# first boot attempt to boot the entry
if [ "${boot_success}" = "1" -o "${boot_indeterminate}" = "1" ];
then
    set menu_hide_ok=1
else
    set menu_hide_ok=0
fi
...
### END /etc/grub.d/41_custom ###
```

- Some things to note about the `grub.cfg` file:
 - Do not edit this file. The file is generated by the `grub2-mkconfig` command using template scripts from the `/etc/grub.d` directory and settings from the `/etc/default/grub` file.
 - Options passed to the kernel are set in the `kernelopts` variable.
 - Installed kernels are not listed in the file, but in files located in `/boot/loader/entries` instead.
 - The menu of available kernels is created by the `blscfg` module.
 - Other boot options are managed by this file, such as the option to hide the grub boot menu if the default kernel successfully boot last time. With this option, the administrator would need to press a key to make the boot menu appear.

c. Press “q” to exit the `less` command.

d. Use the `grep` command to list the titles of bootable kernels in the `/boot/loader/entries` files.

```
# grep -h ^title /boot/loader/entries/*
title Oracle Linux Server (0-rescue-9e66...) 8.3
title Oracle Linux Server (4.18.0-240.el8.x86_64) 8.3
title Oracle Linux Server 8 (5.4.17-2011.7.4.el8uek.x86_64)
```

e. Use the `grep` command to list the filenames for the bootable kernels in the `/boot/loader/entries` files.

```
# grep -h ^linux /boot/loader/entries/*
linux /vmlinuz-0-rescue-9e66f5c549854e2691e6866423bb82d8
linux /vmlinuz-4.18.0-240.el8.x86_64
linux /vmlinuz-5.4.17-2011.7.4.el8uek.x86_64
```

f. Use the `grep` command to list the initial RAM disk filenames in the `/boot/loader/entries` files.

```
# grep -h ^initrd /boot/loader/entries/*
initrd /initramfs-0-rescue-9e66f5c549854e2691e6866423bb82d8.img
initrd /initramfs-4.18.0-240.el8.x86_64.img $tuned_initrd
initrd /initramfs-5.4.17-2011.7.4.el8uek.x86_64.img $tuned_in...
```

g. Use the `ls` command to view the kernel files.

```
# ls /boot/vmlinuz*
/boot/vmlinuz-0-rescue-9e66f5c549854e2691e6866423bb82d8
/boot/vmlinuz-4.18.0-240.el8.x86_64
/boot/vmlinuz-5.4.17-2011.7.4.el8uek.x86_64
```

- Note that these file names match with entries in the files in `/boot/loader/entries`.
- For clarification purposes, the “e1” in the file names is lowercase letter E followed by lowercase letter L. The “e1” stands for Enterprise Linux and is used for compatibility with Red Hat Enterprise Linux.

- h. Use the `ls` command to view the initial RAM disk files.

```
# ls /boot/initramfs*
/boot/initramfs-0-rescue-ac34469c2d1349df99cfed6fefef41039.img
/boot/initramfs-4.18.0-240.el8.x86_64.img
/boot/initramfs-5.4.17-2011.7.4.el8uek.x86_64.img
```

- Note that these file names match the `initrd` entries in the `/boot/loader/entries` files.

4. View the kernel boot parameters.

- a. Use the `uname -r` command to determine the loaded kernel version.

```
# uname -r
5.4.17-2011.7.4.el8uek.x86_64
```

- In this example, UEK (`el8uek`) is loaded.

- b. Use the `grep` command to find options specified in the `/boot/loader/entries` files.

```
# grep ^options /boot/loader/entries/*
/boot/...-0-rescue.conf:options $kernelopts
/boot/...el8.x86_64.conf:options $kernelopts $tuned_params
/boot/...el8uek.x86_64.conf:options $kernelopts $tuned_params
```

- The main bootable kernels use two variable for options, `$kernelopts` and `$tuned_params`. The rescue kernel is used for recovery so does not include any tuning parameters.
- As we have not configured `tuned`, the `$tuned_params` variable is empty.

- c. Find the `$kernelopts` value we saw earlier in `/boot/grub2/grub.cfg`.

```
# grep 'set kernelopts' /boot/grub2/grub.cfg
set kernelopts="root=UUID=... ro resume=UUID=... rhgb quiet "
```

- These parameters are written to the `/proc/cmdline` file and viewable after boot.
- The UUID values differ on your system and, therefore, are not displayed.

- d. Use the `cat` command to view the `/proc/cmdline` file.

```
# cat /proc/cmdline
BOOT_IMAGE=(hd0,msdos1)/vmlinuz-5.4.17-2011.7.4.el8uek.x86_64
root=UUID=... ro resume=UUID=... rhgb quiet
```

- Note that the content contains the `vmlinuz` filename from `/boot/loader/entries` file followed by the kernel boot parameters defined in the `$kernelopts` variable.

Practice 5-2: Booting Different Kernels

Overview

In this practice, you change the GRUB 2 configuration file to boot different kernels.

You also modify the options passed to the kernel when it starts.

Assumptions

You are the `root` user on the **host03** VM.

You do not have access to the GRUB boot menu during a reboot.

Tasks

1. Boot the Red Hat Compatible Kernel (RHCK).

- a. Use the `uname -r` command to display the current running kernel.

```
[root@host03 ~]# uname -r  
5.4.17-2011.7.4.el8uek.x86_64
```

- The current running kernel is UEK (el8uek).

- b. Use the `grub2-editenv` command to find the default (saved) kernel entry.

```
# grub2-editenv list  
saved_entry=ac3...-5.4.17-2011.7.4.el8uek.x86_64  
kernelopts=root=UUID=... ro resume=UUID=... rhgb quiet  
boot_success=1
```

- Note that this is the same UEK kernel version.

- c. Use the `grep` command to find the file containing the RHCK kernel details.

```
# grep -l 4.18.0 /boot/loader/entries/*  
/boot/loader/entries/ac3...-4.18.0-240.el8.x86_64.conf
```

- d. Use the `grub2-set-default` command to change the default kernel to boot.
Use the filename you found using the `grep` command above, removing the `.conf` extension.

```
# grub2-set-default ac3...-4.18.0-240.el8.x86_64  
# grub2-editenv list  
saved_entry=ac3...- 4.18.0-240.el8.x86_64  
kernelopts=root=UUID=... ro resume=UUID=... rhgb quiet  
boot_success=1
```

- e. Use the `systemctl reboot` command to reboot your system.

It might take a couple of minutes to complete the reboot.

```
# systemctl reboot  
Connection to host03 closed by remote host.  
Connection to host03 closed.
```

- After you reboot your system, the `ssh` session closes.

- f. After a couple of minutes, reconnect to **host03** using the `ssh` command.

```
[oracle@gateway ~]$ ssh root@host03
Last login...
```

- g. Use the `uname -r` command to display the new running kernel.

```
[root@host03 ~]# uname -r
4.18.0-240.el8.x86_64
```

- If the command returns the UEK kernel version, you may have mistyped the entry in the `grub2-set-default` command above. You can try again if you wish, or continue.

2. Modify the kernel options.

- a. Use the `vi` editor to update the `GRUB_LINUX_CMDLINE` entry in `/etc/default/grub` with the kernel boot parameter `panic=20`.

```
# vi /etc/default/grub
...
GRUB_CMDLINE_LINUX="resume=UUID=... rhgb quiet " (old)
GRUB_CMDLINE_LINUX="resume=UUID=... rhgb quiet panic=20 " (new)
```

- Note: This parameter instructs the kernel to automatically reboot 20 seconds after a fatal error causes it to panic (abort).

- b. Use the `grub2-mkconfig` command to generate `grub.cfg`.

```
# grub2-mkconfig -o /boot/grub2/grub.cfg
Generating grub configuration file ...
done
```

- c. Check the current value for the panic kernel parameter.

- Many kernel parameter values are found in a special file system mounted as `/proc`.

```
# cat /proc/sys/kernel/panic
0
```

- d. Use the `systemctl reboot` command to reboot your system.

```
# systemctl reboot
Connection to host03 closed by remote host.
Connection to host03 closed.
```

- e. After a couple of minutes, reconnect to **host03** using the `ssh` command.

```
[oracle@gateway ~]$ ssh root@host03
Last login...
```

- f. Check the current value for the panic kernel parameter.

```
[root@host03 ~]# cat /proc/sys/kernel/panic
20
```

3. Reset the default kernel to UEK and reboot.

- a. List the kernel files in /boot.

```
# ls /boot/vmlinuz*
/boot/vmlinuz-0-rescue-ac34469c2d1349df99cfed6fefe41039
/boot/vmlinuz-4.18.0-240.el8.x86_64
/boot/vmlinuz-5.4.17-2011.7.4.el8uek.x86_64
```

- b. Use the grubby command to set the default kernel.

```
# grubby --set-default=/boot/vmlinuz-5.4.17-2011.7.4.el8uek.x86_64
The default is /boot/loader/entries/ac3...-5.4.17-2011.7.4...
# grub2-editenv list
saved_entry=ac3...-5.4.17-2011.7.4.el8uek.x86_64
kernelopts=root=UUID=... ro resume=UUID=... rhgb quiet
boot_success=0
boot_indeterminate=0
```

- The `boot_success` value may be zero because you have rebooted several times in quick succession. It does not necessarily mean a failure occurred.

- c. Use the `systemctl reboot` command to reboot your system.

```
# systemctl reboot
Connection to host03 closed by remote host.
Connection to host03 closed.
```

- d. After a couple of minutes, reconnect to **host03** using the `ssh` command.

```
[oracle@gateway ~]$ ssh root@host03
Last login...
```

- e. Use the `uname -r` command to display the new running kernel.

```
[root@host03 ~]# uname -r
5.4.17-2011.7.4.el8uek.x86_64
```

Practice 5-3: Rebuilding the `initramfs`

Overview

In this practice, you rebuild the `initramfs` file for a kernel to include an additional module. This is sometimes required to provide support for hardware during the early stages of the boot process. For the practice, you will add support for FAT file systems to the `initramfs` for the RHCK.

Assumptions

You are the `root` user on the `host03` VM.

Tasks

1. Check whether the FAT file system modules are in the current Initial RAM disk file.
 - a. Use the `ls` command to list the initial RAM disk files.

```
[root@host03 ~]# ls /boot/initramfs*  
/boot/initramfs-0-rescue-ac34469c2d1349df99cfed6fefe41039.img  
/boot/initramfs-4.18.0-240.el8.x86_64.img  
/boot/initramfs-5.4.17-2011.7.4.el8uek.x86_64.img
```

- Note that the `initramfs-4.18.0-240.el8.x86_64.img` file is for the RHCK.

- b. Use the `lsinitrd` command to search the initial RAM disk file for the FAT modules.

```
# lsinitrd /boot/initramfs-4.18.0-240.el8.x86_64.img | grep fat  
#
```

- Note that the `grep` command does not match anything.

- c. Use the `find` command to check the module is available for the RHCK.

```
# find /usr/lib/modules/4.18.0-240.el8.x86_64 -name fat.ko.xz  
/usr/lib/modules/4.18.0-240.el8.x86_64/kernel/fs/fat/fat.ko.xz
```

2. Configure dracut to include the required module.

Use the `vi` editor to create a custom configuration file to add the module.

```
# vi /etc/dracut.conf.d/10-fat.conf  
add_drivers+=" fat "
```

- Note the spaces around the module name. This may be concatenated with other values defined elsewhere.

3. Rebuild the Initial RAM disk file.

- a. Use the `mv` command to back up the existing file in case you have a problem.

```
# mv /boot/initramfs-4.18.0-240.el8.x86_64.img /boot/initramfs-4.18.0-240.el8.x86_64.img_SAV
```

- b. Use the `dracut` utility to re-create `initramfs-4.18.0-240.el8.x86_64.img`.

```
# dracut /boot/initramfs-4.18.0-240.el8.x86_64.img 4.18.0-240.el8.x86_64
```

- The first parameter is the desired file name, `initramfs-4.18.0-240.el8.x86_64.img`.
- The second parameter is the kernel version, `4.18.0-240.el8.x86_64`. By default, `dracut` creates `initramfs` images using the currently booted kernel version. The parameter is necessary if a kernel version different from the currently booted kernel must be created, which is true here.
- This command takes a few seconds to complete.

4. Verify that the new Initial RAM disk file contains the required module.

- a. Use the `ls` command to view the initial RAM disk files.

```
# ls /boot/initramfs*  
/boot/initramfs-0-rescue-ac34469c2d1349df99cfed6fefe41039.img  
/boot/initramfs-4.18.0-240.el8.x86_64.img  
/boot/initramfs-4.18.0-240.el8.x86_64.img_SAV  
/boot/initramfs-5.4.17-2011.7.4.el8uek.x86_64.img
```

- Note that the `initramfs-3.10.0-862.el7.x86_64.img` file now exists again.

- b. Use the `lsinitrd` command to view the contents of the image created by `dracut`.

- You can pipe the output of the command to `less` to view one screen at a time.

```
# lsinitrd /boot/initramfs-4.18.0-240.el8.x86_64.img | grep fat  
...usr/lib/modules/4.18.0-240.el8.x86_64/kernel/fs/fat  
...usr/lib/modules/4.18.0-240.el8.x86_64/kernel/fs/fat/fat.ko.xz
```

5. Use the `exit` command to log off `host03`.

```
# exit  
logout  
Connection to host03 closed.
```

You are now the `oracle` user on `gateway`.

Practices for Lesson 6:
Introduction to `systemd`

Practices for Lesson 6: Overview

Practices Overview

In these practices, you:

- View various `systemd` units
- View the default target unit
- View service units
- Configure a service to start and stop
- Start, stop, and restart services
- Customize service options using Environment Files and `systemd` drop-in files

Practice 6-1: Exploring `systemd` Units

Overview

In this practice, you view `systemd` units. In most cases, partial output is shown and your output can vary. Where needed, scroll through the output using navigation keys such as Page Up and Page Down. Also where needed, enter "q" to terminate output and return to the prompt.

Assumptions

You are the `oracle` user on the **gateway** VM.

Tasks

1. Log on to **host03**.

Use `ssh` to log on to **host03**, providing the root password.

```
[oracle@gateway ~]$ ssh root@host03
root@host03's password:
Last login: ...
```

2. Explore all `systemd` unit files.

Run the `systemctl list-unit-files` command to list all installed unit files.

```
[root@host03 ~]# systemctl list-unit-files
UNIT FILE                                STATE
proc-sys-fs-binfmt_misc.automount       static
...
dev-hugepages.mount                     static
dev-mqueue.mount                        static
...
systemd-ask-password-console.path        static
...
auditd.service                          enabled
...
user.slice                              static
...
sshd.socket                             disabled
...
sssd-kcm.socket                          enabled
...
basic.target                            static
bluetooth.target                        static
...
dnf-makecache.timer                      enabled
...
```

The output is a selection of the different types of `systemd` units:

- `automount`: Provide automount capabilities for on-demand mounting of file systems as well as parallelized boot-up
- `mount`: Mount units control mount points in the file system
- `path`: Can activate services when file system path information changes
- `scope`: Similar to service units but manage foreign processes instead of starting them as well
- `service`: Start and control daemons and the processes they consist of
- `slice`: Use to group units that manage system processes, such as service units and scope units, in a hierarchical tree for resource management purposes
- `socket`: Encapsulate local interprocess communication (IPC) or network sockets in the system, which are useful for socket-based activation
- `target`: Use to group units or to provide well-known synchronization points during boot-up
- `timer`: Use to trigger activation of other units based on timers
- `device`: Expose kernel devices in `systemd` and can also be used to implement device-based activation
- `swap`: Encapsulate memory swap partitions or swap files

3. Explore `mount` units.

- a. Use the `systemctl list-units --type mount` command to list the currently active mount units on your system.

```
# systemctl list-units --type mount
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
-.mount                            loaded active mounted Root mount
boot.mount                         loaded active mounted /boot
dev-hugepages.mount               loaded active mounted Huge Pages File ...
...

10 loaded units listed. Pass --all to see loaded but inactiv...
To show all installed unit files use 'systemctl list-unit-files'
```

- b. View the contents of `/etc/fstab`.

```
# cat /etc/fstab
...
# After editing this file, run 'systemctl daemon-reload' to
# update system units generated from this file.
#
UUID=... /                xfs      defaults      0 0
UUID=... /boot             xfs      defaults      0 0
UUID=... /home             xfs      defaults      0 0
UUID=... none              swap     defaults      0 0
```

- Note there are only four mount entries listed.
 - These entries were customized during the installation and are converted into systemd mount units.
 - Note the comment about regenerating the unit files after making any changes.
- c. Use the `mount` command to list the currently active mounts on your system. See if you can match these to the `systemd` mount units.

```
# mount
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime,sec...
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
devtmpfs on /dev type devtmpfs (rw,nosuid,seclabel,size=19918...
securityfs on /sys/kernel/security type securityfs (rw,nosuid...
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,seclabel)
...
```

4. Explore swap units.

Use the `systemctl list-units --type swap` command to list the currently active swap units on your system.

```
# systemctl list-units --type swap
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
dev-disk-by\...\swap loaded active active  /dev/disk/by-uuid...

1 loaded units listed. Pass --all to see loaded but inactiv...
To show all installed unit files use 'systemctl list-unit-files'
```

- Although we use `/etc/fstab` to configure swap at boot time, `systemd` has specific swap unit files rather than using mount unit files.

5. Explore socket units.

- a. Use the `systemctl list-units --type socket` command to list the currently active socket units on your system.

```
# systemctl list-units --type socket
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
dbus.socket                        loaded active running  D-Bus System...
iscsid.socket                      loaded active running  Open-iSCSI i...
iscsiuio.socket                   loaded active listening Open-iSCSI i...
rpcbind.socket                    loaded active listening RPCbind Serv...
sssd-kcm.socket                   loaded active listening SSSD Kerbero...
systemd-coredump.socket           loaded active listening Process Core...
...
12 loaded units listed. Pass --all to see loaded but inactive...
To show all installed unit files use 'systemctl list-unit-files'
```

- b. Sockets include file system FIFO. Use the `systemctl list-sockets` command to see the paths of the FIFOs monitored by `systemd` and what service they activate.

```
# systemctl list-sockets
LISTEN          UNIT                                ACTIVATES
/dev/rfkill     systemd-rfkill.socket             systemd-rfkill.service
/run/dbus/sys... dbus.socket                       dbus.service
/run/initctl    systemd-initctl.socket            systemd-initctl.service
...
17 sockets listed.
Pass --all to see loaded but inactive sockets, too.
```

- Note the number of sockets may be more than the number of units as one socket unit file can configure multiple sockets if they are managed together.

6. Explore target units.

- a. Use the `systemctl list-units --type target` command to list the currently active targets on a system.

```
# systemctl list-units --type target
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
basic.target                       loaded active active Basic System
cryptsetup.target                  loaded active active Local Encrypted Volumes
getty.target                       loaded active active Login Prompts
local-fs-pre.target                loaded active active Local File Systems (Pre
local-fs.target                    loaded active active Local File Systems
multi-user.target                  loaded active active Multi-User System
network-online.target              loaded active active Network is Online
network-pre.target                 loaded active active Network (Pre)
network.target                    loaded active active Network
nfs-client.target                  loaded active active NFS client services
nss-user-lookup.target             loaded active active User and Group Name...
...
swap.target                        loaded active active Swap
sysinit.target                     loaded active active System Initialization
timers.target                       loaded active active Timers

LOAD    = Reflects whether the unit definition was properly ...
ACTIVE  = The high-level unit activation state, i.e. general...
SUB      = The low-level unit activation state, values depend ...

22 loaded units listed. Pass --all to see loaded but inactiv...
To show all installed unit files use 'systemctl list-unit-files'
```

- b. Run the same command again, but include the `--all` option to see all loaded units, including those that are inactive.

```
# systemctl list-units --type target --all
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
basic.target                       loaded active active Basic System
cryptsetup.target                  loaded active active Local Encrypted Vol...
emergency.target                   loaded inactive dead    Emergency Mode
getty-pre.target                   loaded inactive dead    Login Prompts (Pre)
getty.target                       loaded active active Login Prompts
graphical.target                   loaded inactive dead    Graphical Interface
...

36 loaded units listed.
To show all installed unit files use 'systemctl list-unit-files'
```

- Some of the inactive targets are for different system states, such as emergency, graphical, rescue, and shutdown.

7. View the default target unit.

- a. Use the `systemctl get-default` command to view which target unit is used by default.

```
# systemctl get-default
multi-user.target
```

- The default target unit is represented by the `/etc/systemd/system/default.target` file.

- b. Use the `ls -l` command to list information about the `/etc/systemd/system/default.target` file.

```
# ls -l /etc/systemd/system/default.target
lrwxrwxrwx ... /etc/systemd/system/default.target ->
/lib/systemd/system/multi-user.target
```

- Note that the `default.target` file is a symbolic link to the current default target unit file, `multi-user.target`.

Practice 6-2: Working with `systemd` Service Units

Overview

In this practice, you view service units, view the status of services, and configure a service to start and stop at a specific target state. You will look at the construction of a service unit file and how it can be customized to your requirements.

In some tasks, partial output is shown and your output can or will differ in some cases. Where needed, scroll through the output using navigation keys such as Page Up and Page Down. Also where needed, enter "q" to terminate output and return to the prompt.

Assumptions

You are the `root` user on the **host03** VM.

Tasks

1. Explore service units.
 - a. Use the `systemctl list-units --type service` command to list the currently active services on a system.

```
# systemctl list-units --type service
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
auditd.service                    loaded active running Security Auditing
Service
crond.service                     loaded active running Command Scheduler
dbus.service                      loaded active running D-Bus System Message Bus
...

LOAD    = Reflects whether the unit definition was properly ...
ACTIVE  = The high-level unit activation state, i.e. general...
SUB     = The low-level unit activation state, values depend ...

43 loaded units listed. Pass --all to see loaded but inactive...
To show all installed unit files use 'systemctl list-unit-files'
```

- b. Run the same command again, but include the `--all` option to see all loaded units, including those that are inactive.

```
# systemctl list-units --type service --all
...
113 loaded units listed.
To show all installed unit files use 'systemctl list-unit-files'
```

- c. Use the `systemctl status` command to view detailed information about the `crond.service` unit.

You can omit the `.service` extension.

```
# systemctl status crond
• crond.service - Command Scheduler
   Loaded: loaded (/usr/lib/systemd/system/crond.service;
   enabled; vendor preset: enabled)
   Active: active (running) since date_time; time ago
 Main PID: 808 (crond)
    Tasks: 1 (limit: 24898)
   Memory: 2.4M
   CGroup: /system.slice/crond.service
           └─808 /usr/sbin/crond -n

... run-parts[16349]: (/etc/cron.daily) starting logrotate
... run-parts[16354]: (/etc/cron.daily) finished logrotate
```

- In this example, the `crond` service is loaded and running.
 - The most recent log entries are displayed at the end of the output.
- d. Use the `systemctl is-active` command to check if the `nfs-server` service is running (active) or not running (inactive).

```
# systemctl is-active nfs-server
inactive
```

- e. Use the `systemctl is-enabled` command to check if the `nfs-server` service is enabled or disabled.

```
# systemctl is-enabled nfs-server
disabled
```

- f. Use the `systemctl status` command to view detailed information about the `nfs-server` service.

```
# systemctl status nfs-server
• nfs-server.service - NFS server and services
   Loaded: loaded (/usr/lib/systemd/system/nfs-server.service;
   disabled; vendor preset: disabled)
   Active: inactive (dead)
```

- This confirms that the `nfs-server` service is not running (inactive) and is disabled.

2. Start, stop, enable, and disable a service.

- a. Use the `systemctl start` command to start the `nfs-server` service.

```
# systemctl start nfs-server
```


- b. Use the `systemctl status` command to view detailed information about the `nfs-server` service.

```
# systemctl status nfs-server
● nfs-server.service - NFS server and services
   Loaded: loaded (/usr/lib/systemd/system/nfs-server.service;
 disabled; vendor preset: disabled)
   Active: active (exited) since ...
     Process: 1148 ExecStart=/bin/sh -c if systemctl -q is-active
 gssproxy; then systemctl reload gssproxy ; fi (code=exited,
 status=0/SUCCESS)
     Process: 1135 ExecStart=/usr/sbin/rpc.nfsd (code=exited,
 status=0/SUCCESS)
     Process: 1134 ExecStartPre=/usr/sbin/exportfs -r (code=exited,
 status=0/SUCCESS)
    Main PID: 1148 (code=exited, status=0/SUCCESS)

... systemd[1]: Starting NFS server and services...
... systemd[1]: Started NFS server and services.
```

- Notice the `nfs-server` service is active (running) but is still disabled.
 - This means that the `nfs-server` service does not automatically start on a system reboot.
 - Also notice that the `nfs-server` service does not just start a single process, but has several `ExecStart` entries, which ultimately start the required daemons.
- c. Use the `systemctl enable` command to enable the `nfs-server` service.

```
# systemctl enable nfs-server
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-
server.service → /usr/lib/systemd/system/nfs-server.service.
```

- Notice that the `systemctl enable` command enables a service by creating a symbolic link for the lowest-level system-state target at which the service starts.
 - In this example, the command creates the symbolic link `nfs-server.service` for the `multi-user` target.
- d. Use the `systemctl status` command to view detailed information about the `nfs-server` service.

```
# systemctl status nfs-server
● nfs-server.service - NFS server and services
   Loaded: loaded (/usr/lib/systemd/system/nfs-server.service;
 enabled; vendor preset: disabled)
   Active: active (exited) since ...
   ...
```

- Note that the `nfs-server` service is now enabled.

- e. Use the `ls -l` command to view the contents of the `/etc/systemd/system/multi-user.target.wants` directory.

```
# ls -l /etc/systemd/system/multi-user.target.wants
lrwxrwxrwx ... auditd.service ->
/usr/lib/systemd/system/auditd.service
lrwxrwxrwx ... crond.service ->
/usr/lib/systemd/system/crond.service
...
lrwxrwxrwx ... nfs-server.service ->
/usr/lib/systemd/system/nfs-server.service
...
```

- Notice the symbolic link `nfs-server.service` file in this directory.
 - With the service enabled, the service starts on a system reboot.
- f. Use the `systemctl disable` command to disable the `nfs-server` service.

```
# systemctl disable nfs-server
Removed /etc/systemd/system/multi-user.target.wants/nfs-
server.service.
```

- Notice that the `systemctl disable` command deletes the symbolic link for the service.
- g. Use the `systemctl stop` command to stop the `nfs-server` service.

```
# systemctl stop nfs-server
```

- h. Use the `systemctl status` command to view detailed information about the `nfs-server` service.

```
# systemctl status nfs-server
• nfs-server.service - NFS server and services
   Loaded: loaded (/usr/lib/systemd/system/nfs-server.service;
disabled; vendor preset: disabled)
   Active: inactive (dead) (thawing) since ...
   Process: 1213 ExecStopPost=/usr/sbin/exportfs -f (code=exited,
status=0/SUCCESS)
   Process: 1210 ExecStopPost=/usr/sbin/exportfs -au
(code=exited, status=0/SUCCESS)
   Process: 1208 ExecStop=/usr/sbin/rpc.nfsd 0 (code=exited,
status=0/SUCCESS)
   Main PID: 1148 (code=exited, status=0/SUCCESS)

... systemd[1]: Starting NFS server and services...
... systemd[1]: Started NFS server and services.
... systemd[1]: Stopping NFS server and services...
... systemd[1]: nfs-server.service: Succeeded.
... systemd[1]: Stopped NFS server and services.
```

- This confirms that the `nfs-server` service is not running (inactive) and is disabled.

3. Customize a service unit using an Environment File.

a. View the `crond.service` unit file.

```
# cat /usr/lib/systemd/system/crond.service
[Unit]
Description=Command Scheduler
After=auditd.service nss-user-lookup.target systemd-user-
sessions.service time-sync.target ypbind.service

[Service]
EnvironmentFile=/etc/sysconfig/crond
ExecStart=/usr/sbin/crond -n $CRONDARGS
ExecReload=/bin/kill -HUP $MAINPID
KillMode=process
Restart=on-failure
RestartSec=30s

[Install]
WantedBy=multi-user.target
```

- Notice `EnvironmentFile` points to a file in `/etc/sysconfig`.

b. Use the `vi` command to edit `/etc/sysconfig/crond`, setting the command-line option `-s` for `crond`.

```
# vi /etc/sysconfig/crond
# Settings for the CRON daemon.
# CRONDARGS= : any extra command-line startup arguments for...
CRONDARGS="-s"
```

c. Restart the `crond` service.

```
# systemctl restart crond
```

d. View the status of the `crond` service.

```
# systemctl status crond
● crond.service - Command Scheduler
...
   CGroup: /system.slice/crond.service
           └─25949 /usr/sbin/crond -n -s
...
```

- Notice that the `-s` option now appears on the `crond` process.

4. Customize a service unit using a drop-in file.

- a. Create a directory in `/etc/systemd/system` for the drop-in file.

```
# mkdir /etc/systemd/system/crond.service.d
```

- b. Use the `vi` command to create your new drop-in file.

```
# vi /etc/systemd/system/crond.service.d/10-options.conf
[Service]
EnvironmentFile=-/etc/sysconfig/crond.local
```

- The dash before the file path ensures that `systemd` does not display any error message if the file is missing.

- c. Copy the existing Environment file and modify the option used to `-p`.

```
# cp /etc/sysconfig/crond /etc/sysconfig/crond.local
# vi /etc/sysconfig/crond.local
# Settings for the CRON daemon.
# CRONDARGS= : any extra command-line startup arguments for...
CRONDARGS="-p"
```

- d. View the status of the `crond` service.

```
# systemctl status crond
Warning: The unit file, source configuration file or drop-ins of
crond.service changed on disk. Run 'systemctl daemon-reload' to
reload units.
• crond.service - Command Scheduler
...
   CGroup: /system.slice/crond.service
           └─25949 /usr/sbin/crond -n -s
```

- Notice the warning message and that the `crond` service is still using the `-s` option.

- e. Reload the `systemd` units.

```
# systemctl daemon-reload
```

- f. View the status of the `crond` service.

```
# systemctl status crond
• crond.service - Command Scheduler
...
   Drop-In: /etc/systemd/system/crond.service.d
           └─10-options.conf
...
   CGroup: /system.slice/crond.service
           └─25949 /usr/sbin/crond -n -s
```

- The drop-in file is now identified.
- Note, however, that `crond` is still running with the previous options: `-n -s`

- g. Restart the `crond` service to begin using the new option.

```
# systemctl restart crond
```

- h. View the status of the `crond` service.

```
# systemctl status crond
● crond.service - Command Scheduler
...
Drop-In: /etc/systemd/system/crond.service.d
└─10-options.conf
...
CGroup: /system.slice/crond.service
└─26023 /usr/sbin/crond -n -p
```

- Now the `-p` option is used instead of `-s`.
- Note although both Environment Files were processed, the `CRONDARGS` variable is overridden not concatenated.

5. Use the `exit` command to log off **host03**.

```
# exit
logout
Connection to host03 closed.
```

You are now the `oracle` user on **gateway**.

Practices for Lesson 7:
Managing Time

Practices for Lesson 7: Overview

Practices Overview

In these practices, you configure date and time, enable NTP time synchronization using `chrony`. You also configure automate tasks using both `cron` and `systemd` timers.

Practice 7-1: Configuring Date and Time

Overview

In this practice, you use the `date`, `hwclock`, and `timedatectl` utilities to configure and display date and time information.

Assumptions

You are logged on to **gateway** as the `oracle` user.

Tasks

1. Log on to **host03**.

Use `ssh` to log on to **host03**, providing the root password.

```
[oracle@gateway ~]$ ssh root@host03
root@host03's password:
Last login: ...
```

2. Use the `date` command.

Sample output is displayed.

- a. Run the `date` command with no arguments to display the current date and time.

```
[root@host03 ~]# date
Thu 17 Dec 11:22:05 UTC 2020
```

- b. Run the `date` command with the argument to view the weekday name.

```
# date +%A
Thursday
```

- c. Run the `date` command with the argument to view the full month name.

```
# date +%B
December
```

- d. Use the `date` command to change the date to another date (for example, 19 November, 2015).

You can substitute any date. The example changes the date to 19 November, 2015.

```
# date +%D -s 2015-11-19
11/19/15
```

- e. Use the `date` command to change the date to the correct date.

Substitute today's correct date. The example changes the date to 17 December 2020.

```
# date +%D -s 2020-12-17
12/17/20
```

- Note that we did not specify the time, so it is set to midnight.

- f. Use the `date` command to change the time to the correct time.

Substitute the current UTC time. The example changes the time to 4:14 PM.

```
# date +%T%p -s 11:40:00AM
11:40:00am
```

3. Use the `hwclock` command.

The sample output is displayed.

- a. Run the `hwclock` command with no arguments to display the hardware clock's current date and time.

```
# hwclock
2020-12-17 11:41:36.515681+00:00
```

- b. Use the `hwclock` command with the `-w` option to set the hardware clock to the current system time.

```
# hwclock -w
```

- c. Run the `date` command and the `hwclock` command consecutively to confirm that the date and time settings are the same.

```
# date; hwclock
Thu 17 Dec 11:44:46 UTC 2020
2020-12-17 11:44:46.985362+00:00
```

4. Use the `timedatectl` command.

- a. Run the `timedatectl` command with no arguments to display the system date and time information. The sample output is displayed.

```
# timedatectl
          Local time: Thu 2020-12-17 12:09:00 UTC
    Universal time: Thu 2020-12-17 12:09:00 UTC
           RTC time: Thu 2020-12-17 12:09:00
        Time zone: UTC (UTC, +0000)
System clock synchronized: no
           NTP service: n/a
      RTC in local TZ: no
```

- b. Use the `timedatectl` command with the `set-time` argument to change the date to another date (for example, 19 November, 2015).

You can substitute any date. The example changes the date to 19 November, 2015.

Run the `timedatectl` command afterwards with no arguments to verify that the date changed.

```
# timedatectl set-time 2015-11-19
# timedatectl
      Local time: Thu 2015-11-19 00:00:04 UTC
      Universal time: Thu 2015-11-19 00:00:04 UTC
      RTC time: Thu 2015-11-19 00:00:04
...
```

- c. Use the `timedatectl` command to change the date to the correct date.

Substitute today's correct date. The example changes the date to 17 December 2020.

Run the `timedatectl` command afterwards with no arguments to verify that the date changed.

```
# timedatectl set-time 2020-12-17
# timedatectl
      Local time: Tue 2020-12-17 00:00:01 UTC
      Universal time: Tue 2020-12-17 00:00:01 UTC
      RTC time: Tue 2020-12-17 00:00:01
...
```

- d. Use the `timedatectl` command to change the time to the correct time.

Substitute the current UTC time. The example changes the time to 1:15 PM.

Enter the hour using a 24-hour clock.

Run the `timedatectl` command afterwards with no arguments to verify the time.

```
# timedatectl set-time 13:15:00
# timedatectl
      Local time: Tue 2020-12-17 14:15:01 UTC
      Universal time: Tue 2020-12-17 13:15:01 UTC
      RTC time: Tue 2020-12-17 13:15:01
...
```

5. Log off **host03**.

Use the `logout` command to log out as the `root` user.

```
[root@host03 ~]# logout
Connection to host03 closed.
[oracle@gateway ~]$
```

You are now the `oracle` user on **gateway**.

Practice 7-2: Configuring Chrony

Overview

In this practice, you configure chrony on **host03** to synchronize using NTP with chrony on **gateway**.

Assumptions

- This practice is performed on both **gateway** and **host03**.
- You are the `oracle` user and have a terminal window open on **gateway**.

Tasks

1. View the `chronyd` configuration on **gateway**.

This task is performed on **gateway**.

- a. From **gateway**, use the `systemctl` command to check the status of the chrony daemon, `chronyd`.

```
[oracle@gateway ~]$ systemctl status chronyd
● chronyd.service - NTP client/server
   Loaded: loaded (/usr/lib/systemd/system/chronyd.service;...
   Active: active (running) since ...
   ...
```

- The `chronyd` daemon is running on **gateway**.
- b. Use the `less` command to view the `/etc/chrony.conf` file.

```
$ less /etc/chrony.conf
...
## 2 lines below so can act as NTP server for private-net hosts
local stratum 10
allow 192.0.0.0/8
...
```

- Notice that `chronyd` is configured to act as an NTP server for hosts on the `192.0.0.0/8` network, which is any IPv4 address with the first octet value of 192.
- The local `stratum` value sets the stratum of the **gateway** machine if it is unable to locate any external NTP servers. A stratum value of 10 is the default for this setting and it indicates that the system is not a very reliable time source. Clients of the `chronyd` service on **gateway** will always be assigned a stratum greater than the value on **gateway**.

- c. Use the `chronyc sources` command to display a list of known NTP peers.

```
$ chronyc sources
210 Number of sources = 2
MS Name/IP...    Stratum Poll Reach LastRx Last sample
=====
^? 10.236.1.10      0 10      0      -      +0ns[  +0ns] +/- 0ns
^* 10.236.1.2       2  9     377    414    +11us[ +99us] +/- 16ms
```

- The carat (^) symbol in the output denotes a server.
- The question mark (?) indicates a server that is not responding or is initializing.
- The asterisk (*) indicates the source to which `chronyd` is synchronized.

- d. Use the `chronyc tracking` command to show network time synchronization status.

```
$ chronyc tracking
Reference ID      : 0AEC0102 (10.236.1.2)
Stratum          : 3
Ref time (UTC)   : Thu Dec 17 13:45:51 2020
System time      : 0.000002879 seconds slow of NTP time
Last offset      : -0.000076062 seconds
RMS offset       : 0.000102387 seconds
Frequency        : 11.585 ppm slow
Residual freq    : -0.013 ppm
Skew             : 0.285 ppm
Root delay       : 0.030092506 seconds
Root dispersion  : 0.001567763 seconds
Update interval  : 256.3 seconds
Leap status      : Normal
```

- Note the Stratum number of 3. This is an increment of the stratum in which the source server is located. If no external server can be found, the local stratum number (10) configured in `/etc/chrony.conf` would be used.

2. Log in to **host03**.

The following tasks are performed on **host03**.

Connect to the **host03** guest by using the `ssh` command.

```
[oracle@gateway ~]$ ssh root@host03
root@host03's password:
[root@host03 ~]#
```

3. Configure NTP on **host03** using `chrony`.

- a. Use the `cp` command to make a backup copy of the `/etc/chrony.conf` file.

```
[root@host03 ~]# cp /etc/chrony.conf /etc/chrony.conf_SAV
```

- b. Use the `vi` command to edit the `/etc/chrony.conf` file and make the following changes:

```
# vi /etc/chrony.conf
```

Insert a `#` character for each line beginning with `pool` as shown:

```
#pool 2.pool.ntp.org iburst
```

Add a `server` line below to designate **gateway** (192.0.2.1) as the remote clock synchronization server as shown:

```
server 192.0.2.1
```

The following represents all the changes needed to the `/etc/chrony.conf` file.

The specific changes are in **bold**.

```
#pool 2.pool.ntp.org iburst
server 192.0.2.1
```

4. Start the `chronyd` service and wait for synchronization to complete.

- a. Use the `systemctl` command to start the `chrony` daemon, `chronyd`.

```
# systemctl start chronyd
```

- To check if `chrony` is synchronized, use the `tracking`, `sources`, or `sourcestats` commands as shown below.

- b. Run the `chronyc tracking` command to check `chrony` tracking.

```
# chronyc tracking
Reference ID      : 00000000 ()
Stratum          : 0
Ref time (UTC)   : Thu Jan 01 00:00:00 1970
System time      : 0.000000000 seconds fast of NTP time
Last offset      : +0.000000000 seconds
RMS offset       : 0.000000000 seconds
Frequency        : 0.000 ppm slow
Residual freq    : +0.000 ppm
Skew             : 0.000 ppm
Root delay       : 1.000000000 seconds
Root dispersion  : 1.000000000 seconds
Update interval  : 0.0 seconds
Leap status      : Not synchronised
```

- c. If the previous command shows that time is not synchronized, run the `chronyc waitsync` command to poll until the servers have synchronized.

```
# chronyc waitsync
try: 1, refid: 00000000, correction: 0.000000000, skew: 0.000
try: 2, refid: 00000000, correction: 0.000000000, skew: 0.000
try: 3, refid: 00000000, correction: 0.000000000, skew: 0.000
try: 4, refid: 00000000, correction: 0.000000000, skew: 0.000
try: 5, refid: 00000000, correction: 0.000000000, skew: 0.000
try: 6, refid: 00000000, correction: 0.000000000, skew: 0.000
try: 7, refid: 00000000, correction: 0.000000000, skew: 0.000
try: 8, refid: 00000000, correction: 0.000000000, skew: 0.000
try: 9, refid: 00000000, correction: 0.000000000, skew: 0.000
try: 10, refid: 00000000, correction: 0.000000000, skew: 0.000
try: 11, refid: C0000201, correction: 0.000000000, skew: 750.125
```

- It can take several minutes to synchronize. The command exits once it succeeds.

5. View data about the status of `chronyd` and synchronization.

- a. Run the `chronyc tracking` command again to check chrony time tracking.

```
# chronyc tracking
Reference ID      : C0000201 (_gateway)
Stratum          : 4
Ref time (UTC)   : Thu Dec 17 14:03:24 2020
System time      : 0.000006592 seconds fast of NTP time
Last offset      : +0.000010042 seconds
RMS offset       : 0.000010042 seconds
Frequency        : 11.037 ppm slow
Residual freq    : +0.000 ppm
Skew             : 2.178 ppm
Root delay       : 0.027313061 seconds
Root dispersion  : 0.002577690 seconds
Update interval  : 64.5 seconds
Leap status      : Normal
```

Some of the fields are described:

- **Reference ID:** Refers to the server providing synchronization
- **Stratum:** The distance from a computer attached to a highly accurate reference clock. As **gateway** is in stratum 3, **host03** is in stratum 4.
- **Ref time:** The most recent time `chronyd` utilized a time measurement from the server providing synchronization (time is UTC).

- b. Run the `chronyc sources` command to show information about chronyd time sources.

```
# chronyc sources
210 Number of sources = 1
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^*_gateway                   3      6      77      16  +6995ns[ +16us] +/- 16ms
```

- c. Run the `chronyc sources` command with the `-v` option (verbose) to provide column descriptions.

```
# chronyc sources -v
210 Number of sources = 1
.-- Source mode '^' = server, '=' = peer, '#' = local clock.
/ .- Source state '*' = current synced, '+' = combined , '-' = not combined,
| /  '?' = unreachable, 'x' = time may be in error, '~' = time too variable.
||
||                                     .- xxxx [ yyyy ] +/- zzzz
||      Reachability register (octal) -.          | xxxx = adjusted offset,
||      Log2(Polling interval) --.      |          | yyyy = measured offset,
||                                     \      |          | zzzz = estimated error.
||                                     |      |          |
||                                     |      |          |
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^*_gateway                   3      6     177     99  +6995ns[ +16us] +/- 16ms
```

- For more information about chrony, see the files in `/usr/share/doc/chrony`. Also see the `chronyc(1)` man page.

- d. Run the `chronyc sourcestats` command to view statistics around calculations for time sources.

```
# chronyc sourcestats
210 Number of sources = 1
Name/IP Address          NP  NR  Span Frequency Freq Skew Offset Std Dev
=====
_gateway                 7   5   451    +0.001    0.222   +12ns   12us
```

- e. Run the command again but include the `-v` option to obtain column descriptions.

```
# chronyc -n sourcestats -v
210 Number of sources = 1
.-- Number of sample points in measurement set.
/ .- Number of residual runs with same sign.
| /  .- Length of measurement set (time).
| | /      .- Est. clock freq error (ppm).
| | | /      .- Est. error in freq.
| | | | /      .- Est. offset.
| | | | |      On the -.
| | | | |      samples. \
| | | | |
Name/IP Address          NP  NR  Span Frequency Freq Skew Offset Std Dev
=====
_gateway                 11  5   710    -0.020    0.147  -1173ns   24us
```


6. Stop chrony on the **host03** VM because we do not need it in other practices.
 - a. Use the `systemctl` command to stop the chrony daemon, `chronyd`.

```
# systemctl stop chronyd
```

- b. Run the `chronyc tracking` command.

```
# chronyc tracking
506 Cannot talk to daemon
```

- Notice `chronyc` cannot talk to the `chronyd` daemon.

Practice 7-3: Automating Tasks with `cron`

Overview

In this practice, you use the `crontab` command and crontab files to automate tasks.

Assumptions

You are the `root` user on `host03`.

Tasks

1. Create a crontab for the `root` user.

- a. Use the `crontab -l` command to list the contents of your crontab file.

```
[root@host03 ~]# crontab -l
no crontab for root
```

- b. Use the `ls` command to view the contents of the `/var/spool/cron` directory.

```
# ls /var/spool/cron
```

- Note that there are no files in this directory.

- c. Use the `crontab -e` command to create a crontab job that runs the `ls` command every minute.

```
# crontab -e
* * * * * ls
```

- The `crontab -e` command uses the `vi` editor. Save your changes and exit `vi`.
- The following messages are displayed:
no crontab for root - using an empty one
crontab: installing new crontab

- d. Use the `ls` command to view the contents of the `/var/spool/cron` directory.

```
# ls /var/spool/cron
root
```

- Now there is a `root` file in the `/var/spool/cron` directory.

- e. Use the `cat` command to display the contents of the `/var/spool/cron/root` file.

```
# cat /var/spool/cron/root
* * * * * ls
```

- f. Use the `tail` command to view the results of your job which are sent to the cron log.

```
# tail /var/log/cron
...
... CROND[1342]: (root) CMD (ls)
... CROND[1340]: (root) CMDOUT (anaconda-ks.cfg)
... CROND[1368]: (root) CMD (ls)
... CROND[1366]: (root) CMDOUT (anaconda-ks.cfg)
```

- It is best to include stdout and stderr redirection with your command in crontab.
- Without redirection, output from cron jobs is typically emailed to the user. However, we do not have an email program (for example, `postfix`) installed, so `syslog` is used.

- g. Use the `crontab -r` command to remove crontab.

```
# crontab -r
# crontab -l
no crontab for root
```

- h. Use the `ls` command to view the contents of the `/var/spool/cron` directory.

```
# ls /var/spool/cron
```

- Note that the `root` file has been removed from this directory.

2. Create a system crontab that runs a job as the `oracle` user.

- a. Use the `cat` command to show the contents of `/etc/crontab`.

```
# cat /etc/crontab
SHELL=/bin/bash
PATH=/sbin:/bin:/usr/sbin:/usr/bin
MAILTO=root

# For details see man 4 crontabs

# Example of job definition:
# .----- minute (0 - 59)
# | .----- hour (0 - 23)
# | | .----- day of month (1 - 31)
# | | | .----- month (1 - 12) OR jan,feb,mar,apr ...
# | | | | .---- day of week (0 - 6) (Sunday=0 or 7) OR...
# | | | | |
# * * * * * user-name command to be executed
```

- Notice the format is very similar to the user crontab, but with the addition of a `user-name` field.

- b. Use the `vi` command to create a system crontab file that runs a simple command as `oracle` every other minute.

```
# vi /etc/cron.d/oracle-jobs
*/2 * * * * oracle /bin/touch /tmp/oracle-test
```

- It is good practice to create separate crontab files rather than edit `/etc/crontab`.

- c. Use the `tail` command to watch the cron log for your job running.

```
# tail -f /var/log/cron
... crontab[1389]: (root) DELETE (root)
... crontab[1390]: (root) LIST (root)
... CROND[1415]: (oracle) CMD (/bin/touch /tmp/oracle-test)
```

- You may need to wait a couple of minutes for your job to run

- d. Check the file was created in `/tmp`.

```
# ls -l /tmp/oracle-test
-rw-r--r--. 1 oracle oracle 0 Dec 17 15:55 /tmp/oracle-test
```

- The timestamp on the file should be updated every two minutes.

- e. Delete the `/etc/cron.d/oracle-jobs` file.

```
# rm /etc/cron.d/oracle-jobs
rm: remove regular file '/etc/cron.d/oracle-jobs'? y
```

Practice 7-4: Automating Tasks Using `systemd` Timers

Overview

In this practice, you use `systemd` timers to automate tasks.

Assumptions

You are the `root` user on `host03`.

Tasks

1. Examine installed `systemd` timers.

- a. Use the `systemctl list-units` command to list the installed timer units.

```
# systemctl list-units --type timer
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
dnf-makecache.timer                loaded active waiting dnf makecache --timer
systemd-tmpfiles-c...loaded active waiting Daily Cleanup...
unbound-anchor.timer               loaded active waiting daily update of...

LOAD    = Reflects whether the unit definition was properly...
ACTIVE  = The high-level unit activation state, i.e. genera...
SUB      = The low-level unit activation state, values depen...

3 loaded units listed. Pass --all to see loaded but inacti...
To show all installed unit files use 'systemctl list-unit-files'
```

- b. Use the `systemctl list-unit-files` command, as suggested.

```
# systemctl list-unit-files --type timer
UNIT FILE                          STATE
chrony-dnssrv@.timer               disabled
dnf-makecache.timer                enabled
fstrim.timer                       disabled
systemd-tmpfiles-clean.timer       static
unbound-anchor.timer               enabled
```

- The disabled timers did not appear in the previous command output.

- c. Use the `systemctl list-timers` command to view more detail about the timers.

```
# systemctl list-timers
```

For each timer the following are shown:

- NEXT: The date and time the timer is next due to trigger
- LEFT: The amount of time until the timer is next due to trigger
- LAST: The date and time the timer last triggered
- PASSED: The amount of time since the timer last triggered
- UNIT: The name of the timer unit
- ACTIVATES: The unit (typically a service) the timer runs when it is triggered

- d. Use the `cat` command to examine the `systemd-tmpfiles-clean` timer.

```
# cat /usr/lib/systemd/system/systemd-tmpfiles-clean.timer
...
[Unit]
Description=Daily Cleanup of Temporary Directories
Documentation=man:tmpfiles.d(5) man:systemd-tmpfiles(8)

[Timer]
OnBootSec=15min
OnUnitActiveSec=1d
```

- By default, the timer will start a service of the same name when it triggers.
- The `OnBootSec` directive starts the service 15 minutes after the system boots.
- The `OnUnitActiveSec` directive starts the service one day after it was last started.

- e. Use the `cat` command to examine the `fstrim` timer.

```
# cat /usr/lib/systemd/system/fstrim.timer
[Unit]
Description=Discard unused blocks once a week
Documentation=man:fstrim

[Timer]
OnCalendar=weekly
AccuracySec=1h
Persistent=true

[Install]
WantedBy=timers.target
```

- The `OnCalendar` directive starts the service once a week.
- The `AccuracySec` directive means the service will be started within 1 hour of becoming due.

2. Create a custom `systemd` timer.

- a. Use the `vi` command to create a simple service that updates a file's timestamp.

```
# vi /etc/systemd/system/alive.service
[Unit]
Description=Touch a file to prove system is alive

[Service]
Type=oneshot
ExecStart=/bin/touch /tmp/alive-check
```

- Note that custom files are placed in `/etc/systemd` and not `/usr/lib/systemd`.
- The type is `oneshot` so that the service is marked started after `/bin/touch` exits.

- b. Use the `vi` command to create a timer for our alive service.

```
# vi /etc/systemd/system/alive.timer
[Unit]
Description=Run alive service at every fifth minute

[Timer]
OnCalendar=*:0/5:00
AccuracySec=10s

[Install]
WantedBy=timers.target
```

- c. Use the `systemctl start` command to activate our new timer.

```
# systemctl start alive.timer
```

- Note: If you want the timer to start after a reboot, you should also enable it.

- d. Use the `systemctl status` command to check the status of our service and timer.

```
# systemctl status alive.service
• alive.service - Touch a file to prove system is alive
  Loaded: loaded (/etc/systemd/system/alive.service; static;...
  Active: inactive (dead)

# systemctl status alive.timer
• alive.timer - Run alive service at every fifth minute
  Loaded: loaded (/etc/systemd/system/alive.timer; static; ...
  Active: active (waiting) since ...
  Trigger: ... 1min 40s left

... systemd[1]: Started Run alive service at every fifth minute.
```

- Note: Your output will be different if the timer has already triggered.

- e. If you needed to edit your timer or service files you may see the message *"Warning: The unit file, source configuration file or drop-ins of alive.timer changed on disk. Run 'systemctl daemon-reload' to reload units."* If so, run `systemctl daemon-reload`.

```
# systemctl daemon-reload
```

- f. Repeat the `systemctl status` command or use `systemctl list-timers` until the timer triggers. Once triggered, you will see the alive service has been activated.

```
# systemctl status alive.service
• alive.service - Touch a file to prove system is alive
  Loaded: loaded (/etc/systemd/system/alive.service; static;...
  Active: inactive (dead) since ...
  Process: 1318 ExecStart=/bin/touch /tmp/alive-check (code=e...
  Main PID: 1318 (code=exited, status=0/SUCCESS)
...
```

- Note: The service is inactive (dead) because the touch command just takes a moment to complete.

- g. Use the `ls` command to check the service touched the file.

```
# ls -l /tmp/alive-check
-rw-r--r--. 1 root root 0 Dec 18 11:05 /tmp/alive-check
```

- Note the timestamp will update each time the timer activates the service.

- h. Use the `systemctl stop` command to stop the timer.

```
# systemctl stop alive.timer
```

3. Run a command at a future time using `systemd-run`.

- a. Use the `systemd-run` command to touch a file at a specified time.

```
# systemd-run --on-active=90 --timer-property=AccuracySec=100ms
/bin/ps -f
Running timer as unit: run-rID.timer
Will run service as unit: run-rID.service
```

- Transient timer and service units are created for the command.
- Note you will need these unit names to check the result of the request.

- b. Use the `journalctl` command to view log messages for the timer.

```
# journalctl -u run-rID.timer
-- Logs begin at date, end at date. --
... systemd[1]: Started /bin/ps -f.
... systemd[1]: run-rID.timer: Succeeded.
```

- Substitute the value for *ID* as reported by `systemd-run` in the previous step.
- You may need to wait a couple of minutes and repeat the command before you see the Succeeded message.

- c. Use the `journalctl` command to view log messages for the service.

```
# journalctl -u run-rID.service
-- Logs begin at date, end at date. --
...systemd[1]: Started /bin/ps -f.
...ps[1268]: UID  PID  PPID  C  STIME TTY      TIME CMD
...ps[1268]: root   1    0   0 10:55 ?       00:00:02 /usr/lib/s...
...ps[1268]: root   2    0   0 10:55 ?       00:00:00 [kthreadd]
...ps[1268]: root   3    2   0 10:55 ?       00:00:00 [rcu_gp]
...
```

- As it was not redirected elsewhere, the output of the `ps` command appears in the log records.

4. Use the `exit` command to log off **host03**.

```
# exit
logout
Connection to host03 closed.
```

You are now the `oracle` user on **gateway**.

**Practices for Lesson 8:
Kernel and Device
Configuration**

Practices for Lesson 8: Overview

Practices Overview

In these practices, you work with loadable kernel modules and some special file systems that give access to the kernel configuration.

You also investigate how special files are created for device access as the system boots.

Practice 8-1: Using Loadable Kernel Modules

Overview

In this practice, you perform the following tasks:

- Use kernel module utilities to list modules.
- Get detailed information about modules.
- Load and unload kernel modules.
- Explore the kernel module configuration directory.

Assumptions

- You are the `oracle` user on **gateway**.
- Sample output is provided. Kernel modules shown might differ from the actual output.

Tasks

1. Log on to **host03**.

Use `ssh` to log on to **host03**, providing the root password.

```
[oracle@gateway ~]$ ssh root@host03
root@host03's password:
Last login: ...
```

2. List kernel modules.

- a. Use the `lsmod` command to list kernel modules currently loaded into the kernel.

Your output may be different.

```
[root@host03 ~]# lsmod
Module                               Size  Used by
...
nft_fib_inet                        16384   1
nft_fib_ipv4                        16384   1 nft_fib_inet
nft_fib_ipv6                        16384   1 nft_fib_inet
...
```

- The “Size” column displays the amount of memory the module uses.
- The “Used by” column gives the total number of instances of the module in addition to any modules that depend on the module, with the names of modules that depend on the given module shown last.
- In the sample output, the `nft_fib_inet` module depends on the `nft_fib_ipv4` and `nft_fib_ipv6` modules.

- b. Use the `modinfo` command to list detailed information about a specific kernel module (for example, `nls_utf8`).

```
# modinfo nls_utf8
filename:          /lib/modules/5.4.17-
2011.7.4.el8uek.x86_64/kernel/fs/nls/nls_utf8.ko.xz
license:           Dual BSD/GPL
srcversion:        1979FB9C37ACEACE1F7BC35
...
```

- Notice that the kernel modules are loaded from the `/lib/modules/kernel_version/kernel` directory.

- c. Use the `uname -r` command to display the kernel version.

```
# uname -r
5.4.17-2011.7.4.el8uek.x86_64
```

- d. List the kernel modules for the kernel version, piping the output to the `less` command.

```
# ls -R /lib/modules/$(uname -r)/kernel | less
/lib/modules/5.4.17-2011.7.4.el8uek.x86_64/kernel:
arch
crypto
drivers
fs
kernel
lib
mm
net
sound
virt
vmlinux.ctfa

/lib/modules/5.4.17-2011.7.4.el8uek.x86_64/kernel/arch:
x86

/lib/modules/5.4.17-2011.7.4.el8uek.x86_64/kernel/arch/x86:
crypto
events
kernel
kvm
...
```

- Scroll through the output using navigation keys such as Page Up and Page Down.
- The actual kernel modules have a `.ko.xz` (kernel object with `xz` compression) extension.
- Enter "q" to return to the prompt.

3. Load kernel modules.

- a. Determine whether the `btrfs` kernel module is currently loaded.

```
# lsmod | grep btrfs
```

- In this example, the `btrfs` kernel module is not loaded.

- b. Use the `modprobe` command to load the `btrfs` kernel module.

```
# modprobe -v btrfs
insmod /lib/modules/5.4.17-
2011.7.4.el8uek.x86_64/kernel/lib/raid6/raid6_pq.ko.xz
insmod /lib/modules/5.4.17-
2011.7.4.el8uek.x86_64/kernel/lib/zstd/zstd_compress.ko.xz
insmod /lib/modules/5.4.17-
2011.7.4.el8uek.x86_64/kernel/lib/zstd/zstd_decompress.ko.xz
insmod /lib/modules/5.4.17-
2011.7.4.el8uek.x86_64/kernel/crypto/xor.ko.xz
insmod /lib/modules/5.4.17-
2011.7.4.el8uek.x86_64/kernel/fs/btrfs/btrfs.ko.xz
```

- The preceding example includes the `-v` (verbose) option.
 - Note that modules `raid6_pq`, `zstd_compress`, `zstd_decompress`, and `xor` are loaded in addition to the `btrfs` module. These modules must be loaded to support features of the `btrfs` file system.
 - Also note that `modprobe` uses the `insmod` command to load the modules.
- c. Confirm that the `btrfs` module is loaded along with the additional modules.

The sample output is provided. Yours might be different.

```
# lsmod | grep btrfs
btrfs                1253376  0
xor                   24576  1 btrfs
zstd_decompress       73728  1 btrfs
zstd_compress        155648  1 btrfs
raid6_pq              114688  1 btrfs
libcrc32c             16384  4 nf_conntrack,nf_nat,btrfs,xfs
```

- Note that the `btrfs`, `xor`, `zstd_decompress`, `zstd_compress`, and `raid6_pq` modules are loaded.
- The output also shows that the `btrfs` module depends on the `libcrc32c` module which was already loaded because it is also needed by the `nf_conntrack`, `nf_nat`, and `xfs` modules.

4. List module dependencies.

Kernel module dependencies are in `/lib/modules/kernel_version/modules.dep`.

a. Use the `grep` command to list the kernel module dependencies for `nfs`.

Include the `-w` option for `grep` to search for the whole word “`nfs`”.

```
# grep -w btrfs.ko.xz /lib/modules/$(uname -r)/modules.dep
kernel/fs/btrfs/btrfs.ko.xz: kernel/crypto/xor.ko.xz
kernel/lib/zstd/zstd_decompress.ko.xz
kernel/lib/zstd/zstd_compress.ko.xz
kernel/lib/raid6/raid6_pq.ko.xz kernel/lib/libcrc32c.ko.xz
```

- Note that the dependencies listed in the `modules.dep` file for `btrfs` correspond with the modules loaded when running the `modprobe -v btrfs` command.

5. Unload kernel modules.

Use the `modprobe -r` command to unload the `nfs` kernel module.

```
# modprobe -rv btrfs
rmmod btrfs
rmmod raid6_pq
rmmod zstd_compress
rmmod zstd_decompress
rmmod xor
```

- Do not unload or experiment with any other modules because you could potentially damage the operating system or leave the virtual machine in an unusable state.
- The preceding example includes the `-v` (verbose) option.
- Note that `modprobe -r` uses the `rmmod` command to remove the modules.
- Also note that the dependent modules are removed when possible. In this example, the `libcrc32c` module was not removed because other modules depend on it.

Practice 8-2: Exploring the `/proc` File System

Overview

In this practice, you explore the `proc` file system (directory), view various files and directories that represent the current state of the kernel, and change the value of current settings. Where needed, scroll through output using navigation keys such as Page Up and Page Down. Also where needed, enter "q" to terminate output and return to the prompt.

Assumptions

You are the `root` user on the `host03` VM.

Tasks

1. Explore the `proc` file system (directory).
 - a. Use the `ls` command to display the contents of the `/proc` directory.

Sample output is shown.

```
# ls /proc
1   17   29527   29749   341   497   acpi      kcore      slabinfo
10  18   29594   29763   382   5     buddyinfo keys       softirqs
11  19   29615   29770   39    507   bus       key-users  stat
...
```

- Numbered entries are directories referring to process IDs.
- b. Use the `ls` command to display entries without numerical names.

```
# ls -ld /proc/[a-z]* | less
dr-xr-xr-x.  ... root root  ... /proc/acpi
-r--r--r--.  ... root root  ... /proc/buddyinfo
dr-xr-xr-x.  ... root root  ... /proc/bus
-r--r--r--.  ... root root  ... /proc/cgroups
-r--r--r--.  ... root root  ... /proc/cmdline
-r--r--r--.  ... root root  ... /proc/console
-r--r--r--.  ... root root  ... /proc/cpuinfo
...
```

- Notice that some entries are files and some entries are directories.
- c. View the `meminfo` file to display information about RAM.

```
# less /proc/meminfo
MemTotal:      4018704 kB
MemFree:       3468368 kB
...
```

- d. View the `cpuinfo` file to display information about the processors used by your system.

```
# less /proc/cpuinfo
processor      : 0
vendor_id     : GenuineIntel
cpu_family    : 6
model         : 62
...
```

- e. View the `devices` file to display information about the various character and block devices currently configured.

```
# less /proc/devices
Character devices:
 1 mem
 4 /dev/vc/0
 4 tty
...
Block devices:
 2 fd
 9 md
11 sr
...
```

2. Explore process directories in the `proc` file system.
- a. Use the `ls` command to display only the directories in `/proc` with numerical names. Change to the `/proc` directory first. Sample output is shown.

```
# cd /proc
# ls -d [0-9]*
1      1305 16430 221 29452 29672 29737 29846 315 422 ...
10     1389 17      23 29517 29680 29741 29853 333 423 ...
11      14 18      24 29527 29684 29747 29883 336 428 ...
...
```

- b. Show that one process directory exists for each process running on your system.

```
# ls -d [0-9]* | wc -l
135
# ps -e | wc -l
136
```

- The `ps` command produces one more entry due to the column headings.

- c. View the contents of the process directory for PID=1.

```
# ls -l /proc/1
-r--r--r--. ... root root ... arch_status
dr-xr-xr-x. ... root root ... attr
-r----- . ... root root ... auxv
-r--r--r--. ... root root ... cgroup
--w----- . ... root root ... clear_refs
-r--r--r--. ... root root ... cmdline
-rw-r--r--. ... root root ... comm
-rw-r--r--. ... root root ... coredump_filter
-r--r--r--. ... root root ... cpuset
lrwxrwxrwx. ... root root ... cwd -> /
-r----- . ... root root ... environ
lrwxrwxrwx. ... root root ... exe -> /usr/lib/systemd/...
dr-x----- . ... root root ... fd
...
```

- Notice that some entries are files, some entries are directories, and some entries are symbolic links.

- d. Use the `less` command to display the status of PID=1.

```
# less /proc/1/status
Name:   systemd
Umask:  0000
State:  S (sleeping)
Tgid:   1
Ngid:   0
Pid:    1
PPid:   0
...
```

3. Change the values of kernel settings.

- a. Use the `cat` command to check the status of IP forwarding.

```
# cat /proc/sys/net/ipv4/ip_forward
0
```

- If the value is 0, IP forwarding is disabled; if set to 1, forwarding is enabled.
 - IP forwarding allows you to set up a Linux router or gateway, for example.
- b. If the value of IP forwarding is currently set to 1, use the `echo` command to set the value of IP forwarding to 0 and then change it back to 1. If the value of IP forwarding is currently set to 0, simply change it to 1 by only running the second `echo` command followed by the `cat` command.

```
# echo 0 > /proc/sys/net/ipv4/ip_forward
# cat /proc/sys/net/ipv4/ip_forward
0
# echo 1 > /proc/sys/net/ipv4/ip_forward
# cat /proc/sys/net/ipv4/ip_forward
1
```

- c. Use the `cat` command to view the local port range used by TCP and UDP traffic.

```
# cat /proc/sys/net/ipv4/ip_local_port_range
32768    60999
```

- d. If the port range is not 32768 60999, use the `echo` command to define the local port range used by TCP and UDP traffic to be 32768 60999.

```
# echo "32768 60999" > /proc/sys/net/ipv4/ip_local_port_range
```

- Notice that the first local port and the last local port allowed are surrounded by quotation marks.

Practice 8-3: Exploring the `sysfs` File System

Overview

In this practice, you explore the `sysfs` file system. You view the virtual block devices and virtual interfaces, and determine which power states are supported on your system.

Assumptions

You are the `root` user on the **host03** VM.

Tasks

1. Explore the `/sys` directory.
 - a. Use the `ls` command to display the contents of the `/sys` directory.

```
# ls -l /sys
drwxr-xr-x. ... block
drwxr-xr-x. ... bus
drwxr-xr-x. ... class
drwxr-xr-x. ... dev
drwxr-xr-x. ... devices
drwxr-xr-x. ... firmware
drwxr-xr-x. ... fs
drwxr-xr-x. ... hypervisor
drwxr-xr-x. ... kernel
drwxr-xr-x. ... module
drwxr-xr-x. ... power
```

- b. Use the `ls` command to display the virtual disk block (`xvd`) devices on your system.

```
# ls -l /sys/block | grep xvd
lrwxrwxrwx. ... xvda -> ../devices/vbd-51712/block/xvda
lrwxrwxrwx. ... xvdb -> ../devices/vbd-51728/block/xvdb
lrwxrwxrwx. ... xvdd -> ../devices/vbd-51760/block/xvdd
```

- c. Traverse the `/sys/bus` directory and display the virtual interface (`vif`) devices.

```
# cd /sys/bus/xen/devices
# ls vif*
vif-0:
devtype driver modalias net nodename power state
subsystem uevent

vif-1:
devtype driver modalias net nodename power state
subsystem uevent
```

- d. Display the operational state, MAC address, and MTU of `eth0`.

```
# cd vif-0/net/eth0
# pwd
/sys/bus/xen/devices/vif-0/net/eth0
# ls
...
address  device  flags  link_mode  phys_port_id ...
...
# cat operstate
up
# cat address
00:16:3e:02:01:01
# cat mtu
1500
```

- e. Determine which power states are supported.

```
# cd /sys/power
# ls
disk  image_size  mem_sleep...  state ...
# cat state
freeze  mem  disk
```

The possible states and their meanings are as follows:

- `mem`: The following possible strings in the `/sys/power/mem_sleep` file determine action when `mem` is written to `/sys/power/state`:
 - `s2idle`: Suspend-To-Idle
 - `shallow`: Power-On Suspend
 - `deep`: Suspend-To-RAM
 - `standby`: Power-On Suspend
 - `freeze`: Suspend-To-Idle
 - `disk`: Suspend-To-Disk
 - See the following for more information about Power Management Sleep States:
<https://www.kernel.org/doc/Documentation/power/states.txt>.
- f. View the `/sys/fs/xfs` directory to display the xfs file systems currently configured.

```
# ls /sys/fs/xfs
stats  xvda1  xvda2  xvda3
```

Practice 8-4: Using the `sysctl` Utility

Overview

In this practice, you use the `sysctl` utility and view the `/etc/sysctl.d/99-sysctl.conf` configuration file.

Assumptions

You are the `root` user on the **host03** VM.

Tasks

1. Use the `sysctl` utility.
 - a. Use the `sysctl` command to disable IP forwarding.

```
# cat /proc/sys/net/ipv4/ip_forward
1
# sysctl -w net.ipv4.ip_forward=0
net.ipv4.ip_forward = 0
# cat /proc/sys/net/ipv4/ip_forward
0
```

- Setting the `ip_forward` variable to 0 disables IP forwarding.

- b. Use the `sysctl` command to enable IP forwarding.

```
# sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
# cat /proc/sys/net/ipv4/ip_forward
1
```

- Setting the `ip_forward` variable to 1 enables IP forwarding.

- c. Use the `sysctl -a | less` command to display the current kernel settings, piping the output to the `less` command.

```
# sysctl -a | less
...
debug.exception-trace = 1
...
dev.raid.speed_limit_min = 1000
dev.scsi.logging_level = 0
...
fs.inode-nr = 30454      493
fs.inode-state = 30454  493      0      0      0      0      0
...
kernel.sched_child_runs_first = 0
kernel.sched_latency_ns = 12000000
...
net.ipv4.route.gc_thresh = -1
net.ipv4.route.gc_timeout = 300
...
net.ipv6.neigh.default.mcast_solicit = 3
net.ipv6.neigh.default.proxy_delay = 80
...
net.netfilter.nf_log.0 = NONE
net.netfilter.nf_log.1 = NONE
...
vm.page-cluster = 3
vm.panic_on_oom = 0
...
```

- Scroll through the output using navigation keys such as Page Up and Page Down. Enter "q" to return to the prompt.

2. Use the `less` command to view the contents of the `/usr/lib/sysctl.d/50-default.conf` file.

```
# less /usr/lib/sysctl.d/50-default.conf
# This file is part of systemd.
...
# To override settings in this file, create a local file in /etc
# (e.g. /etc/sysctl.d/90-override.conf), and put any assignments
# there.

# System Request functionality of the kernel (SYNC)
...
kernel.sysrq = 16
```

- Some `sysctl` settings can be found in `/usr/lib/sysctl.d/name.conf` files.
 - Do not modify files in the `/usr/lib/sysctl.d` directory.
 - As noted in the file comments, changes should be added to files in `/etc/sysctl.d/name.conf` with a name that is lexically processed after the one found in `/usr/lib/sysctl.d`.
3. Use the `cat` command to view the contents of the `/etc/sysctl.d/99-sysctl.conf` file.

```
# cat /etc/sysctl.d/99-sysctl.conf
# sysctl settings are defined through files in
# /usr/lib/sysctl.d/, /run/sysctl.d/, and /etc/sysctl.d/.
#
# Vendors settings live in /usr/lib/sysctl.d/.
# To override a whole file, create a new file with the same in
# /etc/sysctl.d/ and put new settings there. To override
# only specific settings, add a file with a lexically later
# name in /etc/sysctl.d/ and put new settings there.
#
# For more information, see sysctl.conf(5) and sysctl.d(5).
```

- Changes made using either `echo` or `sysctl` are lost when the system is rebooted.
- To preserve custom settings, add them to a `/etc/sysctl.d/name.conf` file.
- Values that are added to these files take effect each time the system boots.
- To immediately enable changes added to files in `/etc/sysctl.d/name.conf`, run `sysctl -p /etc/sysctl.d/name.conf`, providing the file name.

Practice 8-5: Exploring and Configuring udev Rules

Overview

In this practice, you:

- Explore udev files and directories
- Query the udev database
- Create a udev rule to create a symbolic link to a device

Assumptions

You are the `root` user on **host03**.

Tasks

1. Explore udev.
 - a. Use the `rpm -ql` command to view the files included with the `systemd-udev` RPM package. There are many, so you may want to pipe the result through `less`.

```
# rpm -ql systemd-udev | less
/etc/kernel
/etc/kernel/install.d
/etc/modules-load.d
/etc/udev
/etc/udev/hwdb.bin
/etc/udev/hwdb.d
/etc/udev/rules.d
...
```

- b. Use the `ls` command to view existing udev rules files in the `/usr/lib/udev/rules.d` and `/etc/udev/rules.d` directories.

Sample output is shown.

```
# ls /usr/lib/udev/rules.d
10-dm.rules              66-kpartx.rules
11-dm-parts.rules        68-del-part-nodes.rules
13-dm-disk.rules         69-bcache.rules
...
# ls /etc/udev/rules.d
70-persistent-ipoib.rules  99-vmware-scsi-timeout.rules
```

- Files on your system might be different. The `/etc/udev/rules.d` directory may be empty.

- c. Use the `less` command to view the `/usr/lib/udev/rules.d/50-udev-default.rules` file.

Page through the file. Press `q` to return to the command prompt.

Note the operators:

`==`: Compare for equality

`=:` Assign a value to a key

`+=`: Add the value to the current values for the key

```
# less /lib/udev/rules.d/50-udev-default.rules
# do not edit this file, it will be overwritten on update

# run a command on remove events
ACTION=="remove", ENV{REMOVE_CMD}!="", RUN+=" $env{REMOVE_CMD}"
ACTION=="remove", GOTO="default_end"

SUBSYSTEM=="virtio-ports", KERNEL=="vport*", ATTR{name}=="?*",
SYMLINK+="virtio-ports/$attr{name}"

# select "system RTC" or just use the first one
SUBSYSTEM=="rtc", ATTR{hctosys}=="1", SYMLINK+="rtc"
SUBSYSTEM=="rtc", KERNEL=="rtc0", SYMLINK+="rtc",
OPTIONS+="link_priority=-100"

SUBSYSTEM=="usb", ENV{DEVTYPE}=="usb_device",
IMPORT{builtin}="usb_id", IMPORT{builtin}="hwdb --subsystem=usb"
ENV{MODALIAS}!="", IMPORT{builtin}="hwdb --
subsystem=$env{SUBSYSTEM}"

ACTION!="add", GOTO="default_end"

SUBSYSTEM=="tty", KERNEL=="ptmx", GROUP="tty", MODE="0666"
SUBSYSTEM=="tty", KERNEL=="tty", GROUP="tty", MODE="0666"
SUBSYSTEM=="tty", KERNEL=="tty[0-9]*", GROUP="tty", MODE="0620"
SUBSYSTEM=="tty", KERNEL=="sclp_line[0-9]*", GROUP="tty",
MODE="0620"
...
```

2. Query the udev database.

Sample output is shown.

- a. Use the `udevadm` command to query the Udev database for all device information for `/dev/xvdd`.

```
# udevadm info --query=all --name=/dev/xvdd
P: /devices/vbd-51760/block/xvdd
N: xvdd
E: DEVNAME=/dev/xvdd
E: DEVPATH=/devices/vbd-51760/block/xvdd
E: DEVTYPE=disk
E: MAJOR=202
E: MINOR=48
E: SUBSYSTEM=block
E: TAGS=:systemd:
E: USEC_INITIALIZED=75531
```

- b. Use the `udevadm` command to query the Udev database for the device path of `/dev/xvdd`.

```
# udevadm info --query=path --name=/dev/xvdd
/devices/vbd-51760/block/xvdd
```

- c. Use the `udevadm` command to print all `sysfs` properties of `/dev/xvdd`.

```
# udevadm info --attribute-walk --name=/dev/xvdd
Udevadm info starts with the device specified by the devpath and
then walks up the chain of parent devices. It prints for every
device found, all possible attributes in the udev rules key
format. A rule to match, can be composed by the attributes of
the device and the attributes from one single parent device.

looking at device '/devices/vbd-51760/block/xvdd':
    KERNEL=="xvdd"
    SUBSYSTEM=="block"
    DRIVER==" "
    ATTR{alignment_offset}=="0"
    ATTR{capability}=="10"
...
looking at parent device '/devices/vbd-51760':
    KERNELS=="vbd-51760"
    SUBSYSTEMS=="xen"
    DRIVERS=="vbd"
    ATTRS{devtype}=="vbd"
    ATTRS{nodename}=="device/vbd/51760"
    ATTRS{state}=="Connected"
```

3. Create a symbolic link to a device node.

- a. Use the `vi` editor to create the `/etc/udev/rules.d/10-local.rules` file as follows, using the `KERNEL` and `SUBSYSTEM` values from the previous `udevadm info --attribute-walk` command.

```
# vi /etc/udev/rules.d/10-local.rules
KERNEL=="xvdd", SUBSYSTEM=="block", SYMLINK="my_disk"
```

- The `SYMLINK` directive names the new symlink for the device.

- b. Run the `udevadm trigger` command to manually force `udev` to trigger rules.

```
# udevadm trigger
```

- c. Use the `ls -l` command to list the `/dev/my*` devices.

```
# ls -l /dev/my*
lrwxrwxrwx. 1 root root 4 ... /dev/my_disk -> xvdd
```

- Note that `/dev/my_disk` is a symlink to `/dev/xvdd`.

- d. Use the `udevadm info` command to query the `udev` database for the symlinks for `/dev/xvdd`.

```
# udevadm info --query=symlink --name=/dev/xvdd
my_disk
```

4. Remove the `/dev/my_disk` symlink.

- a. Use the `rm` command to remove the `/etc/udev/rules.d/10-local.rules` file.

```
# rm /etc/udev/rules.d/10-local.rules
rm: remove regular file '/etc/udev/rules.d/10-local.rules'? y
```

- b. Run the `udevadm trigger` command to manually force `Udev` to trigger rules.

```
# udevadm trigger
```

- c. Use the `ls` command to list the `/dev/my*` devices.

```
# ls /dev/my*
ls: cannot access /dev/my*: No such file or directory
```

- Note that `/dev/my_disk` no longer exists.

5. Use the `exit` command to log off `host03`.

```
# exit
logout
Connection to host03 closed.
```

You are now the `oracle` user on `gateway`.

Practices for Lesson 9: Package Management

Practices for Lesson 9: Overview

Practices Overview

In these practices, you:

- Use the `rpm` utility to perform software package management
- Access the Oracle Linux yum server
- Create a local `yum` repository and use the `dnf` utility
- Install and use packages from a `dnf` module

Practice 9-1: Using the rpm Utility

Overview

In this practice, you use the `rpm` utility to query installed packages, install packages, upgrade packages, and remove packages.

Assumptions

- You are logged on to **gateway** as the `oracle` user.
- Sample output is provided. Package names and versions shown might differ from the actual output.

Tasks

1. Log on to **host02**.

Connect to **host02** using the `ssh` command.

```
[oracle@gateway ~]$ ssh root@host02
Password:
[root@host02 ~]#
```

2. Query packages by using `rpm`.

- a. Query all installed packages.

```
# rpm -qa
libldb-2.1.3-2.el8.x86_64
tzdata-2020a-1.el8.noarch
rpm-plugin-selinux-4.14.3-4.el8.x86_64
xkeyboard-config-2.28-1.el8.noarch
passwd-0.80-3.el8.x86_64
pcre2-10.32-2.el8.x86_64
python3-linux-procfs-0.6.2-2.el8.noarch
basesystem-11-5.el8.noarch
dbus-glib-0.110-2.el8.x86_64
glibc-langpack-en-2.28-127.0.1.el8.x86_64
...
```

- b. Query whether a specific package (`bash`, for example) is installed.

```
# rpm -q bash
bash-4.4.19-12.el8.x86_64
```

- c. Get detailed information about the `bash` package.

```
# rpm -qi bash
Name       : bash
Version    : 4.4.19
Release    : 12.el8
Architecture: x86_64
Install Date: ...
Group      : Unspecified
Size       : 6861588
License    : GPLv3+
...
```

- d. List the files in the `bash` package.

```
# rpm -ql bash
/etc/skel/.bash_logout
/etc/skel/.bash_profile
/etc/skel/.bashrc
/usr/bin/alias
/usr/bin/bash
...
```

- e. Perform a reverse search. That is, determine what package the `/etc/sysconfig` file belongs to.

```
# rpm -qf /etc/sysconfig
filesystem-3.8-3.el8.x86_64
```

- f. List the configuration files associated with the `cups` package.

```
# rpm -qc tuned
/etc/tuned/active_profile
/etc/tuned/bootcmdline
/etc/tuned/post_loaded_profile
/etc/tuned/profile_mode
/etc/tuned/tuned-main.conf
```

3. Install packages by using `rpm`.

- a. Use the `mkdir -p` command to create the `/localrepo/media` directory. Then use the `mount` command to mount the Oracle Linux installation media, available in the `/stage/ISO` directory.

```
# mkdir -p /localrepo/media
# mount -o ro /stage/ISO/OracleLinux-R8-U3-x86_64-dvd.iso
/localrepo/media
```

- b. Add the following entry to the end of the `/etc/fstab` file to mount the ISO if the system is rebooted (the `/etc/fstab` file is discussed in the lesson titled “Partitions, File Systems and Swap”):

```
# vi /etc/fstab
...
/stage/ISO/ /localrepo/media iso9660 ro,nosuid,nodev 0 0
```

- c. Use the `cd` command to change to the `/localrepo/media` directory.

```
# cd /localrepo/media
```

- d. Use the `ls` command to list the contents of the directory.

```
# ls
AppStream BaseOS EFI EULA extra_files.json GPL ...
```

- Notice the `AppStream` and `BaseOS` subdirectories. These contain packages from their corresponding repositories.

- e. Use the `cd` command to change to the `BaseOS/Packages` directory, and then list the contents of the directory.

```
# cd BaseOS/Packages
# ls
aajohan-comfortaa-fonts-3.001-2.el8.noarch.rpm
acl-2.2.53-1.el8.x86_64.rpm
acpica-tools-20180629-3.el8.x86_64.rpm
...
zlib-devel-1.2.11-16.el8_2.i686.rpm
zlib-devel-1.2.11-16.el8_2.x86_64.rpm
zsh-5.5.1-6.el8_1.2.x86_64.rpm
```

- f. Verify that the `zsh` package (Z-Shell) is not already installed.

```
# zsh
-bash: zsh: command not found
# rpm -q zsh
package zsh is not installed
```

- In this example, the `zsh` package is not installed.

- g. Verify whether a GPG key has been imported.

```
# rpm -q gpg-pubkey
gpg-pubkey-...
```

- In this case, a GPG key has been imported. If you do not see `gpg-pubkey-...` as a response, you will get a `NOKEY` warning when installing `zsh` in the next step. This is not a concern for the purposes of this practice, and a GPG key will be imported in a later task.

- h. Install the `zsh` package by using `rpm`.

```
# rpm -Uvh zsh-5.5.1-6.el8_1.2.x86_64.rpm
warning: zsh-5.5.1-6.el8_1.2.x86_64.rpm: Header V3 RSA/SHA256
Signature, key ID ad986da3: NOKEY
Verifying...                               ##### [100%]
Preparing...                               ##### [100%]
Updating / installing...
 1:zsh-5.5.1-6.el8_1.2  ##### [100%]
```

- i. Verify that the `zsh` package is now installed.

```
# rpm -q zsh
zsh-5.5.1-6.el8_1.2.x86_64
```

- j. Run the `zsh` command and then display the process ID of `zsh`.

```
[root@host02 Packages]# zsh
[root@host02]/localrepo/media/BaseOS/Packages# ps
  PID  TTY          TIME CMD
 12047 pts/0      00:00:00 bash
 12616 pts/0      00:00:00 zsh
 12635 pts/0      00:00:00 ps
```

- k. Use the `exit` command to log out of `zsh`.

```
[root@host02]/localrepo/media/BaseOS/Packages # exit
[root@host02 Packages]# ps
  PID  TTY          TIME CMD
 12047 pts/0      00:00:00 bash
 12697 pts/0      00:00:00 ps
```

4. Remove packages by using `rpm`.

- a. Remove the `zsh` package.

```
# rpm -e zsh
```

- b. Verify that the `zsh` package has been removed.

```
# rpm -q zsh
package zsh is not installed
# zsh
-bash: /usr/bin/zsh: No such file or directory
```

Practice 9-2: Accessing the Oracle Linux Yum Server

Overview

In this practice, you review the configuration required to access the Oracle Linux yum server from the **host02** VM. You then simulate upgrading your system (**you do not actually perform the upgrade**) from the Oracle Linux yum server.

Assumptions

- You are logged in using `ssh` as the `root` user on **host02**.
- You are the `oracle` user on **gateway**.

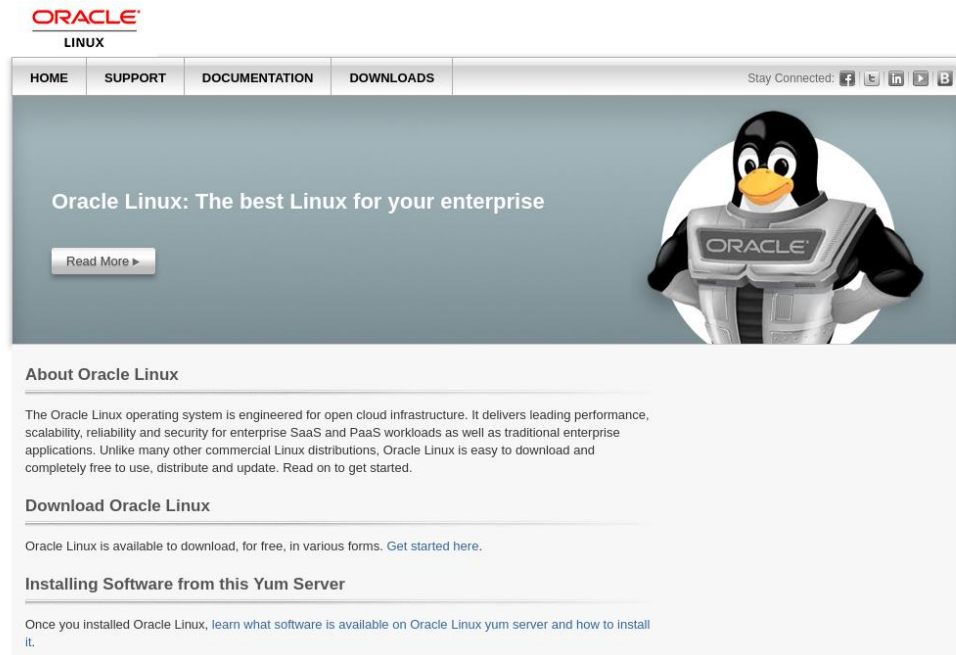
Tasks

1. Access the Oracle Linux yum server URL.
 - a. On **gateway**, open Firefox by double-clicking the **Firefox** icon on the desktop.



- b. In the browser, enter the URL for the Oracle Linux yum server:
<https://yum.oracle.com>.

The page looks like the following:



- The page contains a link to steps to manually configure your system to receive updates and additional software from the Oracle Linux yum server if necessary.

- c. Scroll down if needed. Click "learn what software is available..." in the paragraph which begins "Once you have installed Oracle Linux..." to view the information.

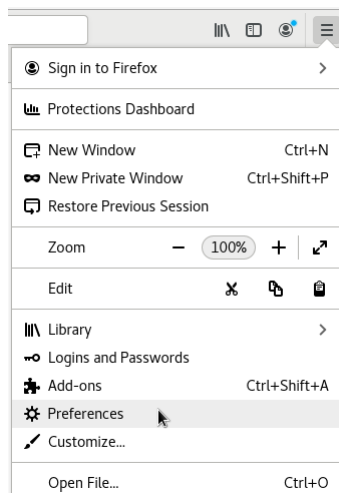
Installing Software from Oracle Linux Yum Server

Oracle Linux yum server hosts many different types of software in repositories for which the configuration is installed and updated via release packages. Below is a list of available software and the corresponding release packages to configure yum.

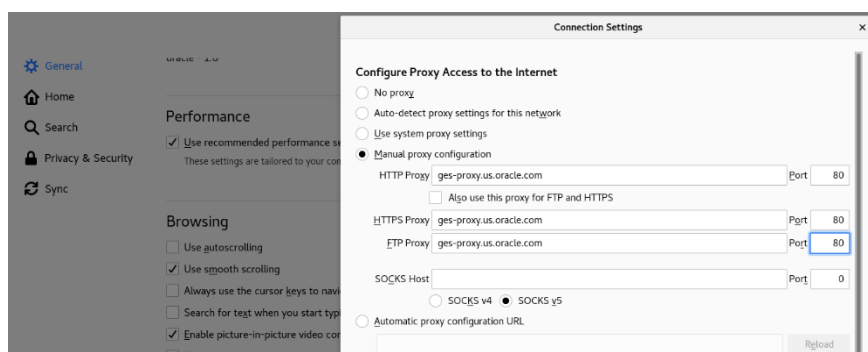
Available Software

Software	Release RPM
Oracle Linux, UEK & Addons. In addition: - For OL7: Virtualization tools, Oracle® Container Runtime for Docker and Container Services for use with Kubernetes, etc. - For OL8: BaseOS, Appstream	oraclelinux-release-el8, oraclelinux-release-el7, oraclelinux-release-el6
Oracle Linux patch repositories (for Oracle Cloud Infrastructure customers only)	oraclelinux-patchonly-release-el7, oraclelinux-patchonly-release-el6
Software Collection Library for Oracle Linux	oracle-softwarecollection-release-el7, oracle-softwarecollection-release-el6

- This is for information only; you do not need to run any commands from this page.
2. Determine the **HTTP Proxy** server on **gateway**.
 - a. Click the menu icon in the upper-right corner of the browser window and select **Preferences** to display the Firefox Preferences window.



- b. Scroll to the bottom of the **General** section in the Firefox Preferences where you will find the **Network Settings** subsection.
 - c. Click the **Settings** button. The Connection Settings dialog box appears.



- d. Note the **HTTP Proxy** setting.
 - In this example, the setting is `ges-proxy.us.oracle.com` on Port 80.
- e. Click **Cancel** to close the **Connection Settings** window.
- f. Click **X** in the Firefox **Preferences** tab to close it.
- g. Click **X** in the upper-right corner of the browser window to close the browser on gateway.

All remaining commands are issued from **host02**.

3. Set the proxy variable in the `/etc/dnf/dnf.conf` file.

- a. Back up `/etc/dnf/dnf.conf` first.

```
[root@host02 ...]# cp /etc/dnf/dnf.conf /etc/dnf/dnf.conf_SAV
```

- b. Use the `vi` command to add the proxy variable definition to the `/etc/dnf/dnf.conf` file.

- Add the entry at the bottom of the `/etc/dnf/dnf.conf` file.
- This proxy variable is needed for `dnf` to access the Oracle Linux yum server from within the practice environment network.

```
# vi /etc/dnf/dnf.conf
proxy=http://ges-proxy.us.oracle.com:80
```

4. Review the Oracle Linux yum server repository file on **host02**.

- a. Use the `cd` command to change to the `/etc/yum.repos.d` directory. Use the `ls` command to display the contents of the directory.

```
# cd /etc/yum.repos.d
# ls
oracle-linux-ol8.repo  uek-ol8.repo
```

- Note that these files already exist in this directory as part of the system installation.

- b. Use the `less` command to view the `oracle-linux-ol8.repo` file.

```
# less oracle-linux-ol8.repo
[ol8_baseos_latest]
...
enabled=1
...
[ol8_appstream]
...
enabled=1
...
```

- Your system is preconfigured to access the latest Oracle Linux 8 repository on the Oracle Linux yum server, `[ol8_baseos_latest]`, and the Oracle Linux 8 app stream repository, `[ol8_appstream]`.
- These are enabled by the `enabled=1` setting.
- All other repositories are disabled by the `enabled=0` setting.

- c. Use the `cat` command to view the `uek-ol8.repo` file.

```
# cat oracle-linux-ol8.repo
[ol8_UEKR6]
name=Latest Unbreakable Enterprise Kernel Release 6 for...
baseurl=https://yum$ociregion.oracle.com/repo/OracleLinux/OL8...
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-oracle
gpgcheck=1
enabled=0

[ol8_UEKR6_RDMA]
name=Oracle Linux 8 UEK6 RDMA ($basearch)
baseurl=https://yum$ociregion.oracle.com/repo/OracleLinux/OL8...
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-oracle
gpgcheck=1
enabled=0
```

- Your system is preconfigured to access the latest UEK R6 kernel for Oracle Linux 8 repository on the Oracle Linux yum server, `[ol8_UEKR6]`.
- This is enabled by the `enabled=1` setting.
- All other repositories are disabled by the `enabled=0` setting.

- d. Run the `dnf repolist` command to list the configured repositories.

```
# dnf repolist
repo id          repo name
ol8_UEKR6        Latest Unbreakable Enterprise Kernel Releas...
ol8_appstream    Oracle Linux 8 Application Stream (x86_64)
ol8_baseos_latest Oracle Linux 8 BaseOS Latest (x86_64)
```

- Note that the three repositories correspond to the three “`enabled=1`” settings in the `*.repo` files.

5. Run the `dnf check-upgrade` command to display available updates.

DO NOT APPLY ANY UPDATES!

```
# dnf check-upgrade
Oracle Linux 8 BaseOS Latest (x86_64)...
Oracle Linux 8 Application Stream (x86_64)...
Latest Unbreakable Enterprise Kernel Release 6 for Oracle...
Last metadata expiration check: ...

NetworkManager.x86_64 1:1.26.0-12.0.1.el8_3 ol8_baseos_latest
...
tzdata.src             2020f-1.el8             ol8_appstream
xfsprogs.x86_64        5.4.0-1.0.1.el8         ol8_UEKR6
```

- You should see that new packages are available to be updated or installed.
- ***Do not use `dnf upgrade` to apply updates. It could take several hours to do so.***

Practice 9-3: Creating a Local Yum Repository

Overview

In this practice, you configure a `repo` file to access the packages on the Oracle Linux install DVD. You then install and use the `createrepo` utility to create a local `yum` repository for additional `rpm` files copied from **gateway**, and create a `repo` file to enable the local repository.

Assumptions

- You are the `root` user on the **host02** VM.
- The Oracle Linux `dvd.iso` image is mounted on `/localrepo/media/`.
- This local `yum` repository is needed in subsequent practices.

Tasks

1. Disable the Oracle Linux `yum` server repositories.

- a. Run the `dnf clean all` command to clean up the `dnf` cache.

```
# dnf clean all
32 files removed
```

- b. Use the `dnf config-manager` command to disable all three enabled repositories.

```
# dnf config-manager --disable ol8_baseos_latest
# dnf config-manager --disable ol8_appstream
# dnf config-manager --disable ol8_UEKR6
```

- Note that the commands produce no output if successful.
- c. To ensure all repositories are disabled, use the `grep` command and search for the string `"enabled"` in the `/etc/yum.repos.d/*.repo` files.

```
# grep enabled /etc/yum.repos.d/*.repo
enabled=0
enabled=0
enabled=0
enabled=0
enabled=0
enabled=0
enabled=0
enabled=0
```

- In this example, all repositories are disabled (`enabled=0`).
- Repeat the above tasks, if necessary, to ensure that all repositories are disabled.

2. Ensure that the Oracle Linux `dvd.iso` image is mounted on `/localrepo/media`, as set up previously.
 - a. Use the `df` command to display the mounted file systems.

```
# df -h
Filesystem              Size  Used Avail Use% Mounted on
...
/dev/sr0                 8.7G  8.7G    0 100% /localrepo/media
...
```

- Note that the Oracle Linux media is still mounted on `/localrepo/media`.
- b. Verify this by using the `ls` command to list the contents of `/localrepo/media`.

```
# ls /localrepo/media
AppStream  BaseOS  EFI  EULA  extra_files.json  GPL  images
isolinux  media.repo  RELEASE-NOTES-en  RELEASE-NOTES-en.html
RPM-GPG-KEY  RPM-GPG-KEY-oracle  TRANS.TBL
```

- Note that the contents of the directory reflect the Oracle Linux ISO contents, as seen previously.
3. Create local repositories for the DVD contents.
 - a. Use the `vi` command to create a `repo` file `/etc/yum.repos.d/local.repo`.

```
# vi /etc/yum.repos.d/local.repo
[ol8_baseos_local]
name=Oracle Linux 8 BaseOS Local ($basearch)
baseurl=file:///localrepo/media/BaseOS
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-oracle
gpgcheck=1
enabled=1

[ol8_appstream_local]
name=Oracle Linux 8 Application Stream Local ($basearch)
baseurl=file:///localrepo/media/AppStream
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-oracle
gpgcheck=1
enabled=1
```

- b. Use the `dnf repoinfo` command to verify the new repositories.

```
# dnf repoinfo
...
Repo-id           : ol8_appstream_local
Repo-name         : Oracle Linux 8 Application Stream Local
                   (x86_64)
...
Repo-filename     : /etc/yum.repos.d/local.repo

Repo-id           : ol8_baseos_local
Repo-name         : Oracle Linux 8 BaseOS Local (x86_64)
...
Repo-filename     : /etc/yum.repos.d/local.repo
Total packages: 7,480
```

4. Copy extra packages from **gateway** machine.
- a. Open a new terminal window, if required, on **gateway** to upload the files.
- b. Use the `scp` command to check whether the `createrepo_c` package is installed.

```
[oracle@gateway ~]$ scp -r /stage/extras root@host02:/localrepo
root@host02's password:
...
```

- c. You can close the terminal window on **gateway** now and return to **host02**.

5. Install tools to create a local repository.

Use the `dnf` command to install the `createrepo_c` package and its dependencies.

```
[root@host02 ~]# dnf install -y createrepo_c
Last metadata expiration check: ...
Dependencies resolved.
=====
Package           Arch... Version           Repository...
=====
Installing:
createrepo_c      x86_64  0.15.11-2.el8     ol8_appstream_local...
Installing dependencies:
createrepo_c-libs x86_64  0.15.11-2.el8     ol8_appstream_local...
drpm              x86_64  0.4.1-3.el8       ol8_appstream_local...

Transaction Summary
=====
Install 3 Packages
...
Complete!
```

- Note that the packages are installed from the `ol8_appstream_local` repo.

6. Create the local repository.

- a. Change to the `/localrepo/extras` directory.

```
# cd /localrepo/extras
# ls
Packages
```

- b. Use the `createrepo` command to create a repository of the current directory.

- This command takes several minutes to complete.
- Include the “.” argument to represent the current directory.

```
# createrepo .
Directory walk started
Directory walk done - 12 packages
Temporary output repo path: ../.repodata/
Preparing sqlite DBs
Pool started (with 5 workers)
Pool finished
```

- c. View the results of the `createrepo` command.

```
# pwd
/localrepo/extras
# ls -l
drwx-r-x-r-x.    ...    Packages
drwx-r-x-r-x.    ...    repodata
```

- Notice that the **repodata** directory has been created.

- d. View the contents of the `repodata` directory.

```
# ls -l repodata
-rw-r--r--    ... -other.sqlite.bz2
-rw-r--r--    ... -other.xml.gz
-rw-r--r--    ... -filelists.xml.gz
-rw-r--r--    ... -primary.xml.gz
-rw-r--r--    ... -primary.sqlite.bz2
-rw-r--r--    ... -filelists.sqlite.bz2
-rw-r--r--    ... repomd.xml
```

- e. Use the `cd` command to change to the `yum` repository directory. Use the `vi` editor to update the `local.repo` file, adding the entry for this new repository to the end.

```
# cd /etc/yum.repos.d
# vi local.repo
...
gpgcheck=1
enabled=1

[extras]
name=Oracle Linux Extras
baseurl=file:///localrepo/extras
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-oracle
gpgcheck=1
enabled=1
```

- f. In this example, the packages are signed with the Oracle GPG key installed in `/etc/pki/rpm-gpg`. If the packages were signed with another key, it could be installed using the `rpm --import` command. For example:

```
# rpm --import RPM-GPG-KEY-thirdparty
```

- Do not run this command.

- g. Use the `dnf repoinfo` command to list the configured repositories.

- Note that the `extras` repository is now listed.

```
# dnf repoinfo extras
...
Repo-id           : extras
Repo-name         : Oracle Linux Extras
Repo-status       : enabled
...
Repo-filename     : /etc/yum.repos.d/local.repo
Total packages: ...
```

Practice 9-4: Using the `dnf` Utility

Overview

In this practice, you use the `dnf` utility to list both installed packages and packages available to be installed, install a package, check for packages with updates available, and remove a package. You install the *Security Tools* package group and verify the availability of tools from the group. Finally, you remove packages from the *Security Tools* package group.

Assumptions

- This practice assumes that you have a local `yum` repository configured.
- You are the `root` user on **host02**.

Tasks

The following tasks are completed using the **host02** VM.

1. List packages by using `dnf`.
 - a. List all packages installed on your system and all packages available in all configured repositories. You may wish to pipe the output through `less`.

```
[root@host02 ~]# dnf list
Last metadata expiration check: ...
Installed Packages
NetworkManager.x86_64      1:1.26.0-8.0.1.el8   @anaconda
...
yum.noarch                 4.2.23-4.el8         @anaconda
zlib.x86_64                1.2.11-16.el8_2      @anaconda
Available Packages
CUnit.i686                 2.1.3-17.el8         ol8_appstream_local
CUnit.x86_64               2.1.3-17.el8         ol8_appstream_local
...
zsh.x86_64                 5.5.1-6.el8_1.2      ol8_baseos_local
zsh-html.noarch            5.5.1-6.el8_1.2      ol8_appstream_local
zziplib.i686               0.13.68-8.el8        ol8_appstream_local
zziplib.x86_64             0.13.68-8.el8        ol8_appstream_local
zziplib-utils.x86_64      0.13.68-8.el8        ol8_appstream_local
```

- Notice that some packages were installed during installation (`@anaconda`) and some are available in the new repositories (`ol8_baseos_local` and `ol8_appstream_local`).

- b. List only the installed packages.

```
# dnf list installed
Installed Packages
NetworkManager.x86_64      1:1.26.0-8.0.1.el8 @anaconda
...
yum.noarch                 4.2.23-4.el8        @anaconda
zlib.x86_64                1.2.11-16.el8_2     @anaconda
```

- c. List only the packages that are available to be installed from enabled repositories.

```
# dnf list available
Last metadata expiration check: ...
Available Packages
CUnit.i686                 2.1.3-17.el8        ol8_appstream_local
CUnit.x86_64               2.1.3-17.el8        ol8_appstream_local
...
```

- d. List the name of the package to which the `/etc/sysconfig/crond` file belongs.

```
# dnf provides /etc/zlogin
Last metadata expiration check: ...
zsh-5.5.1-6.el8_1.2.x86_64 : Powerful interactive shell
Repo                        : ol8_baseos_local
Matched from:
Filename                   : /etc/zlogin
```

2. Install packages by using `dnf`.

Install the `zsh` package. Answer **y** when prompted.

```
# dnf install zsh
Last metadata expiration check: ...
Dependencies resolved.
=====
Package                Architecture Version           Repository...
=====
Installing:
  zsh                   x86_64          5.5.1-6.el8_1.2  ol8_baseos_local

Transaction Summary
=====
Install 1 Package

Total size: 2.9 M
Installed size: 7.2 M
Is this ok [y/N]: y
Downloading Packages:
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing           :                               1/1
  Installing          : zsh-5.5.1-6.el8_1.2.x86_64    1/1
  Running scriptlet: zsh-5.5.1-6.el8_1.2.x86_64    1/1
  Verifying           : zsh-5.5.1-6.el8_1.2.x86_64    1/1

Installed:
  zsh-5.5.1-6.el8_1.2.x86_64

Complete!
```

3. Update the packages by using `dnf`.

Check which installed packages have updates available.

```
# dnf check-update
Last metadata expiration check: ...
```

- In this case, there are no updates available as we only have the local repos enabled.

4. Remove the packages by using `dnf`.

Remove the `zsh` package. Answer **y** when prompted.

```
# dnf remove zsh
Dependencies resolved.
=====
Package Architecture Version Repository Size
=====
Removing:
zsh x86_64 5.5.1-6.el8_1.2 @ol8_baseos_local 7.2M

Transaction Summary
=====
Remove 1 Package

Freed space: 7.2 M
Is this ok [y/N]: y
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing : 1/1
  Running scriptlet: zsh-5.5.1-6.el8_1.2.x86_64 1/1
  Erasing : zsh-5.5.1-6.el8_1.2.x86_64 1/1
  Running scriptlet: zsh-5.5.1-6.el8_1.2.x86_64 1/1
  Verifying : zsh-5.5.1-6.el8_1.2.x86_64 1/1

Removed:
zsh-5.5.1-6.el8_1.2.x86_64

Complete!
```

5. Install the *Security Tools* package group.

Use the `dnf` command to install the *Security Tools* package group in order to use the OpenSCAP security tools (covered in another course). Answer **y** when prompted.

```
# dnf group install "Security Tools"
Last metadata expiration check: ...
Dependencies resolved.
=====
Package                Arch...  Version                Repository...
=====
Installing group/module packages:
scap-security-guide    noarch   0.1.50-14.0.1.el8     ol8_appstre...
Installing dependencies:
GConf2                 x86_64   3.2.6-22.el8          ol8_appstre...
libxslt                x86_64   1.1.32-5.0.1.el8      ol8_baseos_...
openscap               x86_64   1.3.3-5.0.1.el8       ol8_appstre...
openscap-scanner       x86_64   1.3.3-5.0.1.el8       ol8_appstre...
psmisc                 x86_64   23.1-5.el8            ol8_baseos_...
xml-common             noarch   0.6.3-50.el8          ol8_baseos_...
Installing Groups:
Security Tools

Transaction Summary
=====
Install 7 Packages

Total size: 8.0 M
Installed size: 148 M
Is this ok [y/N]: y
Downloading Packages:
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
...
Complete!
```

6. Verify that the OpenSCAP tools have installed successfully.

Using the `oscap` command, query details from the Security Guide.

```
# oscap info /usr/share/xml/scap/ssg/content/ssg-ol8-xccdf.xml
Document type: XCCDF Checklist
Checklist version: 1.1
Imported: 2020-11-05T22:15:46
Status: draft
Generated: 2020-11-05
Resolved: true
Profiles:
    Title: Criminal Justice Information Services (CJIS) ...
        Id: cjis
    Title: Unclassified Information in Non-federal Infor...
        Id: cui
    Title: [DRAFT] Australian Cyber Security Centre (ACS...
        Id: e8
    Title: Health Insurance Portability and Accountabili...
        Id: hipaa
    Title: [DRAFT] Protection Profile for General Purpos...
        Id: ospp
    Title: PCI-DSS v3.2.1 Control Baseline Draft for Ora...
        Id: pci-dss
    Title: Standard System Security Profile for Oracle L...
        Id: standard
    Title: [DRAFT] DISA STIG for Oracle Linux 8
        Id: stig
Referenced check files:
    ssg-ol8-oval.xml
        system: http://oval.mitre.org/XMLSchema/oval...
    ssg-ol8-ocil.xml
        system: http://scap.nist.gov/schema/ocil/2
        https://linux.oracle.com/security/oval/com.oracle.el...
        system: http://oval.mitre.org/XMLSchema/oval...
```

- The list shows various security profiles that could be evaluated.

7. Remove packages from the *Security Tools* package group.

Use the `dnf` command to remove packages from the *Security Tools* package group.

```
# dnf group remove "Security Tools"
Dependencies resolved.
=====
Package                Arc...  Version                Repository...
=====
Removing:
scap-security-guide    noarch  0.1.50-14.0.1.el8     @ol8_appstre...
Removing unused dependencies:
GConf2                 x86_64  3.2.6-22.el8           @ol8_appstre...
libxslt                x86_64  1.1.32-5.0.1.el8      @ol8_baseos_...
openscap               x86_64  1.3.3-5.0.1.el8       @ol8_appstre...
openscap-scanner       x86_64  1.3.3-5.0.1.el8       @ol8_appstre...
psmisc                 x86_64  23.1-5.el8             @ol8_baseos_...
xml-common             noarch  0.6.3-50.el8           @ol8_baseos_...
Removing Groups:
Security Tools

Transaction Summary
=====
Remove  7 Packages

Freed space: 148 M
Is this ok [y/N]: y
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
...
Complete!
```

Practice 9-5: Using dnf Modules

Overview

In this practice, you work with modules using the local `yum` repository set up earlier. You review the versions of software available in a module before installing one and testing it to confirm that it is installed correctly.

Assumptions

- You are the `root` user on **host02**.

Tasks

1. View the available version(s) of software packaged in a module.
 - a. Use the `dnf list` command to view the version of Ruby available to install by default.

```
[root@host02 ~]# dnf list ruby
Last metadata expiration check: ...
Available Packages
ruby.i686      2.5.5-106.module+el8.3.0+7756+e45777e9  ol8_app...
ruby.x86_64    2.5.5-106.module+el8.3.0+7756+e45777e9  ol8_app...
```

- Note that version 2.5.5 of ruby is available for installation.
- b. Use the `dnf module list` command to view the Ruby module streams available.

```
# dnf module list ruby
Last metadata expiration check: ...
Oracle Linux 8 Application Stream Local (x86_64)
Name  Stream  Profiles  Summary
ruby  2.5 [d]  common [d]  An interpreter of object-oriented...
ruby  2.6      common [d]  An interpreter of object-oriented...
ruby  2.7      common [d]  An interpreter of object-oriented...

Hint: [d]efault, [e]nabled, [x]disabled, [i]nstalled
```

- Note the `[d]` next to Stream 2.5 identifies this as the default version to install.

- c. Use the `dnf module enable` command to select version 2.7 of Ruby.

```
# dnf module enable ruby:2.7
Last metadata expiration check: ...
Dependencies resolved.
=====
Package           Architecture      Version           Repository        Size
=====
Enabling module streams:
  ruby                                           2.7

Transaction Summary
=====

Is this ok [y/N]: y
Complete!
```

- d. Use the `dnf list` command to view the version of Ruby now available to install.

```
# dnf list ruby
Last metadata expiration check: ...
Available Packages
ruby.i686          2.7.1-133.module+el8.3.0+7760+537395ec  ol8_app...
ruby.x86_64        2.7.1-133.module+el8.3.0+7760+537395ec  ol8_app...
```

- Note that version 2.7.1 of Ruby is now available for installation.

- e. Use the `dnf module reset` command to reset the Ruby module to its defaults.

```
# dnf module reset ruby
Last metadata expiration check: ...
Dependencies resolved.
=====
Package           Architecture      Version           Repository        Size
=====
Resetting modules:
  ruby

Transaction Summary
=====

Is this ok [y/N]: y
Complete!
```

2. Install a specific module stream.

- a. Use the `dnf module install` command to install Ruby from the 2.6 stream.

```
# dnf module install ruby:2.6
Last metadata expiration check: ...
Dependencies resolved.
=====
Package                ... Version ...
=====
Installing group/module packages:
 ruby                  ... 2.6.3-106.module+el8.1.0+5406+ce01f9b9...
Installing dependencies:
 ruby-libs              ... 2.6.3-106.module+el8.1.0+5406+ce01f9b9...
 rubygem-irb            ... 1.0.0-106.module+el8.1.0+5406+ce01f9b9...
 rubygem-json           ... 2.1.0-106.module+el8.1.0+5406+ce01f9b9...
 rubygem-psych          ... 3.1.0-106.module+el8.1.0+5406+ce01f9b9...
Installing weak dependencies:
 rubygem-bigdecimal... 1.4.1-106.module+el8.1.0+5406+ce01f9b9...
...
Installing module profiles:
 ruby/common
Enabling module streams:
 ruby                                                         2.6

Transaction Summary
=====
Install 11 Packages

Total size: 4.6 M
Installed size: 15 M
Is this ok [y/N]: y
Downloading Packages:
...
Installed:
...
Complete!
```

- Note that there are several dependencies for Ruby 2.6. Specific versions of the dependent packages are included in the module stream.

- b. Use the `dnf module list` command to view the Ruby module status.

```
# dnf module list ruby
Last metadata expiration check: ...
Oracle Linux 8 Application Stream Local (x86_64)
Name           Stream   Profiles      Summary
ruby           2.5 [d]   common [d]     An interpreter of...
ruby           2.6 [e]   common [d] [i] An interpreter of...
ruby           2.7      common [d]     An interpreter of...

Hint: [d]efault, [e]nabled, [x]disabled, [i]nstalled
```

- Note the 2.6 stream is marked [e] for enabled and [i] for installed.

- c. Use the `ruby` command to verify the version of Ruby installed on the system.

```
# ruby --version
ruby 2.6.3p62 (2019-04-16 revision 67580) [x86_64-linux]
```

3. Remove the packages installed from the module and reset the module to its defaults.

- a. Use the `dnf module remove` command to uninstall packages from the Ruby module.

```
# dnf module remove ruby
Last metadata expiration check: ...
Dependencies resolved.

=====
Package           ... Version ...
=====

Removing:
ruby               ... 2.6.3-106.module+el8.1.0+5406+ce01f9b9...
Removing unused dependencies:
ruby-libs          ... 2.6.3-106.module+el8.1.0+5406+ce01f9b9...
rubygem-bigdecimal... 1.4.1-106.module+el8.1.0+5406+ce01f9b9...
...
Disabling module profiles:
ruby/common

Transaction Summary
=====
Remove 11 Packages

Freed space: 15 M
Is this ok [y/N]: y
...
Removed:
...
Complete!
```


- b. Use the `dnf module list` command to view the Ruby module status.

```
# dnf module list ruby
Last metadata expiration check: ...
Oracle Linux 8 Application Stream Local (x86_64)
Name           Stream  Profiles  Summary
ruby           2.5 [d]   common [d] An interpreter of object-...
ruby           2.6 [e]   common [d] An interpreter of object-...
ruby           2.7     common [d] An interpreter of object-...

Hint: [d]efault, [e]nabled, [x]disabled, [i]nstalled
```

- Note that stream 2.6 is no longer installed but is still enabled.

- c. Use the `dnf module reset` command to reset the Ruby module to its defaults.

```
# dnf module reset ruby
Last metadata expiration check: ...
Dependencies resolved.

=====
Package           Architecture      Version           Repository        Size
=====
Resetting modules:
  ruby

Transaction Summary
=====

Is this ok [y/N]: y
Complete!
```

- d. Use the `dnf module list` command to verify the Ruby module has been reset.

```
# dnf module list ruby
Last metadata expiration check: ...
Oracle Linux 8 Application Stream Local (x86_64)
Name           Stream  Profiles  Summary
ruby           2.5 [d]   common [d] An interpreter of object-...
ruby           2.6     common [d] An interpreter of object-...
ruby           2.7     common [d] An interpreter of object-...

Hint: [d]efault, [e]nabled, [x]disabled, [i]nstalled
```

4. Use the `exit` command to log off **host02**.

```
# exit  
logout  
Connection to host02 closed.
```

You are now the `oracle` user on **gateway**.

Practices for Lesson 10:
Oracle Ksplice

Practices for Lesson 10: Overview

Practices Overview

In these practices, you:

- Install the Ksplice Uptrack offline client and kernel updates to **host02**
- View the effective kernel version after updates are applied
- List the active Oracle Ksplice updates in your running kernel
- Watch a demonstration of the Ksplice online client

Practice 10-1: Installing the Ksplice Offline Client and Kernel Updates

Overview

In this practice, you:

- Install the Ksplice Uptrack Offline Client packages on **host02**
- Install a kernel update package on **host02**
- View the effective kernel version after updates have been installed
- List the active Oracle Ksplice updates in your running kernel

Assumptions

- You are the `oracle` user on **gateway**.
- You created the local yum repository in the lesson *Package Management*.

Tasks

1. Log on to **host02**.

Connect to **host02** using the `ssh` command.

```
[oracle@gateway ~]$ ssh root@host02
Password:
[root@host02 ~]#
```

2. View the packages available on **host02**.

As the `root` user on **host02**, use the `dnf list` command to show available Ksplice packages.

```
[root@host02 ~]# dnf list available ksplice* uptrack*
Last metadata expiration check: ...
Available Packages
ksplice.x86_64                                1.0.48-1.el8      extras
ksplice-core0.x86_64                        1.0.48-1.el8      extras
ksplice-known-exploit-detection.x86_64      1.0.48-1.el8      extras
ksplice-offline.x86_64                      1.0.48-1.el8      extras
ksplice-tools.x86_64                        1.0.48-1.el8      extras
uptrack.noarch                              1.2.71-0.el8      extras
uptrack-offline.noarch                      1.2.71.of...      extras
uptrack-updates-5.4.17-2011.7.4.el8uek...  20201217-0        extras
```

- The `uptrack-offline` package enables offline Ksplice kernel updates and the `uptrack-updates` package provides the Ksplice updates for UEK version 5.4.17-2011.7.4.
- The `ksplice-offline` package enables offline Ksplice userspace updates, but also requires Ksplice versions of the libraries to be installed first. This practice will only configure kernel updates using `uptrack-offline`.

3. Install Ksplice uptrack kernel updates.

a. Use the `dnf` command to install the Ksplice uptrack Offline Client package.

The `uptrack-offline` package is the Ksplice uptrack Offline Client.

Note: The `uptrack-offline` package will not install if the `uptrack` package is already installed.

```
# dnf install uptrack-offline
Last metadata expiration check: ...
Dependencies resolved.
=====
Package                        Arch    Version                        Repository
=====
Installing:
uptrack-offline                noarch  1.2.71.offline-0.el8         extras...
Installing dependencies:
perl-Carp                      noarch  1.42-396.el8                 ol8_bas...
perl-Data-Dumper               x86_64  2.167-399.el8                ol8_bas...
...
Installing weak dependencies:
perl-IO-Socket-IP              noarch  0.39-5.el8                   ol8_app...
...
Enabling module streams:
perl                           5.26
perl-IO-Socket-SSL              2.066
perl-libwww-perl                6.34

Transaction Summary
=====
Install  47 Packages

Total size: ...
Installed size: ...
Is this ok [y/N]: y
Downloading Packages:
...
Now run 'uptrack-upgrade' to protect your kernel
...
Complete!
```

- The Ksplice uptrack Offline Client package is installed successfully.

- b. Use the `dnf` command to install the `uptrack-updates` package.

```
# dnf install uptrack-updates-$(uname -r)
Last metadata expiration check: ...
Dependencies resolved.
=====
Package                                     Arch    Version...
=====
Installing:
uptrack-updates-5.4.17-2011.7.4.el8uek.x86_64 noarch 2020121...

Transaction Summary
=====
Install 1 Package

Total size: 5.7 M
Installed size: 6.2 M
Is this ok [y/N]: y
Downloading Packages:
...
Running scriptlet: uptrack-updates-5.4.17-2011.7.4.el8u... 1/1
The following steps will be taken:
Install [r0ls89zr] CVE-2020-25211: Denial-of-service in Netfi...
Install [qljv2whb] CVE-2020-13974: Integer overflow in virtua...
Install [d0cdgsm7] CVE-2019-19770: use-after-free in the debu...
Install [3p9ublu0] CVE-2020-12114: Denial-of-service due to m...
Install [csvhhave] CVE-2020-14390: Memory corruption when res...
...
Installing [je4tdeii] Disable infiniband completion queue tim...
Your kernel is fully up to date.
Effective kernel version is 5.4.17-2036.101.2.el8uek
...
Complete!
```

- All kernel updates are applied.

- c. Use the `uname -r` command to display the version of the running kernel.

```
# uname -r
5.4.17-2011.7.4.el8uek.x86_64
```

- d. Use the `uptrack-uname -r` command to display the effective kernel version.

```
# uptrack-uname -r
5.4.17-2036.101.2.el8uek.x86_64
```

- The effective kernel version is now up-level from the original kernel version.

- e. Use the `uptrack-show` command to list the active Oracle Ksplice updates in your running kernel.

```
# uptrack-show
Installed updates:
[r0ls89zr] CVE-2020-25211: Denial-of-service in Netfilter due...
[q1jv2whb] CVE-2020-13974: Integer overflow in virtual termin...
...
[gk9u6qfv] CVE-2020-25656: Use-after-free in console subsystem.
[gc7ubjk9] CVE-2020-27675: Race condition when reconfiguring ...
[b4vmuq2h] CVE-2020-12352: Information leak when handling AMP...
[n27ynl2p] CVE-2020-25668: Race condition when sending ioctls...
[je4tdeii] Disable infiniband completion queue time stamping.

Effective kernel version is 5.4.17-2036.101.2.el8uek
```

4. Remove Ksplice uptrack kernel updates.

- a. Use the `uptrack-remove` command to remove Ksplice update `gc7ubjk9`.

Answer **y** when prompted to go ahead.

```
# uptrack-remove gc7ubjk9
The following steps will be taken:
Remove [je4tdeii] Disable infiniband completion queue time st...
Remove [n27ynl2p] CVE-2020-25668: Race condition when sending...
Remove [b4vmuq2h] CVE-2020-12352: Information leak when handl...
Remove [gc7ubjk9] CVE-2020-27675: Race condition when reconfi...

Go ahead [y/N]? y
Removing [je4tdeii] Disable infiniband completion queue time ...
Removing [n27ynl2p] CVE-2020-25668: Race condition when sendi...
Removing [b4vmuq2h] CVE-2020-12352: Information leak when han...
Removing [gc7ubjk9] CVE-2020-27675: Race condition when recon...
Effective kernel version is 5.4.17-2036.100.6.1.el8uek
```

- Note that the effective kernel version reverted to an earlier release (5.4.17-2036.100.6.1.el8uek) but it is still more recent than the original kernel: 5.4.17-2011.7.4.el8uek.
- Note that all patches released since the one selected for removal were also removed. Patches are cumulative, you are not able to choose a set of individual patches from those available.

- b. Use the `uptrack-remove --all` command to remove all Ksplice updates.
Answer **y** when prompted to go ahead.

```
# uptrack-remove --all
The following steps will be taken:
Remove [gk9u6qfv] CVE-2020-25656: Use-after-free in console s...
Remove [c56a10z3] CVE-2020-25704: Denial-of-service in the pe...
...
Remove [r0ls89zr] CVE-2020-25211: Denial-of-service in Netfil...

Go ahead [y/N]? y
Removing [gk9u6qfv] CVE-2020-25656: Use-after-free in console...
...

Effective kernel version is 5.4.17-2011.7.4.el8uek
```

- Note that the effective kernel version reverted to the original kernel: 5.4.17-2011.7.4.el8uek.

5. Remove the Ksplice packages from **host02**.

Use the `dnf erase` command to remove the packages just installed.

```
# dnf erase -y uptrack-offline
Dependencies resolved.
...
Removing:
  uptrack-offline          noarch 1.2.71.offline-0.el8   @extras...
Removing dependent packages:
  uptrack-updates-5.4.17-2011.7.4.el8uek.x86_64
                                noarch 20201217-0      @extras...
Removing unused dependencies:
...
Complete!
```

- Note that the `uptrack-updates-5.4.17-2011.7.4.el8uek.x86_64` package is removed because it depends on the `uptrack-offline` package.

6. Log off **host02** in preparation for the next practice.

Use the `logout` command to close the `ssh` connection to **host02**.

```
[root@host02 ~]# logout
Connection to host02 closed.
[oracle@gateway ~]$
```

You are now the `oracle` user on **gateway**.

Practice 10-2: Demonstration of Ksplice Online Updates

Overview

This video demonstrates the use of the Ksplice online client.

Assumptions

You are the `oracle` user on **gateway**.

Tasks

1. A video is available...

Practices for Lesson 11: User and Group Administration

Practices for Lesson 11: Overview

Practices Overview

In these practices, you:

- Create user and group accounts
- Describe the benefits of user private groups
- Configure password aging
- Restrict the use of the `su` command
- Allow the use of the `sudo` command

Practice 11-1: Administering User Accounts

Overview

In this practice, you use command-line utilities to create new user accounts, view files that are updated when adding a new user, modify a user account, set a password for the new user, and log in as the new user.

Assumptions

You are the `oracle` user on **gateway**.

Tasks

1. Log in to **host03**.

Use the `ssh` command to log in to **host03** as the `root` user.

Provide the `root` password when prompted.

```
[oracle@gateway ~]$ ssh root@host03
root@host03's password:
Last login: ...
```

2. Add a user.

- a. Use the command-line utility to add the `student1` user.

```
[root@host03# useradd student1
```

- b. Use the `tail` command to view the new `student1` entry in the `/etc/passwd` file.

```
# tail /etc/passwd
...
oracle:x:1000:1000:Oracle Student:/home/oracle:/bin/bash
...
student1:x:1001:1001::/home/student1:/bin/bash
```

- Note that the new user's UID and GID are incremented by one compared to those of the `oracle` user.
- Note that a home directory was created for the new user (`/home/student1`).
- Note that the default shell for the new user is `/bin/bash`.

- c. Use the `ls` command to verify that the new user's home directory was created.

```
# ls /home
oracle  student1
```

- A home directory was created because `CREATE_HOME` in `/etc/login.defs` is set to `yes`.

- d. View the value of `CREATE_HOME` in `/etc/login.defs`.

```
# grep CREATE_HOME /etc/login.defs
CREATE_HOME    yes
```

- e. View the default settings for a new user, stored in `/etc/default/useradd`.

```
# cat /etc/default/useradd
# useradd defaults file
GROUP=100
HOME=/home
INACTIVE=-1
EXPIRE=
SHELL=/bin/bash
SKEL=/etc/skel
CREATE_MAIL_SPOOL=yes
```

- Note that the `SKEL` directive is set to `/etc/skel`.

- f. View the contents of `/etc/skel`.

```
# ls -lA /etc/skel
-rw-r--r--. 1 root root ... .bash_logout
-rw-r--r--. 1 root root ... .bash_profile
-rw-r--r--. 1 root root ... .bashrc
```

- g. View the contents of the new user's home directory.

```
# ls -lA /home/student1
-rw-r--r--. 1 student1 student1 ... .bash_logout
-rw-r--r--. 1 student1 student1 ... .bash_profile
-rw-r--r--. 1 student1 student1 ... .bashrc
```

- Note that the contents of `SKEL` (`/etc/skel`) are copied to the new user's home directory with ownership set to user `student1` and group `student1`.
- h. View the new `student1` entry in the `/etc/group` file.

```
# tail /etc/group
...
oracle:x:1000:oracle
...
student1:x:1001:
```

- Because Oracle Linux uses a user private group (UPG) scheme, a new private group (`student1`, `GID=1001`) was created when the `student1` user was created.
- i. View the new `student1` entry in the `/etc/shadow` file.

```
# tail /etc/shadow
...
oracle:$6$...::0:99999:7:::
...
student1:!!:18632:0:99999:7:::
```

- The field with the number `18632` will be a higher value for your user because it represents the date the user password was last set or the user was created.

- j. View the new `student1` entry in the `/etc/gshadow` file.

```
# tail /etc/gshadow
...
oracle:!:
...
student1:!:
```

- k. Add a new user with the following characteristics:

Username = `student2`

UID = 1055

GECOS information = "Oracle Student2"

Default shell = `/bin/sh` (Bourne shell)

```
# useradd -u 1055 -c "Oracle Student2" -s /bin/sh student2
```

- l. View the new `student2` entry in `/etc/passwd`.

```
# tail -2 /etc/passwd
student1:x:1001:1001::/home/student1:/bin/bash
student2:x:1055:1055:Oracle Student2:/home/student2:/bin/sh
```

- Note that the UID and GID are 1055.
- Note the GECOS information.
- Note the default shell.

3. Modify a user account.

- a. Use the `usermod` command to modify GECOS information for the `student1` user as follows:

```
# usermod -c "Oracle Student1" student1
```

- b. View the `student1` entry in the `/etc/passwd` file.

```
# grep student1 /etc/passwd
student1:x:1001:1001:Oracle Student1:/home/student1:/bin/bash
```

4. Assign a password to the new user.

- a. Use the `tail` command to view the last few entries in the `/etc/shadow` file.

```
# tail /etc/shadow
...
oracle:$6$...:0:99999:7::
...
student1:!:18632:0:99999:7::
student2:!:18632:0:99999:7::
```

- Note that the "!" in the `student1` and `student2` records indicate that no password has been assigned (and that the accounts are locked).

- b. Use the `passwd` command to create a password (of `password`) for the `student1` user.

Ignore the “BAD PASSWORD” warning, continuing to use `password` as the password.

```
# passwd student1
Changing password for user student1.
New password: password
BAD PASSWORD: The password fails the dictionary check - it is
based on a dictionary word
Retype new password: password
passwd: all authentication tokens updated successfully.
```

- c. View the `/etc/shadow` file.

```
# tail /etc/shadow
...
oracle:$6$...:0:99999:7:::
...
student1:$6$...:18632:0:99999:7:::
student2:!!:18632:0:99999:7:::
```

- Now the “!!” for `student1` has been replaced with a hashed password value.
- Note that the `student2` account is still locked because a password has not been assigned.

5. Log in as a new user.

- a. Log out of **host03** using the `logout` command.

```
[root@host03 ~]# logout
Connection to host03 closed.
[oracle@gateway ~]$
```

- b. Attempt to log in as `student2` (**Oracle Student2**).

```
[oracle@gateway ~]$ ssh student2@host03
student2@host03's password: password
Permission denied, please try again.
student2@host03's password: letmein
Permission denied, please try again.
student2@host03's password: ENTER
student2@host03: Permission denied (publickey,gssapi-
keyex,gssapi-with-mic,password).
[oracle@gateway ~]$
```

- You are presented with a **password:** prompt.
- Because no password has been assigned to `student2`, the account is locked.
- Regardless of what you enter as a password, you cannot log in as `student2`.
- After three attempts you are disconnected.

- c. Log in as `student1` using `ssh`.

```
[oracle@gateway ~]$ ssh student1@host03  
student1@host03's password: password  
[student1@host03 ~]$
```

- d. From the terminal window, use the `whoami` command to verify that you are logged in as `student1`.

```
$ whoami  
student1
```

- e. Use the `logout` command to log out as the `student1` user. Use the `ssh` command to log in to `host03` as the `root` user.

```
[student1@host03 ~]$ logout  
Connection to host03 closed.  
[oracle@gateway ~]$ ssh root@host03  
root@host03's password:  
[root@host03 ~]#
```

Practice 11-2: Administering Group Accounts

Overview

In this practice, you create a new group account and add users to this new group.

Tasks

1. Add a group.
 - a. Use the `groupadd` command without any options to add the `students` group.

```
# groupadd students
```

- b. View the new `students` entry in the `/etc/group` file.

```
# tail /etc/group
...
student1:x:1001:
student2:x:1055:
students:x:1056:
```

- Note that the GID for the new group is incremented by one.

2. Add users to the new group.
 - a. Use the `usermod` command to add the `student1` and `student2` users to the `students` group.

```
# usermod -aG 1056 student1
# tail -1 /etc/group
students:x:1056:student1
# usermod -aG 1056 student2
# tail -1 /etc/group
students:x:1056:student1,student2
```

- Note that both `student1` and `student2` have a secondary group membership in the `students` group.

- b. View the primary group membership for `student1` and `student2`.

```
# grep student /etc/passwd
student1:x:1001:1001:Oracle Student1:/home/student1:/bin/bash
student2:x:1055:1055:Oracle Student2:/home/student2:/bin/sh
```

- Note that the `student1` primary group is still 1001.
- Note that the `student2` primary group is still 1055.

Practice 11-3: Implementing User Private Groups

Overview

In this practice, you use the UPG scheme to give different users write access to files in a single directory.

Tasks

1. Set up a shared `/students` directory.

- a. Use the `mkdir` command to create the `/students` directory.

```
# mkdir /students
```

- b. Use the `chgrp` command to change the group for this `/students` directory to the `students` group.

```
# ls -ld /students
drwxr-xr-x. 2 root  root ... /students
# chgrp -R students /students
# ls -ld /students
drwxr-xr-x. 2 root  students ... /students
```

- Note that the group was set to `root` before issuing the `chgrp` command.
- The `-R` option (recursive) sets the new group for files and directories within `/students`.

- c. Use the `chmod` command to set the `setgid` bit on the directory, and give write permissions to the group.

```
# chmod -R 2775 /students
# ls -ld /students
drwxrwsr-x. 2 root  students ... /students
```

- Note the new group permissions on the `/students` directory.

2. Create files in the `/students` directory as the `oracle` user.

- a. Use the `usermod` command to add the `oracle` user to the `students` group.

```
# usermod -aG students oracle
```

- b. Use the `su` command to become the `oracle` user.

```
# su - oracle
$ whoami
oracle
```

- c. Use the `groups` command to display group membership.

```
$ groups
oracle students
```

- Note that the `oracle` user belongs to two groups, `oracle` and `students`.

- d. Use the `cd` command to change to the `/students` directory.

```
$ cd /students
```

- e. Use the `touch` command to create a file in the `/students` directory.

```
$ touch oracle_file
```

- f. Use the `ls` command to display the permissions and ownership of the new file.

```
$ ls -l oracle_file
-rw-rw-r--. 1 oracle  students ... oracle_file
```

- Note that the permissions are read/write for the `students` group.

3. Create and edit files within the shared directory as a different user.

- a. Use the `su` command to become the `student1` user. The password is `password`.

```
$ su - student1
Password: password
$ whoami
student1
```

- b. Use the `groups` command to display group membership.

```
$ groups
student1  students
```

- Note that the `student1` user belongs to two groups, `student1` and `students`.

- c. Use the `cd` command to change to the `/students` directory.

```
$ cd /students
```

- d. Use the `touch` command to create a file in the `/students` directory.

```
$ touch student1_file
```

- e. Use the `ls` command to display the permissions and ownership of the files.

```
$ ls -l
-rw-rw-r--. 1 oracle  students ... oracle_file
-rw-rw-r--. 1 student1 students ... student1_file
```

- Note that the permissions are read/write for the `students` group.

- f. Use the `touch` command to update the time stamp on the `oracle_file`.

```
$ touch oracle_file
```

- The `touch` command updates the time stamp, which implies write permissions on the file.

- g. Use the `exit` command twice to log out as the `student1` user, and then as the `oracle` user.

```
$ exit
logout
$ whoami
oracle
$ exit
logout
# whoami
root
```

Practice 11-4: Configuring Password Aging

Overview

In this practice, you modify the password aging parameters for a user.

Tasks

1. View the password aging information.

- a. List password aging information in `/etc/shadow` for the `student1` user.

```
# grep student1 /etc/shadow
student1:$6$. . . :18632:0:99999:7:::
```

- This example gives 18632 days since the password changed (counted in days since Jan 1, 1970). This differs depending on when you created the account.
 - 0 is the number of days that need to pass before the password must be changed by the user (0 means *never*).
 - 99999 is the maximum number of days since the password changed that the password can be used. After this number of days, the password needs to be changed by the user.
 - 7 is the number of days before the `expire` date that the user is warned about the pending password change requirement. If the password is not changed after this number of days, the user account is locked.
- b. Use the `chage` command to view password aging information for the `student1` user. Use the `-l` (list) option with the command.

```
# chage -l student1
Last password change           : Jan 05, 2021
Password expires                : never
Password inactive              : never
Account expires                : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days or warning before password expires: 7
```

- The `chage` command allows you to change user password expiry information.
- Reminder: You can edit the `/etc/login.defs` file to change the default password expiry information for all new users.

2. Modify the password aging parameters.

a. Make the following changes to password aging for the `student1` user:

Change the minimum password age value to 14.

Change the maximum password age value to 30.

```
# chage student1
Changing the aging information for student1
Enter the new value, or press ENTER for the default
    Minimum Password Age [0]: 14
    Maximum Password Age [99999]: 30
    Last Password Change (YYYY-MM-DD) [2021-01-05]: ENTER
    Password Expiration Warning [7]: ENTER
    Password Inactive [-1]: ENTER
    Account Expiration Date (YYYY-MM-DD) [-1]: ENTER
```

- If the minimum password age is 14, the user cannot change their password for 14 days. When the maximum password age is 30, the user has 16 days remaining before having to change their password.

b. View the results of the change for `student1` in `/etc/shadow`.

```
# grep student1 /etc/shadow
student1:$6$...:18632:14:30:7:::
```

c. Use the `chage` command to view password aging information for the `student1` user.

```
# chage -l student1
Last password change           : Jan 05, 2021
Password expires                : Feb 04, 2021
Password inactive              : never
Account expires                : never
Minimum number of days between password change : 14
Maximum number of days between password change : 30
Number of days or warning before password expires: 7
```

Practice 11-5: Restricting the Use of the `su` Command

Overview

In this practice, you limit the use of the `su` command to members of the `wheel` group.

Tasks

1. Add the `oracle` user to the `wheel` group.

- a. Use the `grep` command to display the `wheel` entry in the `/etc/group` file.

```
# grep wheel /etc/group
wheel:x:10:
```

- b. Use the `usermod` command to add the `oracle` user to the `wheel` group. Rerun the `grep` command to verify that the modification occurred.

```
# usermod -aG wheel oracle
# grep wheel /etc/group
wheel:x:10:oracle
```

- Note the `oracle` user is now a member of the `wheel` group.

2. Restrict the use of the `su` command to members of the `wheel` group.

The procedure shown next involves the modification of a Pluggable Authentication Module (PAM) configuration file. PAM is covered in another course.

Use the `vi` editor to uncomment (remove the `#` character) the following entry in the `/etc/pam.d/su` file (be sure to modify the line containing the `required` keyword):

```
# vi /etc/pam.d/su
...
#auth      required    pam_wheel.so    use_uid      (old entry)
auth       required    pam_wheel.so    use_uid      (new entry)
...
```

- With this entry uncommented, only members of the `wheel` group can use the `su` command (with the exception of the `root` user).

3. Verify that only member of the `wheel` group can use the `su` command.

- a. As the `root` user, use the `su - student1` command to become the `student1` user.

```
# su - student1
```

- The `su` command is successful because you are the `root` user.

- b. Use the `whoami` command to confirm you are the `student1` user. Use the `su` command to become the `root` user. Provide the password when prompted.

```
$ whoami
student1
$ su -
Password:
su: Permission denied
```

- The `su` command fails because your system is configured to allow only members of the `wheel` group to use the `su` command.

- c. Use the `exit` command to log out as the `student1` user. Use the `whoami` command to confirm you are the `root` user.

```
$ exit
logout
# whoami
root
```

- d. Use the `su` command to become the `oracle` user. Use the `whoami` command to confirm you are the `oracle` user.

```
# su - oracle
$ whoami
oracle
```

- e. Use the `su` command to become the `root` user. Provide the password when prompted.

```
$ su -
Password:
...
# whoami
root
```

- The `su` command succeeded this time because the `oracle` user is a member of the `wheel` group.

4. Remove restriction of the `su` command to members of the `wheel` group.

Use the `vi` editor to comment out (insert the `#` character) the following entry in the `/etc/pam.d/su` file:

```
# vi /etc/pam.d/su
...
auth      required    pam_wheel.so    use_uid        (old entry)
#auth      required    pam_wheel.so    use_uid        (new entry)
...
```

- With this entry commented out, all users can use the `su` command.

Practice 11-6: Allowing the Use of the `sudo` Command

Overview

In this practice, you run administrative commands with and without `sudo` with the `oracle` user. You work with the `/etc/sudoers` file to understand the interaction of entries in this file along with the `oracle` user's membership in the `wheel` group.

Tasks

1. Attempt to add a new user with and without `root` privileges.
 - a. Use the `exit` command to log out of the `root` user. Then run the `whoami` command to verify that you are the `oracle` user.

```
# exit
logout
$ whoami
oracle
```

- b. Use the `useradd` command to add the `new_user` user.

```
$ useradd new_user
useradd: Permission denied.
useradd: cannot lock /etc/passwd; try again later.
```

- Note that you do not have permission to add a new user.
- c. Insert the `sudo` command before the previous `useradd` command to add the `new_user` user. Enter the password when prompted.

```
$ sudo useradd new_user

We trust you have received the usual lecture from the local
System Administrator. It usually boils down to these three
things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

[sudo] password for oracle:
```

- Note that the command succeeds because the `oracle` user is a member of the `wheel` group and the following `%wheel` entry exists in the `/etc/sudoers` file:

```
%wheel    ALL=(ALL)        ALL
```

- This entry allows any member of the `wheel` group to execute any command, when preceded by `sudo`.

- d. Use the `grep` command to search for `new_user` in the `/etc/passwd` file.

```
$ grep new_user /etc/passwd
new_user:x:1056:1057::/home/new_user:/bin/bash
```

2. Edit the `/etc/sudoers` file to deny `sudo` authorization for the `oracle` user.

- a. Use the `exit` command to log out of the `oracle` user. Then run the `whoami` command to verify that you are the `root` user.

```
$ exit
# whoami
root
```

- b. Use the `visudo` command to edit the `/etc/sudoers` file and comment out the first `%wheel` entry by inserting the `#` character at the beginning of the line, as shown. This prevents members of the `wheel` group from executing commands using `sudo`, in the absence of other entries in the `/etc/sudoers` file that provide `sudo` authorization.

```
# visudo
...
%wheel    ALL=(ALL)        ALL        (old entry)
#%wheel   ALL=(ALL)        ALL        (new entry)
...
```

- This command opens the `/etc/sudoers` file with the `vi` editor but also locks the file.

- c. After commenting out the first `%wheel` entry, save the file and exit `visudo`.

Note: There is a line containing `%wheel` that has `NOPASSWD` in it. Do not modify this line.

- d. Use the `su` command to become the `oracle` user again. Use the `whoami` command to confirm you are the `oracle` user.

```
# su - oracle
$ whoami
oracle
```

3. Attempt to remove the `new_user` user.

- a. Use the `userdel` command to remove the user.

```
$ userdel new_user
userdel: Permission denied.
userdel: cannot lock /etc/passwd; try again later.
```

- Note that you do not have permission to delete a user.

- b. Insert the `sudo` command before the `userdel` command to delete the `new_user` user.

```
$ sudo userdel new_user
[sudo] password for oracle:
oracle is not in the sudoers file. This incident will be reported.
```

- The `sudo` command fails this time because the `%wheel` entry is commented out in the `/etc/sudoers` file, and there is no separate `oracle` user entry in that file either.
 - The password prompt may not appear on your system, depending on the time since the `oracle` user issued a `sudo` command.
 - The attempt to issue this administrator command without proper authorization is reported in the `/var/log/secure` file.
- c. Take a look at the log entry in `/var/log/secure`. Use the `exit` command to return to the `root` user.

```
$ exit
# whoami
root
```

- d. Use the `tail` command to view the end of the `/var/log/secure` file.

```
# tail /var/log/secure
...
... sudo[PID]: oracle : user NOT in sudoers ...
...
```

- Note the "sudo" entry for the attempted use of `/sbin/userdel` in the `/var/log/secure` file.

4. Add the `oracle` user to the `/etc/sudoers` file.

- a. Use the `visudo` command to edit the `/etc/sudoers` file.

```
# visudo
```

- b. Add the following line to `/etc/sudoers`, which allows the `oracle` user to use `sudo` to run the `userdel` and `systemctl` commands as the `root` user. Add this line after the line specifying "`root ALL=(ALL) ALL`". After adding the line, save the file and exit `visudo`.

```
oracle ALL=(ALL) /usr/sbin/userdel, /usr/bin/systemctl
```

5. Remove the user `new_user` by using `sudo`.

- a. Use the `su` command to become the `oracle` user again. Use the `whoami` command to confirm you are the `oracle` user.

```
# su - oracle
$ whoami
oracle
```

- b. Use the `sudo userdel` command to delete the `new_user` user.

```
$ sudo userdel new_user  
[sudo] password for oracle:
```

- No error message implies the command was successful.

- c. Use the `grep` command to search for `new_user` in the `/etc/passwd` file.

```
$ grep new_user /etc/passwd
```

- The `new_user` has been removed from `/etc/passwd`.

6. Run another command, which requires administrative privileges.

- a. Use the `systemctl` command to restart the `rsyslog` service.

```
$ systemctl restart rsyslog
```

- As a GUI is not being used, the following text messages will appear:

```
=== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ===  
Authentication is required to manage system services or units.  
Authenticating as: Oracle Student (oracle)  
Password:  
==== AUTHENTICATION COMPLETE ====
```

- If the user `oracle` is not a member of the `wheel` group, authentication is then required for the `root` user, requiring the `root` password.

- b. Use `sudo` to run the `systemctl` command again, to restart the `rsyslog` service.

```
$ sudo systemctl restart rsyslog
```

- Note that there was no authentication message, showing that the command succeeded without additional authentication this time, as a result of using `sudo`.
- Also note that you may not have been prompted to enter the `oracle` user's password. When a user has been authenticated, the user may then use `sudo` without a password for a short period of time. (The default timeout is 5 minutes.)
- Do not perform this, but to override the default 5-minute timeout, set the "timestamp_timeout" value in `/etc/sudoers` to change the default.

7. Restore system changes and return to **gateway**.

- a. Use the `exit` command to return to the `root` user.

```
$ exit  
# whoami  
root
```

- b. Remove the `oracle` user from the `wheel` group.

```
# gpasswd -d oracle wheel  
Removing user oracle from group wheel
```

- c. Remove the `students` group.

```
# groupdel students
```

- d. Remove the `student1` and `student2` users.

```
# userdel student1
# userdel student2
```

- e. Remove the home directories for `student1`, `student2`, and `new_user`. Change directories to the `/home` directory first.

```
# cd /home
# /bin/rm -r student1 student2 new_user
```

- f. Remove the mail files for `student1`, `student2`, and `new_user`. Change directories to the `/var/mail` directory first. Change directories to the `root` user's home directory afterward.

```
# cd /var/mail
# /bin/rm student1 student2 new_user
# cd
```

- g. Remove the `oracle` user entry and uncomment the `%wheel` line in the `/etc/sudoers` file using `visudo`.

```
# visudo
```

- h. Delete the following line:

```
oracle ALL=(ALL) /usr/sbin/userdel, /usr/bin/systemctl
```

- i. Uncomment the first line containing `%wheel` to be as follows (remove the `#` from the beginning of the line), and then save the file and exit `visudo`:

```
...
#%wheel ALL=(ALL)    ALL            (old entry)
%wheel ALL=(ALL)    ALL            (new entry)
...
```

- Note: There is a line containing `%wheel` that has `NOPASSWD` in it. Do not modify this line.

8. Log off `host03`.

Use the `exit` command to log out as the `root` user.

```
[root@host03 ~]# exit
logout
Connection to host03 closed.
[oracle@gateway ~]$
```

You are now the `oracle` user on `gateway`.

Practices for Lesson 12
Partitions, File Systems, and
Swap

Practices for Lesson 12: Overview

Practices Overview

In these practices, you:

- Display the partition table, list the mounted file systems, and display the swap space configured during installation
- Partition disk devices, and create and mount file systems on the partitions
- Increase the amount of swap space by creating a swap file

Practice 12-1: Listing the Current Disk Partitions

Overview

In this practice, you examine the disk partitioning, mount points for file systems, and swap space that were configured during installation.

Assumptions

You are the `oracle` user on `gateway`.

Tasks

1. Log in to **host03**.

Connect to the **host03** guest by using the `ssh` command.

```
[oracle@gateway ~]$ ssh root@host03
root@host03's password:
[root@host03 ~]#
```

2. Review the disk partition tables created during installation.

Use the `fdisk` command to display the disk partition tables.

```
# fdisk -l | grep /dev
Disk /dev/xvda: 30.5 GiB, 32768000000 bytes, 64000000 sectors
/dev/xvda1 *          2048  2099199  2097152      1G 83 Linux
/dev/xvda2           2099200 52430847 50331648     24G 83 Linux
/dev/xvda3           52430848 56625151  4194304      2G 83 Linux
/dev/xvda4           56625152 63999999  7374848     3.5G  5 Extended
/dev/xvda5           56627200 62918655  6291456      3G 82 Linux swap /
Solaris
Disk /dev/xvdb: 5 GiB, 5299999744 bytes, 10351562 sectors
Disk /dev/xvdd: 5 GiB, 5299999744 bytes, 10351562 sectors
```

- This output lists the three storage devices:
 - `/dev/xvda`, 30.5 GiB (approximately) in size
 - `/dev/xvdb`, 5 GiB (approximately) in size
 - `/dev/xvdd`, 5 GiB (approximately) in size
- The `*` indicates that the first partition on `xvda` contains Boot files.
- The output also indicates that the fourth partition on `xvda` is an Extended partition.
 - A maximum of four primary partitions can be placed on an MS-DOS format disk.
 - One of the four partitions can be designated as an Extended partition.
 - This Extended partition can then be subdivided into multiple logical partitions.
 - The `/dev/xvda5` is a logical partition, which is designated as a Linux swap partition.

3. Review the file systems and mount points configured during installation.

Use the `df` command to list the mounted partitions. You will see some additional file systems listed, the order of the devices might be different, and some numbers might vary.

```
# df -h
Filesystem      Size  Used Avail Use% Mounted on
...
/dev/xvda2      24G   1.8G   23G   8% /
/dev/xvda3      2.0G   47M   2.0G   3% /home
/dev/xvda1     1014M  237M   778M  24% /boot
...
```

- The first partition on hard drive xvda (`xvda1`) contains boot files and is mounted on `/boot`.
- The second partition on hard drive xvda (`xvda2`) is the root file system and is mounted on `/`.
- The third partition on hard drive xvda (`xvda3`) is for user home directories and is mounted on `/home`.

4. Review the swap space configured during installation. Your output might vary slightly.

- a. Use the `swapon` command to display the swap space.

```
# swapon -s
Filename                Type              Size      Used    Priority
/dev/xvda5              partition        3145724    0        -2
```

- The fifth partition (the first logical partition) on hard drive xvda (`xvda5`) is swap space.

- b. Use the `cat` command to show that the `/proc/swaps` file provides the same information.

```
# cat /proc/swaps
Filename                Type              Size      Used    Priority
/dev/xvda5              partition        3145724    0        -2
```

Practice 12-2: Partitioning a Storage Device

Overview

In this practice, you create a partition by using the `fdisk` utility, and create a second partition by using the `parted` utility.

Assumptions

You are the `root` user on `host03`.

Tasks

1. Partition a storage device by using `fdisk`.

- a. Use the `fdisk` command to partition `/dev/xvdb`.

```
# fdisk /dev/xvdb
Welcome to fdisk (util-linux 2.32.1).
...
Command (m for help):
```

- b. Display the `fdisk` menu.

```
Command (m for help): m

Help:
...
Generic
  d  delete a partition
  F  list free unpartitioned space
  l  list known partition types
  n  add a new partition
  p  print the partition table
  t  change a partition type
  v  verify the partition table
  i  print information about a partition
...
Save & Exit
  w  write table to disk and exit
  q  quit without saving changes
...
Command (m for help):
```

- c. Add a new primary partition, giving the partition number 1.

```
Command (m for help): n
Partition type:
   p   primary partition (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
```

- A maximum of four primary partitions can be created in an MS-DOS partition table.
- One of the four partitions can be designated as an extended partition. This partition can then be subdivided into multiple logical partitions.

- d. Continue adding the new partition using the following parameters:

```
First sector (2048-10351561, default 2048): ENTER
Last sector, +sectors or +size{K,M,G,T,P} (2048-10351561,
default 10351561): 2100000

Created a new partition 1 of type 'Linux' and of size 1 GiB.
```

- e. Display the new partition table.

```
Command (m for help): p
Disk /dev/xvdb: 5 GiB, 5299999744 bytes, 10351562 sectors
...
Disklabel type: dos
...
Device      Boot Start      End Sectors Size Id Type
/dev/xvdb1           2048 2100000 2097953   1G 83 Linux
```

- f. Save the new partition table.

```
Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

- g. Use the `fdisk` command to list the partition table on `/dev/xvdb`.

```
# fdisk -l /dev/xvdb
Disk /dev/xvdb: 5 GiB, 5299999744 bytes, 10351562 sectors
...
Device      Boot Start      End Sectors Size Id Type
/dev/xvdb1           2048 2100000 2097953   1G 83 Linux
```

2. Partition a storage device by using parted.

- a. Use the parted command to partition /dev/xvdd.

```
# parted /dev/xvdd
GNU Parted 3.2
Using /dev/xvdd
Welcome to GNU Parted! Type 'help' to view a list of commands.
(parted)
```

- b. Enter help to view a list of commands.

```
(parted) help
align-check TYPE N      check partition N for TYPE(min|opt) ...
help [COMMAND]          print general help, or help on COMMAND
mklabel,mktable LABEL-TYPE ...
mkpart PART-TYPE [FS-TYPE] START END ...
name NUMBER NAME        name partition NUMBER as NAME
...
```

- c. Enter print to print the partition table.

```
(parted) print
Error: /dev/xvdd: unrecognised disk label
Model: Xen Virtual Block Device (xvd)
Disk /dev/xvdd: 5300MB
Sector size (logical/physical): 512B/512B
Partition Table: unknown
Disk Flags:
```

- Note the "unrecognised disk label" error.
- Note that the partition table is "unknown".

- d. Get help for the mkpart command.

```
(parted) help mkpart
mkpart PART-TYPE [FS-TYPE] START END      make a partition
PART-TYPE is one of: primary, logical, extended
FS-TYPE is one of: btrfs, nilfs2, ext4, ext3, ext2, fat32...
START and END are disk locations, such as 4GB or 10%...
'mkpart' makes a partition without creating a new file
system on the partition. FS-TYPE may be specified to set
an appropriate partition ID.
```

- e. Use the mkpart command to create a partition.

```
(parted) mkpart
Error: /dev/xvdd: unrecognized disk label
```

- The mkpart command fails due to an unrecognized disk label.

- f. Get help for the `mklabel` command.

```
(parted) help mklabel
mklabel, mktable LABEL-TYPE      create a new disklabel
(partition table)

LABEL-TYPE is one of: atari, aix, amiga, bsd, dvh, gpt, mac,
msdos, pc98, sun, loop
```

- g. Use the `mklabel` command to create a new disk label of type `gpt`.

```
(parted) mklabel gpt
```

- h. Enter `print` to print the partition table.

```
(parted) print
Model: Xen Virtual Block Device (xvd)
Disk /dev/xvdd: 5300MB
Sector size (logical/physical): 512B/512B
Partition Table: gpt
Disk Flags:

Number  Start  End  Size  File system  Name  Flags
```

- Note that the partition table is now “`gpt`” and there is no error message.

- i. Use the `mkpart` command to create a partition using the following parameters:

```
(parted) mkpart
Partition name? []? ENTER
Filesystem type? [ext2] ENTER
Start? 0
End? 20%
Warning: The resulting partition is not properly aligned for
best performance:
34s % 2048s != 0s
Ignore/Cancel? i
```

- The `mkpart` command works after using the `mklabel` command.

- j. Print the partition table.

```
(parted) print
Model: Xen Virtual Block Device (xvd)
Disk /dev/xvdd: 5300MB
Sector size (logical/physical): 512B/512B
Partition Table: gpt
Disk Flags:

Number  Start      End      Size      File system  Name  Flags
1       17.4kB     1060MB   1060MB    ext2
```

- k. Exit the parted utility.

```
(parted) quit
Information: You may need to update /etc/fstab
```

- l. Use the `fdisk` command to list the partition table on `/dev/xvdd`.

```
# fdisk -l /dev/xvdd
Disk /dev/xvdd: 5 GiB, 5299999744 bytes, 10351562 sectors
...
Disk label type: gpt
...
Device      Start      End Sectors  Size Type
/dev/xvdd1   34 2070312 2070279 1010.9M Linux filesystem
```

3. Display the major and minor numbers.

- a. Display the partitions in `/proc/partitions`. Your output might vary.

```
# cat /proc/partitions
major minor #blocks name

 11         0    9019392 sr0
202         0   32000000 xvda
202         1    1048576 xvda1
202         2    25165824 xvda2
202         3     2097152 xvda3
202         4           1 xvda4
202         5     3145728 xvda5
202        16     5175781 xvdb
202        17     1048976 xvdb1
202        48     5175781 xvdd
202        49    1035139 xvdd1
```

- The kernel uses the major and minor numbers to access a device.

- b. Use the `ls` command to list the major and minor numbers for devices in `/dev`.

```
# ls -l /dev/xvd*
brw-rw----. 1 root disk 202, 0 date_time /dev/xvda
brw-rw----. 1 root disk 202, 1 date_time /dev/xvda1
brw-rw----. 1 root disk 202, 2 date_time /dev/xvda2
brw-rw----. 1 root disk 202, 3 date_time /dev/xvda3
brw-rw----. 1 root disk 202, 4 date_time /dev/xvda4
brw-rw----. 1 root disk 202, 5 date_time /dev/xvda5
brw-rw----. 1 root disk 202, 16 date_time /dev/xvdb
brw-rw----. 1 root disk 202, 17 date_time /dev/xvdb1
brw-rw----. 1 root disk 202, 48 date_time /dev/xvdd
brw-rw----. 1 root disk 202, 49 date_time /dev/xvdd1
```

Practice 12-3: Creating ext4 File Systems

Overview

In this practice, you create ext4 file systems on the new partitions, mount the file systems, and update the file system mount table.

Assumptions

You are the root user on host03.

Tasks

1. Make an ext4 file system on /dev/xvdb1.
 - a. Use the mkfs command to make an ext4 file system on /dev/xvdb1.

```
# mkfs -t ext4 /dev/xvdb1
mke2fs 1.45.6 (20-Mar-2020)
Discarding device blocks: done
Creating filesystem with 262144 4k blocks and 65664 inodes
Filesystem UUID: e08552ce-f27d-42dd-87e6-2e907697a0ca
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done
```

- b. Use the blkid command to display the attributes of the /dev/xvdb1 block device.

```
# blkid /dev/xvdb1
/dev/xvdb1: UUID="..." BLOCK_SIZE="4096" TYPE="ext4"
PARTUUID="..."
```

- UUID values are generally not shown throughout this practice to save space.

2. Create an ext4 file system on /dev/xvdd1.
 - a. Use the mkfs.ext4 command to create an ext4 file system on /dev/xvdd1 and assign a label of Test.

```
# mkfs.ext4 -L Test /dev/xvdd1
mke2fs 1.45.6 (20-Mar-2020)
Creating filesystem with 258784 4k blocks and 64768 inodes
Filesystem UUID: c3e39bc2-7156-48c7-81ed-afb4c16dfcee
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376
...
```


- b. Display the attributes of the `/dev/xvdd1` block device.

```
# blkid /dev/xvdd1
/dev/xvdd1: LABEL="Test" UUID="..." BLOCK_SIZE="4096"
TYPE="ext4" PARTUUID="..."
```

3. Set the file system label on `/dev/xvdb1`.

- a. Use the `e2label` command to set the file system label on `/dev/xvdb1` to `Dev`.
Remember that Linux is case-sensitive. Use uppercase `Dev` as shown.

```
# e2label /dev/xvdb1 Dev
```

- b. Display the attributes of all the block devices.

```
# blkid
/dev/xvdb1: LABEL="Dev" UUID="..." BLOCK_SIZE="4096" TYPE="e...
/dev/xvdd1: LABEL="Test" UUID="..." BLOCK_SIZE="4096" TYPE="e...
/dev/sr0: BLOCK_SIZE="2048" UUID="..." LABEL="OL-8-3-0-BaseOS...
/dev/xvda1: UUID="..." BLOCK_SIZE="512" TYPE="xfs" PARTUUID="..."
/dev/xvda2: UUID="..." BLOCK_SIZE="512" TYPE="xfs" PARTUUID="..."
/dev/xvda3: UUID="..." BLOCK_SIZE="512" TYPE="xfs" PARTUUID="..."
/dev/xvda5: UUID="..." TYPE="swap" PARTUUID="..."
```

- Note that both `/dev/xvdb1` and `/dev/xvdd1` now have labels.

4. Mount the file systems.

- a. Use the `mkdir` command to create mount points.

Be careful with the case-sensitive names: `/Dev` is not the same as `/dev`!

```
# mkdir /Test /Dev
```

- b. Use the `mount` command to mount `/dev/xvdb1` on `/Dev`.

```
# mount /dev/xvdb1 /Dev
```

- c. Mount `/dev/xvdd1` on `/Test`.

```
# mount /dev/xvdd1 /Test
```

- d. Use the `df` command to display the mounted file systems. Your output might vary.

```
# df -h
Filesystem      Size  Used Avail Use% Mounted on
...
/dev/xvda2       24G   1.8G   23G   8% /
/dev/xvda3       2.0G    47M   2.0G   3% /home
/dev/xvda1      1014M   237M   778M  24% /boot
...
/dev/xvdb1       976M   2.6M   907M   1% /Dev
/dev/xvdd1       979M   2.5M   910M   1% /Test
```

- e. Use the `mount` command to display the mounted file systems. Your output might be ordered differently.

```
# mount
...
/dev/xvda2 on / type xfs (rw,relatime,seclabel,attr2,inode64,...
...
/dev/xvda1 on /boot type xfs (rw,relatime,seclabel,attr2,inod...
/dev/xvda3 on /home type xfs (rw,relatime,seclabel,attr2,inod...
...
/dev/xvdb1 on /Dev type ext4 (rw,relatime,seclabel)
/dev/xvdd1 on /Test type ext4 (rw,relatime,seclabel)
```

- Note that the file systems are mounted read/write (`rw`) by default.
- f. Display the mounts in `/proc/mounts`.

```
# cat /proc/mounts
sysfs /sys sysfs rw,seclabel,nosuid,nodev,noexec,relatime 0 0
proc /proc proc rw,nosuid,nodev,noexec,relatime 0 0
...
/dev/xvdb1 /Dev ext4 rw,seclabel,relatime 0 0
/dev/xvdd1 /Test ext4 rw,seclabel,relatime 0 0
```

5. Update the file systems mount table.

Use the `vi` editor to add the following entries to the end of `/etc/fstab`:

```
# vi /etc/fstab
LABEL=Dev    /Dev    ext4    defaults    0 0
LABEL=Test   /Test   ext4    defaults    0 0
```

- After updating `/etc/fstab`, the new file systems mount after a reboot.
 - You can also run the `mount -a` command to mount all file systems in `/etc/fstab`.
6. Mount all file systems in `/etc/fstab`.
- a. Use the `umount` command to unmount the file systems on `/Dev` and `/Test`.

```
# umount /Dev /Test
```

- b. Use the `df` command to display the mounted file systems. Your output might vary.

```
# df -h
Filesystem      Size  Used Avail Use% Mounted on
...
/dev/xvda2       24G   1.8G   23G    8% /
/dev/xvda3       2.0G    47M   2.0G    3% /home
/dev/xvda1      1014M   237M   778M   24% /boot
...
```

- Note that `/Dev` and `/Test` are not listed.

- c. Use the `mount -a` command to mount all file systems in `/etc/fstab`.

```
# mount -a
```

- d. Use the `df` command to display the mounted file systems. Your output might vary.

```
# df -h
Filesystem      Size  Used Avail Use% Mounted on
...
/dev/xvda2       24G   1.8G   23G   8% /
/dev/xvda3       2.0G   47M   2.0G   3% /home
/dev/xvda1      1014M  237M   778M  24% /boot
...
/dev/xvdb1       976M   2.6M   907M   1% /Dev
/dev/xvdd1       979M   2.5M   910M   1% /Test
```

- Note that `/Dev` and `/Test` are now listed.

7. Create files on the new file systems.

This task verifies that the new file systems are usable.

- a. Use the `vi` command to create a file on `/Dev`.

- Name the file whatever you want. This example names the file `dev_file`.

```
# vi /Dev/dev_file
insert any text you want
```

- b. Use the `ls` command to list the contents of `/Dev`.

```
# ls -la /Dev
total 24
drwxr-xr-x.  3 root root  4096 date_time .
dr-xr-xr-x. 20 root root   263 date_time ..
-rw-r--r--.  1 root root    25 date_time dev_file
drwx-----.  2 root root 16384 date_time lost+found
```

- The `lost+found` directory is automatically created when you make an `ext` file system. This directory is used by the `fsck` command to repair a damaged file system. You do not need to manually interact with the `lost+found` directory.
 - Note that `dev_file` exists on `/Dev`.
- c. Use the `cp` command to copy the `/Dev/dev_file` file to `/Test/test_file`.

```
# cp /Dev/dev_file /Test/test_file
```

- d. Use the `ls` command to list the contents of `/Test`.

```
# ls -la /Test
total 24
drwxr-xr-x.  3 root root  4096 date_time .
dr-xr-xr-x. 20 root root   263 date_time ..
drwx-----.  2 root root 16384 date_time lost+found
-rw-r--r--.  1 root root    25 date_time test_file
```

- Note that `test_file` exists on `/Test`.

Practice 12-4: Increasing Swap Space

Overview

In this practice, you increase the amount of swap space by creating, initializing, and enabling a swap file.

Assumptions

You are the `root` user on `host03`.

Tasks

1. Display the current amount of swap space. Your output might vary.
 - a. Use the `swapon` command to display the current amount of swap space.

```
# swapon -s
Filename              Type              Size      Used  Priority
/dev/xvda5            partition        3145724  0      -2
```

- b. Display the amount of swap space used by viewing the `/proc/meminfo` file.

```
# grep -i swap /proc/meminfo
SwapCached:           0 kB
SwapTotal:            3145724 kB
SwapFree:              3145724 kB
```

- c. Use the `free` command to display the amount of swap space.

```
# free
              total            used            free              ...
Swap          3145724              0          3145724
```

2. Create and initialize a swap file.
 - a. Use the `dd` command to create a 1 GiB swap file, `/swapfile`.

```
# dd if=/dev/zero of=/swapfile bs=1024 count=1048576
1048576+0 records in
1048576+0 records out
1073741824 bytes (1.1 GB, 1.0 GiB) copied...
```

- b. Use the `chmod` command to protect the swap file.

```
# chmod 600 /swapfile
```

- c. Use the `mkswap` command to initialize the swap file.

```
# mkswap /swapfile
Setting up swspace version 1, size = 1024 MiB (1073737728...
no label, UUID=...
```

3. Enable swapping on the swap file.

- a. Use the `swapon` command to enable swapping on the swap file.

```
# swapon /swapfile
```

- b. Display the updated amount of swap space. Your output might vary.

```
# swapon -s
Filename                Type                Size      Used    Priority
/dev/xvda5              partition          3145724  0       -2
/swapfile               file              1048572  0       -3

# grep -i swap /proc/meminfo
SwapCached:              0 kB
SwapTotal:               4194296 kB
SwapFree:                 4194296 kB

# free
              total            used            free              ...
...
Swap:         4194296              0         4194296
```

Practice 12-5: Removing Partitions and Additional Swap Space

Overview

In this practice, you delete the partitions on the `/dev/xvdb` and `/dev/xvdd` devices, and remove the additional swap file in preparation for the next lesson.

Assumptions

You are the `root` user on `host03`.

Tasks

1. View the mounted file systems.

Use the `df` command to display the mounted file systems. Your output might vary.

```
# df -h
Filesystem      Size  Used Avail Use% Mounted on
...
/dev/xvda2       24G   2.8G   22G   12% /
/dev/xvda3       2.0G   47M   2.0G    3% /home
/dev/xvda1      1014M  237M   778M   24% /boot
...
/dev/xvdb1       976M   2.6M   907M    1% /Dev
/dev/xvdd1       979M   2.5M   910M    1% /Test
```

- Note that `/dev/xvdd1` is mounted on `/Test` and `/dev/xvdb1` is mounted on `/Dev`.

2. Unmount the file systems created in the practice.

Use the `umount` command to unmount `/Dev` and `/Test`. Use the `cd` command first, to ensure you are in the `root` user's home directory.

```
# cd
# umount /Dev /Test
```

- You cannot delete a partition when a file system is mounted on it.

3. Delete the partitions created earlier.

- a. Use the `fdisk` command to delete the `/dev/xvdb1` partition.

Use the `d` command to delete partition 1.

Use the `p` command to print the partition table and confirm there are no partitions.

Use the `w` command to save the partition table and exit the `fdisk` utility.

```
# fdisk /dev/xvdb
...
Command (m for help): d
Selected partition 1
Partition 1 has been deleted

Command (m for help): p

Disk /dev/xvdb: 5 GiB, 5299999744 bytes, 10351562 sectors
...
    Device Boot          Start            End          Blocks      Id  System

```

Command (m for help): **w**
The partition table has been altered.
Calling `ioctl()` to re-read partition table.
Syncing disks.

- b. In the same way, use the `fdisk` command to delete the `/dev/xvdd1` partition.

```
# fdisk /dev/xvdd
...
Command (m for help): d
Selected partition 1
Partition 1 has been deleted

Command (m for help): p

Disk /dev/xvdd: 5 GiB, 5299999744 bytes, 10351562 sectors
...
Disklabel type: gpt
...
Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```


4. Use `parted` to change the label on `/dev/xvdd` from `gpt` to `msdos`.

Use the `mklabel` command to create an `msdos` label.

Use the `print` command to show the partition table to confirm the label change.

Use the `quit` command to exit the `parted` utility.

```
# parted /dev/xvdd
GNU Parted 3.2
Using /dev/xvdd
Welcome to GNU Parted! Type 'help' to view a list of commands.
(parted) mklabel msdos
Warning: The existing disk label on /dev/xvdd will be destroyed
and all data on this disk will be lost. Do you want to continue?
Yes/No? yes
(parted) print
Model: Xen Virtual Block Device (xvd)
Disk /dev/xvdd: 5300MB
Sector size (logical/physical): 512B/512B
Partition Table: msdos
Disk Flags:

Number  Start  End  Size  Type  File system  Flags

(parted) quit
Information: You may need to update /etc/fstab.
```

- Note: This could also be achieved using `fdisk`.

5. View the `xvd` devices in the `/proc/partitions` file to ensure that the `xvdb1` and `xvdd1` entries no longer exist.

```
# grep xvd /proc/partitions
202      0  31457280 xvda
202      1   1048576 xvda1
202      2  25165824 xvda2
202      3   2097152 xvda3
202      4           1 xvda4
202      5   3143680 xvda5
202     16   5242880 xvdb
202     48   5242880 xvdd
```

6. Use the `vi` command to remove the `Dev` and `Test` entries from the `/etc/fstab` file.

```
# vi /etc/fstab
LABEL=Dev /Dev ext4 acl 0 0
LABEL=Test /Test ext4 defaults 0 0
```

7. Use the `rmdir` command to remove the `/Dev` and `/Test` mount points.

```
# rmdir /Dev /Test
```

8. Use the `swapoff` command to disable swapping to `/swapfile`.

```
# swapoff /swapfile
```

9. Use the `rm` command to remove `/swapfile`.

```
# rm /swapfile
rm: remove regular file '/swapfile'? y
```

10. Log off **host03**.

Use the `logout` command to log out as the `root` user.

```
[root@host03 ~]# logout
Connection to host03 closed.
[oracle@gateway ~]$
```

You are now the `oracle` user on **gateway**.

Practices for Lesson 13: Network Configuration

Practices for Lesson 13: Overview

Practices Overview

In these practices, you:

- Configure the `eth1` network interface by editing network interface configuration files
- Use the `nmcli` utility to configure networking properties
- Use the `nmtui` text-based utility to configure network interfaces
- Use the `ip` utility to manage network links, addresses, and the ARP cache

Practice 13-1: Configuring the `eth1` Network Interface

Overview

In this practice, you:

- Display the available network interfaces on your system
- View the network interface configuration files
- Configure a static IP address for the `eth1` network interface
- Update your `/etc/hosts` file
- Display your route table
- Ensure connectivity to **gateway** and the other VM guests

Assumptions

- You are the `oracle` user on **gateway**.

Tasks

1. Log in to **host03**.
 - a. Connect to the **host03** guest by using the `ssh` command.

```
[oracle@gateway ~]$ ssh root@host03
root@host03's password:
[root@host03 ~]#
```

2. Use the `ip addr` command to display your available network interfaces.

```
# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state ...
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    ...
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:01:01 brd ff:ff:ff:ff:ff:ff
    inet 192.0.2.103/24 brd 192.0.2.255 scope global ...
    ...
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:01:02 brd ff:ff:ff:ff:ff:ff
    ...
```

- Note that you have two Ethernet interfaces (`eth0` and `eth1`) and the loopback interface (`lo`).
 - The `eth0` Ethernet interface has an IP address, but `eth1` does not.
3. View the network interface configuration files.
 - a. Use the `cd` command to change to the `/etc/sysconfig/network-scripts` directory.

```
# cd /etc/sysconfig/network-scripts
```

- b. Use the `ls` command to view the contents of this directory.

```
# ls
ifcfg-eth0
```

- Note that you have a configuration file for `eth0`, `ifcfg-eth0`.
- Note this directory previously contained many shell scripts for managing network interfaces, but these have been removed in Oracle Linux 8. A software package called `network-scripts` is available for systems that depend on the legacy scripts, but its use is discouraged.

- c. Use the `cat` command to view the contents of the `ifcfg-eth0` file.

```
# cat ifcfg-eth0
TYPE=Ethernet
PROXY_METHOD=none
BROWSER_ONLY=no
DEFROUTE=yes
IPV4_FAILURE_FATAL=no
IPV6INIT=yes
IPV6_AUTOCONF=yes
IPV6_DEFROUTE=yes
IPV6_FAILURE_FATAL=no
IPV6_ADDR_GEN_MODE=stable-privacy
NAME=eth0
UUID=...
DEVICE=eth0
ONBOOT=yes
IPADDR=192.0.2.103
PREFIX=24
GATEWAY=192.0.2.1
```

- Note that this Ethernet interface is configured with a static IPv4 address:
 - `IPADDR=192.0.2.103`
 - `PREFIX=24`
- Some network interface configuration file parameters are described in <https://docs.oracle.com/en/operating-systems/oracle-linux/8/network/ol-netconf.html#ol-netconf-about>.

4. Configure `eth1` with a static IP address of `192.168.1.103`.

- a. Copy the configuration file for `eth0` to create the `ifcfg-eth1` file.

```
# cp ifcfg-eth0 ifcfg-eth1
```

- b. Use the `vi` editor to edit the `ifcfg-eth1` file as follows: (The required changes are listed in the following bullets and are in **bold font**.)

Change `DEFROUTE=yes` to **`DEFROUTE=no`**

Change `NAME=eth0` to **`NAME=eth1`**

Delete the `UUID` entry. A new `UUID` will be generated by NetworkManager

Change `DEVICE=eth0` to **`DEVICE=eth1`**

Change `IPADDR=192.0.2.103` to **`IPADDR=192.168.1.103`**

Delete the entry `GATEWAY=192.0.2.1`

```
# vi ifcfg-eth1
TYPE=Ethernet
PROXY_METHOD=none
BROWSER_ONLY=no
DEFROUTE=no
IPV4_FAILURE_FATAL=no
IPV6INIT=yes
IPV6_AUTOCONF=yes
IPV6_DEFROUTE=yes
IPV6_FAILURE_FATAL=no
IPV6_ADDR_GEN_MODE=stable-privacy
NAME=eth1
UUID=d7d5315f-778b-4470-b644-04a35673cdf3
DEVICE=eth1
ONBOOT=yes
IPADDR=192.168.1.103
PREFIX=24
GATEWAY=192.0.2.1
```

5. Update the `/etc/hosts` file to provide hostname to IP address mapping.
- a. Make a copy of the original `/etc/hosts` file.

```
# cp /etc/hosts /etc/hosts_SAV
```

- b. Use the `vi` editor to edit the `/etc/hosts` file, adding the entries in bold:

```
# vi /etc/hosts
127.0.0.1    localhost localhost.localdomain localhost4...
::1         localhost localhost.localdomain localhost6...
192.0.2.1    example.com                gateway
192.0.2.102  host02.example.com            host02
192.0.2.103  host03.example.com            host03
```

6. Use the `nmcli` command to apply the changes to the `eth1` interface.

```
# nmcli c reload
# nmcli c up eth1
Connection successfully activated (D-Bus active path:
/org/freedesktop/NetworkManager/ActiveConnection/2)
```

7. Use the `ip addr` command to display the status of the interfaces.

```
# ip addr
...
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:01:01 brd ff:ff:ff:ff:ff:ff
    inet 192.0.2.103/24 brd 192.0.2.255 scope global ...
...
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.103/24 brd 192.168.1.255 scope global ...
...
```

- Note that both `eth0` and `eth1` now have IP addresses.

8. Display the route table.

Use the `ip route` command to display the route table.

```
# ip route
default via 192.0.2.1 dev enp1s0 proto static metric 100
192.0.2.0/24 dev eth0 proto kernel scope link src 192.0.2.103 metric 100
192.168.1.0/24 dev eth1 proto kernel scope link src 192.168.1.103 metric 101
```

- Note that all packets destined for the `192.168.1` subnet use the `eth1` interface.
- Observe that all packets destined for the `192.0.2` subnet use the `eth0` interface.
- Note that all other packets are routed through the `192.0.2.1` default gateway (`example.com`), via `eth0`.

9. Use the `ping` command to verify that you can communicate to **gateway** and **host02**.

- a. Test connections via the `eth0` network interface.

```
# ping -c 3 gateway
64 bytes from example.com (192.0.2.1):...
# ping -c 3 host02
64 bytes from host02.example.com (192.0.2.102):...
```

- b. Test connections via the `eth1` network interface.

```
# ping -c 3 192.168.1.1
64 bytes from 192.168.1.1:...
# ping -c 3 192.168.1.102
64 bytes from 192.168.1.102:...
```


10. Use the `nmcli` command to bring down the `eth1` interface.

```
# nmcli c down eth1
Connection successfully activated (D-Bus active path:
/org/freedesktop/NetworkManager/ActiveConnection/2)
```

11. Use the `ip addr` command to display your available network interfaces.

```
# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state ...
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    ...
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:01:01 brd ff:ff:ff:ff:ff:ff
    inet 192.0.2.103/24 brd 192.0.2.255 scope global ...
    ...
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:01:02 brd ff:ff:ff:ff:ff:ff
    ...
```

- Note that `eth1` does not have an IP address.

Practice 13-2: Using the `nmcli` Utility

Overview

In this practice, you use the `nmcli` command-line utility to view network status information, change the system host name, change the logging level, disable and enable networking, view connection information, add, edit and delete a connection profile, and view device status information.

Assumptions

You are the `root` user on `host03`.

Tasks

1. Run the `nmcli help` command.

```
# nmcli help
Usage: nmcli [OPTIONS] OBJECT { COMMAND | help }
OPTIONS
  -a, --ask                ask for missing parameters
  -c, --colors auto|yes|no  whether to use colors in output
  -e, --escape yes|no      escape columns separators in values
  ...
OBJECT
  g[eneral]      NetworkManager's general status and operations
  n[etworking]   overall networking control
  r[adio]        NetworkManager radio switches
  c[onnection]   NetworkManager's connections
  d[evice]       devices managed by NetworkManager
  a[gent]        NetworkManager secret agent or polkit agent
  m[onitor]      monitor NetworkManager changes
```

- Note that several options are available:
 - t|--terse: This mode is designed and suitable for script processing.
 - p|--pretty: This mode produces easily readable output with a header.
 - m|--mode tabular|multiline: This produces output in table format or on multiple lines.
- Refer to the `nmcli(1)` man page for a description of all options.
- Note that there are seven different objects for the `nmcli` command.

2. Run the `nmcli` general object commands.

a. Run the `nmcli` general help command.

```
# nmcli general help
Usage: nmcli general { COMMAND | help }

COMMAND := { status | hostname | permissions | logging }
...
```

- Note that the `nmcli` general object provides four commands.

b. Run the `nmcli` general status command.

```
# nmcli general status
STATE      CONNECTIVITY  WIFI-HW  WIFI      WWAN-HW  WWAN
connected  full                enabled  enabled   enabled   enabled
```

- Note that “status” is the default. That is, you can omit this argument.
- Note that the network status is “connected” with “full” connectivity.
- Full connectivity means the host is connected to a network and has full access to the Internet.

c. Use the `systemctl` command to stop the NetworkManager service.

```
# systemctl stop NetworkManager
```

d. Run the `nmcli` general status command.

```
# nmcli general status
Error: NetworkManager is not running.
```

- With the NetworkManager service stopped, an error is returned.

e. Use the `systemctl` command to start the NetworkManager service.

```
# systemctl start NetworkManager
```

f. Run the `nmcli` general hostname command.

```
# nmcli general hostname
host03.example.com
```

- This command reports the host name when the NetworkManager service is running.
- You can also use this `nmcli` general hostname command to change the host name.

g. Run the `nmcli` general hostname command and change the host name to “myhost”.

```
# nmcli general hostname myhost
```

- The host name is stored in the `/etc/hostname` file.

h. Use the `cat` command to view the contents of the `/etc/hostname` file.

```
# cat /etc/hostname
myhost
```

- i. Run the `nmcli general hostname` command and change the host name back to "host03.example.com".

```
# nmcli general hostname host03.example.com
```

- j. Use the `cat` command to view the contents of the `/etc/hostname` file.

```
# cat /etc/hostname
host03.example.com
```

- k. Run the `nmcli general permissions` command.

```
# nmcli general permissions
PERMISSION                                VALUE
...
org.freedesktop.NetworkManager.enable-disable-wimax    yes
org.freedesktop.NetworkManager.enable-disable-wwan      yes
org.freedesktop.NetworkManager.network-control         yes
org.freedesktop.NetworkManager.reload                  yes
org.freedesktop.NetworkManager.settings.modify.global-dns yes
org.freedesktop.NetworkManager.settings.modify.hostname yes
...
```

- In this example, all permissions are set to `yes`, which means that you can enable and disable networking and modify all connections and settings.

- l. Run the `nmcli general logging` command.

```
# nmcli general logging
LEVEL  DOMAINS

INFO   PLATFORM,RFKILL,ETHER,WIFI,BT,MB,DHCP4,DHCP6,PPP,IP4,IP6,
AUTOIP4,DNS,VPN,SHARING,SUPPLICANT,AGENTS,SETTINGS,SUSPEND,CORE,
DEVICE,OLPC,INFINIBAND,FIREWALL,ADSL,BOND,VLAN,BRIDGE,TEAM,CONCH
ECK,DCB,DISPATCH,AUDIT,SYSTEMD,PROXY
```

- With no arguments, this command shows the current logging level by domain.
- In this example, the logging level is `INFO` for all domains.
- Refer to the `NetworkManager.conf(5)` man page for information about logging levels and domain descriptions.

- m. Use the `nmcli general logging` command to change the logging level to `DEBUG` for the `IP4` domain.

```
# nmcli general logging level DEBUG domains IP4
```

- n. Run the `nmcli general logging` command to show the current logging level.

```
# nmcli general logging
LEVEL  DOMAINS
DEBUG  IP4
```

- o. Use the `nmcli general logging` command to change the logging level to `INFO` for the `ALL` domains.

```
# nmcli general logging level INFO domains ALL
```

- This command returns the logging level to the default setting.

3. Run the `nmcli networking` object commands.

- a. Run the `nmcli networking help` command.

```
# nmcli networking help
Usage: nmcli networking { COMMAND | help }

COMMAND := { [ on | off | connectivity ] }

...
```

- Note that the `nmcli networking` object provides three commands.

- b. Run the `nmcli networking` command with no options or arguments to show the networking status.

```
# nmcli networking
enabled
```

- Note that the status is `enabled`.
- Note there are options to turn networking off and on, but as you are connected to **host03** via the network you cannot run these commands or you will lose your connection.

- c. Run the `nmcli networking connectivity` command to show the network connectivity state.

Include the `check` argument.

```
# nmcli networking connectivity check
full
```

- Without the `check` argument, the command displays the most recent known connectivity state without re-checking.
- Connectivity state `full` means the host is connected to a network and has full access to the Internet, based on the result of and configured connectivity checks.
- The package `NetworkManager-config-connectivity-oracle` configures connectivity checking to poll a service at `yum.oracle.com`.

4. Run the `nmcli radio` object commands.

- a. Run the `nmcli radio help` command.

```
# nmcli radio help
Usage: nmcli radio { COMMAND | help }

COMMAND := { all | wifi | wwan }

...
```

- b. Run the `nmcli radio` command with no options or arguments to show the status of the radio switches.

```
# nmcli radio
WIFI-HW  WIFI      WWAN-HW  WWAN
enabled  enabled  enabled  enabled
```

- Note that all switches are enabled.

- c. Run the `nmcli radio wifi off` command to disable the WI-FI radio switch.

```
# nmcli radio wifi off
```

- d. Run the `nmcli radio` command to show the status of the radio switches.

```
# nmcli radio
WIFI-HW  WIFI      WWAN-HW  WWAN
enabled  disabled  enabled  enabled
```

- Note that the WI-FI switch is disabled.

- e. Run the `nmcli radio wifi on` command to enable the WI-FI radio switch.

- Run the `nmcli radio` command to show the status of the radio switches.

```
# nmcli radio wifi on
# nmcli radio
WIFI-HW  WIFI      WWAN-HW  WWAN
enabled  enabled  enabled  enabled
```

- Note that all switches are enabled.

5. Run the `nmcli connection` object commands.

- a. Run the `nmcli connection help` command.

```
# nmcli connection help
Usage: nmcli connection { COMMAND | help }

COMMAND := { show | up | down | add | modify | clone | edit |
delete | monitor | reload | load | import | export }
...
```

- Note that the `nmcli connection` object provides 13 commands.

- b. Run the `nmcli connection show` command.

```
# nmcli connection show
NAME          UUID          TYPE          DEVICE
eth0          ...          ethernet     eth0
eth1          ...          ethernet     eth1
```

- This command lists all the connection profiles.
- Note that the loopback interface does not have a Network Manager connection profile.

- c. Run the `nmcli connection show id eth0` command.

Only a partial output is shown.

```
# nmcli connection show id eth0
connection.id:          eth0
connection.uuid:        ...
...
connection.type:        802-3-ethernet
connection.interface-name: eth0
connection.autoconnect: yes
...
802-3-ethernet.mtu:     auto
...
ipv4.method:            manual
...
ipv4.addresses:         192.0.2.103/24
ipv4.gateway:           192.0.2.1
...
```

- This command shows detailed information for a specific connection, `eth0`.

- d. Run the `nmcli connection show id eth1` command.

Pipe the output to `grep` and search for the string: `autoconnect`.

```
# nmcli connection show id eth1 | grep autoconnect
connection.autoconnect: yes
...
```

- Note that the `autoconnect` parameter is set to `yes`.
- When NetworkManager was restarted in an earlier step, it brought connection `eth1` up because `autoconnect` is set to `yes`.

6. Use the `nmcli connection` object to add a connection profile.

- a. Run the `nmcli connection add` command to add a new connection profile. Use the following parameters:

Connection name (`con-name`): `new_eth1`

Interface name (`ifname`): `eth1`

Type (`type`): `ethernet`

IPv4 address (`ip4`): `192.168.1.113/24`

IPv4 Gateway (`gw4`): `192.168.1.1`

```
# nmcli connection add con-name new_eth1 ifname eth1 type
ethernet ip4 192.168.1.113/24 gw4 192.168.1.1
Connection 'new_eth1' (...) successfully added.
```

- b. Run the `nmcli connection show` command.

```
# nmcli connection show
NAME          UUID          TYPE          DEVICE
eth0          ...           ethernet     eth0
eth1          ...           ethernet     eth1
new_eth1      ...           ethernet     --
```

- Note that the `new_eth1` connection now exists.

- c. Use the `ls` command to view network interface configuration files in the `/etc/sysconfig/network-scripts` directory.

```
# ls /etc/sysconfig/network-scripts
ifcfg-eth0  ifcfg-eth1  ifcfg-new_eth1
```

- Note that the `nmcli connection add` command created a network interface configuration file, `ifcfg-new_eth1`, for the new connection profile, `new_eth1`.

7. Edit parameters of the connection profile interactively.

- a. Run the `nmcli connection edit` command to edit connection parameters of the `new_eth1` connection profile.

This command uses an interactive editor.

```
# nmcli connection edit new_eth1

===| nmcli interactive connection editor |===

Editing existing '802-3-ethernet' connection: 'new_eth0'

Type 'help' or '?' for available commands.
Type 'print' to show all the connection properties.
Type 'describe [<settings>.<prop>]' for detailed property
description.

You may edit the following settings: connection, 802-3-ethernet
(ethernet), 802-1x, dcb, sriov, ethtool, match, ipv4, ipv6, tc,
proxy
nmcli>
```

- b. Enter `help` at the `nmcli>` prompt to list the available commands.

```
nmcli> help
...
set      [<setting>.<prop> <value>]  :: set property value
describe [<setting>.<prop>]          :: describe property
print    [all | <setting>[.<prop>]]  :: print the connection
verify   [all | fix]                :: verify the connection
save     [persistent|temporary]      :: save the connection
...
```


- c. Enter `print` at the `nmcli>` prompt to print the connection profile details.
Only partial output is shown.

```
nmcli> print
=====
                        Connection profile details (new_eth1)
=====
connection.id:                new_eth1
connection.uuid:              ...
...
connection.type:              802-3-ethernet
connection.interface-name:    eth1
connection.autoconnect:       yes
...
-----
...
802-3-ethernet.mac-address:   --
...
802-3-ethernet.mtu:           auto
...
-----
ipv4.method:                  manual
ipv4.dns:                     --
ipv4.dns-search:              --
...
ipv4.addresses:               192.168.1.113/24
ipv4.gateway:                 192.168.1.1
...

```

- Note that the output is similar to the `nmcli connection show id new_eth1` command output.
 - Note that the `connection.autoconnect` parameter is set to `yes`.
- d. Use the `set` command from the `nmcli>` prompt to change the following parameter:
`connection.autoconnect=no`

```
nmcli> set connection.autoconnect no
```

- e. Use the `quit` command from the `nmcli>` prompt to exit the interactive editor.
Answer `n` when prompted. Do not exit the `nmcli` interactive editor.

```
nmcli> quit
The connection is not saved. Do you really want to quit?
(yes/no) [no] n
```

- f. Use the `save` command from the `nmcli>` prompt to save your change.

```
nmcli> save
Connection 'new_eth1' (...) successfully updated.
```

- g. Use the `quit` command from the `nmcli>` prompt to exit the interactive editor.

```
nmcli> quit
```

- h. Run the `nmcli connection show id new_eth1` command.

Pipe the output to `grep` and search for the string: `autoconnect`.

```
# nmcli connection show id new_eth1 | grep autoconnect
connection.autoconnect:      no
...
```

- Note that the `connection.autoconnect` parameter is set to `no`.

8. Modify parameters of the connection profile using command-line instructions.

- a. Run the `nmcli connection modify` command to modify a connection parameter for the `new_eth0` connection.

Set the `ipv4.dns` parameter to `192.168.1.1`.

```
# nmcli connection modify new_eth1 ipv4.dns 192.168.1.1
```

- This command does not use an interactive editor.

- b. Run the `nmcli connection show id new_eth1` command.

Pipe the output to `grep` and search for the string: `ipv4.dns`.

```
# nmcli connection show id new_eth1 | grep ipv4.dns
ipv4.dns:      192.168.1.1
ipv4.dns-search:  --
...
```

- Note that the `ipv4.dns` parameter is set to `192.168.1.1`.

9. Bring the new connection profile online.

- a. Run the `nmcli connection up id new_eth1` command.

```
# nmcli connection up id new_eth1
Connection successfully activated (D-Bus active path:
/org/freedesktop/NetworkManager/ActiveConnection/3)
```

- Note that this is a second profile configure for network interface `eth1`.

- b. Run the `nmcli connection show` command.

```
# nmcli connection show
NAME          UUID          TYPE          DEVICE
eth0          ...           ethernet      eth0
new_eth1      ...           ethernet      eth1
eth1          ...           Ethernet      --
```

- Note that only one profile can be active for a device, so profile `eth1` has been taken down.

10. Delete the connection profile.

- a. Run the `nmcli connection delete` command to delete the `new_eth0` connection profile.

```
# nmcli connection delete new_eth1
Connection 'new_eth1' (...) successfully deleted.
```

- b. Run the `nmcli connection show` command.

```
# nmcli connection show
```

NAME	UUID	TYPE	DEVICE
eth0	...	ethernet	eth0
eth1	...	ethernet	eth1

- Note that the `new_eth1` connection no longer exists.
- Note that the original profile for `eth1` has reactivated because it is set to `autoconnect`.

- c. Use the `ls` command to view network interface configuration files in the `/etc/sysconfig/network-scripts` directory.

```
# ls /etc/sysconfig/network-scripts
ifcfg-eth0  ifcfg-eth1
```

- Note that the network interface configuration file for the `new_eth1` connection profile no longer exists.

11. Run the `nmcli device` object commands.

- a. Run the `nmcli device help` command.

```
# nmcli device help
Usage: nmcli device { COMMAND | help }

COMMAND := { status | show | set | connect | reapply | modify |
disconnect | delete | monitor | wifi | lldp }
...
```

- Note that the `nmcli device` object provides 11 commands.

- b. Run the `nmcli device status` command.

- This command displays the status of all the devices.

```
# nmcli device status
```

DEVICE	TYPE	STATE	CONNECTION
eth0	ethernet	connected	eth0
eth1	ethernet	connected	eth1
lo	loopback	unmanaged	--

- c. Run the `nmcli device disconnect eth1` command.

```
# nmcli device disconnect eth1
Device 'eth1' successfully disconnected.
```

- d. Run the `nmcli device status` command.

The `status` command is the default for the `nmcli device` object and so it is not required.

```
# nmcli device
DEVICE      TYPE          STATE          CONNECTION
eth0        ethernet      connected      eth0
eth1        ethernet      disconnected    --
lo          loopback      unmanaged      --
```

- Note the change in the `eth1` device.

- e. Run the `nmcli device connect eth1` command to reconnect the `eth1` device.

```
# nmcli device connect eth1
Device 'eth1' successfully activated with '...'.
```

Practice 13-3: Using the `nmtui` Utility

Overview

In this practice, you use the `nmtui` text-based utility to view network connections.

Assumptions

You are the `root` user on `host03`.

Tasks

1. Use the `rpm` command to verify that the `NetworkManager-tui` package is installed.

The `nmtui` utility is provided by the `NetworkManager-tui` software package.

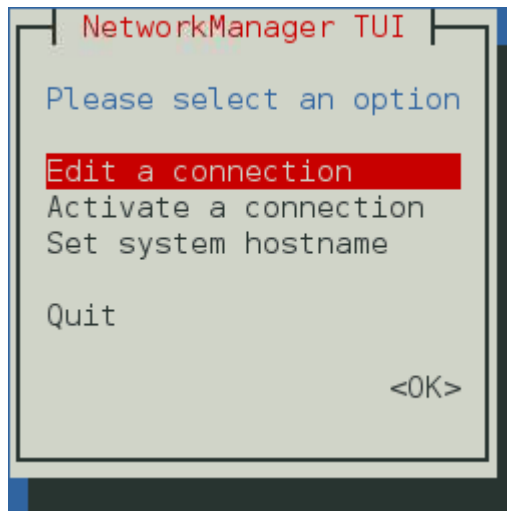
```
# rpm -qa | grep NetworkManager-tui
NetworkManager-tui-...
```

2. Use `nmtui` to configure network interfaces.

- a. Enter `nmtui` from the command line.

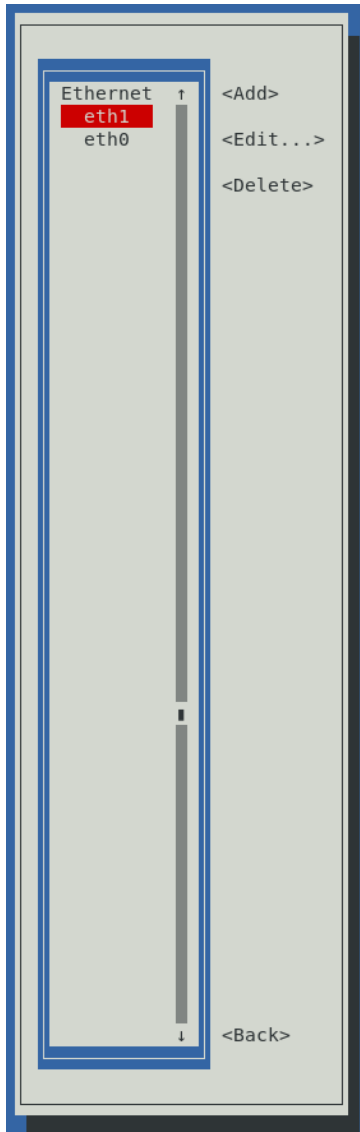
```
# nmtui
```

- b. The screen appears as follows:



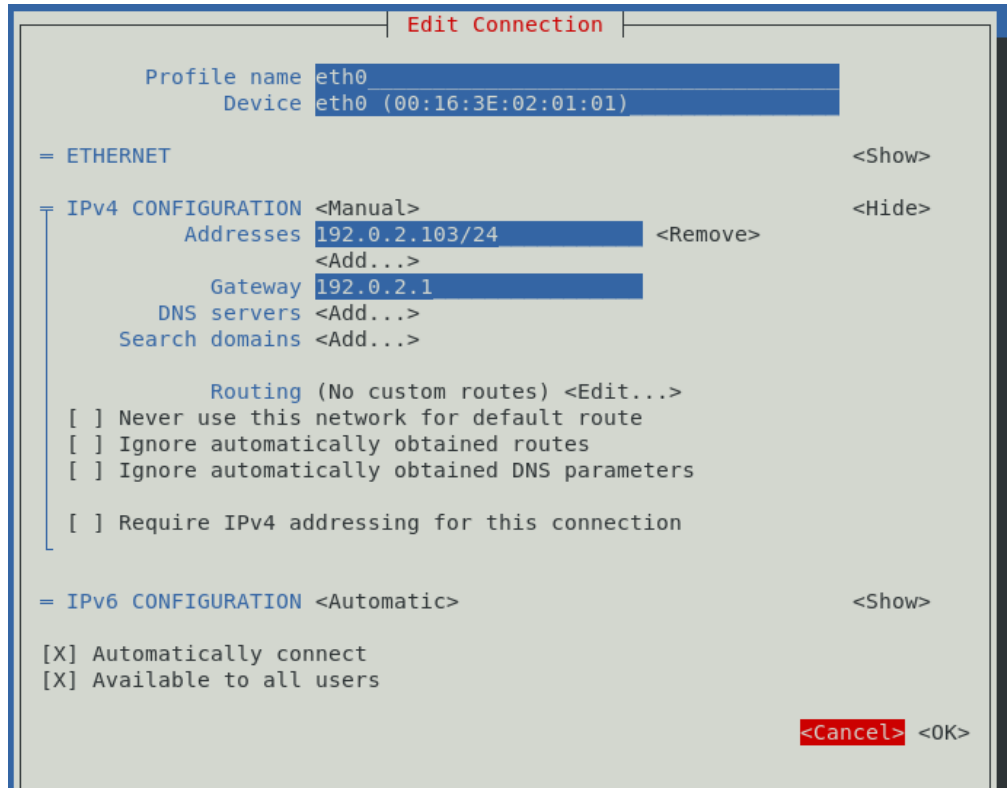
Use the up/down arrows to select **Edit a connection**. Use the Tab key to select **<OK>** and press Enter.

- c. The screen appears as follows:



- d. Use the up/down arrows to select **eth0**. Use the Tab key to select **<Edit...>** and press Enter.

e. The screen appears as follows:



Do not make any changes. After reviewing the information on the screen, use the down arrow key to select **<Cancel>** as shown.

- Note that the information displayed (except the MAC address) is included in the configuration file, `/etc/sysconfig/network-scripts/ifcfg-eth0`.
 - Any updates made from this screen are written to the configuration file.
- f. With **<Cancel>** selected as shown, press Enter.
- g. Use the Tab key to select **<Back>** and then press Enter.
- h. Use the down arrow key to select **<Quit>** and press Enter to exit the `nmtui` utility.

Practice 13-4: Using the `ip` Utility

Overview

In this practice, you use the `ip` utility to view network device information; add, edit, and delete a link; view IP addresses; add and delete an IPv4 address to a network device; and view and flush the ARP cache.

Assumptions

You are the `root` user on **host03**.

Tasks

1. Run the `ip` command without any options or arguments.

Only partial output is shown.

```
# ip
Usage: ip [ OPTIONS ] OBJECT { COMMAND | help }
       ip [ -force ] -batch filename
where OBJECT := { link | address | addrlabel | route | rule |
                  neigh | ntable | tunnel | tuntap | maddress |
                  mroute | mrule | monitor | xfrm | netns |
                  l2tp | fou | macsec | tcp_metrics | token |
                  netconf | ila | vrf | sr | nexthop | mptcp }
...
```

- Note that several options are available:
 - b, -batch *FILENAME*: Read and invoke commands from *FILENAME*.
 - V, -Version: Print the version of the `ip` utility.
 - s, -stats, -statistics: Output more information.
- Refer to the `ip(8)` man page for a description of all options.
- Note that there are 26 different objects for the `ip` command.

2. Run the `ip link` object commands.

A *link* is a network device. The `ip link` command is used to display and configure network devices.

a. Run the `ip link help` command.

Only partial output is shown.

```
# ip link help
Usage: ip link add [link DEV] [ name ] NAME
    ...
    ip link delete { DEVICE | dev DEVICE | ... } ... [ ARGS ]
    ...
    ip link set { DEVICE | dev DEVICE | ... }
                [ { up | down } ]
    ...
    ip link show [ DEVICE | group GROUP ] [up] ...

    ip link xstats type TYPE [ ARGS ]

    ip link afstats [ dev DEVICE ]

    ...
```

- Note that the `ip link` object provides six commands:
 - The six commands are `add`, `delete`, `set`, `show`, `xstats`, and `afstats`.

b. Run the `ip link show` command.

The `show` command is the default for the `ip link` object so it is not required.

```
# ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state ...
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:01:01 brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:01:02 brd ff:ff:ff:ff:ff:ff
```

- This command shows the existing network devices.
- Note that you have three network devices; two Ethernet devices (`eth0` and `eth1`) and one loopback device (`lo`).
- These are the same devices that are listed by the `nmcli device status` command.

- c. Run the `nmcli device status` command.

```
# nmcli device
DEVICE      TYPE          STATE          CONNECTION
eth0        ethernet     connected     eth0
eth1        ethernet     connected     eth1
lo          loopback     unmanaged     --
```

- This command lists the same devices as the `ip link show` command.

3. Use the `ip link` utility to add, edit, and delete a link.

The `ip link add` command adds a virtual link.

The following are examples of virtual link types: `vlan`, `veth`, `vcan`, `dummy`, `ifb`, `macvlan`, `bridge`, `ipoib`, `ip6tnl`, `ipip`, `sit`, or `vxlan`.

Refer to the `ip-link(8)` man page for more information about link types.

VLANs and advanced networking topics are covered in the *Oracle Linux System Administration II* course.

- a. Run the `ip link add` command to add a new link. Use the following parameters:

Physical device to operate on (link): `eth0`

Name (name): `eth0.10`

Type (type): `vlan`

VLAN ID (id): `10`

```
# ip link add link eth0 name eth0.10 type vlan id 10
```

- b. Run the `ip link show` command.

```
# ip link
...
4: eth0.10@eth0: <BROADCAST,MULTICAST> mtu 1500 ...
    link/ether 00:16:3e:02:01:01 brd ff:ff:ff:ff:ff:ff
```

- Note that you now have four network devices, including the new link `eth0.10`.

- c. Run the `nmcli device status` command.

```
# nmcli device
DEVICE      TYPE          STATE          CONNECTION
eth0        ethernet     connected     eth0
eth1        ethernet     connected     eth1
lo          loopback     unmanaged     --
eth0.10     vlan         unmanaged     --
```

- Note the `eth0.10` device is listed, but is not managed by NetworkManager.

- d. Run the `ip link set` command to change device attributes.

Change the MTU for the `eth0.10` device to `1400`.

```
# ip link set eth0.10 mtu 1400
```

- e. Run the `ip link show` command to show the `eth0.10` device.

The `show` command is required when specifying a device as an argument.

```
# ip link show eth0.10
4: eth0.10@eth0: <BROADCAST,MULTICAST> mtu 1400 ...
    link/ether 00:16:3e:02:01:01 brd ff:ff:ff:ff:ff:ff
```

- Note that the MTU is 1400.

- f. Run the `ip link delete` command to delete the `eth0.10` device.

```
# ip link delete eth0.10
```

- g. Run the `ip link show` command.

```
# ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state ...
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:01:01 brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:02:01 brd ff:ff:ff:ff:ff:ff
```

- Note that the `eth0.10` device no longer exists.

4. Run the `ip addr` object commands.

The `ip addr` command is used to display and manage IP addresses on network devices.

- a. Run the `ip addr help` command.

Only partial output is shown.

```
# ip addr help
Usage: ip address {add|change|replace} IFADDR dev IFNAME ...
       ip address del IFADDR dev IFNAME [mngtmpaddr]
       ip address {save|flush} [ dev IFNAME ] [ scope ... ] ...
       ip address [ show [ dev IFNAME ] [ scope SCOPE-ID ] ...
       ip address {showdump|restore}
IFADDR := PREFIX | ADDR peer PREFIX
...
```

- Note that the `ip addr` object provides nine commands:
 - The commands are `add`, `change`, `replace`, `del`, `show`, `save`, `flush`, `showdump`, and `restore`.
 - Refer to the `ip-address(8)` man page for more information on using `ip addr` commands.

- b. Run the `ip addr show` command.

The `show` command is the default for the `ip addr` object so it is not required.

Only partial output is shown.

```
# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state ...
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft ...
    inet6 ...
...
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:01:01 brd ff:ff:ff:ff:ff:ff
    inet 192.0.2.103/24 brd 192.0.2.255 scope global ...
        valid_lft ...
    inet6 ...
...
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.103/24 brd 192.168.1.255 scope global ...
        valid_lft ...
    inet6 ...
```

- This command shows the same information as `ip link` but also shows IP address information.

- c. Run the `ip addr add` command to add a second IP address to `eth1`.

Use `10.1.1.1/24` as the second IPv4 address.

```
# ip addr add 10.1.1.1/24 dev eth1
```

- d. Run the `ip addr show` command to show the `eth1` device.

The `show` command is required when specifying a device as an argument.

```
# ip addr show eth1
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.103/24 brd 192.168.1.255 scope global ...
        valid_lft ...
    inet 10.1.1.1/24 scope global eth1
        valid_lft ...
    inet6 ...
...
```

- Note that the device now has two IPv4 addresses.

- e. Run the `ip addr del` command to delete an IP address from `eth1`.

Delete address `10.1.1.1/24`.

```
# ip addr del 10.1.1.1/24 dev eth1
```

- f. Run the `ip addr show` command to show the `eth1` device.

```
# ip addr show eth1
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.103/24 brd 192.168.1.255 scope global ...
        valid_lft ...
    inet6 ...
...
```

- Note that the `10.1.1.1/24` address no longer exists.

5. Run the `ip neigh` object commands.

The "`ip neigh`" command is used to display and manage ARP cache entries.

- a. Run the `ip neigh help` command.

Only a partial output is shown.

```
# ip neigh help
Usage: ip neigh { add | del | change | replace }
        { ADDR ...
        ip neigh { show | flush } [ proxy ] [ to PREFIX ] ...
...
```

- Note that the `ip neigh` object provides six commands:
 - The commands are `add`, `del`, `change`, `replace`, `show`, and `flush`.

- b. Run the `ip neigh show` command.

The `show` command is the default for the `ip neigh` object so is not required.

```
# ip neigh
192.0.2.102 dev eth0 lladdr 00:16:3e:01:01:01 STALE
192.0.2.1 dev eth0 lladdr 00:21:f6:dc:01:01 REACHABLE
```

- Your output might vary depending on recent network activity.
- c. If no output is produced from the `ip neigh show` command, use the `ping` command to communicate to **gateway** and **host02**.

```
# ping -c 3 gateway
64 bytes from example.com (192.0.2.1)...
# ping -c 3 host02
64 bytes from host02.example.com (192.0.2.102)...
```

- d. Run the `ip neigh show` command again to list ARP cache entries. Your output might vary.

```
# ip neigh
192.0.2.102 dev eth0 lladdr 00:16:3e:01:01:01 REACHABLE
192.0.2.1 dev eth0 lladdr 00:21:f6:dc:01:01 REACHABLE
```

- e. Run the `ip neigh flush` command to clear all entries in the ARP cache.

```
# ip neigh flush all
```

- f. Run the `ip neigh show` command. Your output might vary.

```
# ip neigh
192.0.2.1 dev eth0 lladdr 00:21:f6:dc:01:01 REACHABLE
```

- In this example the entry for **gateway** (192.0.2.1) has reappeared because an `ssh` connection is being used between **gateway** and **host03**.

6. Log off **host03**.

Use the `logout` command to log out as the `root` user.

```
[root@host03 ~]# logout
Connection to host03 closed.
[oracle@gateway ~]$
```

You are now the `oracle` user on **gateway**.

Practices for Lesson 14: Network Security

Practices for Lesson 14: Overview

Practices Overview

In these practices, you:

- Explore and configure `firewalld`
- Configure `nftables`

Assumptions

- You updated the `/etc/hosts` file on **host03** in the practices for the lesson titled “Network Configuration”.
- You created a local Yum repository on **host02** in the practices for the lesson titled “Package Management”.

Practice 14-1: Exploring firewalld

Overview

In this practice, you:

- Start the `firewalld` service if necessary
- Explore `firewalld` zones
- Change the default zone
- Explore **runtime** and **permanent** configuration modes
- Explore `firewalld` services

Assumptions

- You are the `oracle` user on **gateway**.

Tasks

1. Log in to **host02**.

Connect to the **host02** guest by using the `ssh` command.

```
[oracle@gateway ~]$ ssh root@host02
root@host02's password:
[root@host02 ~]#
```

2. If necessary, start the `firewalld` service on **host02**.

- a. Use the `systemctl` command to check if the `firewalld` service is running.

```
# systemctl status firewalld
firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service...)
   Active: active (running) since...
   ...
```

- In this example, `firewalld` is enabled and running.
- b. Use the `firewall-cmd` command-line utility to check if the `firewalld` service is running.

```
# firewall-cmd --state
running
```

- This command provides another method to check the status of `firewalld`.
- c. If the `firewalld` service is not running, use the `systemctl` command to start it and repeat the steps above to confirm it is now running.

```
# systemctl start firewalld
```

3. Explore firewalld zones.

- a. From a command prompt, use the `ls` command to display the contents of the `/usr/lib/firewalld/zones` directory.

```
# ls /usr/lib/firewalld/zones
block.xml  dmz.xml  drop.xml  external.xml  home.xml
internal.xml  nm-shared.xml  public.xml  trusted.xml  work.xml
```

- Note that there is an XML configuration file for each of the 10 predefined zones.

- b. Use the `firewall-cmd` command to display the available zones.

```
# firewall-cmd --get-zones
block dmz drop external home internal nm-shared public trusted
work
```

- Note that this command shows the same 10 predefined zones.

- c. Use the `firewall-cmd` command to identify the active zone for each network interface.

```
# firewall-cmd --get-active-zone
public
interfaces: eth0 eth1
```

- Note that the output of this `firewall-cmd` command confirms that the `eth0` and `eth1` interfaces are bound to the `public` zone.

- d. Use the `less` command to view the main `firewalld` configuration file, `/etc/firewalld/firewalld.conf`.

```
# less /etc/firewalld/firewalld.conf
# firewalld config file

# default zone
# The default zone used if an empty zone string is used.
# Default: public
DefaultZone=public
...
```

- Note that the default zone, `public`, is defined in this configuration file.

- e. Use the `cat` command to view the contents of the `public.xml` zone file:

```
# cat /usr/lib/firewalld/zones/public.xml
<?xml version="1.0" encoding="utf-8"?>
<zone>
  <short>Public</short>
  <description>For use in public areas. You do not trust the
other computers on networks to not harm your computer. Only
selected incoming connections are accepted.</description>
  <service name="ssh"/>
  <service name="dhcpv6-client"/>
  <service name="cockpit"/>
</zone>
```

- Note that the description states, "Only selected incoming connections are accepted."
- In this example, the selected incoming connections (trusted services) are `ssh`, `dhcpv6-client`, and `cockpit`.

4. Change the default zone.

- a. From a command prompt, use the `firewall-cmd` command to display the current default zone.

```
# firewall-cmd --get-default-zone
public
```

- Note that the `public` zone is the default zone.

- b. Use the `firewall-cmd` command to change the default zone to the `work` zone.

```
# firewall-cmd --set-default-zone=work
success
```

- c. Use the `firewall-cmd` command to display the current default zone.

```
# firewall-cmd --get-default-zone
work
```

- Note that the `work` zone is the default zone.

- d. Use the `grep -i` command to search for the string "defaultzone" in the `firewalld` configuration file.

```
# grep -i defaultzone /etc/firewalld/firewalld.conf
DefaultZone=work
```

- Note that the preceding command, which set the default zone, updated the `DefaultZone` setting in the configuration file.

- e. Use the `firewall-cmd` command to identify the active zone for each network interface

```
# firewall-cmd --get-active-zone
work
interfaces: eth0 eth1
```

- Note that the output of this `firewall-cmd` command confirms that the `eth0` and `eth1` interfaces are now bound to the `work` zone.

- f. Use the `firewall-cmd` command to change the default zone back to the `public` zone.

```
# firewall-cmd --set-default-zone=public
success
```

5. Explore **runtime** and **permanent** configuration modes.

- a. From a command prompt, use the `grep` command to search for the “`service`” string in the `/usr/lib/firewalld/zones/public.xml` zone file.

```
# grep service /usr/lib/firewalld/zones/public.xml
<service name="ssh"/>
<service name="dhcpv6-client"/>
<service name="cockpit"/>
```

- Note that these services correspond with the trusted services for the `public` zone.

- b. Use the `firewall-cmd` command to add the `dhcp` service to the `public` zone.

```
# firewall-cmd --add-service=dhcp --zone=public
success
```

- c. Use the `grep` command to search for the string “`service`” in the `/etc/firewalld/zones/public.xml` zone file.

```
# grep service /etc/firewalld/zones/public.xml
<service name="ssh"/>
<service name="dhcpv6-client"/>
<service name="cockpit"/>
```

- Note that this file was not updated. There is no entry for the `dhcp` service.
- Configuration files are not updated in **runtime** configuration mode.
- Configuration changes made in **runtime** configuration mode are lost when the `firewalld` service is restarted.

- d. Use the `firewall-cmd` command to add the `dhcp` service permanently to the `public` zone.

```
# firewall-cmd --add-service=dhcp --zone=public --permanent
success
```

- e. Use the `grep` command to search for the string “`service`” in the `/usr/lib/firewalld/zones/public.xml` zone file.

```
# grep service /usr/lib/firewalld/zones/public.xml
<service name="ssh"/>
<service name="dhcpv6-client"/>
<service name="cockpit"/>
```

- Note that there is no entry for the `dhcp` service.
- Configuration files in `/usr/lib/firewalld` are not updated when changes are made.

- f. Use the `grep` command to search for the string “service” in the `/etc/firewalld/zones/public.xml` zone file.

```
# grep service /etc/firewalld/zones/public.xml
<service name="ssh"/>
<service name="dhcpv6-client"/>
<service name="cockpit"/>
<service name="dhcp"/>
```

- Note that there is an entry for the `dhcp` service.
- The configuration files in `/etc/firewalld` are updated when changes are made.
- Changes made to the files or using `--permanent` take effect after restarting the `firewalld` service or instructing it to reload the files.
- However, as you also ran the command without the option `--permanent`, the changes are already active.

6. Explore `firewalld` services.

- a. Use the `ls` command to display the contents of the `/usr/lib/firewalld/services` directory.

```
# ls /usr/lib/firewalld/services/
amanda-client.xml      isns.xml              radius.xml
amanda-k5-client.xml   jenkins.xml           rdp.xml
amqps.xml              kadmin.xml            redis-sentinel.xml
...
```

- Note that there is an XML configuration file for each of the predefined services.

- b. Use the `firewall-cmd` command to display the available services.

```
# firewall-cmd --get-services
amanda-client amanda-k5-client amqp amqps apcupsd audit bacula
bacula-client bb bgp bitcoin bitcoin-rpc bitcoin-testnet
bitcoin-testnet-rpc bittorrent-lsd ceph ceph-mon cfengine
cockpit condor-collector ctdb dhcp dhcpv6 dhcpv6-client distcc
dns dns-over-tls...
```

- Note that this command shows the same predefined services.

- c. Use the `cat` command to display the contents of the `/usr/lib/firewalld/services/samba.xml` file.

```
# cat /usr/lib/firewalld/services/samba.xml
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>Samba</short>
  <description>This option allows you to access and participate
in Windows file and printer sharing networks. You need the samba
package installed for this option to be useful.</description>
  <port protocol="udp" port="137"/>
  <port protocol="udp" port="138"/>
  <port protocol="tcp" port="139"/>
  <port protocol="tcp" port="445"/>
  <helper name="netbios-ns"/>
</service>
```

- Note that the Samba software requires a number of ports to operate.
- Samba also provides a name service which requires more complex configuration. The helper identified in this file is a module that is used to support this complexity.

Practice 14-2: Configuring `firewalld`

Overview

In this practice, you create a `firewalld` rule to trust NFS.

- You configure **host02** as an NFS server and attempt mounts on NFS client **host03**.
- You create `firewalld` rules to trust NFS so that the NFS client can mount the exported file system with the `firewalld` service running.

Assumptions

- You are already logged in to **host02** as the `root` user by using the `ssh` command.
- This practice is performed on **host02** and **host03**.
- The prompts include either **host02** or **host03** to indicate which system to enter the command from.

Tasks

1. Configure **host02** as an NFS server.

- a. Use the `rpm` command to check if the `nfs-utils` package is installed.

```
[host02]# rpm -q nfs-utils
package nfs-utils is not installed
```

- In this example, the package is not installed.
- b. If `nfs-utils` is not installed, use `dnf` to install it.

```
# dnf install -y nfs-utils
```

- c. Use the `systemctl` command to verify whether the `nfs` service is started.

```
[host02]# systemctl status nfs-server
Loaded: loaded (/usr/lib/systemd/system/nfs-server.service;
disabled; vendor preset: disabled)
Active: inactive (dead)
```

- In this example, the `nfs-server` service is disabled and not running.
- d. Use the `systemctl` command to start the `nfs` service and associated services.

```
[host02]# systemctl start nfs-server
```

- e. Use the `systemctl` command to enable the `nfs-server` service to start at boot time.

```
[host02]# systemctl enable nfs-server
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-
server.service → /usr/lib/systemd/system/nfs-server.service.
```

2. Create and export a file system on **host02**.

- a. On **host02**, use the `mkfs` command to make an `ext4` file system on `/dev/xvdb`.

```
[host02]# mkfs -t ext4 /dev/xvdb
mke2fs 1.45.6 (20-Mar-2020)
Creating filesystem with 1293945 4k blocks and 327680 inodes
Filesystem UUID: 353d005d-9e4a-4957-a4b7-8493c542353a
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

- b. Use the `mkdir` command to create the `/Test` mount point.

```
[host02]# mkdir /Test
```

- c. Use the `mount` command to mount `/dev/xvdb` on `/Test`.

```
[host02]# mount /dev/xvdb /Test
```

- d. Use the `exportfs` command to export `/Test` to all clients with the `rw` and `no_root_squash` options.

```
[host02]# exportfs -o rw,no_root_squash */Test
```

- e. Use the `cat` command to view the `/var/lib/nfs/etab` file on **host02**.

```
[host02]# cat /var/lib/nfs/etab
/Test *(rw,sync,wdelay,hide,nocrossmnt,secure,no_root_squash ...
```

- The `/Test` file system is exported to all client systems.
- The `rw` option allows client systems to make changes to the file system.
- The `no_root_squash` option allows `root` users on client systems to retain `root` privileges on the file system.

- f. Use the `showmount -e` command to display exported file systems.

```
[host02]# showmount -e
Export list for host02.example.com:
/Test *
```

- This confirms that the `/Test` file system is exported to all client systems.

3. Attempt to mount the exported `/Test` file system on **host03**.

- a. Open a second terminal window on **gateway** and use the `ssh` command to connect to **host03** as the `root` user.

```
[oracle@gateway ~]$ ssh root@host03
root@host03's password:
[root@host03 ~]#
```


- b. On **host03**, use the `mkdir` command to create the `/remote_test` directory.

```
[host03]# mkdir /remote_test
```

- c. Use the `mount` command to mount the exported `/Test` file system from **host02** with `rw` and `nosuid` options on to the local mountpoint `/remote_test`.

```
[host03]# mount -t nfs -o rw,nosuid host02:/Test /remote_test
mount.nfs: No route to host
```

- The `rw` option mounts the file system with read/write permissions.
- The `nosuid` option does not allow `setuid` or `setgid` bits to take effect.
- In this example, the `mount` command fails because of the firewall on **host02**.
- It takes time for the `mount` command to time out. Rather than wait, you can press `Ctrl + C` to abort the `mount` command.

4. Enable `nfs` access through the firewall on **host02**.

- a. From **host02**, use the `firewall-cmd` command to add the `nfs` service to the `public` zone.

```
[host02]# firewall-cmd --add-service=nfs --zone=public
success
```

5. Attempt to mount the exported `/Test` file system on **host03**.

- a. Re-issue the `mount` command from the previous step on **host03**.

```
[host03]# mount -t nfs -o rw,nosuid host02:/Test /remote_test
```

- The `mount` command is successful.

- b. On **host03**, use the `df` command to display the mounted file systems.

```
[host03]# df -h
Filesystem      Size  Used Avail Use% Mounted on
...
/dev/xvda2       24G   1.8G   23G   8% /
/dev/xvda3       2.0G   47M   2.0G   3% /home
/dev/xvda1      1014M  237M   778M  24% /boot
...
host02:/Test     4.8G   20M   4.6G   1% /remote_test
```

- Note that the `host02:/Test` file system is mounted on the local file system as `/remote_test`.

6. Unmount the exported file system on **host03**.

- a. Use the `umount` command to unmount `/remote_test` on **host03**.

```
[host03]# umount /remote_test
```

- b. On **host03**, use the `df` command to display the mounted file systems.

```
[host03]# df -h
Filesystem      Size  Used Avail Use% Mounted on
...
/dev/xvda2       24G   1.8G   23G   8% /
/dev/xvda3       2.0G   47M   2.0G   3% /home
/dev/xvda1      1014M  237M   778M  24% /boot
...
```

- Note that the `host02:/Test` file system is no longer mounted on the local file system as `/remote_test`.

7. From **host02**, use the `firewall-cmd` command to permanently trust the `nfs` service.

- a. From a command prompt on **host02**, use the `firewall-cmd` command to permanently trust the `nfs` service.

```
[host02]# firewall-cmd --runtime-to-permanent
Success
```

- Note this makes all runtime changes permanent, which could be useful if making many changes.

- b. Use the `systemctl` command to restart the `firewalld` service on **host02**.

```
[host02]# systemctl restart firewalld
```

8. Attempt to mount the exported `/Test` file system on **host03**.

- a. Re-issue the `mount` command from the preceding step on **host03**.

```
[host03]# mount -t nfs -o rw,nosuid host02:/Test /remote_test
```

- The `mount` command is successful even after the firewall service was restarted.

- b. On **host03**, use the `df` command to display the mounted file systems.

```
[host03]# df -h
Filesystem      Size  Used Avail Use% Mounted on
...
/dev/xvda2       24G   1.8G   23G   8% /
/dev/xvda3       2.0G   47M   2.0G   3% /home
/dev/xvda1      1014M  237M   778M  24% /boot
...
host02:/Test     4.8G   20M   4.6G   1% /remote_test
```

- Note that the `host02:/Test` file system is mounted on the local file system as `/remote_test`.

9. Restore the initial configuration in preparation for the next practice.

- a. Use the `umount` command to unmount `/remote_test` on **host03**.

```
[host03]# umount /remote_test
```

- b. From **host02**, use the `grep` command to search for the string "service" in the `/etc/firewalld/zones/public.xml` file.

```
[host02]# grep service /etc/firewalld/zones/public.xml
<service name="ssh"/>
<service name="dhcpv6-client"/>
<service name="cockpit"/>
<service name="dhcp"/>
<service name="nfs"/>
```

- c. Remove the "dhcp" and "nfs" service entries from the public zone.
You can use the `vi` editor to edit the file and delete the entries, or you can use the `firewall-cmd` command as shown.

```
[host02]# firewall-cmd --permanent --zone=public --remove-
service=nfs
success
[host02]# firewall-cmd --permanent --zone=public --remove-
service=dhcp
success
```

- d. From **host02**, use the `grep` command to search for the "service" string in the `/etc/firewalld/zones/public.xml` file.

```
[host02]# grep service /etc/firewalld/zones/public.xml
<service name="ssh"/>
<service name="dhcpv6-client"/>
<service name="cockpit"/>
```

- Note that the entries for `nfs` and `dhcp` no longer exist.

- e. Use the `firewall-cmd` command to reload the `firewalld` service on **host02**.

```
[host02]# firewall-cmd --reload
```

Practice 14-3: Configuring `nftables`

Overview

In this practice, you use the `nft` command to allow a client system to mount an NFS file system.

Assumptions

- This practice is performed on **host02** and **host03** VMs.
- You are the `root` user on **host02** and **host03**.
- The prompts include either **host02** or **host03** to indicate which system to enter the command from.

Tasks

1. From **host02**, examine how the `firewalld` service uses `nftables`.
 - a. Check the status of the `firewalld` service with the `systemctl` command.

```
[host02]# systemctl status firewalld
• firewalld.service - firewalld - dynamic firewall daemon
  Loaded: loaded (/usr/lib/systemd/system/firewalld.service;
  enabled; vendor preset: enabled)
  Active: active (running) since...
  Docs: man:firewalld(1)
  ...
```

- You can see that `firewalld` is enabled and running.

- b. Use the `nft` command to list the tables created by the `firewalld` service.

```
[host02]# nft list tables
table ip filter
table ip6 filter
table bridge filter
table ip security
table ip raw
table ip mangle
table ip nat
table ip6 security
table ip6 raw
table ip6 mangle
table ip6 nat
table bridge nat
table inet firewalld
table ip firewalld
table ip6 firewalld
```

- c. Use the `nft` command to view the `inet` ruleset created by the `firewalld` service.

```
[host02]# nft list ruleset inet | less
table inet firewallld {
...
  chain filter_INPUT {
    type filter hook input priority filter + 10; policy accept;
    ct state { established, related } accept
    ...
    iifname "lo" accept
    ...
    jump filter_INPUT_ZONES
    ct state { invalid } drop
    reject with icmpx type admin-prohibited
  }
...

  chain filter_INPUT_ZONES {
    iifname "eth0" goto filter_IN_public
    iifname "eth1" goto filter_IN_public
    goto filter_IN_public
  }
...

  chain filter_IN_public {
    ...
    jump filter_IN_public_allow
    ...
    meta l4proto { icmp, ipv6-icmp } accept
  }
...

  chain filter_IN_public_allow {
    tcp dport 22 ct state { new, untracked } accept
    ip6 daddr fe80::/64 udp dport 546 ct state { new,
untracked } accept
    tcp dport 9090 ct state { new, untracked } accept
  }
...
}
```

- Note there are many chains defined. Those shown above reflect the rules for the public zone used by interfaces `eth0` and `eth1`.

2. Disable and stop the `firewalld` service and prepare to use the `nftables` service.

- a. Use the `systemctl` command to disable and stop the `firewalld` service.

```
[host02]# systemctl disable --now firewalld
Removed /etc/systemd/system/multi-
user.target.wants/firewalld.service.
Removed /etc/systemd/system/dbus-
org.fedoraproject.FirewallD1.service.
```

- b. Use the `systemctl` command to mask the `firewalld` service.

```
[host02]# systemctl mask firewalld
Created symlink /etc/systemd/system/firewalld.service →
/dev/null.
```

- Note that masking a service prevents it from being started, because disabled services can still be started manually or by other services.

- c. Re-check the status of the `firewalld` service with the `systemctl` command.

```
[host02]# systemctl status firewalld
• firewalld.service
  Loaded: masked (Reason: Unit firewalld.service is masked.)
  Active: inactive (dead)
...
```

- You can see that `firewalld` is now masked and inactive.

- d. Use the `nft` command to flush any remaining tables left by `firewalld` (avoiding the need to reboot).

```
[host02]# nft flush ruleset
```

- e. Use the `nft` command to create a table called `filter` for both IPv4 and IPv6 traffic.

```
[host02]# nft add table inet filter
```

- f. Use the `nft` command to add a chain called `input` to the table.

```
[host02]# nft add chain inet filter input { type filter hook
input priority 0 \; policy accept \; }
```

- Note for now we will accept all traffic or we will lose our `ssh` connection.

- g. Use the `nft` command to create a rule to accept traffic on the `lo` network interface.

```
[host02]# nft add rule inet filter input iif lo accept
```

- h. Use the `nft` command to create a rule to accept incoming responses to established connections.

```
[host02]# nft add rule inet filter input ct state
established,related accept
```

- i. Use the `nft` command to create a rule to accept incoming `ssh` connections.

```
[host02]# nft add rule inet filter input tcp dport 22 accept
```

- j. Use the `nft` command to create a rule to drop and count all other traffic.

```
[host02]# nft add rule inet filter input counter drop
```

- k. Use the `nft` command to display the ruleset.

```
[host02]# nft list ruleset
table inet filter {
    chain input {
        type filter hook input priority filter; policy accept;
        iif "lo" accept
        ct state established,related accept
        tcp dport 22 accept
        counter packets 0 bytes 0 drop
    }
}
```

3. Attempt to mount the NFS exported file system on **host02** from a remote host, **host03**.

- a. From **host02**, use the `showmount -e` command to display exported file systems.

```
[host02]# showmount -e
Export list for host02.example.com:
/Test *
```

- This confirms that the `/Test` file system is exported to all client systems.

- b. From **host03**, use the `mount` command to mount the exported `/Test` NFS file system from **host02**.

Mount with `rw` and `nosuid` options on the local mountpoint, `/remote_test`.

```
[host03]# mount -t nfs -o rw,nosuid host02:/Test /remote_test
^C
```

- With the `nft` ruleset configured, and no rule to trust `nfs`, the `mount` command fails.
- It takes time for the `mount` command to time out. Rather than wait, you can press `Ctrl + C` to abort the `mount` command.

4. From **host02**, view and modify the `nft` rules.

- a. Run `nft --handle list ruleset` to display the rules with handle numbers.

```
[host02]# nft --handle list ruleset
table inet filter { # handle 24
    chain input { # handle 1
        type filter hook input priority filter; policy accept;
        iif "lo" accept # handle 2
        ct state established,related accept # handle 3
        tcp dport 22 accept # handle 4
        counter packets 58 bytes 9474 drop # handle 7
    }
}
```

- Note the handle number for the `ssh` port 22 rule; in this example handle 4.

- b. Run the `nft` command to insert a rule for `nfs` before the `ssh` rule.

```
[host02]# nft insert rule inet filter input handle 4 tcp dport
nfs accept
```

- Remember that handle number 4 contained a rule to accept `ssh` traffic. Adjust the number if it not the same in your case.

- c. Repeat the preceding step to list the new `nfs` rule.

```
[host02]# nft --handle list ruleset
table inet filter { # handle 24
    chain input { # handle 1
        type filter hook input priority filter; policy accept;
        iif "lo" accept # handle 2
        ct state established,related accept # handle 3
        tcp dport 2049 accept # handle 8
        tcp dport 22 accept # handle 4
        counter packets 66 bytes 10842 drop # handle 7
    }
}
```

- Note that there now is a rule to accept `nfs` traffic.

5. Attempt again to mount the NFS exported file system on **host02** from a remote host, **host03**.

- a. From **host03**, use the `mount` command to mount the exported `/Test` NFS file system from **host02**.

```
[host03]# mount -t nfs -o rw,nosuid host02:/Test /remote_test
```

- With the new `nft` rule configured the `mount` command now succeeds.

- b. On **host03**, use the `df` command to display the mounted file systems.

```
[host03]# df -h
Filesystem      Size  Used Avail Use% Mounted on
...
/dev/xvda2      24G   1.8G   23G   8% /
/dev/xvda3      2.0G   47M   2.0G   3% /home
/dev/xvda1     1014M  237M   778M  24% /boot
...
host02:/Test    4.8G   20M   4.6G   1% /remote_test
```

- Note that the `host02:/Test` file system is mounted on the local file system as `/remote_test`.

- c. From **host03**, use the `umount` command to unmount `/remote_test`.

```
[host03]# umount /remote_test
```


6. From **host02**, configure the `nftables` service to apply the ruleset at boot.

a. Use the `nft` command to save the ruleset to the `/etc/sysconfig/nftables.conf` file.

```
[host02]# nft list ruleset >> /etc/sysconfig/nftables.conf
```

- Note there are some comments in the `/etc/sysconfig/nftables.conf` file that you may want to retain.

b. Use the `nft` command to flush the ruleset before testing the `nftables` service.

```
[host02]# nft flush ruleset
```

c. Use the `nft` command to confirm the ruleset has been removed.

```
[host02]# nft list ruleset
```

d. Use the `systemctl` command to enable and run the `nftables` service.

```
[host02]# systemctl enable --now nftables
Created symlink /etc/systemd/system/multi-
user.target.wants/nftables.service →
/usr/lib/systemd/system/nftables.service.
```

e. Use the `systemctl` command to check the status of the `nftables` service.

```
[host02]# systemctl status nftables
• nftables.service - Netfilter Tables
  Loaded: loaded (/usr/lib/systemd/system/nftables.service;...
  Active: active (exited) since ...
...
... ExecStart=/sbin/nft -f /etc/sysconfig/nftables.conf
Main PID: 2694 (code=exited, status=0/SUCCESS)
...
```

- Note that the service does not stay running, as noted by `active (exited)`.
- It executes the `nft` command to load the `/etc/sysconfig/nftables.conf` file and then exits.
- The exit status of the `nft` command can be seen in the status output, along with log messages generated by the command.

f. Use the `nft` command to confirm that the ruleset has been applied.

```
[host02]# nft list ruleset
table inet filter {
    chain input {
        type filter hook input priority filter; policy accept;
        iif "lo" accept
        ct state established,related accept
        tcp dport 2049 accept
        tcp dport 22 accept
        counter packets 278 bytes 47094 drop
    }
}
```

7. Perform the following tasks on **host02** to reinstate `firewalld` as the active firewall manager.

- a. Disable the `nftables` service.

```
[host02]# systemctl disable --now nftables  
Removed /etc/systemd/system/multi-  
user.target.wants/nftables.service.
```

- b. Start and enable the `firewalld` service.

```
[host02]# systemctl unmask firewalld  
Removed /etc/systemd/system/firewalld.service.  
[host02]# systemctl enable --now firewalld  
Created symlink /etc/systemd/system/dbus-  
org.fedoraproject.FirewallD1.service →  
/usr/lib/systemd/system/firewalld.service.  
Created symlink /etc/systemd/system/multi-  
user.target.wants/firewalld.service →  
/usr/lib/systemd/system/firewalld.service.
```

8. Disable the NFS service and unmount the `/Test` file system.

- a. Use the `systemctl` command to stop and disable the NFS service.

```
# systemctl disable --now nfs-server  
Removed /etc/systemd/system/multi-user.target.wants/nfs-  
server.service.
```

- b. Use the `umount` command to unmount the `/Test` filesystem.

```
# umount /Test
```

9. Log off from **host02** and **host03** in preparation for the next practice.

- a. Use the `logout` command to close the `ssh` connection to **host02**.

```
[host02]# logout  
Connection to host02 closed.
```

You are now the `oracle` user on **gateway**.

- b. Use the `logout` command to close the `ssh` connection to **host03**.

```
[host03]# logout  
Connection to host03 closed.
```

You are now the `oracle` user on **gateway**.

You can close the second terminal window because it is not needed in the next practice.

Practices for Lesson 15:
Oracle on Oracle

Practices for Lesson 15: Overview

Practices Overview

In these practices, you install and run the Oracle Database Pre-Install RPM for Oracle Linux 8. You also configure ASMLib.

The packages you need are included in the extras Yum repository created in an earlier lesson. Normally, you obtain these packages from ULN or, in the case of the `oracle-database-preinstall-19c` package, also from the Oracle Linux yum server. The `oracleasm` package is not available on the Oracle Linux yum server but, in addition to ULN, can be downloaded from the Oracle Technology Network here:

<https://www.oracle.com/linux/downloads/linux-asmlib-v8-downloads.html>

Assumptions

You created a local Yum repository in the practices for the lesson titled “Package Management.”

Practice 15-1: Installing and Running Oracle Database Preinstall

Overview

In this practice, you install the `oracle-database-preinstall-19c` package and view the results.

Assumptions

You are logged in to **gateway** as the `oracle` user.

Tasks

1. Log in to **host02**.

Use the `ssh` command to log in to **host02**. Provide the `root` password when prompted.

```
[oracle@gateway ~]$ ssh root@host02
root@host02's password:
Last login: ...
```

All remaining commands in this practice are entered from **host02**.

2. Install the `oracle-database-preinstall-19c` package on **host02**.

Use the `dnf` command to install the `oracle-database-preinstall-19c` package.

Answer **y** when prompted.

```
# dnf install oracle-database-preinstall-19c
...
Installing:
  oracle-database-preinstall-19c
                                x86_64 1.0-1.el8          extras    24 k
Installing dependencies:
  bc                            x86_64 1.07.1-5.el8          ol8_baseos_loc...
  bind-libs                     x86_64 32:9.11.20-5.el8    ol8_appstream_...
...
Install  52 Packages

Total size: 23 M
Installed size: 67 M
Is this ok [y/N]: y
Downloading packages:
...
Installed:
...
  oracle-database-preinstall-19c-1.0-1.el8.x86_64
...
Complete!
```

3. View the results of the installation of the `oracle-database-preinstall-19c` package.
 - a. Use the `rpm -ql` command to list the files provided by the package.

```
# rpm -ql oracle-database-preinstall-19c
/etc/rc.d/init.d/oracle-database-preinstall-19c-firstboot
/etc/security/limits.d/oracle-database-preinstall-19c.conf
/etc/sysconfig/oracle-database-preinstall-19c
/etc/sysconfig/oracle-database-preinstall-19c/oracle-database-
preinstall-19c-verify
/etc/sysconfig/oracle-database-preinstall-19c/oracle-database-
preinstall-19c.param
/usr/bin/oracle-database-preinstall-19c-verify
/var/log/oracle-database-preinstall-19c
/var/log/oracle-database-preinstall-19c/results
```

- b. Use the `less` command to view the contents of the `/etc/security/limits.d/oracle-database-preinstall-19c.conf` file.

```
# less /etc/security/limits.d/oracle-database-preinstall-
19c.conf
# oracle-database-preinstall-19c setting for nofile soft limit
is 1024
oracle    soft    nofile    1024
oracle    hard    nofile    65536

# oracle-database-preinstall-19c setting for nproc soft limit is
16384 # refer orabug15971421 for more info.
oracle    soft    nproc     16384
oracle    hard    nproc     16384

# oracle-database-preinstall-19c setting for stack soft limit is
10240KB
oracle    soft    stack     10240
oracle    hard    stack     32768

# oracle-database-preinstall-19c setting for memlock hard limit
is maximum of 128GB on x86_64 or 3GB on x86 OR 90 % of RAM
oracle    hard    memlock   134217728
oracle    soft    memlock   134217728

# oracle-database-preinstall-19c setting for data soft limit is
'unlimited'
oracle    soft    data      unlimited
oracle    hard    data      unlimited
```

- Note that the shell resource limits for `oracle` are defined in this file.
- Note some content of the file has been omitted for brevity.

- c. Use the `ls -l` command to display the file type of the `/usr/bin/oracle-database-preinstall-19c-verify` file.

```
# ls -l /usr/bin/oracle-database-preinstall-19c-verify
lrwxrwxrwx. ... /usr/bin/oracle-database-preinstall-19c-verify
-> /etc/sysconfig/oracle-database-preinstall-19c/oracle-
database-preinstall-19c-verify
```

- Note that this file is a symbolic link to the `/etc/sysconfig/oracle-database-preinstall-19c/oracle-database-preinstall-19c-verify` file.
- d. Change to the `/etc/sysconfig/oracle-database-preinstall-19c` directory and view the contents of the directory.

```
# cd /etc/sysconfig/oracle-database-preinstall-19c
# ls
oracle-database-preinstall-19c.conf
oracle-database-preinstall-19c.param
oracle-database-preinstall-19c-verify
```

- e. Use the `cat` command to view the contents of the `oracle-database-preinstall-19c.conf` file.

```
# cat oracle-database-preinstall-19c.conf
RUN_ORACLE_DATABASE_PREINSTALL_19C=NO
```

- Note that the value of this parameter is `NO`, meaning do not run the preinstall script again.
 - The preinstall script is run automatically when you install the `oracle-database-preinstall-19c` package.
- f. Use the `less` command to view the contents of the `oracle-database-preinstall-19c.param` file.

```
# less oracle-database-preinstall-19c.param
# Oracle-database Parameter config file
#Kernel parameters
#category:arch:kernel:distribution:parameter:recommended-value

kernelcomment:
kernelcomment:oracle-database-preinstall-19c setting for
fs.file-max is 6815744
kernel:*:*:*:fs.file-max:6815744
kernelcomment:
...

#Oracle OS User limits (domain=oracle/*)
#category:arch:kernel:distribution:type:item:recommended-value
```

```

usercomment:
usercomment:*:oracle-database-preinstall-19c setting for nofile
soft limit is 1024
user:*:*:*:soft:nofile:1024
...

#Kernel boot parameters
#category:arch:processor:kernel:distribution:parameter:recommended-value
boot:x86_64:*:*:*:numa:off
boot:*:*:*:*:transparent_hugepage:never

#Group Names and IDs
#category:arch:processor:kernel:distribution:parameter:recommended-value
group:*:*:*:*:oinstall:54321
group:*:*:*:*:dba:54322
group:*:*:*:*:oper:54323
group:*:*:*:*:backupdba:54324
group:*:*:*:*:dgdba:54325
group:*:*:*:*:kmdba:54326
group:*:*:*:*:racdba:54330

#User name and IDs
#category:arch:processor:kernel:distribution:parameter:recommended-value
username:*:*:*:*:oracle:54321:oinstall,dba,oper,backupdba,dgdba,
kmdba,racdba

```

- Note that this file defines kernel parameters, Oracle OS user limits, kernel boot parameters, group names and IDs, and usernames and IDs.
- g. Use the `less` command to view the contents of the `oracle-database-preinstall-19c-verify` file.

```

# less oracle-database-preinstall-19c-verify
#!/bin/bash
...
# Name: oracle-database-preinstall-verify
# Description: A script to verify and set Oracle Preinstall
settings
# on Oracle Linux.
...

```

- Note that this file is the bash script that modifies settings.

4. View the modifications made by the `oracle-database-preinstall-19c-verify` script.

Note that the `oracle-database-preinstall-19c-verify` script is executed automatically when the `oracle-database-preinstall-19c` package is installed.

- a. Change to the `/var/log/oracle-database-server-19c-preinstall` directory and view the contents of the directory.

```
# cd /var/log/oracle-database-preinstall-19c
# ls -l
drwxr-xr-x. 3 root root ... backup
drwx-----. 2 root root ... results
```

- Note that two directories exist in this directory.
 - The `backup` directory contains copies of files before they were modified.
 - The `results` directory contains the Oracle Database Pre-install log file.
- b. Change to the `results` directory and view the contents of the directory.

```
# cd results
# ls
orakernel.log
```

- c. Use the `less` command to view the `orakernel.log` (sample output shown).

```
# less orakernel.log
Adding group oinstall with gid 54321
Adding group dba with gid 54322
...
User oracle - Already exists. Not creating or modifying.
User creation passed

Saving a copy of the initial sysctl.conf
Verifying kernel parameters as per Oracle recommendations...
Trying to remove instances of - setting for fs.file-max is
Adding fs.file-max = 6815744
...
Setting kernel parameters as per oracle recommendations...
Altered file /etc/sysctl.conf
Saved a copy of the current file in /etc/sysctl.d/99-oracle-
database-preinstall-19c-sysctl.conf
Check /etc/sysctl.d for backups
Verification & setting of kernel parameters passed

Setting user limits using /etc/security/limits.d/oracle-
database-preinstall-19c.conf

Verifying oracle user OS limits as per Oracle recommendations...
```

```

Adding oracle soft nofile 1024
...
Saving a copy of /etc/default/grub file in /etc/default/grub-
initial.orabackup
Saving a copy of /etc/default/grub in /var/log/oracle-database-
preinstall-19c/backup/Jan-07-2021-20-47-29...
Verifying kernel boot parameters as per Oracle
recommendations...
old boot params: "resume=UUID=670a8e47-f268-4961-bc8c-
d80b8959bef6 rhgb quiet"
...
new boot params: "resume=UUID=670a8e47-f268-4961-bc8c-
d80b8959bef6 rhgb quiet numa=off transparent_hugepage=never"
...
Running grub2-mkconfig
...
Trying to add NOZEROCONF parameter...
Taking a backup of existing file to
/etc/sysconfig/network.orabackup
Successfully added parameter NOZEROCONF to
/etc/sysconfig/network
Setting /etc/sysconfig/network parameters passed

Disabling Transparent Hugepages.
Refer Oracle Note:1557478.1

Disabling defrag.
Refer Oracle Note:1557478.1

Taking a backup of old config files under /var/log/oracle-
database-preinstall-19c/backup/Jan-07-2021-20-47-29

```

- Note that the required users and groups are created as needed.
- Note that kernel parameters are set and /etc/sysctl.conf is backed up beforehand, to /etc/sysctl.d/99-initial-sysctl.conf. /etc/sysctl.conf is also copied to /etc/sysctl.d/99-oracle-database-preinstall-19c-sysctl.conf.
- Note that oracle user OS limits are set in /etc/security/limits.d/oracle-database-preinstall-19c.conf.
- Note that kernel boot parameters are set and /etc/default/grub is backed up beforehand, to /etc/default/grub-initial.orabackup.
- While not recorded in the orakernel.log file, the /boot/grub2/grub.cfg file is also backed up.
- Backup files are also placed in the backup/date directory.

- d. Use the `find` command to list some of the files backed up before settings were changed, `*orabackup`.

```
# find / -name "*orabackup"
/boot/grub2/grub.cfg-oracle-database-preinstall-19c.orabackup
/etc/sysconfig/network.orabackup
/etc/default/grub-initial.orabackup
/etc/default/grub-oracle-database-preinstall-19c.orabackup
```

- e. View the contents of the `backup/date` directory.

- A `date` subdirectory is created, which contains the original files.

```
# cd /var/log/oracle-database-preinstall-19c
# ls backup/date
grub  grub.cfg  orakernel.log  sysctl.conf
```

- The `backup/date/grub` file is the same as the `/etc/default/grub-initial.orabackup` file.
- The `backup/date/grub.cfg` file is the same as the `/boot/grub2/grub.cfg-oracle-database-preinstall-19c.orabackup` file.
- The `backup/date/sysctl.conf` file is the same as the `/etc/sysctl.d/99-initial-sysctl.conf` file.

- f. Use the `diff` command to view the changes made in the `grub.cfg` file.

```
# diff /boot/grub2/grub.cfg /boot/grub2/grub.cfg-oracle-
database-preinstall-19c.orabackup
140c140
<  set kernelopts="root=UUID=3bbca2f4-2cf8-43bf-9a38-
69bd7cf2ba87 ro resume=UUID=670a8e47-f268-4961-bc8c-d80b8959bef6
rhgb quiet numa=off transparent_hugepage=never "
---
>  set kernelopts="root=UUID=3bbca2f4-2cf8-43bf-9a38-
69bd7cf2ba87 ro resume=UUID=670a8e47-f268-4961-bc8c-d80b8959bef6
rhgb quiet "
```

- Note that the `numa=off` and `transparent_hugepage=never` boot parameters are added to the `grub.cfg` file.

- g. Use the `wc -l` command to display the number of lines in the `sysctl.conf` files.

```
# wc -l /etc/sysctl.conf /etc/sysctl.d/99-oracle-database-
preinstall-19c-sysctl.conf /etc/sysctl.d/99-initial-sysctl.conf
 53 /etc/sysctl.conf
 53 /etc/sysctl.d/99-oracle-database-preinstall-19c-sysctl.conf
 10 /etc/sysctl.d/99-initial-sysctl.conf
...
```

- Note that 43 new lines are added to `/etc/sysctl.conf`, and this file is copied to `/etc/sysctl.d/99-oracle-database-preinstall-19c-sysctl.conf` during the preinstall RPM installation. These two files are therefore identical.

- h. Use the `diff` command to view the changes made in the `/etc/sysctl.d/99-oracle-database-server-12cR2-preinstall-sysctl.conf` file.

```
# diff /etc/sysctl.d/99-oracle-database-preinstall-19c-  
sysctl.conf /etc/sysctl.d/99-initial-sysctl.conf  
11,53d10  
<  
< # oracle-database-preinstall-19c setting for fs.file-max is  
6815744  
< fs.file-max = 6815744  
...
```

- i. Use the `diff` command to view the changes made in the `/etc/default/grub` file.

```
# diff /etc/default/grub /etc/default/grub-initial.orabackup  
6c6  
< GRUB_CMDLINE_LINUX="resume=UUID=... rhgb quiet numa=off  
transparent_hugepage=never"  
---  
> GRUB_CMDLINE_LINUX="resume=UUID=... rhgb quiet"
```

- Note that the `numa=off` and `transparent_hugepage=never` boot parameters have been added to the `/etc/default/grub` file.

- j. Use the `cat` command to view the limits set in the `/etc/security/limits.d/oracle-database-preinstall-19c.conf` file.

```
# cat /etc/security/limits.d/oracle-database-preinstall-19c.conf  
  
# oracle-database-preinstall-19c setting for nofile soft limit  
is 1024  
oracle    soft    nofile     1024  
  
# oracle-database-preinstall-19c setting for nofile hard limit  
is 65536  
oracle    hard    nofile     65536  
  
# oracle-database-preinstall-19c setting for nproc soft limit is  
16384  
# refer orabug15971421 for more info.  
oracle    soft    nproc      16384  
...
```

Practice 15-2: Preparing Disks for ASM Use

Overview

In this practice, you:

- Create one partition using the entire disk on `/dev/xvdb`
- Create one partition using the entire disk on `/dev/xvdd`

Assumptions

You are the `root` user on **host02**.

Tasks

1. Use the `fdisk` command to create a single primary partition on `/dev/xvdb`, using the entire disk as shown.

```
# fdisk /dev/xvdb
...
Command (m for help): n
Partition type:
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): ENTER
Using default response p
Partition number (1-4, default 1): ENTER
First sector (2048-10351561, default 2048): ENTER
Using default value 2048
Last sector, +sectors or +size{K,M,G,T,P} (2048-10351561,
default 10351561): ENTER

Created a new partition 1 of type 'Linux' and of size 5 GiB.

Command (m for help): w
The partition table has been altered.
Syncing disks.
```

- You may see the warning *The old ext4 signature will be removed by a write command* because this device was used for a file system in the *Network Security* practices.

2. Use the `fdisk` command to create a single primary partition on `/dev/xvdd`, using the entire disk as shown.

```
# fdisk /dev/xvdd
...
Command (m for help): n
Partition type:
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): ENTER
Using default response p
Partition number (1-4, default 1): ENTER
First sector (2048-10351561, default 2048): ENTER
Using default value 2048
Last sector, +sectors or +size{K,M,G,T,P} (2048-10351561,
default 10351561): ENTER

Created a new partition 1 of type 'Linux' and of size 5 GiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

Practice 15-3 Installing and Configuring ASMLib

Overview

In this practice, you:

- Install the `oracleasmlib` package
- Install the `oracleasm-support` package
- Configure ASMLib
- Load and initialize the ASMLib driver
- Mark disk partitions for ASM use
- View information about the ASM disk partitions

Assumptions

You are the `root` user on `host02` VM.

Tasks

1. Install the `oracleasm` packages.
 - a. Use the `dnf` command to install the `oracleasmlib` package from the `extras` repo.

```
# dnf install -y oracleasmlib
...
Installed:
  oracleasmlib-2.0.17-1.el8.x86_64

Complete!
```

- b. Use the `dnf` command to install the `oracleasm-support` package.

```
# dnf install -y oracleasm-support
...
Created symlink /etc/systemd/system/multi-
user.target.wants/oracleasm.service →
/usr/lib/systemd/system/oracleasm.service.

Verifying      : oracleasm-support-2.1.12-1.el8.x86_64    1/1

Installed:
  oracleasm-support-2.1.12-1.el8.x86_64

Complete!
```

2. Use the `oracleasm` utility to configure ASMLib.

a. Run the `oracleasm -h` command to display the usage and commands.

```
# oracleasm -h
Usage: oracleasm [--exec-path=<exec_path>] <command> [ <args> ]
      oracleasm -exec-path
      oracleasm -h
      oracleasm -V

The basic oracleasm commands are:
  configure      Configure the Oracle Linux ASMLib driver
  init           Load and initialize the ASMLib driver
  exit           Stop the ASMLib driver
  scandisks      Scan the system for Oracle ASMLib disks
  status         Display the status of the Oracle ASMLib ...
  listdisks      List known Oracle ASMLib disks
  listiids       List the iid files
  deleteiids     Delete the unused iid files
  querydisk      Determine if a disk belongs to Oracle AS...
  createdisk      Allocate a device for Oracle ASMLib use
  deletedisk     Return a device to the operating system
  renamedisk     Change the label of an Oracle ASMLib disk
  update-driver  Download the latest ASMLib driver
```

b. Use the `oracleasm configure -i` command to configure the ASMLib driver.

```
# oracleasm configure -i
Configuring the Oracle ASM library driver.

This will configure the on-boot properties of the Oracle ASM
library driver. The following questions will determine whether
the driver is loaded on boot and what permissions it will have.
The current values will be shown in brackets ('[]'). Hitting
<ENTER> without typing an answer will keep that current value.
Ctrl-C will abort.

Default user to own the driver interface []: oracle
Default group to own the driver interface []: dba
Start Oracle ASM library driver on boot (y/n) [n]: y
Scan for Oracle ASM disks on boot (y/n) [y]: ENTER
Writing Oracle ASM library driver configuration: done
```


- c. Use the `oracleasm init` command to load and initialize the ASMLib driver.

```
# oracleasm init
Creating /dev/oracleasm mount point: /dev/oracleasm
Loading module "oracleasm": oracleasm
Configuring "oracleasm" to use device physical block size
Mounting ASMLib driver filesystem: /dev/oracleasm
```

- d. Use the `oracleasm configure` command without the `-i` option.

```
# oracleasm configure
ORACLEASM_ENABLED=true
ORACLEASM_UID=oracle
ORACLEASM_GID=dba
ORACLEASM_SCANBOOT=true
ORACLEASM_SCANORDER=""
ORACLEASM_SCANEXCLUDE=""
ORACLEASM_SCAN_DIRECTORIES=""
ORACLEASM_USE_LOGICAL_BLOCK_SIZE="false"
```

- This command shows the current configuration.

- e. Use the `oracleasm status` command.

```
# oracleasm status
Checking if ASM is loaded: yes
Checking if /dev/oracleasm is mounted: yes
```

- f. Use the `oracleasm createdisk` command to mark `/dev/xvdb1` for ASM use.
Give the disk a label of `VOL1`.

```
# oracleasm createdisk VOL1 /dev/xvdb1
Writing disk header: done
Instantiating disk: done
```

- g. Use the `oracleasm createdisk` command to mark `/dev/xvdd1` for ASM use.
Give the disk a label of `VOL2`.

```
# oracleasm createdisk VOL2 /dev/xvdd1
Writing disk header: done
Instantiating disk: done
```

3. View ASM disks.

- a. Use the `ls` command to display a long list of the `/dev/oracleasm/disks` directory.

```
# ls -l /dev/oracleasm/disks
...
brw-rw----. 1 oracle dba ... VOL1
brw-rw----. 1 oracle dba ... VOL2
```

- b. Use the `oracleasm listdisks` command to list the disk names of marked ASMLib disks.

```
# oracleasm listdisks
VOL1
VOL2
```

- c. Use the `oracleasm scandisks` command to enable cluster nodes to identify which shared disks have been marked as ASMLib disks on another node.

```
# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...
```

- d. Use the `oracleasm querydisk` command to determine whether a disk name or disk device is being used by ASMLib.

The first command uses the disk name as an argument.

The second command uses the device name as an argument.

```
# oracleasm querydisk VOL1
Disk "VOL1" is a valid ASM disk
# oracleasm querydisk /dev/xvdd1
Device "/dev/xvdd1" is marked an ASM disk with the label "VOL2"
```

Practice 15-4 Reverting Changes Made to host02

Overview

In this practice, you:

- Unmark ASM disks, returning them to the system
- Shut down and unload the ASMLib driver
- Remove the `oracleasm-support`, `oracleasm-lib`, and `oracle-database-preinstall-19c` packages
- Delete the `/dev/xvdb1` and `/dev/xvdd1` partitions
- Log off from **host02**

Assumptions

You are the `root` user on **host02** VM.

Tasks

1. Unmark ASM disks, returning them to the system, and shut down and unload the ASMLib driver.

- a. Use the `oracleasm deletedisk` command to unmark the ASM disk VOL1, returning it to the system.

```
# oracleasm deletedisk VOL1
Clearing disk header: done
Dropping disk: done
```

- b. Use the `oracleasm deletedisk` command to unmark the ASM disk VOL2, returning it to the system.

```
# oracleasm deletedisk VOL2
Clearing disk header: done
Dropping disk: done
```

- c. Use the `oracleasm exit` command to shut down and unload the ASMLib driver.

```
# oracleasm exit
Unmounting ASMLib driver filesystem: /dev/oracleasm
Unloading module "oracleasm": oracleasm
```

2. Remove `oracle` packages.

- a. Use the `dnf` command to remove the `oracleasm-support` package.

```
# dnf remove -y oracleasm-support oracleasm-lib
Removed:
  oracleasm-support-2.1.12-1.el8.x86_64      oracleasm-lib-2.0.17-
  1.el8.x86_64
Complete!
```

- b. Use the `dnf` command to remove the `oracle-database-preinstall-19c` package.

```
# dnf remove -y oracle-database-preinstall-19c
...
Removing:
  oracle-database-preinstall-19c
                                x86_64 1.0-1.el8      @extras          60 k
Removing unused dependencies:
  bc                            x86_64 1.07.1-5.el8    @ol8_baseos_local 236 k
...
Complete!
```

3. Delete the `/dev/xvdb1` and `/dev/xvdd1` partitions.

- a. Use the `fdisk` command to delete the `/dev/xvdb1` partition.

Use the `d` command to delete partition 1.

Use the `p` command to print the partition table and confirm there are no partitions.

Use the `w` command to save the partition table and exit the `fdisk` utility.

```
# fdisk /dev/xvdb
...
Command (m for help): d
Selected partition 1
Partition 1 is deleted

Command (m for help): p

Disk /dev/vdb: 5 GiB, 5299999744 bytes, 10351562 sectors
...

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

- b. Use the `fdisk` command to delete the `/dev/xvdd1` partition.
- Use the `d` command to delete partition 1.
- Use the `p` command to print the partition table and confirm there are no partitions.
- Use the `w` command to save the partition table and exit the `fdisk` utility.

```
# fdisk /dev/xvdd
...
Command (m for help): d
Selected partition 1
Partition 1 is deleted

Command (m for help): p

Disk /dev/vdb: 5 GiB, 5299999744 bytes, 10351562 sectors
...

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

4. Log off from **host02** in preparation for the next practice.
- Use the `logout` command to close the `ssh` connection to **host02**.

```
# logout
Connection to host02 closed.
```

You are now the `oracle` user on **gateway**.

**Practices for Lesson 16:
System Monitoring and
Management**

Practices for Lesson 16: Overview

Practices Overview

In these practices, you:

- Use the `sosreport` utility to collect system information
- Use standard Linux utilities to monitor system resource usage
- Use OSWatcher Black Box and OSWatcher Analyzer

Practice 16-1: Using `sosreport` to Collect System Information

Overview

In this practice, you:

- Use the `sosreport` utility to collect system information
- Extract the compressed TAR file and view the collected information
- View the status of the `sosreport` plug-ins

Assumptions

You are the `oracle` user on **gateway**.

Tasks

1. Log in to **host02**.

Use the `ssh` command to log in to **host02**. Provide the `root` password when prompted.

```
[oracle@gateway ~]$ ssh root@host02
root@host02's password:
Last login: ...
```

2. Ensure the required packages are installed.

- a. Use the `rpm` command to verify that the `sos` package is installed.

```
# rpm -q sos
package sos is not installed
```

- In this example, the package is not installed.

- b. Use the `rpm` command to verify that the `tar` package is installed.

```
# rpm -q tar
package tar is not installed
```

- In this example, the package is not installed.

- c. Use the `dnf` command to install the missing packages.

```
# dnf install -y sos tar
...
Installing:
  sos      noarch      3.9.1-6.0.1.el8      ol8_baseos_local      549 k
  tar      x86_64      2:1.30-5.el8         ol8_baseos_local      838 k
Installing dependencies:
  bzip2    x86_64      1.0.6-26.el8         ol8_baseos_local      60 k
...
Complete!
```

3. Run the `sosreport` command.

Press **Enter** when prompted to continue.

Press **Enter** when prompted to enter your first initial and last name.

Enter number **1** as the case number for which you are generating the report.

```
# sosreport
sosreport (version 3.9)

This command will collect diagnostic and configuration
information from this Oracle Linux system and installed
applications.

An archive containing the collected information will be
generated in /var/tmp/sos... and may be provided to a Oracle
America support representative.

Any information provided to Oracle America will be treated in
accordance with the published support policies at:

    http://linux.oracle.com/

The generated archive may contain data considered sensitive and
its content should be reviewed by the originating organization
before being passed to any third party.

No changes will be made to system configuration.

Press ENTER to continue, or CTRL-C to quit.
ENTER

Please enter the case id that you are generating this report for
[]: 1
...
Running plugins. Please wait ...
...
Creating compressed archive...

Your sosreport has been generated and saved in:
    /var/tmp/sosreport-host02-...tar.xz

Size      ... MiB
Owner     root
md5       ...

Please send this file to your support representative.
```

4. View the `sosreport` file.

- a. Use the `cd` command to change to the `/var/tmp` directory.

```
# cd /var/tmp
```

- b. Use the `ls` command to display a long listing of the `sos*` files.

```
# ls -l sos*
-rw----- . root root ... sosreport-host02...tar.xz
-rw-r--r-- . root root ... sosreport-host02...tar.xz.md5
```

- Note the two `sosreport` files, one with the `.xz` extension and one with the `.md5` extension.
- The `.xz` file is the compressed data file.

- c. Use the `tar` command to extract the `.tar.xz` file.

```
# tar xJvf sosreport-host02...tar.xz
...
```

- Note the `J` option specifies that the archive should be decompressed using `xz`.
- Note that the `tar.xz` file is extracted in a `sosreport-host02...` directory.

- d. Use the `cd` command to change to the `sosreport-host02...` directory.

You can use the `TAB` key to help complete the directory pathname.

```
# cd sosreport-host02...
```

- e. Use the `ls` command to display a long listing of the `sosreport-host02...` directory.

```
# ls -l
...
dr-xr-xr-x    ...      boot
lrwxrwxrwx    ...      chkconfig -> sos_commands/services/...
lrwxrwxrwx    ...      date -> sos_commands/general/date
...
drwxr-xr-x    ...      etc
lrwxrwxrwx    ...      free -> sos_commands/memory/free
lrwxrwxrwx    ...      hostname -> sos_commands/host/hostname
...
dr-xr-xr-x    ...      lib -> usr/lib
...
```

- Note that a number of directories that contain data collected from the system exist.
 - Note that a number of symbolic links that contain the output of several status-related commands exist.
- f. (Optional) Examine some of the files to confirm they contain the corresponding data. Use familiar commands such as `cat`, `less`, and `grep`.

5. Use the `sosreport -l` command to list the plug-ins.

There are many plug-ins, so you may prefer to pipe the output through `less`.

```
# sosreport -l | less
sosreport (version 3.9)

The following plugins are currently enabled:

alternatives      System alternatives
anaconda          Anaconda installer
anacron           Anacron job scheduling service
...

The following plugins are currently disabled:

...

acpid             inactive        ACPI daemon information
activemq          inactive        ActiveMQ message broker
apache           inactive        Apache http daemon
...

The following options are available for ALL plugins:
timeout          -1             Timeout in seconds for...
postproc         True          Enable post-processing...

The following plugin options are available:

boot.all-images  off           collect lsinitrd for all images
dnf.history      off          captures transaction history
...
```

Practice 16-2: Using Standard Linux Performance Monitoring Tools

Overview

In this practice, you use standard Linux system resource monitoring utilities to observe the following:

- CPU statistics
- Memory statistics
- Disk I/O statistics
- Network statistics

Assumptions

- You are the `root` user on **host02**.
- The output shown in the tasks is only a sample. Your output will be different.

Tasks

1. From **gateway**, open a second terminal window.

Use the `ssh` command to log in to **host02**. Provide the `root` password when prompted.

```
[oracle@gateway ~]$ ssh root@host02
root@host02's password:
Last login: ...
```

- You are now logged on to **host02** from two different terminal windows.

2. Install additional system monitoring tools.

Use the `dnf` command to install the `sysstat` package.

```
# dnf install -y sysstat
...
Installed:
  lm_sensors-libs-3.4.0-21.20180522git70f7e08.el8.x86_64
  sysstat-11.7.3-5.el8.x86_64

Complete!
```

3. Observe the CPU statistics.

- a. In one window, use the `top` command to display CPU usage and load averages.

```
# top
```

- The `top` command also monitors process statistics and memory usage.

- b. In the second window, run the following command to generate a system load:

```
# dd if=/dev/zero of=/dev/null bs=1024
```

c. Example output of `top` is shown as follows:

top - 13:40:16 up 3:31, 2 users, load average: 1.02, 1.02, 0.82										
Tasks: 159 total, 3 running, 156 sleeping, 0 stopped, 0 zombie										
%Cpu(s): 12.1 us, 42.4 sy, 0.0 ni, 45.5 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st										
MiB Mem : 3927.2 total, 2948.4 free, 318.6 used, 660.2 buff/cache										
MiB Swap: 3070.0 total, 3070.0 free, 0.0 used. 3538.2 avail Mem										
PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+ COMMAND
11604	root	20	0	7324	828	764	R	94.1	0.0	22:30.90 dd
11675	root	20	0	65384	4436	3804	R	5.9	0.1	0:00.01 top
1	root	20	0	182004	16324	9372	S	0.0	0.4	0:07.22 systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.02 kthreadd
3	root	0	-20	0	0	0	I	0.0	0.0	0:00.00 rcu_gp
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00 rcu_par_gp
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00 kworker/0+

- In this example, CPU usage is moderate as indicated by the 49.3 id (idle) statistic.
- The biggest consumer of the CPU is the `dd` process, which has a PID of 11604.
- Load average is the average number of processes in the run queue or the number of processes waiting to run on the CPU.
 - The load average over the last 1 minute is 0.95, over the last 5 minutes is 0.46, and over the last 15 minutes is 0.18.
 - A high load average would be an indication that your system does not have sufficient CPU capacity.
- Press `q` to quit the `top` command.

- d. Use the `mpstat -P ALL 1` command to display CPU statistics.

```
# mpstat -P ALL 1
```

Linux 5.4.17-2011.7.4.el8uek.x86_64 (host02.example.com) 08/01/21 _x86_64_ (2 CPU)											
13:56:03	CPU	%usr	%nice	%sys	%iowait	%irq	%soft	%steal	%guest	%gnice	%idle
13:56:04	all	8.00	0.00	42.00	0.00	0.00	0.00	0.00	0.00	0.00	50.00
13:56:04	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
13:56:04	1	16.00	0.00	84.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
13:56:04	CPU	%usr	%nice	%sys	%iowait	%irq	%soft	%steal	%guest	%gnice	%idle
13:56:05	all	7.04	0.00	43.22	0.00	0.00	0.00	0.00	0.00	0.00	49.75
13:56:05	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
13:56:05	1	14.00	0.00	86.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

- The `1` argument provides statistics at one-second intervals.
 - The `-P ALL` option provides statistics for each individual processor and globally for all processors.
 - In this example, CPU usage is moderate as indicated by the `50.00 %idle` statistic.
 - Note `%usr`, which is the percentage of CPU used while executing at the user level.
 - Note `%sys`, which is the percentage of CPU used while executing at the system (kernel) level. This does not include time spent servicing hardware and software interrupts.
 - Press `Ctrl + C` after viewing a few intervals.
- e. Use the `vmstat 1` command to display CPU statistics.

```
# vmstat 1
```

procs		-----memory-----				---swap--		-----io----		-system--		-----cpu-----				
r	b	swpd	free	buff	cache	si	so	bi	bo	in	cs	us	sy	id	wa	st
3	0	0	3019612	1288	674772	0	0	20	17	103	38	1	6	92	0	0
2	0	0	3019636	1288	674772	0	0	0	0	1114	123	7	44	50	0	0
1	0	0	3019636	1288	674772	0	0	0	0	1070	89	8	43	50	0	0
2	0	0	3019384	1288	674772	0	0	0	0	1085	93	7	43	50	0	0
1	0	0	3019384	1288	674772	0	0	0	0	1090	112	8	43	50	0	0
1	0	0	3019384	1288	674772	0	0	0	0	1106	109	7	44	50	0	0
1	0	0	3019384	1288	674772	0	0	0	0	1071	90	7	43	50	0	0
1	0	0	3019352	1288	674772	0	0	0	0	1127	152	7	43	50	0	0
1	0	0	3019352	1288	674772	0	0	0	0	1079	106	8	43	50	0	0

- This command is primarily intended for monitoring virtual memory statistics.
- The `1` argument provides statistics at one-second intervals.
- CPU usage is higher when `id` (`%idle`) statistics are lower.
- High run queue length (`r`) statistics are an indication of CPU saturation.
- Press `Ctrl + C` after viewing a few intervals.

- f. Use the `sar -u 1` command to display systemwide CPU usage.

```
# sar -u 1
Linux 5.4.17-2011.7.4.el8uek.x86_64 (host02.example.com) 08/01/21 _x86_64_ (2 CPU)

14:06:31      CPU      %user      %nice      %system      %iowait      %steal      %idle
14:06:32    all        7.04        0.00       42.71        0.00        0.50       49.75
14:06:33    all        7.50        0.00       42.50        0.00        0.00       50.00
14:06:34    all        6.47        0.00       43.78        0.00        0.00       49.75
14:06:35    all        6.06        0.00       44.44        0.00        0.00       49.49
14:06:36    all        6.53        0.00       43.22        0.00        0.00       50.25
14:06:37    all        6.50        0.00       43.50        0.00        0.00       50.00
14:06:38    all        6.50        0.00       43.50        0.00        0.00       50.00
^C
Average:    all        6.66        0.00       43.38        0.00        0.07       49.89
```

- Note that `sar` data is collected periodically and is displayed by default. However, because the `sysstat` package has only just been installed there will be little, if any, data. The `1` argument causes `sar` to display current data at 1-second intervals.
- In this example, CPU usage is largely split between `%system` and `%idle` because one out of two CPUs is busy.
- An Average line is shown when pressing Ctrl + C to stop the sampling.

- g. Use the `sar -q 1` command to display run queue length and load averages.

```
# sar -q 1
Linux 5.4.17-2011.7.4.el8uek.x86_64 (host02.example.com) 08/01/21 ...

14:14:54      runq-sz    plist-sz    ldavg-1    ldavg-5    ldavg-15    blocked
14:14:55          1        184        1.00        1.00        1.00          0
14:14:56          1        184        1.00        1.00        1.00          0
14:14:57          1        184        1.00        1.00        1.00          0
14:14:58          1        184        1.00        1.00        1.00          0
14:14:59          1        184        1.00        1.00        1.00          0
14:15:00          1        184        1.00        1.00        1.00          0
14:15:01          1        184        1.00        1.00        1.00          0
^C
Average:          1        184        1.00        1.00        1.00          0
```

- The CPU is not saturated when there is a low run queue length (`runq-sz`) and low load averages (last 1 minute: `ldavg-1`, last 5 minutes: `ldavg-5`, and last 15 minutes: `ldavg-15`).
- A run queue size greater than the number of CPUs on your system is usually indicative of a CPU bottleneck.

- i. Use the `iostat` command to display CPU usage.

```
# iostat
Linux 5.4.17-2011.7.4.el8uek.x86_64 (host02.example.com)...
```

avg-cpu:	%user	%nice	%system	%iowait	%steal	%idle
	1.91	0.01	10.62	0.02	0.00	87.44

Device	tps	kB_read/s	kB_wrtn/s	kB_read	kB_wrtn
xvda	1.63	29.16	30.38	436098	454343
xvdb	0.08	2.02	0.14	30173	2061
scd0	0.06	3.45	0.00	51662	0
xvdc	0.07	1.68	0.14	25165	2057

- This command is primarily used for monitoring system I/O device loads.
- In this example, the average `%idle` statistic for the CPU is 87.44%.

4. Observe the memory statistics.

- a. Use the `top` command to display memory usage.

```
# top
```

A sample output of `top` is shown as follows:

```
top - 13:40:16 up 3:31, 2 users, load average: 1.02, 1.02, 0.82
Tasks: 159 total, 3 running, 156 sleeping, 0 stopped, 0 zombie
%Cpu(s): 12.1 us, 42.4 sy, 0.0 ni, 45.5 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3927.2 total, 2948.4 free, 318.6 used, 660.2 buff/cache
MiB Swap: 3070.0 total, 3070.0 free, 0.0 used. 3538.2 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
11604	root	20	0	7324	828	764	R	94.1	0.0	22:30.90	dd
11675	root	20	0	65384	4436	3804	R	5.9	0.1	0:00.01	top
1	root	20	0	182004	16324	9372	S	0.0	0.4	0:07.22	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.02	kthreadd
3	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_gp
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_par_gp
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0+

- The upper section of `top` displays memory statistics:
 - `MiB Mem`: This line reflects how much physical memory your system has: how much is free, how much is used, and how much is associated with `buff/cache`.
 - `MiB Swap`: This line reflects how much swap memory your system has, how much is used, and how much is free.
 - When a computer runs out of physical memory and starts using swap space, its performance deteriorates dramatically.
 - If you run out of swap, you will most likely crash your programs or the OS.
- The lower section of `top` displays a list of processes sorted by CPU usage, with the top consumer of CPU listed first.

- b. Sort by memory usage, press F or f to change the sort field, and then use the up/down arrow keys to highlight either of the following:

%MEM = Memory usage (RES)

RES = Resident size (kb)

Press the **s** key to set the selected sort field.

Press **q** or **Esc** to exit the field management window and return to the **top** display.

```
top - 14:24:52 up 4:16, 2 users, load average: 1.03, 1.03, 1.00
Tasks: 161 total, 2 running, 159 sleeping, 0 stopped, 0 zombie
%Cpu(s): 6.7 us, 43.5 sy, 0.0 ni, 49.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3927.2 total, 2944.0 free, 319.6 used, 663.6 buff/cache
MiB Swap: 3070.0 total, 3070.0 free, 0.0 used. 3537.2 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
771	root	20	0	290492	41404	18196	S	0.0	1.0	0:01.22	firewalld
775	root	20	0	224860	40724	39064	S	0.0	1.0	0:02.93	sssd_nss
862	root	20	0	414432	28544	15248	S	0.0	0.7	0:03.71	tuned
1077	rpcuser	20	0	71724	27672	2684	S	0.0	0.7	0:00.00	rpc.statd
728	polkitd	20	0	1631140	22776	17356	S	0.0	0.6	0:00.18	polkitd
841	root	20	0	391784	18672	16268	S	0.0	0.5	0:00.82	NetworkManager
1	root	20	0	182004	16324	9372	S	0.0	0.4	0:07.42	systemd
764	root	20	0	221244	15056	12500	S	0.0	0.4	0:00.49	sssd_be
724	root	20	0	214792	14348	12316	S	0.0	0.4	0:00.12	sssd

- Press **q** to quit the **top** command.

- c. Use the **vmstat 1** command to display memory statistics.

```
# vmstat 1
```

procs		-----memory-----				---swap--		-----io----		-system--		-----cpu-----				
r	b	swpd	free	buff	cache	si	so	bi	bo	in	cs	us	sy	id	wa	st
3	0	0	3019612	1288	674772	0	0	20	17	103	38	1	6	92	0	0
2	0	0	3019636	1288	674772	0	0	0	0	1114	123	7	44	50	0	0
1	0	0	3019636	1288	674772	0	0	0	0	1070	89	8	43	50	0	0
2	0	0	3019384	1288	674772	0	0	0	0	1085	93	7	43	50	0	0
1	0	0	3019384	1288	674772	0	0	0	0	1090	112	8	43	50	0	0
1	0	0	3019384	1288	674772	0	0	0	0	1106	109	7	44	50	0	0
1	0	0	3019384	1288	674772	0	0	0	0	1071	90	7	43	50	0	0
1	0	0	3019352	1288	674772	0	0	0	0	1127	152	7	43	50	0	0
1	0	0	3019352	1288	674772	0	0	0	0	1079	106	8	43	50	0	0

- The **1** argument provides statistics at one-second intervals.
- The important memory statistics are:
 - swpd** – The amount of virtual memory used
 - free** – The amount of idle memory
 - si** – The amount of memory swapped in from disk (per second)
 - so** – The amount of memory swapped out to disk (per second)
- In this example, the system has sufficient free memory and is not swapping.
- Press **Ctrl + C** after viewing a few intervals.

- d. Use the `free -m` command to display memory statistics.

```
# free -m
      total        used        free      shared  buff/cache   available
Mem:   3927         319        2944           8         663        3537
Swap:  3069           0        3069
```

- This example uses the `-m` option to display amounts in megabytes.
- This command displays the total amount of free and used physical and swap memory in your system.

- e. Use the `sar -r 1` command to display memory usage statistics. Partial output is shown as follows:

```
# sar -r 1
Linux 5.4.17-2011.7.4.el8uek.x86_64 (host02.example.com)...

14:52:17      kbmemfree  kbavail  kbmempused  %memused  kbbuffers  kbcached  kbcommit ...
14:52:18      3015064  3622484  1006372     25.03      2324    523524  397488 ...
14:52:19      3015064  3622484  1006372     25.03      2324    523524  397488 ...
14:52:20      3015096  3622516  1006340     25.02      2324    523524  397488 ...
14:52:21      3015096  3622516  1006340     25.02      2324    523524  397488 ...
14:52:22      3015096  3622516  1006340     25.02      2324    523524  397488 ...
14:52:23      3015096  3622516  1006340     25.02      2324    523524  397488 ...
^C

Average:      3015085  3622505  1006351     25.02      2324    523524  397488 ...
```

- Observe the `%memused` value, which is the percentage of used memory.

- f. Use the `sar -B 1` command to display memory paging statistics. Partial output is shown as follows:

```
# sar -B 1
Linux 5.4.17-2011.7.4.el8uek.x86_64 (host02.example.com)...

14:59:16      pgpgin/s pgpgout/s   fault/s  majflt/s  pgfree/s pgscank/s pgscand/s ...
14:59:17         0.00      0.00     17.00      0.00       1.00      0.00      0.00 ...
14:59:18         0.00      0.00      0.00      0.00       0.00      0.00      0.00 ...
14:59:19         0.00      0.00      0.00      0.00       1.00      0.00      0.00 ...
14:59:20         0.00      0.00      0.00      0.00       0.00      0.00      0.00 ...
14:59:21         0.00      0.00      0.00      0.00       1.00      0.00      0.00 ...
14:59:22         0.00      0.00      0.00      0.00     13.00      0.00      0.00 ...
14:59:23         0.00      0.00      0.00      0.00       1.00      0.00      0.00 ...
^C

Average:         0.00      0.00      2.43      0.00      2.43      0.00      0.00 ...
```

- Observe `pgscank/s`, which is the number of pages scanned by the `kswapd` daemon per second, and `pgscand/s`, which is the number of pages scanned directly per second.

- g. Use the `sar -W 1` command to display memory swapping statistics.

```
# sar -W 1
Linux 5.4.17-2011.7.4.el8uek.x86_64 (host02.example.com) ...

15:01:34      pswpin/s pswpout/s
15:01:35          0.00      0.00
15:01:36          0.00      0.00
15:01:37          0.00      0.00
15:01:38          0.00      0.00
15:01:39          0.00      0.00
15:01:40          0.00      0.00
15:01:41          0.00      0.00
^C

Average:          0.00      0.00
```

- The `pswpin/s` value is the number of swap pages the system swapped in per second, and `pswpout/s` is the number of swap pages the system swapped out per second.

- h. Use the `cat` command to view the contents of `/proc/meminfo`.

```
# cat /proc/meminfo
MemTotal:      4021436 kB
MemFree:       3015064 kB
MemAvailable:  3622488 kB
Buffers:       2324 kB
Cached:        523528 kB
SwapCached:    0 kB
Active:        356804 kB
Inactive:      291036 kB
Active(anon):  124388 kB
Inactive(anon): 6428 kB
Active(file):  232416 kB
Inactive(file): 284608 kB
Unevictable:   0 kB
Mlocked:       0 kB
SwapTotal:     3143676 kB
SwapFree:      3143676 kB
Dirty:         0 kB
Writeback:     0 kB
AnonPages:     122000 kB
Mapped:        83772 kB
Shmem:         8828 kB
...
```

5. Observe the disk I/O statistics.

- a. Use the `iostat -x` command to display extended I/O statistics for devices. This can be combined with the `-s` (short) option to fit more easily on a narrow screen as shown here:

```
# iostat -sx
Linux 5.4.17-2011.7.4.el8uek.x86_64 (host02.example.com)...
```

avg-cpu:	%user	%nice	%system	%iowait	%steal	%idle
	2.90	0.01	17.15	0.02	0.01	79.92

Device	tps	kB/s	rqm/s	await	aqu-sz	areq-sz	%util
xvda	1.32	47.76	1.33	1.09	0.00	36.13	0.07
xvdb	0.06	1.72	0.00	0.30	0.00	27.79	0.00
xvdd	0.05	1.46	0.00	0.27	0.00	27.47	0.00
scd0	0.05	2.76	0.00	2.77	0.00	58.71	0.00

- Observe the `%util` value, which is the percentage of CPU time during which I/O requests were issued to the device.
 - Device saturation occurs when this value is close to 100%.
 - Observe the `avgqu-sz` value, which is the average queue length of the requests that were issued to the device.
 - If average queue length is greater than 1, it is an indication of device I/O saturation.
- b. Use the `sar -d 1 1` command to display I/O statistics for devices.

```
# sar -d 1 1
Linux 5.4.17-2011.7.4.el8uek.x86_64 (host02.example.com)...
```

15:24:10	DEV	tps	rkB/s	wkB/s	areq-sz	aqu-sz	await	svctm	%util
15:24:11	dev202-0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
15:24:11	dev202-16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
15:24:11	dev202-48	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
15:24:11	dev11-0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Average:	DEV	tps	rkB/s	wkB/s	areq-sz	aqu-sz	await	svctm	%util
Average:	dev202-0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Average:	dev202-16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Average:	dev202-48	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Average:	dev11-0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

- This command also provides `%util` and `avgqu-sz` (`aqu-sz`) statistics.
- The `1 1` arguments cause `sar` to produce output after 1 second 1 time.

6. Observe network statistics.

a. Use the `ip -s link` command to observe network statistics.

```
# ip -s link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state...
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    RX: bytes  packets  errors  dropped overrun mcast
    3636      42        0       0        0        0
    TX: bytes  packets  errors  dropped carrier collsns
    3636      42        0       0        0        0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc...
    link/ether 00:16:3e:01:01:01 brd ff:ff:ff:ff:ff:ff
    RX: bytes  packets  errors  dropped overrun mcast
    1394573    19704    0       0        0        0
    TX: bytes  packets  errors  dropped carrier collsns
    4018993    8853     0       0        0        0
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc...
    link/ether 00:16:3e:01:01:02 brd ff:ff:ff:ff:ff:ff
    RX: bytes  packets  errors  dropped overrun mcast
    732027     10943    0       0        0        0
    TX: bytes  packets  errors  dropped carrier collsns
    938        13       0       0        0        0
```

- The number of bytes received and bytes transmitted on each interface is provided by the `RX: bytes` and `TX: bytes` values.
- This command also provides the number of packets transmitted and received, errors, dropped packets, overruns, and collisions.
- Frames are dropped, and the overrun counter is incremented when the capacity of the interface is exceeded.

b. Use the `nstat -as` command to observe network statistics.

- If the `ip` command shows an excessive amount of errors, more information can be found by examining the `nstat -as` output. Partial output is shown as follows:

```
# nstat -as
#kernel
IpInReceives          11972          0.0
IpInDelivers           9602          0.0
IpOutRequests          9311          0.0
IpOutNoRoutes          160           0.0
TcpActiveOpens         4             0.0
TcpPassiveOpens        7             0.0
TcpInSegs              9558          0.0
TcpOutSegs             9953          0.0
TcpRetransSegs         14            0.0
...
```

- This command displays summary statistics for each protocol. Statistics with a zero value are omitted; use the `-z` option to include these.
- Observe the number of segments retransmitted (`TcpRetransSegs`) as an indicator of network interface saturation.
- Many performance problems associated with the network involve retransmission of TCP packets.

Practice 16-3: Installing and Using OSWatcher

Overview

In this practice, you install and run the OSWatcher service on **host02**. After sufficient data has been collected, you archive this to analyze offline (on **gateway**).

Assumptions

- You are the `root` user on **host02**.
- You are still running the `dd` command in a second terminal window.
- The `oswatcher` software exists in the `extras` Yum repository on **host02**.

Tasks

1. From **host02**, use the `dnf` command to install the `oswatcher` package.

```
# dnf install -y oswatcher
...
Installed:
  net-tools-2.0-0.52.20160912git.el8.x86_64
  oswatcher-8.3.0-5.el8.noarch

Complete!
```

- Note `oswatcher` uses some monitoring tools from the `net-tools` package.

2. Run the `oswatcher` service.

- a. View the `/etc/sysconfig/oswatcher` file.

```
# cat /etc/sysconfig/oswatcher
# The directory where oswatcher logs should be kept
DATADIR=/var/log/oswatcher
# The interval (in seconds) between runs runs of statistics...
INTERVAL=30
# The maximum age (in hours) of the various log files
MAXAGE=48
# An optional program used to compress the log files
ZIP=gzip
```

- Note samples will be taken every 30 seconds.

- b. Use the `systemctl` command to check the status of the `oswatcher` service.

```
# systemctl status oswatcher
• oswatcher.service - OSWatcher:set of scripts used to ...
   Loaded: loaded (/usr/lib/systemd/system/oswatcher.service;
   enabled; vendor preset: disabled)
   Active: inactive (dead)
```

- Note the service is enabled but is not currently active.

- c. Use the `systemctl` command to start the `oswatcher` service.

```
# systemctl start oswatcher
```

- d. Use the `systemctl` command to check the status of the `oswatcher` service.

```
# systemctl status oswatcher
...
Active: active (running) since ...
...
CGroup: /system.slice/oswatcher.service
├─12700 /bin/sh /usr/sbin/OSWatcher 30 48 gzip /va...
└─12773 sleep 5

... Started OSWatcher:set of scripts used to periodically col...
... Info...Zip option IS specified.
... Info...OSW will use gzip to compress files.
... Changing current working directory to /var/log/oswatcher
```

- Note the options seen in `/etc/sysconfig/oswatcher` are being used.
- e. Let the service run for a few minutes to collect some data.
- f. After a few minutes have passed, use the `systemctl` command to stop the `oswatcher` service.

```
# systemctl stop oswatcher
```

3. View the data collection directories.

Use the `cd` command to change to the `/var/log/oswatcher/archive` directory, and then use the `ls` command with the `--file-type` option to view the contents of the directory.

```
# cd /var/log/oswatcher/archive
# ls --file-type
oswarp/          oswmeminfo/      oswpidstat/      oswslabinfo/
oswcpuinfo/      oswmpstat/       oswpidstatd/     oswtop/
oswifconfig/     oswnetstat/      oswprvtnet/      oswvmstat/
oswiostat/       oswnfsiostat/    oswps/           oswxentop/
```

- The directory contains subdirectories for each data collector.
4. View the `oswiostat` directory.
- a. Use the `cd` command to change to the `oswiostat` directory, and then use `ls` to view the contents of the directory.

```
# cd oswiostat
# ls
host02.example.com_iostat...
```

- b. Use the `less` command to view the file.

```
# less host02...
Linux OSWbb v8.3.0
zzz ***...
...
```

- Note that this file contains the output of the `iostat -x` command.
- The `iostat` command ran at 30-second intervals (the value of `INTERVAL`).
- Each interval begins with `zzz ***` characters followed by a time stamp.

5. View the `oswmpstat` directory.

- a. Use the `cd` command to change to the `oswmpstat` directory, and then use `ls` to view the contents of the directory.

```
# cd ../oswmpstat
# pwd
/var/log/oswatcher/archive/oswmpstat
# ls
host02.example.com_mpstat...
```

- b. Use the `less` command to view the file.

```
# less host02...
Linux OSWbb v8.3.0
zzz ***...
...
```

- Note that this file contains the output of the `mpstat` command.
- The `mpstat` command ran at 30-second intervals (the value of `INTERVAL`).
- Each interval begins with `zzz ***` characters followed by a time stamp.

6. View the `oswprvtnet` directory.

Use the `cd` command to change to the `oswprvtnet` directory, and then use `ls` to view the contents of the directory.

```
# cd ../oswprvtnet
# pwd
/var/log/oswatcher/archive/oswprvtnet
# ls
```

- Note that this directory is empty.
- This directory would contain the status of RAC private networks.
- You need to manually create the `private.net` file to run `traceroute` commands.

7. View the `oswslabinfo` directory.

- a. Use the `cd` command to change to the `oswslabinfo` directory, and then use `ls` to view the contents of the directory.

```
# cd ../oswslabinfo
# pwd
/var/log/oswatcher/archive/oswslabinfo
# ls
host02.example.com_slabinfo...
```

- b. Use the `less` command to view the file.

```
# less host02...
zzz ***...
slabinfo - version: 2.1
...
```

- Note that this file contains periodic contents of the `/proc/slabinfo` file.

8. View the `oswvmstat` directory.

- a. Use the `cd` command to change to the `oswvmstat` directory, and then use `ls` to view the contents of the directory.

```
# cd ../oswvmstat
# pwd
/var/log/oswatcher/archive/oswvmstat
# ls
host02.example.com_vmstat...
```

- b. Use the `less` command to view the file.

```
# less host02...
Linux OSWbb v8.3.0 host02.example.com
SNAP_INTERVAL 30
CPU_CORES 2
OSWBB_ARCHIVE_DEST /var/log/oswatcher/archive
zzz ***...
...
```

- Note that this file contains periodic output of the `vmstat` command.

9. View the `oswmeminfo` directory.

- a. Use the `cd` command to change to the `oswmeminfo` directory, and then use `ls` to view the contents of the directory.

```
# cd ../oswmeminfo
# pwd
/var/log/oswatcher/archive/oswmeminfo
# ls
host02.example.com_meminfo...
```

- b. Use the `less` command to view the file.

```
# less host02...
zzz ***...
MemTotal: ...
MemFree: ...
...
```

- Note that this file contains periodic contents of the `/proc/meminfo` file.

10. View the `oswnetstat` directory.

- a. Use the `cd` command to change to the `oswnetstat` directory, and then use `ls` to view the contents of the directory.

```
# cd ../oswnetstat
# pwd
/var/log/oswatcher/archive/oswnetstat
# ls
host02.example.com_netstat...
```

- b. Use the `less` command to view the file.

```
# less host02...
Linux OSWbb v8.3.0
zzz ***...
...
```

- Note that this file contains periodic output of the legacy `netstat` (from `net-tools`) command.

11. View the `oswps` directory.

- a. Use the `cd` command to change to the `oswps` directory, and then use `ls` to view the contents of the directory.

```
# cd ../oswps
# pwd
/var/log/oswatcher/archive/oswps
# ls
host02.example.com_ps...
```

- b. Use the `less` command to view the file.

```
# less host02...
Linux OSWbb v8.3.0

zzz ***...
...
```

- Note that this file contains periodic output of the `ps` command.

12. View the `oswtop` directory.

- a. Use the `cd` command to change to the `oswtop` directory, and then use `ls` to view the contents of the directory.

```
# cd ../oswtop
# pwd
/var/log/oswatcher/archive/oswtop
# ls
host02.example.com_top...
```

- b. Use the `less` command to view the file.

```
# less host02...
Linux OSWbb v8.3.0
zzz ***...
...
```

- Note that this file contains periodic output of the `top` command.

13. View the `oswifconfig` directory.

- a. Use the `cd` command to change to the `oswifconfig` directory, and then use `ls` to view the contents of the directory.

```
# cd ../oswifconfig
# pwd
/var/log/oswatcher/archive/oswifconfig
# ls
host02.example.com_ifconfig...
```

- b. Use the `less` command to view the file.

```
# less host02...
Linux OSWbb v8.3.0
zzz ***...
...
```

- Note that this file contains periodic output of the legacy `ifconfig` command.

14. View the `oswcpuinfo` directory.

- a. Use the `cd` command to change to the `oswcpuinfo` directory, and then use `ls` to view the contents of the directory.

```
# cd ../oswcpuinfo
# pwd
/var/log/oswatcher/archive/oswcpuinfo
# ls
host02.example.com_cpuinfo... host02.example.com_lscpu...
```

- b. Use the `less` command to view the `cpuinfo` file.

```
# less host02.example.com_cpuinfo...
processor           : 0
vendor_id          : GenuineIntel
cpu family         : 6
model              : 62
model name         : Intel(R) Xeon(R) CPU E5-2697 v2 @ 2.70GHz
...
```

- Note that this file contains the contents of `/proc/cpuinfo`. It is only written once.

- c. Use the `less` command to view the `lscpu` file.

```
# less host02.example.com_lscpu...
Architecture:      x86_64
CPU op-mode(s):    32-bit, 64-bit
Byte Order:        Little Endian
CPU(s):            2
On-line CPU(s) list: 0,1
Thread(s) per core: 1
Core(s) per socket: 2
Socket(s):         1
NUMA node(s):      1
Vendor ID:         GenuineIntel
```

- Note that this file contains the output of the `lscpu` command. It is only written once.

15. Terminate the `dd` command started previously.

In the second terminal window on **host02**, in which you initiated the following `dd` command, press `Ctrl + C` to terminate the command.

```
# dd if=/dev/zero of=/dev/null bs=1024
CTRL-C
```

16. Log off from **host02** in preparation for the next practice.

In each terminal window, use the `logout` command to close the `ssh` connection to **host02**.

```
# logout
Connection to host02 closed.
```

You are now the `oracle` user on **gateway**.

Practice 16-4: Using OSWatcher Analyzer

Overview

In this practice, you perform the following:

- Unpack the OSWatcher Black Box software to a local directory.
- Download the OSWatcher data to **gateway**.
- Start OSWatcher Analyzer (OSWbba) on **gateway**.
- View CPU and Memory Graphs from OSWbba.
- Use OSWbba to analyze the data.
- View the analysis report.

Assumptions

- You are the `oracle` user on **gateway**.
- You completed the previous practice *Installing and Using OSWatcher*.

Tasks

1. Unpack the OSWatcher Black Box software to the `~/oswbb` directory.

Use the `cd` command to ensure you are in the home directory. Then use the `tar` command to unpack the software archive.

```
[oracle@gateway ~]$ cd
[oracle@gateway ~]$ tar xf /stage/oswbb840.tar
```

- The software is extracted into a directory called `oswbb`.

2. Download the OSWatcher data from **host02**.

Use the `scp` command to download the data from **host02**.

```
$ scp -r root@host02:/var/log/oswatcher/archive ~/oswbb
root@host02's password:
host02.example.com_ps_21.01.11.0900.dat...
...
```

3. Start OSWatcher Analyzer (OSWbba) on **host02**.

- a. Use the `java -version` command to display the Java version number.

```
$ java -version
openjdk version "1.8.0_282"
...
```

- In this example, version `1.8.0_282` is installed.

- b. Use the `java -jar` command to display the OSWbba menu. First, ensure that you are in the directory where OSWbba is installed (`~/oswbb`).

```
$ cd ~/oswbb
$ java -jar oswbba.jar -i archive
...
Enter 1 to Display CPU Process Queue Graphs
Enter 2 to Display CPU Utilization Graphs
Enter 3 to Display CPU Other Graphs
Enter 4 to Display Memory Graphs
Enter 5 to Display Disk IO Graphs
...
Enter X to Export Parsed Data to Flat File
Enter S to Analyze Subset of Data(Changes analysis dataset...
Enter A to Analyze Data
Enter D to Generate DashBoard

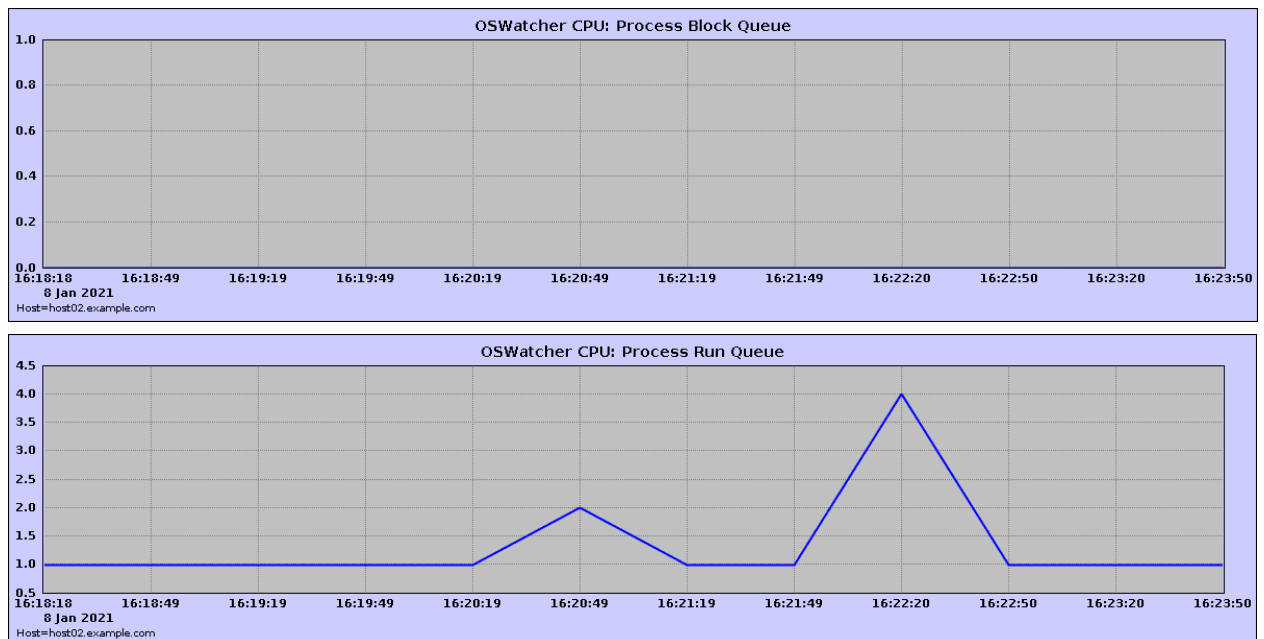
Enter Q to Quit Program

Please Select an Option:
```

4. View CPU and Memory Graphs. Sample output is shown as follows. Your graphs will be different.

- a. From the OSWbba menu, select 1 to display the two CPU Process Queue graphs: Process Block Queue and Process Run Queue.

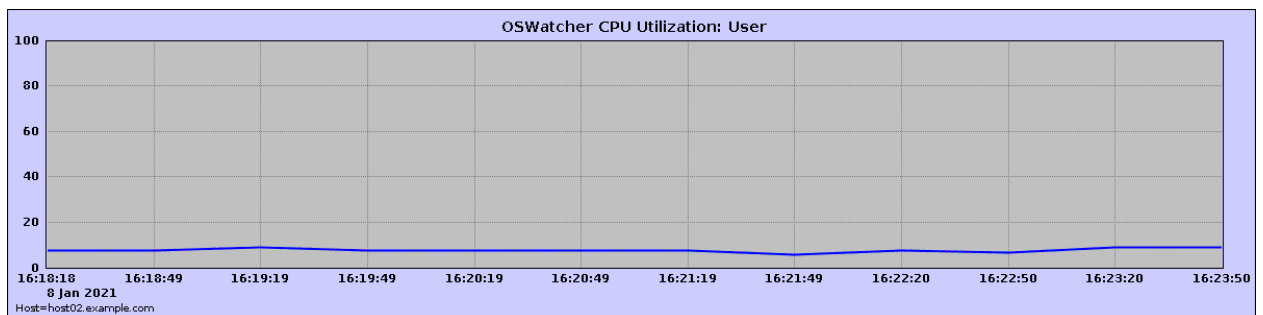
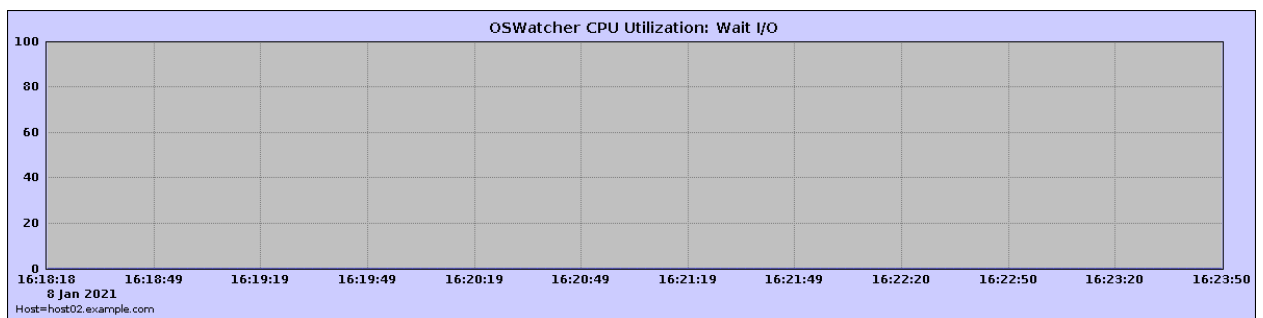
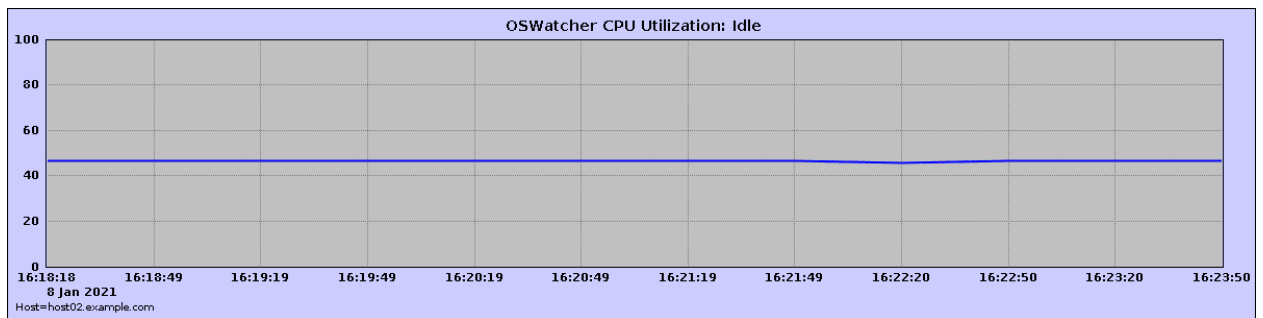
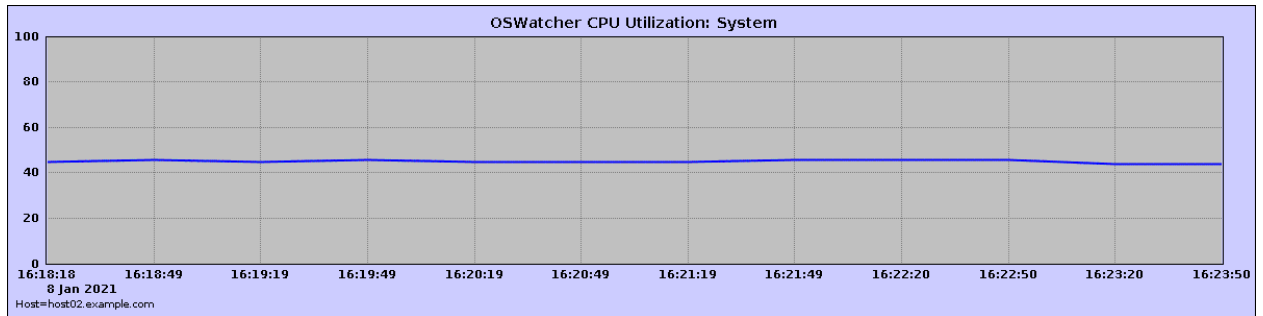
View each graph by dragging and aligning the graphs to see them, as desired.



- After viewing the graphs, click the terminal window to bring it to the foreground, if necessary, and select R from the menu to remove the graphs.

- b. From the OSWbba menu, select 2 to display the four CPU Utilization graphs: System, Idle, Wait I/O, and User.

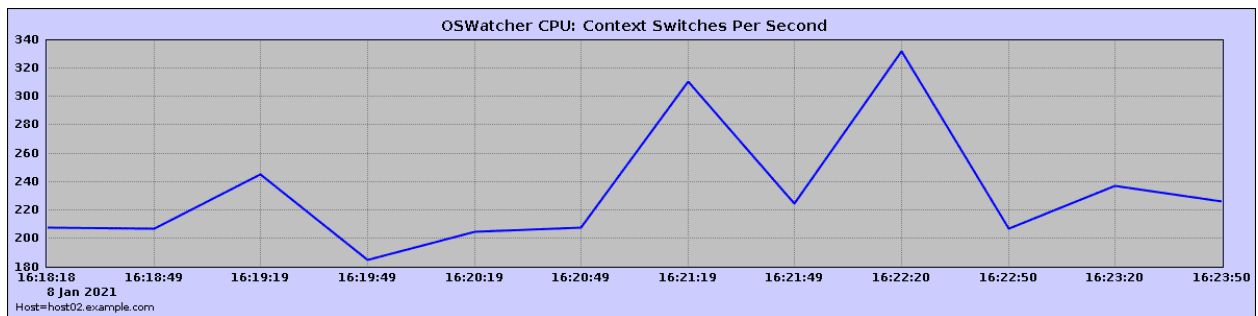
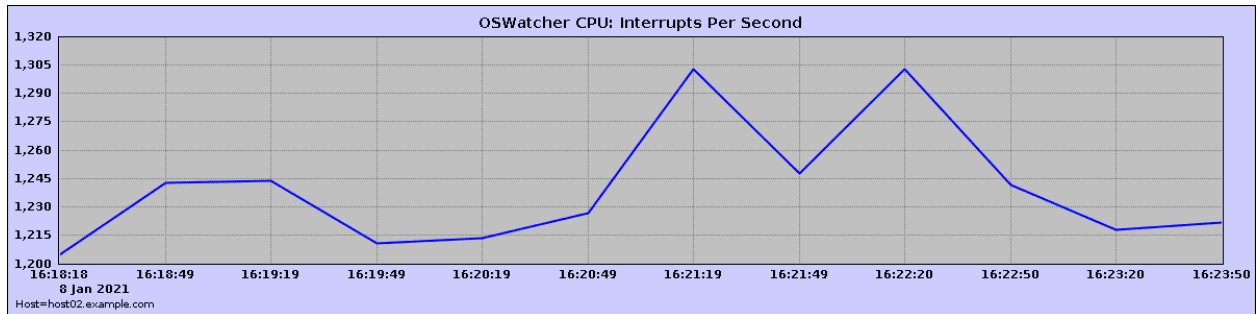
View each graph by dragging and aligning the graphs to see them, as desired.



- After viewing the graphs, click the terminal window to bring it to the foreground, if necessary, and select R from the menu to remove the graphs.

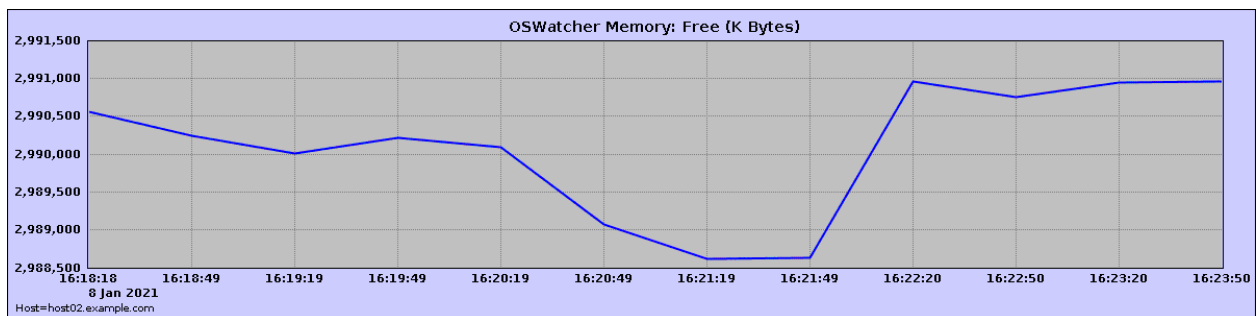
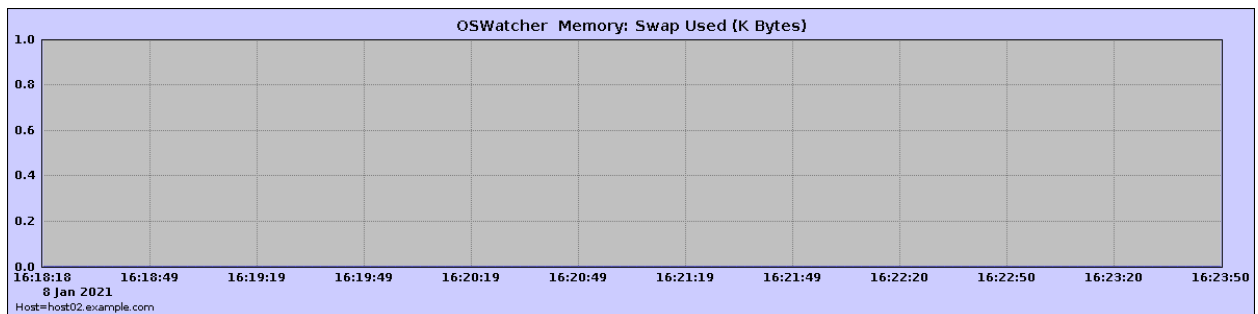
- c. From the OSWbba menu, select 3 to display the two CPU Other graphs: Interrupts Per Second and Context Switches Per Second.

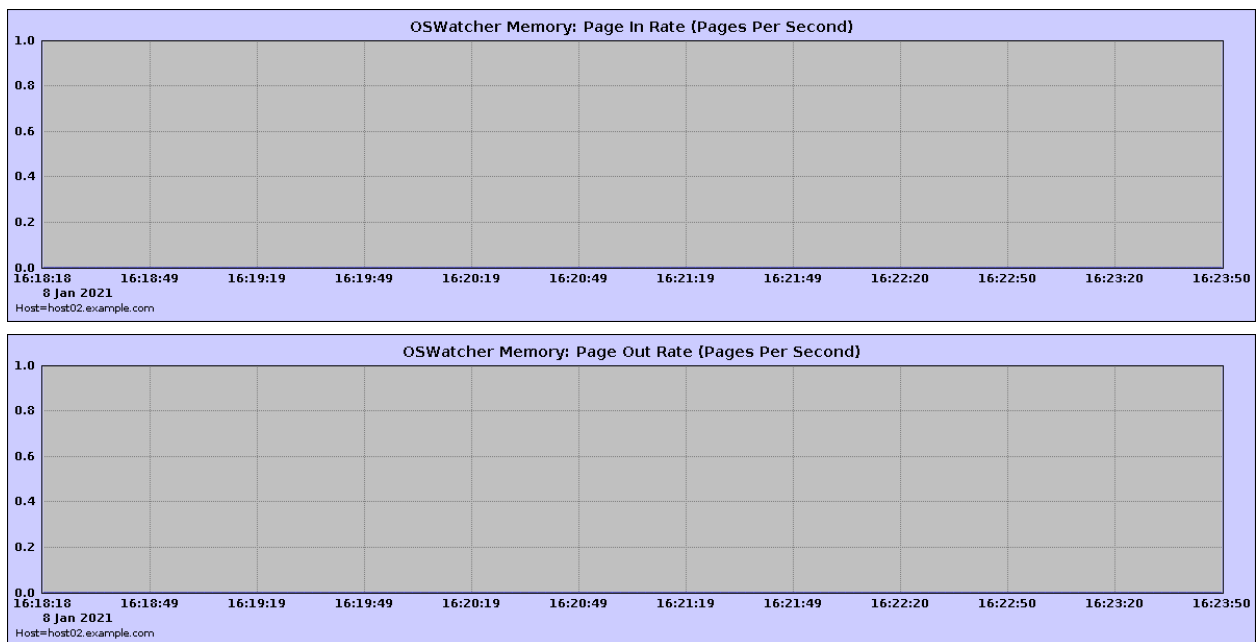
View each graph by dragging and aligning the graphs to see them, as desired.



- After viewing the graphs, click the terminal window to bring it to the foreground, if necessary, and select R from the menu to remove the graphs.
- d. From the OSWbba menu, select 4 to display the four Memory Graphs: Swap Used, Free, Page In Rate, and Page Out Rate.

View each graph by dragging and aligning the graphs to see them, as desired.





- After viewing the graphs, click the terminal window to bring it to the foreground, if necessary, and select R from the menu to remove the graphs.
5. Run the OSWbba analyzer and view the report.
 - a. From the OSWbba menu, select A to analyze the collected data and produce a report. Provide a file name or press Enter to accept the default name.

```
...
Enter A to Analyze Data
...
Please Select an Option:a
Enter a unique analysis directory name or enter <CR> to accept
default name: ENTER
A new analysis file analysis/host02.../analysis.txt has been
created.
```

- This message “A new analysis file analysis/host02.../analysis.txt has been created” appears, and the menu is then redisplayed.
- You could alternatively run the analyzer from the command line by using the following command: (Do not run this command; this is information only.)

```
$ java -jar oswbba.jar -i ~/oswbb/archive -A
```

- b. Exit the OSWbba menu by selecting the Q option to quit.
- c. From the **host02** command line, change to the directory where the `analysis.txt` file is located and use the `ls` command to display the analysis file in the `~/oswbb/analysis/host02...` directory.

```
$ cd ~/oswbb/analysis/host02*
$ ls
analysis.txt
```

- The directory name where the `analysis.txt` file is found will vary.

- d. Use the `less` command to view the `analysis.txt` file.

Partial contents are shown below, showing initial information, followed by an outline of the contents of the file.

```
# less analysis.txt
This report is best viewed in a fixed font editor like
textpad...

OSWatcher Analyzer

Input Archive:      archive
Archive Source Dest: /var/log/oswatcher/archive
Archive Begin Time: ...
Archive End Time:   ...
Analysis Begin Time: ...
Analysis End Time:  ...
Hostname:           HOST02.EXAMPLE.COM
OS Version:         Linux
OSW Version:        v8.3.0
Snapshot Freq:      30
CPU COUNT:          2

#####
# Contents Of This Report:
#
# Section 1: System Status
# Section 2: System Slowdowns
#   Section 2.1: System Slowdown RCA Process Level Ordered By
#   Impact
# Section 3: System General Findings
# Section 4: CPU Detailed Findings
#   Section 4.1: CPU Run Queue:
#   Section 4.2: CPU Utilization: Percent Busy
#   Section 4.3: CPU Utilization: Percent Sys
# Section 5: Memory Detailed Findings
#   Section 5.1: Memory: Process Swap Queue
#   Section 5.2: Memory: Scan Rate
#   Section 5.3  Memory: Page In:
#   Section 5.4  Memory: Page Out:
#   Section 5.5  Memory: Page Tables (Linux only):
#   Section 5.6: Top 5 Memory Consuming Processes Beginning
#   Section 5.7: Top 5 Memory Consuming Processes Ending
```

```

# Section 6: Disk Detailed Findings
#   Section 6.1: Disk Percent Utilization Findings
#   Section 6.2: Disk Service Times Findings
#   Section 6.3: Disk Wait Queue Times Findings
#   Section 6.4: Disk Throughput Findings
#   Section 6.5: Disk Reads Per Second
#   Section 6.6: Disk Writes Per Second
#   Section 6.7: Disk Percent CPU waiting on I/O
# Section 7: Network Detailed Findings
#   Section 7.1 Network Data Link Findings
#   Section 7.2: Network IP Findings
#   Section 7.3: Network UDP Findings
#   Section 7.4: Network TCP Findings
#   Section 7.5: Network Private Interconnect Findings
# Section 8: Process Detailed Findings
#   Section 8.1: PS Process Summary Ordered By Time
#   Section 8.2: PS for Processes With Status = D or T Ordered
By Time
#   Section 8.3: PS for (Processes with CPU > 0) When System
Idle CPU < 30% Ordered By Time
#   Section 8.4: Top VSZ Processes Increasing Memory Per
Snapshot
#   Section 8.5: Top RSS Processes Increasing Memory Per
Snapshot
#   Section 8.6: New Processes Created Since Last Snapshot
#   Section 8.7: Exiting Processes Since Last Snapshot
#
...

```

e. After viewing the analysis file, press `q` to exit the `less` command.

6. Remove the `oswbb` files on **gateway**.

Use the `/bin/rm -r` command to remove `oswbb` and its contents. Use the `cd` command to change to the home directory of the `oracle` user first.

```

$ cd
$ /bin/rm -r oswbb

```


Practices for Lesson 17: System Logging

Practices for Lesson 17: Overview

Practices Overview

In these practices, you configure system logging, use `rsyslog` templates to format log messages, install and run `logwatch`, view the `systemd-journald` journal, and configure persistent `systemd-journald` storage.

Assumptions

You created a local Yum repository in the practices for the lesson titled “Package Management”.

Practice 17-1: Configuring System Logging

Overview

In this practice, you view the system logging configuration file, modify the file, and observe the impact of the modifications. You also configure log file rotation.

Assumptions

You are the `oracle` user on **gateway**.

Tasks

1. Log in to **host02**.

Use the `ssh` command to log in to **host02** as the `root` user.

Provide the `root` password when prompted.

```
[oracle@gateway ~]$ ssh root@host02
root@host02's password:
Last login: ...
```

All remaining commands in this practice are entered from **host02**.

2. Explore the configuration file `/etc/rsyslog.conf`.

Use the `less` command to view the system logging configuration file.

```
# less /etc/rsyslog.conf
...
#### MODULES ####
...
module(load="imuxsock"      # provides support for local system...
        SysSock.Use="off") # Turn off message reception via lo...
                           # local messages are retrieved thro...
module(load="imjournal"     # provides access to the ...
        StateFile="imjournal.state") # File to store the posit...
...
#### GLOBAL DIRECTIVES ####

# Where to place auxiliary files
global(workDirectory="/var/lib/rsyslog")

# Use default timestamp format
module(load="builtin:omfile" Template="RSYSLOG_TraditionalFil...

# Include all config files in /etc/rsyslog.d/
include(file="/etc/rsyslog.d/*.conf" mode="optional")
```

```

#### RULES ####

# Log all kernel messages to the console.
# Logging much else clutters up the screen
#kern.*                                /dev/console

# Log anything (except mail) of level info or higher.
# Don't log private authentication messages!
*.info;mail.none;authpriv.none;cron.none  /var/log/messages

# The authpriv file has restricted access.
authpriv.*                                /var/log/secure

# Log all the mail messages in one place.
mail.*                                    -/var/log/maillog

# Log cron stuff
cron.*                                    /var/log/cron

...

```

- Use the up-arrow and down-arrow keys to view the various sections of the file.
- The **MODULES** section uses the `module()` function to load the modules.
- The **GLOBAL DIRECTIVES** section specifies configuration options.
- The **RULES** section defines a selector (`facility.priority`) and an action.
- Press the `q` key to exit the `less` command.

3. Change the action for `cron` logging.

- Use the `ls` command to list the `cron*` files in the `/var/log` directory.

```

# ls /var/log/cron*
/var/log/cron
/var/log/cron-2021...
...

```

- You might not have any rotated log files, which are files with a date stamp.
- Use the `vi` editor to modify the system logging configuration file. Change the action for `cron` logging to log in to a different log file: `/var/log/cron_new`.

```

# vi /etc/rsyslog.conf
...
# Log cron stuff
cron.*                /var/log/cron                (old entry)
cron.*                /var/log/cron_new            (new entry)

```

- Use the `systemctl` command to restart the `rsyslog` service.

```

# systemctl restart rsyslog

```

- d. Use the `crontab -e` command to create a `cron` job that runs the `ls` command every minute.

```
# crontab -e
* * * * * ls
```

- e. Use the `ls` command to list the `cron*` files in the `/var/log` directory.

```
# ls /var/log/cron*
/var/log/cron
/var/log/cron-2018...
...
/var/log/cron_new
```

- Note the new log file, `cron_new`.

- f. Use the `tail` command to view the last two lines in the `cron` log file.

```
# tail -2 /var/log/cron
Jan 11 11:29:01 host02 CROND[7690]: (root) CMDOUT (anaconda-k...
Jan 11 11:29:01 host02 CROND[7690]: (root) CMDOUT (Ksplice)
```

- This is sample output only.

- g. Use the `head` command to view the first entries in the `cron_new` log file.

```
# head /var/log/cron_new
Jan 11 11:28:44 host02 crontab[7687]: (root) BEGIN EDIT (root)
Jan 11 11:28:52 host02 crontab[7687]: (root) REPLACE (root)
...
```

- Note from the timestamps that the new log entries are being written to `cron_new`.

- h. Use the `vi` editor to modify the system logging configuration file. Change the action for `cron` logging back in to the original log file.

```
# vi /etc/rsyslog.conf
...
# Log cron stuff
cron.*                /var/log/cron_new          (old entry)
cron.*                /var/log/cron          (new entry)
```

- i. Use the `systemctl` command to restart the `rsyslog` service.

```
# systemctl restart rsyslog
```

- j. Use the `tail` command to ensure that `cron` is now logging in to `/var/log/cron`.
This is sample output only.

```
# tail -1 /var/log/cron
Jan 11 11:35:01 host02 CROND[842]: (root) CMD (ls)
# tail -1 /var/log/cron_new
Jan 11 11:34:01 host02 CROND[758]: (root) CMD (ls)
```

- Note that the entry in `cron` has a later timestamp than the entry in `cron_new`.
- You might need to wait a minute for the next `cron` job to run before this is the case.

- k. Use the `rm` command to delete the `cron_new` log file.

```
# rm /var/log/cron_new
rm: remove regular file '/var/log/cron_new'? y
```

- l. Use `crontab -r` to remove the `crontab`.

```
# crontab -r
```

4. Configure `rsyslog` to log debug messages.

- a. Use the `vi` editor to create a custom system logging configuration file `/etc/rsyslog.d/local.conf`. Add an entry to log all debug messages to `/var/log/debug`.

```
# vi /etc/rsyslog.d/local.conf
*.debug /var/log/debug
```

- b. Use the `systemctl` command to restart the `rsyslog` service.

```
# systemctl restart rsyslog
```

- c. Use the `logger` command to generate an **informational** log message.

```
# logger -p info "This is an info-priority message"
```

- `Logger` is an interface to the `syslog(3)` system log module.
- `Logger` makes entries in the system log.

- d. Use the `tail` command to view the log files.

```
# tail /var/log/messages
...
... root[7791]: This is an info-priority message
# tail /var/log/debug
...
... root[7791]: This is an info-priority message
```

- Note that the message was written to both log files.

- e. Use the `logger` command to generate a **debug-level** log message.

```
# logger -p debug "This is a debug-priority message"
```

- f. Use the `tail` command to view the log files.

```
# tail /var/log/messages
...
# tail /var/log/debug
...
... root[7791]: This is an info-priority message
... root[7796]: This is a debug-priority message
```

- Note that the debug-level message was written only to `/var/log/debug`.

- g. Use the `rm` command to delete `/etc/rsyslog.d/local.conf`.

```
# rm /etc/rsyslog.d/local.conf
rm: remove regular file '/etc/rsyslog.d/local.conf'? y
```

- h. Use the `systemctl` command to restart the `rsyslog` service.

```
# systemctl restart rsyslog
```

5. Configure log file rotation.

- a. Use the `ls` command to view the contents of the `/var/log` directory.

This is sample output only.

```
# ls /var/log/messages*
/var/log/messages      /var/log/messages-2021...
...

# ls /var/log/maillog*
/var/log/maillog       /var/log/maillog-2021...
...

# ls /var/log/cron*
/var/log/cron          /var/log/cron-2021...
...
```

- Note that some files in `/var/log` have numbers at the end of the file name.
 - These numbers represent a rotated log with the time stamp added to the log file name.
 - You might not have log files with a time stamp appended to the file name. It depends on how long your system has been running.
- b. Use the `vi` editor to modify the `/etc/logrotate.conf` configuration file. Change the frequency of the default log file rotation from **weekly** to **daily**.

```
# vi /etc/logrotate.conf
...
# rotate log files weekly
weekly                                (old entry)
daily                               (new entry)
```

- Your log files now rotate daily after making this change.

Practice 17-2: Using rsyslog Templates

Overview

In this practice, you use `rsyslog` templates to format `rsyslog` output.

Assumptions

You are the `root` user on `host02`.

Tasks

1. Adding a template definition to the `/etc/rsyslog.conf` file.
 - a. Use the `vi` editor to edit the `/etc/rsyslog.conf` file and define a template.
Add the template definition line shown.
Do not exit the `vi` editor.

```
# vi /etc/rsyslog.conf
...
$template class, "MyMessage: %msg%\n"
```

- This entry creates a template named `class`.
- b. Continue editing `/etc/rsyslog.conf` and create a log file that uses the template.
Add the new line (shown in **bold**) after the entry that defined the template.
Save and exit the `vi` editor after adding the new entries to the file.

```
...
$template class, "MyMessage: %msg%\n"

*. *    /var/log/class.log;class
```

- This entry writes all messages to the `/var/log/class.log` file and formats the entries by using the `class` template.
- c. After saving the changes to `/etc/rsyslog.conf`, use the `systemctl` command to restart the `rsyslog` service.

```
# systemctl restart rsyslog
```

- d. Use the `cat` command to view the `/var/log/class.log` file.

```
# cat /var/log/class.log
...
MyMessage: Stopping System Logging Service...
MyMessage: [origin software="rsyslogd" swVersion="8.1911.0-6.el8" x-pid="1193" x-info="https://www.rsyslog.com"] exiting on signal 15.
...
```

- Note that all entries are preceded by the text `"MyMessage:"` followed by the actual message, as defined in the `class` template.

2. Modify the `class` template to include additional properties.

- a. Use the `vi` editor and change the `class` template definition in `/etc/rsyslog.conf`, as shown:

```
# vi /etc/rsyslog.conf
...
$template class, "Time: %timestamp%, Facility: %syslogfacility-
text%, Priority: %syslogpriority-text%, Hostname: %hostname%,
Message: %msg%\n"

*. *    /var/log/class.log;class
```

- b. After saving the changes to `/etc/rsyslog.conf`, use the `systemctl` command to restart the `rsyslog` service.

```
# systemctl restart rsyslog
```

- c. Use the `cat` command to view the `/var/log/class.log` file.

```
# cat /var/log/class.log
...
Time: Nov 14 13:27:55, Facility: daemon, Priority: info,
Hostname: host02, Message: Stopping System Logging Service...
...
```

- Note that the newest entries now include the `Time`, `Facility`, `Priority`, `Hostname`, and `Message` properties, as defined in the `class` template.

3. Restore default configuration.

- a. Remove the template-related entries added to the `/etc/rsyslog.conf` file.
- b. After removing the template entries, use the `systemctl` command to restart the `rsyslog` service.

```
# systemctl restart rsyslog
```

- c. Use the `rm` command to remove the `/var/log/class.log` file.

```
# rm /var/log/class.log
rm: remove regular file '/var/log/class.log'? y
```

Practice 17-3: Using logwatch

Overview

In this practice, you install the `logwatch` package, view the main configuration file (the `cron` file), and run the `logwatch` utility from the command line.

Assumptions

You are the `root` user on `host02`.

Tasks

1. Install the `logwatch` package.
 - a. Use the `rpm` command to determine whether the `logwatch` package is already installed.

```
# rpm -q logwatch
package logwatch is not installed
```

- In this example, the `logwatch` package is not installed.

- b. Use the `dnf` command to install the `logwatch` package.

```
# dnf install -y logwatch
...
Installed:
  logwatch-7.4.3-9.el8.noarch
  mailx-12.5-29.el8.x86_64
  perl-Date-Manip-6.60-2.el8.noarch
  perl-Sys-CPU-0.61-14.el8.x86_64
  perl-Sys-MemInfo-0.99-6.el8.x86_64
...
Complete!
```

2. View `logwatch` files.
 - a. Use the `rpm` command to list all `logwatch` files.
Only partial output is displayed.

```
# rpm -ql logwatch
...
/etc/cron.daily/0logwatch
...
/etc/logwatch/conf/logwatch.conf
...
/usr/sbin/logwatch
...
/usr/share/logwatch/default.conf/logwatch.conf
...
```


- b. Use the `less` command to view the `logwatch` default configuration file.

```
# less /usr/share/logwatch/default.conf/logwatch.conf
...
```

- Note various configurable items such as the following:

- LogDir
- TmpDir
- MailTo
- MailFrom
- Range
- Detail
- Service

- c. Use the `cat` command to view the `logwatch` cron file.

```
# cat /etc/cron.daily/0logwatch
#!/bin/sh

#Set logwatch location
LOGWATCH_SCRIPT="/usr/sbin/logwatch"
#Add options to this line. Most options should be defined in
/etc/logwatch/conf/logwatch.conf,
#but some are only for the nightly cronrun such as --output mail
and should be set here.
#Other options to consider might be "--format html" or "--encode
base64", man logwatch for more details.
OPTIONS="--output mail"

#Call logwatch
$LOGWATCH_SCRIPT $OPTIONS

exit 0
```

3. Run `logwatch` from the command line.

- a. Run the `logwatch --help` command to view the `logwatch` help.

```
# logwatch --help
Usage: /usr/sbin/logwatch [--detail <level>] [--logfile <name>]
      [--output <output_type>]
...
```

- b. Run `logwatch` with the `range` option and a `date_range` parameter of `today`. This will process log data from the current day.

Sample output is displayed.

```
# logwatch --range today

##### Logwatch 7.4.3 (04/27/16) #####
      Processing Initiated: Mon Jan...
      Date Range Processed: today
...
      Logfiles for Host: host02.example.com
#####
...
-----SSHD Begin -----

SSHD Started: 2 Time(s)

Users logging in through sshd:
    root:
        192.0.2.1 (gateway.example.com): 7 times
...
##### Logwatch End #####
```

4. Remove the `logwatch` package.

Use the `dnf` command to remove the `logwatch` package.

```
# dnf remove -y logwatch

...
Removed:
  logwatch-7.4.3-9.el8.noarch
  mailx-12.5-29.el8.x86_64
  perl-Date-Manip-6.60-2.el8.noarch
  perl-Sys-CPU-0.61-14.el8.x86_64
  perl-Sys-MemInfo-0.99-6.el8.x86_64
...
Complete!
```

Practice 17-4: Using systemd-journald

Overview

In this practice, you use the `journalctl` command to query the `systemd` journal, view `systemd-journald` metadata, and enable persistent `systemd-journald` storage. Your log output might vary.

Assumptions

You are the `root` user on `host02`.

Tasks

1. Use the `journalctl` command to query the `systemd` journal.
 - a. Run `journalctl` with no options or arguments.

```
# journalctl
-- Logs begin at ..., end at ...
... kernel: Linux version 5.4.17-2011.7.4.el8uek.x86_64 (mock...
... kernel: Command line: BOOT_IMAGE=(hd0,msdos1)/vmlinuz-5.4...
... kernel: x86/fpu: x87 FPU will use FXSAVE
...
... kernel: integrity: Unable to open file: /etc/keys/x509_im...
...
```

- Note that all log data is displayed, including rotated logs.
 - Note that entries with error priority and higher are in red.
 - Note that entries with notice and warning priority are in bold font.
- b. Run the `journalctl -h` command to display usage and query options.

```
# journalctl -h
journalctl [OPTIONS...] [MATCHES...]
Query the journal.
Options:
    --system           Show only the system journal
    --user             Show only the user journal for the ...
    -M --machine=CONTAINER Operate on local container
    -S --since=DATE    Show entries not older than the spe...
    -U --until=DATE    Show entries not newer than the spe...
    ...
    -k --dmesg         Show kernel message log from the cu...
    -u --unit=UNIT     Show logs from the specified unit
    ...
    -p --priority=RANGE Show entries with the specified pr...
    ...
```

- c. Run the `journalctl -r` command to display the newest log entries first.

```
# journalctl -r
-- Logs begin at ..., end at ...
... systemd[1]: rsyslog.service: Succeeded.
... rsyslogd[41620]: [origin software="rsyslogd"
swVersion="8.1911.0-6.el8" x-pid="41620" x-
info="https://www.rsyslog.com"] exiting on signal 15.
... systemd[1]: Stopping System Logging Service...
... root[41628]: This is a debug-priority message
...
```

- d. Run the `journalctl -n 3` command to display the three newest log entries.

```
# journalctl -n 3
-- Logs begin at ..., end at ...
... CROND[44756]: (root) CMD (run-parts /etc/cron.hourly)
... run-parts[44759]: (/etc/cron.hourly) starting 0anacron
... run-parts[44765]: (/etc/cron.hourly) finished 0anacron
```

- Using the `-n number` option displays a specific number of the most recent log entries.

- e. Run the `journalctl -p err` command to display log entries with a priority of `err`, `crit`, `alert`, or `emerg`.

```
# journalctl -p err
-- Logs begin at ..., end at ...
kernel: integrity: Unable to open file: /etc/keys/x509_ima.de...
kernel: vbd vbd-51744: 19 xenbus_dev_probe on device/vbd/51744
```

- You can use the `-p` option to display log entries of any priority. Valid priorities are `debug`, `info`, `notice`, `warning`, `err`, `crit`, `alert`, and `emerg`.
- Without specifying a range with the `"-p"` option, log entries at the given level and those with greater significance are shown (a range has the form `FROM..TO`, for example, `crit..alert`).

- f. Run the `journalctl -u sshd` command to display log entries associated with the `crond` unit.

```
# journalctl -u sshd
-- Logs begin at ..., end at ...
... systemd[1]: Starting OpenSSH server daemon...
... sshd[753]: Server listening on 0.0.0.0 port 22.
... sshd[753]: Server listening on :: port 22.
... systemd[1]: Started OpenSSH server daemon.
```

- You can use the `-u` option to display log entries for any `systemd` unit.

2. View systemd-journald metadata.

The `systemd-journald` daemon adds structured metadata to the messages, which assists in troubleshooting.

- a. Type the `journalctl -N` command to display the metadata fields used in the current journal.

Pipe the output through the `sort` command, because the default order may seem random.

```
# journalctl -N | sort
_AUDIT_LOGINUID
_AUDIT_SESSION
AVAILABLE
AVAILABLE_PRETTY
_BOOT_ID
...
UNIT_RESULT
USER_ID
USER_INVOCATION_ID
USERSPACE_USEC
USER_UNIT
```

- b. Run the `journalctl -o verbose` command to display log entries in the verbose format.

```
# journalctl -o verbose
-- Logs begin at ..., end at ...
...
    _SOURCE_MONOTONIC_TIMESTAMP=0
    _TRANSPORT=kernel
    PRIORITY=5
    SYSLOG_FACILITY=0
    SYSLOG_IDENTIFIER=kernel
    MESSAGE=Linux version 5.4.17-2011.7.4.el8uek.x86_64 (mock...
    _BOOT_ID=6f6931d9206549f48fff2b6492f90095
...
```

- The `verbose` format shows the metadata fields and values for journal entries.
- You can use the `-o` option to display log entries in any supported format. Supported format options are `short`, `short-full`, `short-iso`, `short-iso-precise`, `short-precise`, `short-monotonic`, `short-unix`, `verbose`, `export`, `json`, `json-pretty`, `json-see`, `cat`, and `with-unit`.

- c. Run the `journalctl -F _UID` command to display unique values for the `_UID` metadata field.

Sample output is shown.

```
# journalctl -F _UID
1000
995
998
81
0
```

- d. Run the `journalctl _UID=value` command to show only log entries that match the condition.

This example uses 995 as *value*. Substitute a valid `UID` value output from the previous command.

```
# journalctl _UID=995
-- Logs begin at ..., end at ...
chronyd[710]: Frequency 0.000 +/- 1000000.000 ppm read from /...
chronyd[710]: Using right/UTC timezone to obtain leap second ...
```

3. Enable persistent `systemd-journald` storage.

- a. Use the `mount` command, pipe the output to `grep`, and search for the string “run”.

```
# mount | grep run
tmpfs on /run type tmpfs (rw,nosuid,nodev,seclabel,mode=755)
...
```

- Note that the file system type mounted on `/run` is `tmpfs`.

- b. Use the `ls -l` command to view the `/run/log/journal/` directory.

```
# ls -l /run/log/journal
drwxr-s---+ 2 root systemd-journal 60 timestamp ...
```

- By default, journal data is stored in this directory.
- Note the timestamp on this directory.

- c. Use the `mkdir -p` command to make the `/var/log/journal` directory.

```
# mkdir -p /var/log/journal
```

- d. Use the `systemctl` command to restart the `systemd-journald` service.

```
# systemctl restart systemd-journald
```

- e. Use the `ls -l` command to view the `/var/log/journal` directory.

```
# ls -l /var/log/journal
drwxr-xr-x. 2 root root 28 timestamp ...
```

- Journal data is now stored in this directory.
- Note the timestamp on this directory is more recent than the timestamp on `/run/log/journal`.

4. Log off from **host02** in preparation for the next practice.

In the terminal window, use the `logout` command to close the `ssh` connection to **host02**.

```
# logout
Connection to host02 closed.
```

You are now the `oracle` user on **gateway**.

Practices for Lesson 18: Troubleshooting

Practices for Lesson 18: Overview

Practices Overview

In these practices, you perform a variety of troubleshooting exercises and fix some common problems. Specific problems you encounter include the following:

- System utilities are not producing expected output.
- A `cron` job fails to run.
- A user cannot log in.
- File system does not mount.
- You cannot `ping` remote hosts.
- You cannot log in to remote hosts using `ssh`.
- Log file is not getting updated.

Each practice:

- Tells you what problem the activity is designed to simulate
- Tells you how to set up the problem
- Gives you some hints or things to check
- Has an associated solution that provides steps to diagnose and resolve the problem

Each solution:

- Includes only the steps to resolve the problem
- Does not include steps to set up the problem

Appendix:

- An appendix exists, which provides the source code for the executables that cause the problems that you need to troubleshoot and fix.

Practice 18-1: Transferring Utilities from gateway

Overview

In this practice, you transfer executable files hosted on **gateway** to **host03**. You are instructed to run one of these executables that causes the problem that you need to troubleshoot and fix.

Assumptions

You are logged in to **gateway** as the `oracle` user.

Tasks

1. Use the `scp` command to copy files from **gateway** to **host03**.

Provide the `root` password when prompted.

```
[oracle@gateway]$ scp /stage/ts_scripts/* root@host03:~
root@host03's password:
eight.x                100%   11KB   11.0KB/s   00:00
five.x                 100%   11KB   11.1KB/s   00:00
four.x                 100%   11KB   11.0KB/s   00:00
seven.x                100%   11KB   11.3KB/s   00:00
six.x                  100%   11KB   11.1KB/s   00:00
three.x                100%   11KB   11.0KB/s   00:00
two.x                  100%   11KB   11.1KB/s   00:00
```

2. Log in to **host03**.

Use the `ssh` command to log in to **host03** as the `root` user.

Provide the `root` password when prompted.

```
[oracle@gateway ~]$ ssh root@host03
root@host03's password:
Last login: ...
```

Practice 18-2: System Utility Commands Fail

Overview

In this practice, you note that some of the system utilities are not producing the expected output. You diagnose and fix this problem.

Assumptions

You are the `root` user on `host03`.

Tasks

1. Execute the `two.x` program from the `root` user's home directory.

```
# cd
# pwd
/root
# ./two.x
```

2. Run some tools and note the errors.

- a. Run the `df` command.

```
# df
df: cannot read table of mounted file systems: No such file or
directory
```

- b. Run the `lsmod` command.

```
# lsmod
libkmod: kmod_module_new_from_loaded: could not open
/proc/modules: No such file or directory
Error: could not get list of modules: No such file or directory
```

- c. Run the `swapon` command to display the current swap devices.

```
# swapon
```

- Note that none of these tools produce the expected output.

3. Diagnose and fix the problem with the “status” tools.

Review the lesson titled “Kernel and Device Configuration”.

- Commands that are failing are commands that provide information about the current state of the kernel.
- Which file system contains a hierarchy of special files that represent the current state of the kernel?
- How do you check if this file system is mounted?
- If this file system is not mounted, how do you mount it?

4. Rerun some of the earlier commands that failed, to ensure that the problem is fixed.
 - a. Run the `df` command.
 - b. Run the `lsmod` command.
 - c. Run the `swapon` command.

Solution 18-2: System Utility Commands Fail

Steps

1. Some of the output indicates a problem with the `proc` file system. Diagnose the problem.
 - a. Use the `ls` command to display the contents of `/proc`.

```
# ls /proc
```

- No output from this command suggests the `proc` file system is not mounted.

- b. Run the `mount` command to display mounted file systems.

```
# mount
```

```
mount: failed to read mtab: No such file or directory
```

- c. Run the `df -h` command to display mounted file systems.

```
# df -h
```

```
df: cannot read table of mounted file systems: No such file or directory
```

2. Fix the problem.

It appears that the `/proc` file system is not mounted.

Use the `mount -t proc proc /proc` command to mount the `proc` file system.

```
# mount -t proc proc /proc
```

- The `-t` option instructs the `mount` command to treat the file system type as `proc`.

3. Rerun some of the earlier “status” commands to ensure that the problem is fixed.
 - a. Run the `df` command.
 - b. Run the `lsmod` command.
 - c. Run the `swapon` command.
 - d. Run the `mount` command.

Practice 18-3: A cron Job Fails to Run

Overview

In this practice, you diagnose and fix a problem that is preventing a `cron` job from running.

Assumptions

You are the `root` user on `host03`.

Tasks

1. Create a `crontab` for the `root` user.
 - a. Use the `crontab -e` command to create a `crontab` job that runs the `vmstat` command every minute.

```
# crontab -e
* * * * * vmstat
```

- The `crontab -e` command uses the `vi` editor. Save your changes and exit `vi`.
- b. After one minute, use the `tail` command to view the last few lines in the `/var/log/cron` log file.

```
# tail /var/log/cron
...
... host03 CROND[...] (root) CMD (vmstat)
```

- Note that there is an entry in this file that states the date and time that the `cron` daemon ran the `vmstat` command.
 - The output from `cron` jobs is also in the log because **host03** does not have email enabled.
2. Execute the `three.x` program from the `root` user's home directory.

```
# cd
# pwd
/root
# ./three.x
```

3. Edit the `crontab` for the `root` user.
 - a. Use the `crontab -e` command to edit the `crontab` job and replace `vmstat` with `iostat`.

```
# crontab -e
* * * * * vmstat (old entry)
* * * * * /usr/sbin/iostat (new entry)
```

- b. After one minute, use the `tail` command to view the last few lines in the `/var/log/cron` log file.

```
# tail /var/log/cron
...
```

- Note that there is not an entry in this file stating that the `ifstat` command ran.
4. Diagnose and fix the problem of the `cron` job failing to run.

Review the lesson titled “Managing Time”.

- Does the `root` user have a `crontab` entry?
- Is the `root` user’s `crontab` entry valid?
- Are the permissions correct on the `root` user’s `crontab` file?
- Is the `cron` daemon running?

5. Verify that the `cron` job is running every minute.

Use the `tail` command to view `/var/log/cron` and ensure the `cron` job is running every minute.

6. Remove the `root` user’s `crontab` before continuing to the next practice.

```
# crontab -r
```


Solution 18-3: A cron Job Fails to Run

Steps

1. Diagnose the problem of the `cron` job failing to run.

- a. Use the `crontab -l` command to list the `crontab` entry.

```
# crontab -l
* * * * * /usr/sbin/iftstat
```

- Note that the `crontab` entry exists and the format is valid.

- b. Use the `ls -l` command to view the root user's `crontab` entry in the `/var/spool/cron` directory.

```
# ls -l /var/spool/cron
-rw----- . root root ... root
```

- The `root` file exists with the proper permissions (read-write for the owner).

- c. Use the `cat` command to view the `/var/spool/cron/root` file.

```
# cat /var/spool/cron/root
* * * * * /usr/sbin/iftstat
```

- This confirms the `crontab` entry is valid.

- d. Use the `systemctl` command to view the status of the `crond` daemon.

```
# systemctl status crond
crond.service - Command Scheduler
   Loaded: loaded (/usr/lib/systemd/system/crond.service: ...)
   ...
   Active: inactive (dead) ...
   ...
```

- Note that the `crond` service is not running. This is the cause of the problem.

2. Fix the problem of the `cron` job failing to run.

Use the `systemctl` command to start the `crond` daemon.

```
# systemctl start crond
```

3. Verify that the `cron` job is running every minute.

After one minute, use the `tail` command to view the last few lines in the `/var/log/cron` log file.

```
# tail /var/log/cron
...
date_time host03 CROND[...] (root) CMD (/usr/sbin/iftstat)
date_time host03 CROND[...] (root) CMDOUT (#kernel)
date_time host03 CROND[...] (root) CMDOUT (Interface ...
...
```

- Note that there is now an entry in this file that states the date and time that the `cron` daemon ran the `iftstat` command along with the command output.

4. Remove the `root` user's `crontab` before continuing to the next practice.

```
# crontab -r
```

Practice 18-4: A User Cannot Log In

Overview

In this practice, you diagnose and fix a problem that is preventing a user from being able to log in.

Assumptions

You are the `root` user on the **host03** VM.

Tasks

1. Add a new user.
 - a. Use the `useradd john` command to add a user `john`.

```
# useradd john
```

- b. Use the `passwd john` command to assign a password (`password`) to the user `john`. Disregard the “BAD PASSWORD” warning.

```
# passwd john
Changing password for user john.
New password: password
BAD PASSWORD: The password fails the dictionary check - it is
based on a dictionary word
Retype new password: password
passwd: all authentication tokens updated successfully.
```

2. Verify you can log in as user `john`.
 - a. Use the `exit` command to log out as user `root`.

```
# exit
logout
Connection to host03 closed.
```

- The `ssh` connection closes.

- b. From **gateway**, use the `ssh` command to log in to **host03** as user `john`. Password is `password`. Verify you can successfully log in as user `john`.

```
[oracle@gateway ~]$ ssh john@host03
john@host03's password: password
$ pwd
/home/john
$ whoami
john
```

- You can successfully log in as user `john`.

- c. Use the `exit` command to log out as user `john`.

```
$ exit
logout
Connection to host03 closed.
```

- The `ssh` connection closes.

3. From **gateway**, log in to **host03** as `root`.

- a. Use the `ssh` command to log in to **host03**.

Provide the password when prompted.

```
[oracle@gateway ~]$ ssh root@host03
root@host03's password:
```

- b. Run the `four.x` program from the `root` user's home directory.

```
# cd
# pwd
/root
# ./four.x
```

- c. Use the `exit` command to log out.

```
# exit
logout
Connection to host03 closed.
```

- The `ssh` connection closes.

4. From **gateway**, log in to **host03** as user `john`.

Use the `ssh` command to log in to **host03**.

```
[oracle@gateway ~]$ ssh john@host03
john@host03's password: password
Permission denied, please try again.
john@host03's password: CTRL-C
```

- Password is `password`.
- Note you cannot log in as user `john`.
- When prompted a second time for a password, press `Ctrl + C`.

5. Diagnose and fix the problem of user `john` not being able to log in.

Review the lesson titled “User and Group Administration”.

- Which configuration file contains usernames? Is the user `john` present in this file?
- Which configuration file contains user passwords? Is the user `john` present in this file?
- If entries are missing in these files, how do you re-create the entries?
- If passwords are corrupted or forgotten, how do you re-create passwords?

6. Verify that you can successfully log in to **host03** as user `john` using `ssh` and then log out from **host03**.

7. Log in to **host03** as `root` using `ssh` in preparation for the next practice.
8. As `root` on **host03**, delete user `john`.

```
# userdel john
```

Solution 18-4: A User Cannot Log In

Steps

1. Diagnose and fix the problem of user `john` not being able to log in.
 - a. From **gateway**, log in to **host03** as the `root` user. Provide the password when prompted.

```
[oracle@gateway ~]$ ssh root@host03
root@host03's password:
```

- b. Use the `grep` command to search for `john` in the `/etc/passwd` file.

```
# grep john /etc/passwd
john:x:1001:1001::/home/john:/bin/bash
```

- Note the `john` entry is present.

- c. Use the `grep` command to search for `john` in the `/etc/shadow` file.

```
# grep john /etc/shadow
#john:$6$...:0:99999:7:::
```

- Note the `john` entry is present but is commented out (`#` sign).

- d. Use the `vi` editor to modify `/etc/shadow`. Remove the `#` character from the beginning of the `john` line.

```
# vi /etc/shadow
...
#john:$6$...:0:99999:7:::      (old entry)
john:$6$...:0:99999:7:::      (new entry)
```

- Because `/etc/shadow` is a read-only file, use `:wq!` to write, save, and exit `vi` after removing the `#` character.

Alternatively, you could run the following command to re-create a password entry in the `/etc/shadow` file for user `john`.

```
# passwd john
Changing password for user john.
...
```

- Note that this command will not retain the password aging settings for the user.

- e. Use the `exit` command to log out as user `root`.

```
# exit
logout
Connection to host03 closed.
```

- The `ssh` connection closes.

2. Verify you can log in as user `john`.

a. From **gateway**, use the `ssh` command to log in to **host03** as user `john`.

Password is `password`. Verify you can successfully log in as user `john`.

```
[oracle@gateway ~]$ ssh john@host03
john@host03's password: password
$ pwd
/home/john
$ whoami
john
```

- Note that you can now log in as user `john`.

b. Use the `exit` command to log out as user `john`.

```
$ exit
logout
Connection to host03 closed.
```

c. Use the `ssh` command to log in to **host03** as `root` in preparation for the next practice.

Provide the password when prompted.

```
[oracle@gateway ~]$ ssh root@host03
root@host03's password:
```

d. As `root` on **host03**, delete user `john`.

```
# userdel john
```

Practice 18-5: A File System Does Not Mount

Overview

In this practice, you diagnose and fix an unmountable file system.

Assumptions

You are the `root` user on the **host03** VM.

Tasks

1. Create a partition on `/dev/xvdb`.
 - a. Use the `fdisk -l` command to display the partition table on `/dev/xvdb`.

```
# fdisk -l /dev/xvdb

Disk /dev/xvdb: 5 GiB, 5299999744 bytes, 10351562 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
...
```

- Note that there are no partitions on `/dev/xvdb`.

- b. Use the `fdisk` command to partition `/dev/xvdb`.

```
# fdisk /dev/xvdb
...
Command (m for help):
```

- c. Add a new **primary** partition and assign it the number **1**.

```
Command (m for help): n
Partition type:
   p   primary partition (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): ENTER
Using default response p
Partition number (1-4, default 1): ENTER
```

- d. Continue adding the new partition, using the entire disk as follows:

```
First sector (2048-10351561, default 2048): ENTER
Last sector, +sectors or +size{K,M,G,T,P} (2048-10351561,
default 10351561): ENTER
Created a new partition 1 of type 'Linux' and of size 5 GiB.
```


- e. If an existing file system signature is found, answer yes to remove it.

```
Partition #1 contains a ext4 signature.

Do you want to remove the signature? [Y]es/[N]o: y

The signature will be removed by a write command.
```

- f. Save the new partition table.

```
Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

2. Make a new file system on /dev/xvdb1.

Use the `mkfs` command to make an **ext4** file system on /dev/xvdb1.

```
# mkfs -t ext4 /dev/xvdb1
mke2fs 1.45.6 (20-Mar-2020)
Creating filesystem with 1293689 4k blocks and 323840 inodes
...
```

3. Mount the file system.

- a. Use the `mkdir` command to create /Test.

```
# mkdir /Test
```

- b. Use the `mount` command to mount /dev/xvdb1 on /Test.

```
# mount /dev/xvdb1 /Test
```

4. Copy files to /Test.

- a. Use the `cp` command to copy /boot/init* to /Test. Use the `ls` command to display the contents of the /Test directory.

```
# cp /boot/init* /Test
# ls /Test
initramfs-0-rescue-ac34469c2d1349df99cfed6fefe41039.img
initramfs-4.18.0-240.el8.x86_64.img
initramfs-4.18.0-240.el8.x86_64.img_SAV
initramfs-5.4.17-2011.7.4.el8uek.x86_64.img
lost+found
```

5. Unmount the file system on /dev/xvdb1.

Use the `umount` command to unmount /Test.

```
# umount /Test
```

6. Execute the `five.x` program from the `root` user's home directory.

```
# cd
# pwd
/root
# ./five.x
```

7. Attempt to mount the file system on `/dev/xvdb1` on `/Test`.

Use the same `mount` command used previously.

```
# mount /dev/xvdb1 /Test
mount: wrong fs type, bad option, bad superblock on /dev/xvdb1,
missing codepage or helper program, or other error
```

- Note that the `mount` command fails.

8. Diagnose and fix the problem with the file system.

Review the lesson titled *Partitions*, “File Systems, and Swap”.

- How do you specify the file system type with the `mount` command?
- What commands are available to check and repair file systems?

9. Ensure you can mount the file system on `/Test`.

Ensure the files (`initramfs*`) exist on `/Test`.

Note: The following is a clean-up task. Perform this task only after the File System problem is fixed.

10. After ensuring the problem is fixed, restore **host03** to its beginning state.

- a. Use the `umount` command to unmount `/Test`.

This is necessary before the associated partition can be deleted. Ensure you are in the `root` user's home directory first.

```
# cd
# umount /Test
```

- b. Use the `fdisk` command to delete the `/dev/xvdb1` partition.
- Use the `d` command to delete partition 1.
- Use the `p` command to print the partition table and confirm there are no partitions.
- Use the `w` command to save the partition table and exit the `fdisk` utility.

```
# fdisk /dev/xvdb
...
Command (m for help): d
Selected partition 1
Partition 1 is deleted

Command (m for help): p

Disk /dev/xvdb: 5 GiB, 5299999744 bytes, 10351562 sectors
...

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

- c. View the devices in the `/proc/partitions` file to ensure that the `xvdb1` entry no longer exists.

```
# cat /proc/partitions
major minor #blocks name

 11         0    9019392 sr0
202         0   32000000 xvda
202         1    1048576 xvda1
202         2   25165824 xvda2
202         3    2097152 xvda3
202         4         1 xvda4
202         5    3145728 xvda5
202        16    5175781 xvdb
202        48    5175781 xvdd
```

- d. Use the `rmdir` command to remove the `/Test` mount point.

```
# rmdir /Test
```

Solution 18-5: A File System Does Not Mount

Steps

1. Diagnose the problem.

- a. Run the `mount` command but include the `-t ext4` option to specify the file system type.

```
# mount -t ext4 /dev/xvdb1 /Test
mount: wrong fs type, bad option, bad superblock on /dev/xvdb1,
missing codepage or helper program, or other error.
```

- b. Run the `dmesg | tail` command.

```
# dmesg | tail
...
[ ...] EXT4-fs (xvdb1): VFS: Can't find ext4 filesystem
```

- Note the VFS error: “Can’t find ext4 filesystem” on `xvdb1`.

- c. Use the `tail` command to view the last lines in the `/var/log/messages` file.

```
# tail /var/log/messages
...
... kernel: EXT4-fs (xvdb1): VFS: Can't find ext4 filesystem
...
```

- Note the error message written to the `/var/log/messages` file is similar to the error message given by the `dmesg` command.

- d. Use the `tune2fs -l /dev/xvdb1` command to list the contents of the file system superblock.

```
# tune2fs -l /dev/xvdb1
tune2fs 1.45.6 (20-Mar-2020)
tune2fs: Bad magic number in super-block while trying to open
/dev/xvdb1
```

- Note the “superblock” error.

- e. Use the `dumpe2fs /dev/xvdb1` command to dump file system information.

```
# dumpe2fs /dev/xvdb1
dumpe2fs 1.45.6 (20-Mar-2020)
dumpe2fs: Bad magic number in super-block while trying to open
/dev/xvdb1
Couldn't find valid filesystem superblock
```

- Note this command reports the same “superblock” error on `/dev/xvdb1`.

2. Fix the file system.

a. Use the `fsck /dev/xvdb1` command to check (and repair) the file system.

- Press **Enter** to accept the default **y** (yes) answer.

```
# fsck /dev/xvdb1
fsck from util-linux 2.32.1
e2fsck 1.45.6 (20-Mar-2020)
ext2fs_open2: Bad magic number in super-block
fsck.ext2: Superblock invalid, trying backup blocks...
/dev/xvdb1 was not cleanly unmounted, check forced.
Resize inode not valid. Recreate<y>? ENTER
Pass 1: Checking inodes, blocks, and sizes
Pass 2: Checking directory structure
Pass 3: Checking directory connectivity
Pass 4: Checking reference counts
Pass 5: Checking group summary information
Block bitmap differences: +(98304--98936) +(163840--164472)...
Fix<y>? ENTER
Free blocks count wrong for group #0 (24000, counted=24001).
Fix<y>? ENTER
Free blocks count wrong for group #1 (32135, counted=1137).
Fix<y>? ENTER
Free blocks count wrong for group #2 (32768, counted=23667).
Fix<y>? ENTER
Free blocks count wrong (1251914, counted=1211816).
Fix<y>? ENTER
Free inodes count wrong for group #0 (8085, counted=8081).
Fix<y>? ENTER
Free inodes count wrong (323829, counted=323825).
Fix<y>? ENTER
Padding at end of inode bitmap is not set. Fix<y>? ENTER

/dev/xvdb1: ***** FILE SYSTEM WAS MODIFIED *****
/dev/xvdb1: 15/323840 files (13.3% non-contiguous), 81873/129...
```

b. Run the `fsck /dev/xvdb1` command a second time to check the file system.

```
# fsck /dev/xvdb1
fsck from util-linux 2.32.1
e2fsck 1.45.6 (20-Mar-2020)
/dev/xvdb1: clean, 15/323840 files, 81873/1293689 blocks
```

- Note that the file system is fixed (clean).

3. Mount the file system.

- a. Use the `mount` command to mount `/dev/xvdb1` on `/Test`.

```
# mount /dev/xvdb1 /Test
```

- The `mount` command is successful. The `fsck` command fixed the corrupt superblock.

- b. Use the `ls` command to list the contents of the `/Test` directory.

```
# ls /Test
initramfs-0-rescue-ac34469c2d1349df99cfed6fefe41039.img
initramfs-4.18.0-240.el8.x86_64.img
initramfs-4.18.0-240.el8.x86_64.img_SAV
initramfs-5.4.17-2011.7.4.el8uek.x86_64.img
lost+found
```

- Note that all the original files are present in the directory.

4. After ensuring the problem is fixed, restore **host03** to its beginning state.

- a. Use the `umount` command to unmount `/Test`.

This is necessary before the associated partition can be deleted. Ensure you are in the `root` user's home directory first.

```
# cd
# umount /Test
```

- b. Use the `fdisk` command to delete the `/dev/xvdb1` partition.

Use the `d` command to delete partition 1.

Use the `p` command to print the partition table and confirm there are no partitions.

Use the `w` command to save the partition table and exit the `fdisk` utility.

```
# fdisk /dev/xvdb
...
Command (m for help): d
Selected partition 1
Partition 1 is deleted

Command (m for help): p

Disk /dev/xvdb: 5 GiB, 5299999744 bytes, 10351562 sectors
...

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

- c. View the devices in the `/proc/partitions` file to ensure that the `xvdb1` entry no longer exists.

```
# cat /proc/partitions
major minor  #blocks  name

   11         0    9019392  sr0
  202         0   32000000  xvda
  202         1    1048576  xvda1
  202         2   25165824  xvda2
  202         3    2097152  xvda3
  202         4         1    xvda4
  202         5    3145728  xvda5
  202        16    5175781  xvdb
  202        48    5175781  xvdd
```

- d. Use the `rmdir` command to remove the `/Test` mount point.

```
# rmdir /Test
```

Practice 18-6: Network Connectivity Problem

Overview

In this practice, you diagnose and fix a network connectivity problem.

Assumptions

You are the `root` user on **host03**.

Tasks

1. Verify the network interface is configured properly.

Use the `ping` command to contact **gateway** and all addresses on **host03**. Press `Ctrl + C` to abort the commands after connectivity is confirmed.

```
# ping gateway
64 bytes from example.com (192.0.2.1): icmp_seq=1 ttl=64 ...
64 bytes from example.com (192.0.2.1): icmp_seq=1 ttl=64 ...
64 bytes from example.com (192.0.2.1): icmp_seq=1 ttl=64 ...
CTRL-C
# ping host03
64 bytes from host03.example.com (192.0.2.103): icmp_seq=1 ...
64 bytes from host03.example.com (192.0.2.103): icmp_seq=1 ...
64 bytes from host03.example.com (192.0.2.103): icmp_seq=1 ...
CTRL-C
# ping 192.168.1.103
64 bytes from 192.168.1.103: icmp_seq=1 ttl=64 time=0.046 ms
64 bytes from 192.168.1.103: icmp_seq=2 ttl=64 time=0.046 ms
64 bytes from 192.168.1.103: icmp_seq=3 ttl=64 time=0.049 ms
CTRL-C
```

2. **Important!** If the ping of address 192.168.1.103 does not succeed, do not continue with this practice. This address should have been configured as part of an earlier practice.
3. Execute the `six.x` program from the `root` user's home directory.

```
# cd
# pwd
/root
# ./six.x
```

4. The `six.x` program may appear not to finish. Your `ssh` session is not responding. Force a disconnect of the `ssh` session using the key sequence tilde period (`~.`).

```
# ./six.x
~.
[oracle@gateway ~]$
```


5. Attempt to contact host03.

Use the `ping` command to contact **host03**.

```
# ping host03
PING host03.example.com (192.0.2.103) 56(84) bytes of data.
From gateway... (192.0.2.1) ... Destination Host Unreachable
CTRL-C
```

- Press Ctrl + C to abort the command.
 - The remote system appears no longer reachable.
6. Diagnose and fix the network connectivity problem.
Review the lesson titled “Network Configuration”.
 - What are the IP addresses of **host03**?
 - Do you have another route to **host03**?
 - How do you display the configuration of your network?
 - Are the network interfaces up?
 - How do you view the route table?
 - In which files do you configure network interfaces?
 - Are these network interface configuration files configured properly?
 - Is the NetworkManager service running?
 7. Verify network connectivity to the remote hosts is working.
 - a. Run the `ping gateway` command from **host03**.
 - b. Run the `ping host03` command from **gateway**.

Solution 18-6: Network Connectivity Problem

Steps

1. Find another way to connect to **host03**.

Use the `ssh` command to log in to **host03** as the `root` user, using the IP address `192.168.1.103`. Accept the host and enter the password when prompted.

```
[oracle@gateway ~]$ ssh root@192.168.1.103
The authenticity of host '192.168.1.103 (192.168.1.103)' can't
be established.
ECDSA key fingerprint is SHA256:...
Are you sure you want to continue connecting
(yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.103' (ECDSA) to the list
of known hosts.
root@192.168.1.103's password:
[root@host03 ~]#
```

2. Diagnose the network connectivity problem.
 - a. Use the `ip addr` command to display the configuration of your network interfaces.

```
# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state ...
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    ...
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:01:01 brd ff:ff:ff:ff:ff:ff
    inet 192.0.3.103/24 brd 192.0.3.255 scope global ...
    ...
    inet6 ...
    ...
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc ...
    link/ether 00:16:3e:02:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.103/24 brd 192.168.1.255 scope global ...
    ...
    inet6 ...
    ...
```

- The output shows that the `eth0` interface is `UP` and has an IP address of `192.0.3.103`.

- b. Use the `ip route` command to display the route table.

```
# ip route
default via 192.0.2.1 dev enp1s0 proto static metric 102
192.0.2.1 dev eth0 proto static scope link metric 102
192.0.3.0/24 dev eth0 proto kernel scope link src 192.0.3.103
metric 102
192.168.1.0/24 dev eth1 proto kernel scope link src
192.168.1.103 metric 101
```

- The route table indicates the route to the 192.0.3.0 network is through `eth0`.

- c. Use the `cat` command to view the contents of the `/etc/hosts` file.

```
# cat /etc/hosts
...
192.0.2.1      example.com          gateway
192.0.2.102   host02.example.com        host02
192.0.2.103   host03.example.com        host03
```

- The contents of the `/etc/hosts` file indicate the IP address should be configured with 192.0.2 network addresses and not 192.0.3.

3. Fix the network connectivity problem.

- a. Use the `vi` editor to modify `/etc/sysconfig/network-scripts/ifcfg-eth0` and change the `IPADDR` directive as shown.

```
# vi /etc/sysconfig/network-scripts/ifcfg-eth0
...
IPADDR=192.0.3.103          (old value)
IPADDR=192.0.2.103          (new value)
...
```

- b. Use the `nmcli` command to reload the configuration and restart the `eth0` interface.

```
# nmcli c reload
# nmcli c up eth0
Connection successfully activated (D-Bus active path: /org/fr...
```

4. Verify network connectivity to **gateway** is working.

Use the `ping` command to contact **gateway**. Use `Ctrl + C` to abort the command after connectivity is confirmed.

```
# ping gateway
64 bytes from example.com (192.0.2.1): icmp_seq=1 ttl=64 ...
64 bytes from example.com (192.0.2.1): icmp_seq=1 ttl=64 ...
64 bytes from example.com (192.0.2.1): icmp_seq=1 ttl=64 ...
CTRL-C
```

- Network connectivity to **gateway** works.

5. Verify network connectivity to **host03** from **gateway** is working through the default network.
 - a. Log out of **host03** using the `exit` command to return to the `oracle` user on **gateway**.

```
# exit
logout
Connection to 192.168.1.103 closed.
[oracle@gateway ~]$
```

- b. Use the `ping` command to contact **host03**. Use Ctrl + C to abort the command after connectivity is confirmed.

```
# ping host03
PING host03.example.com (192.0.2.103) 56(84) bytes of data.
64 bytes from host03.example.com (192.0.2.103): icmp_seq=1 ...
64 bytes from host03.example.com (192.0.2.103): icmp_seq=2 ...
CTRL-C
```

- Network connectivity to **host03** works.

Practice 18-7: Remote Access Problem

Overview

In this practice, you diagnose and fix a remote access problem.

Assumptions

- You are the `oracle` user on **gateway**.
- This practice is performed on both **host02** and **host03**.
- The prompts include either **host02** or **host03** to indicate which system to enter the command from.

Tasks

1. Log in to **host03**.

Use the `ssh` command to log in to **host03** as the `root` user.

Provide the `root` password when prompted.

```
[oracle@gateway ~]$ ssh root@host03
root@host03's password:
Last login: ...
```

2. Log in to **host02**.

- a. Open a second terminal window on **gateway**.

Stay connected to **host03** from another terminal window.

- b. In the terminal window, use the `ssh` command to log in to **host02** as the `root` user.

Provide the `root` password when prompted.

```
[oracle@gateway ~]$ ssh root@host02
root@host02's password:
Last login: ...
```

3. Ensure remote connectivity is working from **host03** to **host02**.
 - a. As the `root` user on **host03**, confirm that you can use the `ssh` command to connect to **host02**.

Answer **yes** to continue connecting, when prompted.

Provide the `root` password when prompted.

Use the `hostname` command to confirm that you did connect.

```
[host03]# ssh host02
The authenticity of host 'host02 (192.0.2.102)' can't be
established.
ECDSA key fingerprint is ...
Are you sure you want to continue connecting
(yes/no/[fingerprint])? yes
Warning: Permanently added 'host02,192.0.2.102' (ECDSA) to the
list of known hosts.
root@host02's password:
Last login: ...
[host02]# hostname
host02.example.com
```

- b. Use the `logout` command to log off **host02**.

```
[host02]# logout
Connection to host02 closed.
```

4. Ensure remote connectivity is working from **host02** to **host03**.
 - a. As the `root` user on **host02**, confirm that you can use the `ssh` command to connect to **host03**.

Provide the `root` password when prompted.

Use the `hostname` command to confirm that you did connect.

```
[host02]# ssh host03
The authenticity of host 'host03 (192.0.2.103)' can't be
established.
ECDSA key fingerprint is ...
Are you sure you want to continue connecting
(yes/no/[fingerprint])? yes
Warning: Permanently added 'host03,192.0.2.103' (ECDSA) to the
list of known hosts.
root@host03's password:
Last login: ...
[host03]# hostname
host03.example.com
```

- b. Use the `logout` command to log off **host03**.

```
[host03]# logout
Connection to host03 closed.
```

5. From **host03**, execute the `seven.x` program from the `root` user's home directory.

```
[host03]# cd
[host03]# pwd
/root
[host03]# ./seven.x
```

6. Ensure remote connectivity is working from **host03** to **host02**.

- a. As the `root` user on **host03**, confirm that you can use the `ssh` command to connect to **host02**.

```
[host03]# ssh host02
root@host02's password:
Last login: ...
[host02]# hostname
host02.example.com
```

- b. Use the `logout` command to log off **host02**.

```
[host02]# logout
Connection to host02 closed.
```

7. Ensure remote connectivity is working from **host02** to **host03**.

As the `root` user on **host02**, confirm that you can use the `ssh` command to connect to **host03**.

```
[host02]# ssh host03
ssh: connect to host host03 port 22: No route to host
```

- Note the `ssh` command fails.

8. Diagnose and fix the remote connectivity problem.

- a. Think about what you can and what you cannot do:
- You can `ssh` from **host03** to **host02**.
 - You cannot `ssh` from **host02** to **host03**.
 - Can you `ping` **host03** from **host02**? If so, then there is nothing wrong with the network interface configuration.
- b. Review the lesson titled “OpenSSH”.
- Ensure the `sshd` service is configured properly and that the service is running.
 - Note the error message, “ssh: connect to host host03 port 22: No route to host.”
 - View the `/var/log/messages` log file for entries related to `ssh` or port 22.
 - View the `/var/log/secure` log file for entries related to `ssh` or port 22.
- c. Review the lesson titled “Network Security”.
- Is the firewall prohibiting a connection on port 22?

9. After fixing the problem, ensure you can `ssh` from **host02** to **host03**.

Solution 18-7: Remote Access Problem

Steps

1. Attempt to diagnose the problem by testing network connectivity.

From **host02**, use the `ping host03` command to test network connectivity to **host03**.

```
[host03]# ping host03
64 bytes from host03.example.com (192.0.2.103): icmp_seq=1...
64 bytes from host03.example.com (192.0.2.103): icmp_seq=2...
64 bytes from host03.example.com (192.0.2.103): icmp_seq=3...
CTRL-C
```

- Note the `ping` command verifies that network connectivity to **host03** exists.

2. Attempt to diagnose the problem by checking the `sshd` configuration.

- a. From **host03**, use the `systemctl` command to view the status of the `sshd` service.

```
[host03]# systemctl status sshd
sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; vendor preset: enabled)
   Active: active (running) since ...
   ...
```

- The `sshd` service is running.

- b. From **host03**, use the `grep` command to search for "ssh" in the `/var/log/messages` file.

Repeat the command but search for `port 22`.

```
[host03]# grep ssh /var/log/messages
...
[host03]# grep "port 22" /var/log/messages
```

- Note that neither command returned output that indicates the cause of the problem.

- c. From **host03**, use the `grep` command to search for “ssh” in the `/var/log/secure` file.

Repeat the command but search for “port 22”.

Enter the `date` command and note the date and time.

```
[host03]# grep ssh /var/log/secure
...
[host03]# grep "port 22" /var/log/secure
...
[host03]# date
date time
```

- Note that the timestamps on the log file entries do not correspond to the time of the problem.
 - Note that there are several “pam” messages related to the `sshd` service.
 - Pluggable Authentication Modules (PAM) is covered in another course.
 - PAM configuration could cause this problem in the real world, but because PAM has not been covered in this course, you can conclude PAM is not the cause of the problem.
3. Attempt to diagnose the problem by checking the firewall (`firewalld` and `nftables`) configuration.

- a. From **host03**, use the `systemctl` command to view the status of the `nftables` service.

```
[host03]# systemctl status nftables
• nftables.service - Netfilter Tables
  Loaded: loaded (/usr/lib/systemd/system/nftables.service;
disabled; ...
  Active: inactive (dead)
  Docs: man:nft(8)
```

- Note that the `nftables` service is not enabled on **host03**.
- b. From **host03**, use the `systemctl` command to view the status of the `firewalld` service.

```
[host03]# systemctl status firewalld
firewalld.service - firewalld - dynamic firewall daemon
  Loaded: loaded (/usr/lib/systemd/system/firewalld.service...
  Active: active (running) since ...
  ...
```

- Note that the `firewalld` service is running.

- c. From **host03**, use the `firewall-cmd` command to list the trusted services.

```
[host03]# firewall-cmd --list-services
cockpit dhcpv6-client
```

- Note that only the `cockpit` and `dhcpv6-client` services are trusted.
- Note that the `ssh` service is not trusted.
 - This is the cause of the problem.
- To fix the problem, either stop the `firewalld` service or trust the `ssh` service.

4. Fix the remote connectivity problem by trusting the `ssh` service when `firewalld` is running.

- a. From **host03**, use the `firewall-cmd` command to trust the `ssh` service.

```
[host03]# firewall-cmd --add-service=ssh
success
```

- If the service had been removed from the saved configuration, the command should be repeated with the `--permanent` option.

- b. From **host03**, use the `firewall-cmd` command to list the trusted services.

```
[host03]# firewall-cmd --list-services
cockpit dhcpv6-client ssh
```

- Note that the `ssh` service is now trusted.

5. Ensure remote connectivity is working.

- a. From **host02**, confirm that you can use the `ssh` command to connect to **host03**.

Provide the `root` password when prompted.

Use the `hostname` command to confirm that you did connect.

```
[host02]# ssh host03
root@host03's password:
Last login: ...
[host03]# hostname
host03.example.com
```

- b. Use the `logout` command to log off **host03**.

```
[host03]# logout
Connection to host03 closed.
```

- You are now the `root` user on **host02**.

Practice 18-8: Log File Is Not Getting Updated

Overview

In this practice, you diagnose and fix a problem with a log file not getting updated.

Assumptions

- You are the `root` user on both **host02** and **host03**.
- This practice is performed on both **host02** and **host03**.
- The prompts include either **host02** or **host03** to indicate which system to enter the command from.

Tasks

1. From **host03**, set the `oracle` user's password to `oracle`.

Use the `passwd oracle` command to set the password.

Ignore the "BAD PASSWORD" message.

```
[host03]# passwd oracle
Changing password for user oracle.
New password: oracle
BAD PASSWORD: The password is shorter than 8 characters
Retype new password: oracle
passwd: all authentication tokens updated successfully.
```

2. From **host02**, confirm you can use the `ssh` command to log in to **host03** as the `oracle` user.

- a. Use the `ssh` command to log in to **host03** as the `oracle` user.

Use the `hostname` command to verify you were able to log in. Provide the password when prompted.

```
[host02]# ssh oracle@host03
oracle@host03's password:
Last login: ...
[host03]$ hostname
host03.example.com
```

- b. Use the `exit` command to log off **host03**.

```
[host03]$ exit
logout
Connection to host03 closed.
```

3. From **host03**, save a copy of the `/var/log/secure` file.

- a. Use the `cp` command to copy `/var/log/secure` to `~/secure_before`.

```
[host03]# cp /var/log/secure ~/secure_before
```

- You review the differences in these files later in this practice.

4. From **host03**, configure the `pam_nologin` authentication module to prevent non-root login.

PAM is covered in another course.

- For the purposes of this practice, PAM checks for the existence of the `/etc/nologin` file, and if the file exists, remote logins by non-root users are denied and the content of this file is displayed as an error message.

Use the `vi` editor and create the `/etc/nologin` file with the following contents:

```
[host03]# vi /etc/nologin
No logins allowed at this time.
```

5. From **host02**, attempt to log in to **host03** as the `oracle` user.

Use the `ssh` command to connect to **host03**. Provide the password when prompted.

```
[host02]# ssh oracle@host03
oracle@host03's password:
No logins allowed at this time.

Connection closed by 192.0.2.103 port 22
```

- Note that the connection is denied by the PAM authentication module.

6. From **host03**, view the new entries in the `/var/log/secure` log file.

Use the `diff` command to view the differences in the `/var/log/secure` log file and the copy of the log file you made previously.

```
[host03]# diff /var/log/secure ~/secure_before
< ... [sshd[...]: Failed password for oracle from 192.0.2.102
port ... ssh2
< ... [sshd[...]: fatal: Access denied for user oracle by PAM
account configuration [preauth]
```

- Note that two entries were written to this file when the `oracle` user attempted to `ssh` from **host02** to **host03**.
- Also note that one entry specifically references PAM account configuration.

7. From **host03**, execute the `eight.x` program from the `root` user's home directory.

```
[host03]# cd
[host03]# pwd
/root
[host03]# ./eight.x
```

8. From **host03**, save a copy of the `/var/log/secure` file again.

Use the `cp` command to copy `/var/log/secure` to `~/secure_before`.

Answer `y` to overwrite the file.

```
[host03]# cp /var/log/secure ~/secure_before
cp: overwrite '/root/secure_before'? y
```

- You review the differences in these files later in this practice.

9. From **host02**, attempt to log in to **host03** as the `oracle` user.

Use the `ssh` command to connect to **host03**. Provide the password when prompted.

```
[host02]# ssh oracle@host03
oracle@host03's password:
No logins allowed at this time.

Connection closed by 192.0.2.103 port 22
```

- Note that the connection is denied by the PAM authentication module.

10. From **host03**, view the new entries in the `/var/log/secure` log file.

Use the `diff` command to view the differences in the `/var/log/secure` log file and the copy of the log file you made previously.

```
[host03]# diff /var/log/secure ~/secure_before
```

- Note that there are no differences in the files.
- The `/var/log/secure` log file is not getting updated as expected.

11. Diagnose and fix the problem of the log file not getting updated.

Review the lesson titled “System Logging”.

- Is the logging daemon running?
- Is logging configured for the `/var/log/secure` log file?

12. After fixing the problem, ensure the `/var/log/secure` log file is getting updated when you attempt to `ssh` from **host02** to **host03** as the `oracle` user.

13. Return **host03** to its original state.

- a. Use the `rm` command to remove the `/etc/nologin` file.

```
# rm /etc/nologin
rm: remove regular file '/etc/nologin'? y
```

- b. Use the `rm` command to remove the `~/secure_before` file.

```
# rm ~/secure_before
rm: remove regular file '/root/secure_before'? y
```

Solution 18-8: Log File Is Not Getting Updated

Steps

1. Diagnose the cause of the logging problem.
 - a. As the **root** user on **host03**, view the status of the **rsyslog** service.

```
[host03]# systemctl status rsyslog
rsyslog.service - System Logging Service
   Loaded: loaded (/usr/lib/systemd/system/rsyslog.service...)
   Active: active (running) since ...
...
```

- The **rsyslogd** service is running.

- b. Use the **grep** command to search for **/var/log/secure** in the **rsyslog** configuration file.

```
[host03]# grep /var/log/secure /etc/rsyslog.conf
#authpriv.*                                /var/log/secure
```

- Note the configuration file contains an entry for **/var/log/secure**; however, the entry is commented out (preceded by a **#** sign).

2. Fix the problem.

- a. Use the **vi** editor to remove the **#** from the beginning of the line containing **/var/log/secure**.

```
[host03]# vi /etc/rsyslog.conf
...
#authpriv.*                /var/log/secure            (old entry)
authpriv.*                  /var/log/secure            (new entry)
...
```

- b. Use the **systemctl** command to restart the **rsyslog** service.

```
[host03]# systemctl restart rsyslog
```

3. Verify the **/var/log/secure** log file is getting updated.

- a. From **host03**, use the **cp** command to copy **/var/log/secure** to **~/secure_before**.

Answer **y** to overwrite the file.

```
[host03]# cp /var/log/secure ~/secure_before
cp: overwrite '/root/secure_before'? y
```

- b. From **host02**, use the `ssh` command to connect to **host03** as the `oracle` user. Provide the password when prompted.

```
[host02]$ ssh oracle@host03
oracle@host03's password:
No logins allowed at this time.

Connection closed by 192.0.2.103 port 22
```

- Note that the connection is denied by the PAM authentication module.

- c. From **host03**, use the `diff` command to view the differences in the `/var/log/secure` log file and the copy, `~/secure_before`.

```
[host03]# diff /var/log/secure ~/secure_before
< ... [sshd[...]: Failed password for oracle from 192.0.2.102
port ... ssh2
< ... [sshd[...]: fatal: Access denied for user oracle by PAM
account configuration [preauth]
```

- Note that entries are now being written to the `/var/log/secure` file as expected.

4. Return **host03** to its original state.

- a. Use the `rm` command to remove the `/etc/nologin` file.

```
# rm /etc/nologin
rm: remove regular file '/etc/nologin'? y
```

- b. Use the `rm` command to remove the `~/secure_before` file.

```
# rm ~/secure_before
rm: remove regular file '/root/secure_before'? y
```

Appendix A: Source Code for Problem-Causing Executables

Overview

In this appendix, the source code for the executables that cause the problems is given.

Practice 19-2: System Utility Commands Fail

The “two.x” script runs the following `umount` command to unmount the `proc` file system:

```
#!/bin/bash
umount -l /proc
```

Practice 19-3: cron Job Fails to Run

The “three.x” script stops the `crond` service by running the following command:

```
#!/bin/bash
systemctl stop crond > /dev/null
```

Practice 19-4: User Cannot Log In

The “four.x” script inserts the “#” sign at the beginning of the `john` line in the `/etc/shadow` file after backing up the original `/etc/shadow` file:

```
#!/bin/bash
cp /etc/shadow /etc/shadow.save
sed -i -e '/john/s/^/#/' /etc/shadow
```

Practice 19-5: File System Troubleshooting

The “five.x” script runs the following `dd` command to corrupt the file system superblock.

```
#!/bin/bash
dd if=/dev/zero of=/dev/xvdb1 bs=1024 skip=1000 count=300 2> /dev/null
```

Practice 19-6: Network Connectivity Problem

The “six.x” script changes the IP address of `eth0` on **host03** from `192.0.2.103` to `192.0.3.103` by running the following commands after backing up the original `ifcfg-eth0` file:

```
#!/bin/bash
cd /etc/sysconfig/network-scripts
/bin/cp ifcfg-eth0 ~/ifcfg-eth0.save
sed -e 's/2.103/3.103/' ifcfg-eth0 > ifcfg-eth0.new
/bin/mv ifcfg-eth0.new ifcfg-eth0
cd
nmcli c reload > /dev/null 2>&1
nmcli c up eth0 > /dev/null 2>&1
```


Practice 19-7: Remote Access Problem

The “seven.x” script removes the `firewalld` rule on **host03**, which trusts the `sshd` service, by running the following command:

```
#!/bin/bash
firewall-cmd --remove-service=ssh > /dev/null
```

Practice 19-8: Log File Is Not Getting Updated

The “eight.x” script inserts the “#” sign at the beginning of the `secure` line in the `/etc/rsyslog.conf` file and restarts the `rsyslog` service by running the following commands after backing up the original `rsyslog.conf` file:

```
#!/bin/bash
cp /etc/rsyslog.conf /etc/rsyslog.conf.save
sed -i -e '/secure/s/^/#/' /etc/rsyslog.conf
systemctl restart rsyslog > /dev/null
```

